Causal Inference

A Pokemon Approach

Kanto Region Edition

From Undergraduate Foundations to PhD-Level Frontiers

A comprehensive textbook teaching causal inference through the adventure of a Pokemon trainer's journey across the Kanto region.

Eight Chapters. Eight Badges. Eight Methods.

Pallet Town · Pewter City · Cerulean City · Vermilion City
Celadon City · Fuchsia & Cinnabar · Saffron City · Indigo Plateau

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Causal Inference: A Pokemon Approach

Chapter 1: Pallet Town — What Is Causal Inference?

Professor Oak

The Kanto region — your journey begins in Pallet Town.

You wake up in a small house at the edge of a quiet town. Sunlight filters through the curtains. Downstairs, your mother is watching the morning news — something about a new discovery on Route 1. Today is the day you've been waiting for: Professor Oak has invited you to his laboratory to receive your first Pokemon. As you step outside, the cool Pallet Town air carries the sound of an argument drifting from the road ahead.

Two trainers stand nose to nose near the town gate. One insists that Charmander is the key to becoming Champion; the other swears that Squirtle guarantees more Gym Badges. A small crowd has gathered, each person armed with anecdotes and statistics pulled from the latest issue of Pokemon Trainer Monthly.

You make your way to the lab, where Professor Oak is calibrating a Pokedex. His grandson Blue is already there, arms crossed, leaning against the wall. "Professor," Blue says, "the data is clear. Trainers who pick Squirtle average 6.8 Gym Badges. Charmander trainers average only 5.9. Squirtle is obviously superior."

Oak sets down the Pokedex. He looks at you, then back at Blue. "Is it, though?" He pauses. "If I gave this young trainer Charmander instead of Squirtle, would their journey turn out differently? And how could we ever know?"

Blue rolls his eyes. Oak smiles. "That," the Professor says, "is the question that will occupy us for the rest of this book."

1.1 Why Causation Matters More Than Correlation

The Starter Debate

#001 Bulbasaur

#004 Charmander

#007 Squirtle

The three Kanto starters — the choice at the heart of this chapter's causal question.

Every year, a new cohort of trainers walks into Professor Oak's lab and faces the same choice: Bulbasaur, Charmander, or Squirtle. And every year, the same debate rages in the streets of Pallet Town. One camp cites league records showing that Water-type starters are associated with higher badge counts. Another camp points to Championship rosters dominated by Fire-type starters. Both sides wave data. Neither side agrees.

What is going on? The trainers are confusing two fundamentally different ideas: **correlation** and **causation**. This confusion is not unique to Pallet Town. It is, arguably, the most pervasive error in all of human reasoning — and the reason an entire field of study, causal inference, exists.

Correlation: What We See

Let us start with what correlation actually means. Informally, two variables are correlated when knowing the value of one tells you something about the value of the other. If trainers who pick Squirtle tend to earn more badges, then starter choice and badge count are correlated.

> ***Professor Oak Explains: Correlation***
>
> *For two random variables X and Y, the Pearson correlation coefficient is defined as:*
>
> $$\rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}$$
>
> *where $\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$, and σ_X, σ_Y are the standard deviations of X and Y, respectively. A correlation of $\rho = 1$ indicates perfect positive linear association; $\rho = -1$ indicates perfect negative linear association; $\rho = 0$ indicates no linear association.*

Correlation is a property of **observed data**. It tells you what patterns exist in the world as you find it. It does not, by itself, tell you what would happen if you intervened — if you *changed* something.

Consider the following data, collected from the Kanto Trainer Registry:

Starter Pokemon	Mean Badges Earned	Mean Battle Win Rate	n
Bulbasaur	5.4	0.52	312
Charmander	5.9	0.56	287
Squirtle	6.8	0.61	301

Blue looks at this table and declares Squirtle the winner. But should he?

Causation: What Would Happen If

Causation is a claim about what would happen under an **intervention**. To say "Choosing Squirtle *causes* you to earn more badges" is to claim that if we could take a trainer and *force* them to pick Squirtle rather than Charmander — holding everything else constant — that trainer would earn more badges.

This is a much stronger claim than correlation. Correlation says: "Trainers who pick Squirtle happen to earn more badges." Causation says: "If *you* picked Squirtle, *you* would earn more badges." The first is a statement about patterns in data. The second is a statement about a mechanism in the world.

Spurious Associations and Confounders

Why might correlation fail to capture causation? The answer, in a word, is **confounders** — variables that influence both the treatment (starter choice) and the outcome (badge count), creating a spurious association.

Suppose that trainers from Cerulean City — a wealthy, water-themed city — disproportionately choose Squirtle. Cerulean trainers also tend to come from families with more resources: better training equipment, access to expensive held items like Leftovers and Choice Bands, and private tutoring from retired Gym Leaders. These advantages help them earn more badges regardless of which starter they choose.

In this scenario, **wealth** (or equivalently, **hometown advantages**) is a confounder. It affects both starter choice (Cerulean kids pick Water types) and badge count (Cerulean kids have resources that help them win). The observed correlation between Squirtle and badges is partly — perhaps entirely — driven by this confounder.

> ***Blue's Mistake***
>
> *Blue points to the league data and says, "Squirtle trainers earn 0.9 more badges than Charmander trainers on average. Case closed." But Blue is ignoring a critical possibility: wealthy trainers from Cerulean City disproportionately pick Water types and buy better items, hire better tutors, and enter more competitions. The association between Squirtle and badges may have nothing to do with Squirtle itself. Blue has mistaken correlation for causation — a mistake so common it has its own Latin phrase: cum hoc ergo propter hoc ("with this, therefore because of this").*

This is not a contrived example. In observational studies across medicine, economics, and the social sciences, confounders routinely create misleading associations. The history of science is littered with confident causal claims that turned out to be confounded: hormone replacement therapy and heart disease, class size and student achievement, hospital quality and mortality rates.

Pearl's Ladder of Causation

Judea Pearl, a computer scientist and philosopher who has done as much as anyone to formalize causal reasoning, proposed a useful hierarchy for thinking about different types of questions. He calls it the **Ladder of Causation** (Pearl, 2009), and it has three rungs.

Rung 1: Association (Seeing). Questions at this level ask about patterns in observed data. They are answered by conditional probabilities of the form $P(Y \mid X)$.

Pokemon example: "What is the average badge count among trainers who chose Squirtle?" This question can be answered by looking at the Kanto Trainer Registry. No intervention required. You simply filter the data to Squirtle trainers and compute the mean.

Rung 2: Intervention (Doing). Questions at this level ask what would happen if we actively *changed* something in the world. They are expressed using Pearl's *do*-operator: $P(Y \mid \mathrm{do}(X = x))$.

Pokemon example: "If we *assigned* trainers to use Squirtle (regardless of their preferences), what would their average badge count be?" This question cannot be answered from observational data alone. It requires either an experiment (randomly assigning starters) or a set of causal assumptions strong enough to identify the interventional distribution from observational data.

The critical distinction is that $P(Y \mid X = x)$ conditions on *observing* $X = x$ (which may be confounded), while $P(Y \mid \mathrm{do}(X = x))$ conditions on *setting* $X = x$ (which breaks the influence of confounders on X). In the language of graphical models, the do-operator "severs" all arrows pointing into X.

Rung 3: Counterfactual (Imagining). Questions at this level ask about what would have happened in a specific case under a different scenario. They take the form: "Given that we observed $X = x$ and $Y = y$, what would Y have been if X had been x'?"

Pokemon example: "Ash chose Charmander and earned 6 badges. How many badges *would* he have earned if he had chosen Bulbasaur instead?" This is the most demanding type of causal question. It requires reasoning about a specific individual in a specific counterfactual scenario — not just average effects across a population.

Each rung of the ladder strictly subsumes the one below it. You cannot answer Rung 2 questions with Rung 1 data alone, and you cannot answer Rung 3 questions with Rung 2 information alone. This hierarchy explains why so many debates in science (and in Pallet Town) go in circles: people try to answer Rung 2 or Rung 3 questions using Rung 1 data.

Why Causal Claims Need Assumptions Beyond Data

The central lesson of this section — and, in many ways, of this entire book — is that **data alone cannot establish causation**. No matter how large your dataset, no matter how sophisticated your statistical model, you cannot move from Rung 1 to Rung 2 without making assumptions about the underlying causal structure of the world.

This is not a failure of statistics. It is a fundamental feature of the problem. As Holland (1986) put it: "No causation without manipulation." And as Pearl (2009) has argued, causal assumptions are not testable from data alone — they must come from domain knowledge, theory, or experimental design.

The good news is that we have powerful tools for making these assumptions explicit, checking their plausibility, and deriving their implications. That is what the rest of this book is about. We will learn to draw causal diagrams (Directed Acyclic Graphs), to identify when causal effects can be estimated from observational data, and to design experiments and quasi-experiments that allow us to make credible causal claims.

But first, we need to understand the fundamental challenge that makes all of this necessary.

Professor Oak Explains: The Core Insight

The distinction between correlation and causation is not merely philosophical. It has life-or-death practical consequences. Consider a Pokemon Center that notices trainers who use Hyper Potions have higher win rates than those who use regular Potions. Should the Center recommend Hyper Potions to all trainers?

Not necessarily. Trainers who buy Hyper Potions may be wealthier, more experienced, or more dedicated. The observed correlation may reflect these underlying differences, not the effect of the potion itself. Recommending Hyper Potions based on this correlation alone could waste money without improving outcomes.

To make a sound recommendation, the Center needs to estimate the ***causal effect*** *of Hyper Potions on win rates — which requires either a randomized experiment or a careful observational study with credible assumptions about the confounding structure.*

1.2 The Fundamental Problem of Causal Inference

Choosing Your Starter

You stand before three Poke Balls on Oak's table. Your hand hovers. You pick up the one on the left: Charmander. The little Fire-type lizard blinks up at you. Blue smirks and grabs Squirtle. "Good luck with that," he says on his way out.

Professor Oak watches you bond with Charmander. Then he says something that will stay with you for the entire journey: "You've made your choice. And now there is something you will never, ever know: what would have happened if you had chosen Bulbasaur."

This is not a limitation of technology or data. It is not something that could be solved with a better Pokedex or a larger survey. It is a **logical impossibility**. You cannot both choose Charmander and not choose Charmander at the same time. The moment you pick up that Poke Ball, the other two paths through reality cease to be observable.

Holland's Dictum

Donald Rubin, in a series of papers beginning in 1974, formalized the potential outcomes framework that we will study in the next section. Paul Holland, in his landmark 1986 paper "Statistics and Causal Inference," crystallized the core difficulty:

> ***The Fundamental Problem of Causal Inference:*** *It is impossible to observe both potential outcomes for the same unit. At most one potential outcome is ever observed; the other is forever missing.*

Let us make this concrete. Suppose we want to know the causal effect of choosing Charmander (versus Bulbasaur) on the number of Gym Badges a trainer earns. For a specific trainer — call them Ash — there are two potential realities:

1. **Reality A:** Ash chooses Charmander and earns some number of badges. Call this $Y_{\text{Ash}}(\text{Charmander})$.
2. **Reality B:** Ash chooses Bulbasaur and earns some number of badges. Call this $Y_{\text{Ash}}(\text{Bulbasaur})$.

The **individual causal effect** for Ash is the difference:

$$\tau_{\text{Ash}} = Y_{\text{Ash}}(\text{Charmander}) - Y_{\text{Ash}}(\text{Bulbasaur})$$

The problem is that we can observe *at most one* of these two quantities. If Ash chooses Charmander, we observe $Y_{\text{Ash}}(\text{Charmander})$ but $Y_{\text{Ash}}(\text{Bulbasaur})$ is forever unknown. If he chooses Bulbasaur, the reverse is true. We can never observe both.

The Missing Data Perspective

One powerful way to think about this is through the lens of **missing data**. Consider the following table of five trainers who each chose between Charmander and Bulbasaur:

Trainer	Starter Chosen	Badges with Charmander	Badges with Bulbasaur	Individual Effect
Ash	Charmander	6	?	?
Misty	Bulbasaur	?	7	?
Brock	Charmander	8	?	?
Gary	Bulbasaur	?	5	?
Erika	Charmander	4	?	?

Every row has exactly one question mark in the potential outcomes columns. This is not an accident of data collection — it is a structural feature of reality. Each trainer walked one path. The other path is counterfactual: it exists in the realm of "what might have been" but not in the realm of observable fact.

Because each individual causal effect τ_i requires both potential outcomes, and we can observe at most one, we can *never* compute individual causal effects from data. Every single entry in the "Individual Effect" column is unknowable.

Why This Is Not Just a Practical Limitation

It is tempting to think that the fundamental problem is merely a practical inconvenience — that with better technology, we could somehow observe both outcomes. But this is wrong. The problem is not that we lack instruments sensitive enough to measure counterfactual outcomes. The problem is that counterfactual outcomes *do not exist* as observable quantities. They are, by definition, events that did not occur.

You might protest: "Couldn't Ash simply choose Charmander, complete his journey, then go back in time and choose Bulbasaur?" Setting aside the physics, even this thought experiment fails. The Ash who returns to choose Bulbasaur is not the same Ash — he now has memories and experience from his Charmander journey. The unit has changed.

Alternatively: "Couldn't Ash choose Charmander, then on a separate occasion choose Bulbasaur?" Perhaps, but the second journey occurs at a different time, in different conditions, with an older and more experienced trainer. The causal effect of starter choice at time t for person i cannot be identified by observing person i at time $t' \neq t$.

This is why the fundamental problem is truly fundamental. It is not a gap in our data. It is a feature of the logical structure of causation itself.

The Way Forward

If individual causal effects are unknowable, is the entire enterprise hopeless? No. The crucial insight — which we will develop formally in the next section — is that while **individual** causal effects are unidentifiable, **average** causal effects across populations can sometimes be identified under appropriate assumptions. The shift from individual effects to average effects, combined with assumptions about how treatment is assigned, is the foundation of modern causal inference.

When Professor Oak randomly assigns starters to trainers in a controlled experiment, something remarkable happens: the *average* of the missing potential outcomes in the treatment group equals (in expectation) the *average* of the observed outcomes in the control group. Randomization does not solve the fundamental problem for any individual, but it solves it *on average* — and that turns out to be enough for science.

1.3 Potential Outcomes Framework (Rubin Causal Model)

Setup and Notation

We are now ready to formalize the ideas from the previous sections. The framework we will use is the **Rubin Causal Model** (RCM), also known as the **potential outcomes framework**, developed by Donald Rubin (1974, 1978) and extended by many others including Holland (1986) and Imbens and Rubin (2015).

Consider a population of N trainers, indexed by $i = 1, 2, \dots, N$. Each trainer may or may not receive a binary treatment. For concreteness, suppose the treatment is the use of an **Exp. Share** — an item that distributes experience points across a trainer's entire team.

Let $D_i \in \{0, 1\}$ denote the **treatment assignment indicator** for trainer i:

$$D_i = \begin{cases} 1 & \text{if trainer } i \text{ uses Exp. Share} \\ 0 & \text{if trainer } i \text{ does not use Exp. Share} \end{cases}$$

For each trainer i, we define two **potential outcomes**:

- $Y_i(1)$: the number of Gym Badges trainer i would earn **if they used Exp. Share**
- $Y_i(0)$: the number of Gym Badges trainer i would earn **if they did not use Exp. Share**

These potential outcomes exist conceptually for every trainer, regardless of whether they actually receive the treatment. They represent the outcomes that *would* obtain under each treatment condition.

Professor Oak Explains: Potential Outcomes

The notation $Y_i(d)$ *for* $d \in \{0,1\}$ *denotes the outcome that unit* i *would realize if assigned treatment status* d. *Critically, both* $Y_i(1)$ *and* $Y_i(0)$ *are defined for every unit* i, *but at most one of them is ever observed. The pair* $(Y_i(0), Y_i(1))$ *is sometimes called the* ***potential outcome pair*** *for unit* i.

Formally, a potential outcome is a mapping from the treatment space to the outcome space: for each unit i, $Y_i : \{0,1\} \to \mathcal{Y}$, *where* $\mathcal{Y}$ *is the set of possible outcomes (e.g.,* $\{0,1,2,\ldots,8\}$ *for Gym Badges in Kanto).*

Notation Cheat Sheet: Reading Potential Outcomes in Plain English

Every symbol in this chapter has a plain-language translation. Keep this sheet nearby the first few times you read a causal formula — in a month it will feel natural.

Symbol	***Plain-English reading***
i	*A single trainer (the "unit"). Think "trainer number* i*."*
N	*How many trainers are in our study.*
$D_i \in \{0,1\}$	*Did trainer* i *actually get the treatment?* `1` *= yes,* `0` *= no.*
Y_i^{obs}	*The outcome we actually saw for trainer* i *(e.g., they earned 6 badges).*
$Y_i(1)$	*The outcome trainer* i ***would have gotten*** *under treatment — whether or not they were treated.*
$Y_i(0)$	*The outcome trainer* i ***would have gotten*** *without treatment — again, regardless of what happened.*
$\tau_i = Y_i(1) - Y_i(0)$	*Trainer* i*'s personal effect: "how many extra badges treatment gave this specific trainer."*
$\mathbb{E}[\cdot]$	*"The average of …", taken over the whole population (or a group, if we condition).*
$\mathbb{E}[Y_i(1) - Y_i(0)]$	*The ATE — the average personal effect across everyone.*
$\mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1]$	*The ATT — the average personal effect, but only among trainers who actually took the treatment.*
$(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i$	*"Treatment assignment is independent of the potential outcomes." Translation: the type of trainer who gets treated looks just like the type who doesn't.*

If any line in the chapter feels opaque, rewrite it using the right-hand column until the sentence sounds like normal English. That is not a shortcut — that is the real skill.

Individual Treatment Effect

The **individual treatment effect** (ITE) for trainer i is defined as:

$$\tau_i = Y_i(1) - Y_i(0)$$

This is the causal effect of Exp. Share for trainer i specifically. As we discussed in Section 1.2, τ_i is never directly observable because we can only see one of $Y_i(1)$ and $Y_i(0)$.

Causal Estimands

Since individual effects are unobservable, we turn to population-level summaries. The most common causal estimands are:

Average Treatment Effect (ATE):

$$\text{ATE} = \mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)]$$

The ATE is the expected difference in potential outcomes across the entire population. It answers the question: "If we could give every trainer Exp. Share and also observe what would have happened without it, what would the average difference be?"

Average Treatment Effect on the Treated (ATT):

$$\text{ATT} = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1]$$

The ATT restricts attention to those who actually received treatment. It answers: "Among trainers who actually used Exp. Share, what was the average effect?"

Average Treatment Effect on the Control (ATC):

$$\mathrm{ATC} = \mathrm{E}[Y_i(1) - Y_i(0) \mid D_i = 0]$$

The ATC focuses on the untreated. It answers: "Among trainers who did *not* use Exp. Share, how much would they have benefited on average if they had?"

> ***Professor Oak Explains: Why These Estimands Differ***
>
> *In general,* $\mathrm{ATE} \neq \mathrm{ATT} \neq \mathrm{ATC}$. *The three estimands coincide only when treatment effects are homogeneous (the same for everyone) or when treatment assignment is independent of potential outcomes.*
>
> *For example, suppose experienced trainers are more likely to use Exp. Share, and experienced trainers would earn more badges regardless. Then the ATT (average effect among experienced trainers who chose the item) might be smaller than the ATC (average effect among less experienced trainers who would benefit more from the extra experience points). The ATE would be a weighted average of the two.*
>
> *Formally, the ATE relates to the ATT and ATC as follows:*
>
> $$\mathrm{ATE} = P(D = 1) \cdot \mathrm{ATT} + P(D = 0) \cdot \mathrm{ATC}$$

The Observed Outcome: The Switching Equation

In practice, we observe exactly one potential outcome for each trainer. The **observed outcome** is given by the switching equation:

$$Y_i^{obs} = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$$

When $D_i = 1$, this reduces to $Y_i(1)$; when $D_i = 0$, it reduces to $Y_i(0)$. This equation links the theoretical potential outcomes to the data we actually observe.

The Naive Estimator and Selection Bias

The most natural approach to estimating a causal effect is to compare mean outcomes between the treated and control groups:

$$\hat{\Delta}^{naive} = \bar{Y}_{treated} - \bar{Y}_{control} = \mathrm{E}[Y_i^{obs} \mid D_i = 1] - \mathrm{E}[Y_i^{obs} \mid D_i = 0]$$

Substituting the switching equation:

$$\hat{\Delta}^{naive} = \mathrm{E}[Y_i(1) \mid D_i = 1] - \mathrm{E}[Y_i(0) \mid D_i = 0]$$

Now, let us add and subtract $\mathrm{E}[Y_i(0) \mid D_i = 1]$ — the average outcome the treated group *would have had* without treatment:

$$\hat{\Delta}^{naive} = \underbrace{\mathrm{E}[Y_i(1) \mid D_i = 1] - \mathrm{E}[Y_i(0) \mid D_i = 1]}_{\text{ATT}} + \underbrace{\mathrm{E}[Y_i(0) \mid D_i = 1] - \mathrm{E}[Y_i(0) \mid D_i = 0]}_{\text{Selection Bias}}$$

> ***Professor Oak Explains: The Selection Bias Decomposition***
>
> *This is one of the most important equations in causal inference. It says:*
>
> $$\mathrm{E}[Y \mid D = 1] - \mathrm{E}[Y \mid D = 0] = \mathrm{ATT} + \text{Selection Bias}$$
>
> *The naive comparison of means equals the causal effect of interest (ATT)* ***plus*** *a bias term. The selection bias term,* $\mathrm{E}[Y_i(0) \mid D_i = 1] - \mathrm{E}[Y_i(0) \mid D_i = 0]$, *measures the difference in* ***baseline*** *outcomes between the treated and control groups — i.e., how different these groups would have been even in the absence of treatment.*
>
> *Selection bias is zero if and only if* $\mathrm{E}[Y_i(0) \mid D_i = 1] = \mathrm{E}[Y_i(0) \mid D_i = 0]$ *— that is, the treatment and control groups have the same average potential outcome under control. This holds by construction in a randomized experiment, but it is often violated in observational data.*

> ***Blue's Mistake: Ignoring Selection Bias***
>
> *Blue compares mean badge counts between trainers who use Exp. Share and those who don't: "Exp. Share users average 7.2 badges; non-users average 5.8 badges. That's a 1.4-badge effect!" But Blue fails to recognize that trainers who seek out and use Exp. Share are probably more dedicated, strategic, and experienced — they would earn more badges even without the item. Part (perhaps all) of the 1.4-badge gap is selection bias, not a causal effect of the item.*

Worked Example: The Pallet Town Exp. Share Study

Let us work through a complete example. Professor Oak recruits six trainers and records their potential outcomes. (In reality, we could never observe both columns simultaneously; imagine Oak has access to a parallel-universe machine for pedagogical purposes.)

Trainer	D_i	$Y_i(1)$ **(with Exp. Share)**	$Y_i(0)$ **(without)**	$\tau_i = Y_i(1) - Y_i(0)$
Ash	1	7	5	2
Misty	0	8	7	1
Brock	1	6	4	2
Erika	0	5	3	2
Surge	1	8	6	2
Sabrina	0	7	6	1

Step 1: Compute the ATE.

$$\text{ATE} = \frac{1}{6}\sum_{i=1}^{6} \tau_i = \frac{2+1+2+2+2+1}{6} = \frac{10}{6} \approx 1.67$$

Step 2: Compute the ATT.

The treated trainers are Ash, Brock, and Surge:

$$\text{ATT} = \frac{2+2+2}{3} = 2.00$$

Step 3: Compute the ATC.

The control trainers are Misty, Erika, and Sabrina:

$$\text{ATC} = \frac{1+2+1}{3} \approx 1.33$$

Step 4: Verify the ATE decomposition.

$$\text{ATE} = P(D=1)\cdot \text{ATT} + P(D=0)\cdot \text{ATC} = \frac{1}{2}(2.00) + \frac{1}{2}(1.33) = 1.00 + 0.67 = 1.67 \checkmark$$

Step 5: Compute the naive estimator.

In practice, we only observe $Y_i(1)$ for treated trainers and $Y_i(0)$ for control:

$$\bar{Y}_{\text{treated}} = \frac{7+6+8}{3} = 7.00$$

$$\bar{Y}_{\text{control}} = \frac{7+3+6}{3} \approx 5.33$$

$$\hat{\Delta}^{\text{naive}} = 7.00 - 5.33 = 1.67$$

Step 6: Decompose the naive estimator.

$$\text{ATT} = 2.00$$

$$\text{Selection Bias} = E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]$$

$$= \frac{5+4+6}{3} - \frac{7+3+6}{3} = 5.00 - 5.33 = -0.33$$

$$\hat{\Delta}^{\text{naive}} = \text{ATT} + \text{Selection Bias} = 2.00 + (-0.33) = 1.67 \checkmark$$

In this example, the selection bias is negative: the treated trainers actually had *lower* baseline potential outcomes than the controls (perhaps because less naturally talented trainers were the ones who sought out Exp. Share). The naive estimator happens to underestimate the ATT (1.67 vs. 2.00) but still provides a biased picture. Notice also that the naive estimator (1.67) coincidentally equals the ATE in this case; that is a numerical accident, not a general result.

A Second Example: The Potion Study

To build further intuition, consider a study of whether using a Rare Candy before the first Gym battle improves the probability of winning.

Trainer	Used Rare Candy (D_i)	Win with Candy $Y_i(1)$	Win without $Y_i(0)$	τ_i
Red	1	1	1	0
Leaf	1	1	0	1
Silver	0	1	1	0
Kris	0	0	0	0

Here $Y_i \in \{0, 1\}$ (win or lose).

$$\text{ATE} = \frac{0 + 1 + 0 + 0}{4} = 0.25$$

The naive estimator gives:

$$\hat{\Delta}^{\text{naive}} = \frac{1 + 1}{2} - \frac{1 + 0}{2} = 1.0 - 0.5 = 0.5$$

This is double the true ATE! The selection bias is:

$$E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0] = \frac{1 + 0}{2} - \frac{1 + 0}{2} = 0.5 - 0.5 = 0.0$$

Wait — the selection bias is zero, but the naive estimator (0.5) does not equal the ATE (0.25). What happened? Recall that the decomposition shows $\hat{\Delta}^{\text{naive}} = \text{ATT} + \text{Selection Bias}$. Here, $\text{ATT} = \frac{0+1}{2} = 0.5$, and the Selection Bias is 0. The naive estimator correctly estimates the ATT, not the ATE. The ATT and ATE differ because treatment effects are heterogeneous and correlated with treatment assignment: Leaf, who benefits from the Rare Candy ($\tau = 1$), is in the treatment group, while Kris, who does not benefit ($\tau = 0$), is in the control group.

This example illustrates a subtle but critical point: **even with zero selection bias in baseline outcomes, the naive estimator recovers the ATT, not the ATE, and the two may differ when treatment effects are heterogeneous.**

The Assignment Mechanism

The **assignment mechanism** is the process that determines which units receive treatment. Formally, it is the conditional probability:

$$P(D_1, D_2, \ldots, D_N \mid Y_1(0), Y_1(1), Y_2(0), Y_2(1), \ldots, Y_N(0), Y_N(1), X_1, X_2, \ldots, X_N)$$

where X_i are pre-treatment covariates (e.g., trainer experience, hometown, number of Pokemon previously owned).

The assignment mechanism is the key to the entire enterprise of causal inference. Different assignment mechanisms lead to different identification strategies:

- **Randomized experiments** (Chapter 2): D_i is assigned by a coin flip, independent of potential outcomes.
- **Selection on observables** (Chapters 3–5): D_i depends on potential outcomes only through observed covariates X_i.
- **Instrumental variables** (Chapter 6): An external variable Z_i affects D_i but not Y_i directly.
- **Regression discontinuity** (Chapter 7): D_i is determined by whether a running variable crosses a threshold.
- **Difference-in-differences** (Chapter 8): Treatment timing varies across units and periods.

Each subsequent chapter in this book introduces a different identification strategy for different types of assignment mechanisms.

1.4 Counterfactuals

"In Another Timeline, You Picked Bulbasaur…"

You and Charmander take your first steps onto Route 1. The grass rustles. A wild Pidgey appears. As Charmander shoots an Ember, you wonder: *What if I had picked Bulbasaur? Would I be struggling right now? Or would Vine Whip have made this easier?*

This kind of thinking — reasoning about events that did not happen but *could* have — is called **counterfactual reasoning**. It is so natural that we barely notice ourselves doing it. Every trainer who loses a battle thinks, "If only I had used a different move…" Every researcher who observes an outcome wonders, "What would have happened under different conditions?"

Lewis's Possible Worlds

The philosopher David Lewis (1973) developed a formal semantics for counterfactual statements based on the concept of **possible worlds**. In Lewis's framework, a counterfactual statement like "If Ash had chosen Bulbasaur, he would have earned 7 badges" is evaluated by considering the **closest possible world** — the world most similar to the actual world in which the antecedent (Ash chose Bulbasaur) is true, and then checking whether the consequent (7 badges) holds in that world.

The "closeness" of possible worlds is determined by a similarity metric that prioritizes keeping as much of the actual world unchanged as possible. The closest world where Ash chose Bulbasaur is not one where the laws of physics are different or where Kanto has 16 Gyms — it is the world that is identical to ours in every respect except that Ash's hand moved to a different Poke Ball.

This philosophical framework maps directly onto the potential outcomes notation. The counterfactual outcome $Y_{Ash}(\text{Bulbasaur})$ is precisely the outcome in the closest possible world where Ash chose Bulbasaur. The potential outcomes framework and Lewis's possible worlds semantics are, in essence, two different languages for the same underlying concept.

Counterfactuals and Potential Outcomes

The connection between counterfactuals and the Rubin Causal Model is deep and precise:

- Each potential outcome $Y_i(d)$ is a counterfactual: it describes what *would* happen to unit i under treatment d.
- The observed outcome Y_i^{obs} is the one potential outcome that was realized.
- The unobserved potential outcome is the counterfactual: the outcome that *would have been* observed had treatment been different.

When we write $\tau_i = Y_i(1) - Y_i(0)$, we are defining the causal effect as the difference between an actual outcome and a counterfactual outcome (or between two counterfactual outcomes, if treatment has not yet been assigned).

Forward-Looking vs. Backward-Looking Causal Questions

It is useful to distinguish two types of causal questions:

Effects of causes (forward-looking): "What is the effect of Exp. Share on badge count?" This question starts with a cause (Exp. Share) and asks about its effect (badges). The potential outcomes framework is tailor-made for this type of question. We define treatment, specify potential outcomes, and estimate a causal estimand like the ATE.

Causes of effects (backward-looking): "Why did Ash earn only 6 badges? Was it because he chose Charmander?" This question starts with an observed effect (6 badges) and asks which cause is responsible. These questions are fundamentally harder to answer because they require reasoning about specific counterfactual scenarios for specific individuals — Rung 3 of Pearl's Ladder.

Holland (1986) argued that statistics is better suited to "effects of causes" than "causes of effects." The potential outcomes framework is designed for the former. When we estimate the ATE, we are answering: "What is the effect of this cause?" When a trainer wonders why they lost a particular battle, they are asking about the causes of a specific effect — a question that requires individual-level counterfactual reasoning, which is, as we have seen, fundamentally unobservable.

This does not mean backward-looking questions are unimportant. They are central to legal reasoning (Was the exposure the cause of the illness?), policy evaluation (Did this reform cause the observed improvement?), and everyday life (Did choosing Charmander cost me a badge?). But they are harder, and addressing them rigorously requires additional framework — an issue we will return to in later chapters.

> ***Blue's Mistake: Counterfactual Overconfidence***
>
> *After losing to the Elite Four, Blue declares: "I would have won if I had just used my Alakazam instead of Pidgeot in the final battle." This is a specific counterfactual claim about a single unit (Blue) at a single moment. It feels obviously true to Blue — but it is impossible to verify. Perhaps using Alakazam would have prompted the opponent to switch strategies, leading to a different loss. Counterfactual claims about individuals are inherently speculative, no matter how confident they feel. This is the fundamental problem in action.*

1.5 Key Terminology and the Causal Pokedex

Just as every trainer needs a Pokedex to catalog the creatures they encounter, every student of causal inference needs a reference for the key concepts and assumptions that appear throughout the field. In this section, we introduce the essential terminology, with formal definitions and Pokemon-themed examples.

Each entry is classified by:

- **Type:** Whether the concept is an *Estimand* (a quantity we want to estimate), an *Assumption* (a condition we require to hold), or a *Design Element* (a feature of the study).
- **Rarity:** How commonly the term appears in introductory treatments. *Common* terms appear in nearly every paper; *Uncommon* terms appear mainly in more advanced settings; *Rare* terms are specialized.

Entry 1: Treatment

Name	Treatment (Intervention, Exposure)
Type	Design Element
Rarity	Common
Definition	The variable whose causal effect we wish to study. Formally, $D_i \in \{0, 1\}$ in the binary case (or $D_i \in \{0, 1, \dots, K\}$ for multi-valued treatments). The treatment defines the contrast: we compare potential outcomes under different values of D.
Pokemon Example	Using an Exp. Share ($D_i = 1$) versus not using one ($D_i = 0$). Or choosing Charmander vs. Squirtle as a starter. Or training on Route 1 vs. Route 2.

Entry 2: Outcome

Name	Outcome (Response, Dependent Variable)
Type	Design Element
Rarity	Common
Definition	The variable we measure to assess the effect of treatment. Denoted Y_i, with potential outcomes $Y_i(0)$ and $Y_i(1)$.
Pokemon Example	Number of Gym Badges earned, win rate in competitive battles, time to complete the Pokemon League, total Pokemon caught.

Entry 3: Unit

Name	Unit (Observation, Subject)
Type	Design Element
Rarity	Common
Definition	The entity for which we define potential outcomes. Indexed by $i = 1, \dots, N$. Units must be well-defined and distinct.
Pokemon Example	An individual trainer. Could also be a Pokemon, a battle, a Gym, or a region, depending on the research question.

Entry 4: Assignment Mechanism

Name	Assignment Mechanism
Type	Design Element
Rarity	Common
Definition	The process (random or non-random) by which units are assigned to treatment conditions. Formally, the conditional distribution $P(D \mid Y(0), Y(1), X)$. The assignment mechanism is the key object that determines whether and how causal effects can be identified.
Pokemon Example	Professor Oak randomly assigning starters (randomized experiment), trainers self-selecting which held items to use (observational study), or the Pokemon League mandating Exp. Share for trainers above a certain level (regression discontinuity).

Entry 5: Confounder

Name	Confounder (Confounding Variable, Common Cause)
Type	Design Element
Rarity	Common
Definition	A pre-treatment variable that causally affects both the treatment D and the outcome Y. Confounders create spurious associations between D and Y in observational data. Formally, X is a confounder if $X \to D$ and $X \to Y$ (where → denotes a direct causal effect).
Pokemon Example	Trainer experience. Experienced trainers are more likely to use advanced items (Exp. Share) AND more likely to earn more badges regardless. If we do not adjust for experience, the observed association between Exp. Share and badges will overstate the true causal effect.

Entry 6: Mediator

Name	Mediator (Intermediate Variable)
Type	Design Element
Rarity	Uncommon
Definition	A variable on the causal pathway between treatment and outcome: $D \to M \to Y$. A mediator is *caused by* the treatment and itself *causes* the outcome. Controlling for a mediator can block the very causal channel we are trying to measure.
Pokemon Example	Suppose Exp. Share causes Pokemon to level up faster (M), which in turn causes more Gym wins (Y). Pokemon level is a mediator. If we control for Pokemon level, we block the causal effect we are trying to estimate and may find "no effect" of Exp. Share — even if the effect is real.

Entry 7: Collider

Name	Collider
Type	Design Element
Rarity	Uncommon
Definition	A variable that is causally affected by two or more other variables: $D \rightarrow C \leftarrow Y$ (or, more generally, when two variables both have arrows pointing into C). Conditioning on a collider *opens* a spurious path between its causes, creating a non-causal association. This is sometimes called **collider bias** or **Berkson's bias**.
Pokemon Example	Suppose both having a strong starter (D) and natural talent (Y) make a trainer more likely to appear in the Pokemon Hall of Fame (C). If we restrict our analysis to Hall of Fame trainers (conditioning on C), we may find a *negative* association between starter strength and talent — because among Hall of Famers, those with weaker starters must have compensated with more talent. This association is entirely spurious.

Entry 8: Ignorability (Conditional Independence)

Name	Ignorability (Unconfoundedness, Conditional Independence, Exchangeability)
Type	Assumption
Rarity	Common
Definition	Treatment assignment is independent of potential outcomes, conditional on observed covariates. Formally: $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i \mid X_i$. This is also called **conditional exchangeability**: within strata defined by X, treated and control units are exchangeable — as if treatment were randomly assigned.
Pokemon Example	Among trainers with the same experience level, hometown, and number of previously owned Pokemon, whether they use Exp. Share is unrelated to how many badges they would earn with or without it. If this holds, then controlling for these covariates removes all confounding.

Entry 9: Positivity (Overlap)

Name	Positivity (Overlap, Common Support)
Type	Assumption
Rarity	Common
Definition	For every combination of covariate values x in the population, there is a positive probability of receiving each treatment value. Formally: $0 < P(D_i = 1 \mid X_i = x) < 1$ for all x in the support of X. This ensures that for every type of unit, we have both treated and control observations to compare.
Pokemon Example	If *every* trainer from Cerulean City uses Exp. Share and *no* trainer from Lavender Town does, then we cannot compare treated and control trainers within these subgroups. Positivity would be violated. We need some Cerulean trainers who do not use Exp. Share and some Lavender trainers who do.

Entry 10: Consistency (Well-Defined Treatment)

Name	Consistency (Well-Defined Treatment, No Multiple Versions of Treatment)
Type	Assumption
Rarity	Common
Definition	The observed outcome for a treated unit equals their potential outcome under treatment: if $D_i = d$, then $Y_i^{obs} = Y_i(d)$. This requires that the treatment is sufficiently well-defined that there is only one "version" of being treated. Formally: $D_i = d \implies Y_i^{obs} = Y_i(d)$.
Pokemon Example	"Using Exp. Share" must mean the same thing for every trainer. If some trainers equip Exp. Share to their lead Pokemon while others equip it to their weakest, these are different versions of treatment, and the potential outcome $Y_i(1)$ is not well-defined. The treatment must be specified precisely enough that there is a single, unambiguous version.

Entry 11: SUTVA (Stable Unit Treatment Value Assumption)

Name	SUTVA (Stable Unit Treatment Value Assumption)
Type	Assumption
Rarity	Common
Definition	Two components: (1) **No interference** — one unit's treatment does not affect another unit's outcome: $Y_i(D_1, \dots, D_N) = Y_i(D_i)$. (2) **No hidden variations of treatment** — there is only one version of each treatment level (this is closely related to consistency). SUTVA is required for the potential outcomes notation $Y_i(d)$ to even make sense, because it ensures that each unit's outcome depends only on its own treatment assignment.
Pokemon Example	No interference: whether your rival uses Exp. Share should not affect how many badges *you* earn. This might be violated if trainers compete for limited Gym slots, so that one trainer's stronger team (from Exp. Share) makes it harder for others to win. No hidden variations: as above, Exp. Share must be used in a consistent way across trainers.

> ***Professor Oak Explains: Why SUTVA Matters***
> *SUTVA is easy to state but easy to violate. In many real-world settings, units interact: one person's vaccination protects their neighbors (interference); one firm's pricing affects competitors' sales (interference); the same drug may be administered in different dosages (hidden treatment variations). When SUTVA fails, the potential outcomes framework must be generalized, typically by redefining the unit or the treatment. We will revisit interference in Chapter 10.*

Chapter Summary

This chapter introduced the foundational concepts of causal inference. Here are the key takeaways:

- **Correlation is not causation.** Observed associations between variables may be driven by confounders — common causes that affect both treatment and outcome. Moving from correlation to causation requires either experimental design or strong, explicit assumptions.
- **The Fundamental Problem of Causal Inference** (Holland, 1986) is that we can observe at most one potential outcome for each unit. The other potential outcome — the counterfactual — is forever missing. Individual causal effects are therefore unobservable.
- **The Potential Outcomes Framework** (Rubin Causal Model) formalizes causal questions using potential outcomes $Y_i(1)$ and $Y_i(0)$. Key estimands include the ATE, ATT, and ATC. The naive difference-in-means estimator captures the ATT plus selection bias.
- **The selection bias decomposition** shows that $E[Y|D=1] - E[Y|D=0] = ATT + \text{Selection Bias}$. Selection bias arises when treated and control groups differ in their baseline potential outcomes.
- **Pearl's Ladder of Causation** distinguishes three levels of causal reasoning: association (seeing), intervention (doing), and counterfactual (imagining). Each level requires strictly more information than the one below.

- **Counterfactual reasoning** — thinking about what would have happened under different conditions — is the conceptual foundation of causal inference. The potential outcomes framework provides a rigorous mathematical language for counterfactual claims.
- **Key assumptions** for causal identification include ignorability (no unmeasured confounders), positivity (overlap between treatment groups), consistency (well-defined treatment), and SUTVA (no interference between units). Without these assumptions, causal effects cannot generally be identified from data.
- **The assignment mechanism** — the process by which units are assigned to treatment — is the central object of study. Different assignment mechanisms require different identification strategies, which form the subject of the remaining chapters.

Professor Oak's Review Questions

1. **Correlation vs. Causation.** A researcher finds that trainers who carry more Poke Balls tend to catch more Pokemon. Does this mean that buying more Poke Balls causes a trainer to catch more? What confounders might explain this association? Describe a scenario in which the causal effect of additional Poke Balls is zero, yet the correlation is strongly positive.
2. **The Fundamental Problem.** Explain why it is impossible to compute the individual treatment effect $\tau_i = Y_i(1) - Y_i(0)$ for any specific trainer. Why is this a logical impossibility rather than a practical limitation?
3. **Selection Bias.** Trainers who choose to battle in Gyms tend to have stronger Pokemon. A naive comparison shows that Gym-battling trainers earn more money than non-Gym trainers. Decompose this comparison into the ATT and selection bias. Under what conditions would the naive comparison equal the true ATT?
4. **Pearl's Ladder.** Classify each of the following questions according to Pearl's Ladder of Causation (Rung 1, 2, or 3):
    - a. What fraction of trainers who use TMs have a win rate above 60%?
    - b. If we mandated TM use for all trainers, what would the average win rate be?
    - c. Trainer Red used a TM and won. Would he have won without it?
5. **SUTVA.** Give a Pokemon-themed example where SUTVA is likely violated due to interference between units. How would you modify the potential outcomes framework to accommodate this violation?
6. **Estimands.** In a population of 100 trainers, 40 use Exp. Share and 60 do not. The ATT is 2.0 badges and the ATC is 1.0 badge. Compute the ATE. Would you expect the naive estimator to be larger or smaller than the ATE? Explain.

Trainer Challenge Exercises

Exercise 1.1: Potential Outcomes Table. Consider the following (omniscient) data on four trainers and the effect of using a Lucky Egg on total experience points gained (in thousands):

Trainer	D_i	$Y_i(1)$	$Y_i(0)$
Youngster Joey	1	45	30
Lass Anna	0	55	50
Bug Catcher Wade	1	35	20
Hiker Marcos	0	40	35

a. Compute the individual treatment effect τ_i for each trainer.

b. Compute the ATE, ATT, and ATC.

c. Compute the naive estimator $\hat{\Delta}^{\text{naive}}$ using only observed outcomes.

d. Compute the selection bias. Verify that $\hat{\Delta}^{\text{naive}} = \text{ATT} + \text{Selection Bias}$.

e. Is the naive estimator biased upward or downward for the ATT? For the ATE? Explain intuitively why.

Exercise 1.2: Identifying Confounders, Mediators, and Colliders. For each of the following causal structures, identify whether the variable Z is a confounder, mediator, or collider. Draw the causal graph (DAG).

a. Training intensity (D) → Gym badges (Y); Trainer experience (Z) → Training intensity (D); Trainer experience (Z) → Gym badges (Y).

b. Using Exp. Share (D) → Average team level (Z) → Win rate (Y).

c. Type advantage (D) → Making it to the Hall of Fame (Z) ← Strategy skill (Y).

Exercise 1.3: The Pewter City Protein Puzzle (Preview). Nurse Joy in Pewter City claims that trainers who feed their Pokemon Protein earn significantly more badges than those who don't. She shows you data from 500 trainers:

Group	n	Mean Badges
Protein users	200	7.1
Non-users	300	5.4

a. What is the naive estimate of the effect of Protein on badges?

b. Name at least three potential confounders that could explain part or all of this association.

c. Suppose you learn that Protein costs 9,800 Poke Dollars per unit. How does this information help you think about the likely direction of selection bias?

d. Propose a study design that could more credibly estimate the causal effect of Protein. What assignment mechanism would you use? (Don't worry about the details — we will cover this in Chapter 2.)

Exercise 1.4: Counterfactual Reasoning. Trainer Red used a Master Ball to catch Mewtwo and subsequently won the Pokemon League. Consider the counterfactual: "If Red had not caught Mewtwo, he would not have won the League."

a. Formalize this statement using potential outcomes notation. Clearly define the unit, treatment, and outcome.

b. Is this statement testable? Why or why not?

c. Describe a scenario in which the counterfactual is true and a scenario in which it is false (Red would have won anyway). What additional information would help you assess the plausibility of each scenario?

d. Is this an "effect of a cause" or a "cause of an effect" question? Explain.

Further Reading

- **Holland, P. W.** (1986). "Statistics and Causal Inference." *Journal of the American Statistical Association*, 81(396), 945–960. The classic paper that articulated the fundamental problem of causal inference and coined the phrase "no causation without manipulation."
- **Rubin, D. B.** (1974). "Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies." *Journal of Educational Psychology*, 66(5), 688–701. The foundational paper for the potential outcomes framework.
- **Pearl, J.** (2009). *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press. The definitive reference for the structural causal model framework, including the Ladder of Causation and the *do*-calculus.
- **Imbens, G. W., & Rubin, D. B.** (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge University Press. A comprehensive and rigorous textbook on the potential outcomes framework, covering design and analysis of both experiments and observational studies.
- **Hernan, M. A., & Robins, J. M.** (2020). *Causal Inference: What If*. Chapman & Hall/CRC. A modern and accessible textbook that bridges the potential outcomes and graphical models traditions, with extensive examples from epidemiology.
- **Lewis, D.** (1973). *Counterfactuals*. Blackwell. The philosophical foundation for possible worlds semantics and counterfactual reasoning.
- **Morgan, S. L., & Winship, C.** (2015). *Counterfactuals and Causal Inference: Methods and Principles for Social Research* (2nd ed.). Cambridge University Press. An excellent bridge between sociological applications and the formal potential outcomes framework.

Skills to Practice in the Notebook

The companion notebook `notebooks/ch01_pallet_town.ipynb` is where the ideas above become muscle memory. Before moving on to Pewter City, you should be able to do the following without looking back at the chapter:

1. **Load the Kanto Trainer dataset and explore it.** Import `trainers` via `load_trainers()`, inspect its columns, and compute simple group means (for example, average badges by `starter_type`). This is the baseline you will repeatedly compare against causal estimates.

2. **Reproduce "Blue's mistake" numerically.** Compute the naive difference-in-means between two groups (e.g., Squirtle-pickers vs. Charmander-pickers) and then show, using the same dataset, that the groups differ in at least one pre-treatment covariate (experience, wealth, strategy score). Being able to *display* confounding in a two-panel plot is the concrete skill.
3. **Build a potential outcomes table by hand.** Given simulated $Y_i(0)$ and $Y_i(1)$, write code that computes ATE, ATT, ATC, and the naive estimator, and then verifies the selection-bias decomposition $\hat{\Delta}^{\text{naive}} = \text{ATT} + \text{Selection Bias}$ numerically. If your code's ATT plus selection bias does not exactly equal the naive estimate, something is wrong.
4. **Use the `confounder_slider` widget.** Narrow the slider until you are comparing only trainers with nearly identical experience levels. Watch what happens to the naive estimate. You should be able to explain in one sentence *why* it changes.
5. **Complete the three Challenge Exercises in the notebook.** These are the gatekeeper tasks — if you can finish them, you are ready for Chapter 2.
   - **Challenge 1:** Compute ATE and selection bias on a filtered subset (Pewter + Cerulean trainers).
   - **Challenge 2:** Identify a confounder between `cave_training` and `badges`, and explain why it biases the naive estimate.
   - **Challenge 3:** Write `simulate_and_estimate(n, true_ate, confounding_strength, seed)` and plot how bias grows with confounding strength.

You do not need to get these perfect on the first attempt. You *do* need to understand *why* your answer is right when you're done.

Check Your Understanding

Before you leave Pallet Town, walk through these. If you stumble on more than a couple, spend another pass with the chapter and notebook — the rest of the book is built on top of this foundation.

Questions you should be able to answer out loud, without notes:

- What is the difference between correlation and causation? Why does observing the first not justify claiming the second?
- State the Fundamental Problem of Causal Inference in one sentence. Why is it a *logical* impossibility, not a practical one?
- Define $Y_i(1)$, $Y_i(0)$, D_i, and Y_i^{obs}. Which of these can you see in a dataset? Which cannot?
- Write the switching equation $Y_i^{\text{obs}} = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$ and explain what each term "selects" for.
- State the ATE, ATT, and ATC formulas and explain in plain English which population each one refers to.
- Write the selection-bias decomposition and point to which term is the *causal* part and which term is the *bias* part.
- What does it mean for treatment to be *ignorable* given X? Give a Pokemon-themed example where ignorability holds and one where it fails.
- Name the four core assumptions (Ignorability, Positivity, Consistency, SUTVA) and give one way each could break in a Kanto context.
- On Pearl's Ladder, classify: (a) "What fraction of Charmander trainers won Brock's gym?" (b) "If we forced every trainer to use Charmander, what would the win rate be?" (c) "Ash won with Charmander — would he have won with Bulbasaur?"

Tasks you should be able to perform in code:

- Load `trainers` and compute a group-mean comparison (naive estimator) for any pair of groups.
- Simulate a dataset with a known confounder and a known true ATE, then show numerically that the naive estimator is biased.
- Given a potential outcomes table (with both $Y(1)$ and $Y(0)$), compute ATE, ATT, ATC, naive estimator, and selection bias — and verify the decomposition identity.
- Plot how bias changes as confounding strength changes (this is the punchline of Challenge 3 and will matter again in Chapter 3).
- Take any equation from this chapter and rewrite it in plain English using the Notation Cheat Sheet above.

If you can do all of this, you have everything you need for Chapter 2 — where we will learn the one trick that makes the naive estimator *unbiased*: randomization.

Next Time: Pewter City

Charmander grows stronger as you cross Route 1. Viridian City passes in a blur of errands and close encounters with wild Rattata. Ahead lies Pewter City and its famous Rock-type Gym.

At the Pewter City Pokemon Center, you overhear Nurse Joy in a heated discussion with a group of trainers. "I'm tired of arguing about Protein," she says, tapping a clipboard. "Does it actually make Pokemon stronger, or do trainers who buy Protein just happen to be more dedicated? I have an idea for how we could settle this once and for all."

She looks at you. "We're going to run an experiment."

Tomorrow you head to Pewter City, where Nurse Joy has a radical idea for settling the Pewter Protein debate once and for all — a radical idea called **randomization***. You will learn why random assignment is the gold standard for causal inference, how it eliminates selection bias by construction, and what happens when the real world refuses to cooperate.*

Welcome to Chapter 2: Pewter City — Randomized Experiments.

Chapter 2: Pewter City — Randomized Experiments

#095 Onix

Rock-type — Brock's specialty

Brock, Pewter Gym Leader

> *"The best time to plant a tree was twenty years ago. The best way to test a supplement is a randomized experiment." — Professor Oak, Kanto Regional Laboratory*

You have arrived in Pewter City. The rocky terrain gives way to a modest settlement built around the famous Pewter City Gym, home to Brock, the Rock-type specialist. But before you challenge Brock, there is a scientific dispute to settle — one that will teach you why the randomized controlled trial is the gold standard of causal inference.

At the Pewter City Pokemon Center, Nurse Joy has had enough. For weeks, trainers have been filing in after their battles with Brock, some triumphant, others carrying fainted Pokemon. And all of them — winners and losers alike — have an opinion about **Pewter Protein**, a new supplement rumored to boost a Pokemon's Attack stat before battle.

"It definitely works," says one trainer, flexing his Machop's arm. "My Charmander hit 20% harder after one dose!"

"It's a scam," says another, nursing a wounded Pidgey. "I took it and Brock's Onix still destroyed me."

Nurse Joy has data — she has been recording outcomes for every trainer who walks through her doors — but the data is hopelessly confounded. Trainers who *choose* to buy Pewter Protein tend to be wealthier, more experienced, and more likely to have Pokemon with type advantages against Rock. Are they winning because of the Protein, or because they were going to win anyway?

Enter Professor Oak, who arrives in Pewter City for a research consultation.

"Nurse Joy," he says, adjusting his lab coat, "you don't need more data. You need *better* data. You need a **randomized controlled trial**."

2.1 Why Randomization Works

The Coin Flip That Changes Everything

Professor Oak hands Nurse Joy a simple Poke-coin — Magikarp on one side, Gyarados on the other. "For the next 200 trainers who register for a Brock challenge," he instructs, "flip this coin. Magikarp: the trainer gets the real Pewter Protein. Gyarados: they get an identical-looking Placebo Berry — same size, same color, same taste, but no active ingredient. Don't tell them which one they received."

Nurse Joy is skeptical. "That's it? A coin flip decides who gets what?"

"That," Professor Oak replies, "is the most powerful tool in all of causal inference."

Independence of Potential Outcomes and Treatment

Recall from Chapter 1 that every trainer i has two **potential outcomes**: $Y_i(1)$, the outcome they would experience if they received Pewter Protein (treatment), and $Y_i(0)$, the outcome under the Placebo Berry (control). The individual causal effect is $\tau_i = Y_i(1) - Y_i(0)$, and the **Average Treatment Effect (ATE)** is:

$$\tau = E[Y_i(1) - Y_i(0)] = E[Y_i(1)] - E[Y_i(0)]$$

The **fundamental problem of causal inference** (Holland, 1986) tells us we can never observe both $Y_i(1)$ and $Y_i(0)$ for the same trainer. We only see the outcome corresponding to the treatment actually received:

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$$

where $D_i \in \{0, 1\}$ is the treatment indicator.

The question is: when we compare the average outcomes of treated trainers to control trainers, does this comparison identify the ATE? From Chapter 1, we know the naive comparison gives us:

$$E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] = \underbrace{E[Y_i(1) - Y_i(0) \mid D_i = 1]}_{\text{ATT}} + \underbrace{E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]}_{\text{Selection Bias}}$$

When trainers **self-select** into treatment — buying Protein of their own volition — the selection bias term is generally nonzero. But under random assignment, something remarkable happens.

Definition 2.1 (Random Assignment). *Treatment D_i is said to be randomly assigned if the treatment indicator is statistically independent of the potential outcomes:*

$$(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i$$

This means the joint distribution of potential outcomes is the same in the treatment and control groups.

When this independence condition holds, the selection bias vanishes:

$$E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0] = E[Y_i(0)] - E[Y_i(0)] = 0$$

And therefore:

$$E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] = E[Y_i(1)] - E[Y_i(0)] = \tau$$

The simple difference in group means identifies the ATE. No regression. No fancy adjustment. Just a coin flip and a subtraction.

Balancing Observed AND Unobserved Confounders

Why does this work so powerfully? Because the coin doesn't know anything about the trainer. It doesn't know whether they have a Water-type Pokemon (which would be devastating against Brock's Rock-types). It doesn't know their experience level, their number of gym badges, or their secret training regimen in Mt. Moon. It doesn't even know things that *we* can never measure — like the trainer's natural battle instinct, their Pokemon's hidden Individual Values (IVs), or whether they had a good night's sleep.

By assigning treatment entirely at random, we ensure that — **in expectation** — the treatment and control groups are identical on every dimension, observed and unobserved alike. This is worth pausing on, because no other method in this textbook achieves this. Every observational method we will encounter in Chapters 3 through 7 can only address *observed* confounders (or confounders with specific structural properties). Randomization handles them all.

Professor Oak Explains: Why "In Expectation" Matters

Randomization does not guarantee that the treatment and control groups are perfectly balanced in any given experiment. If you flip 200 coins, you might get 105 heads and 95 tails. By chance, the treatment group might have slightly more Water-type trainers. But the expected distribution is balanced, and with a large enough sample, deviations from balance become small and quantifiable. We will make this precise in Section 2.3 when we discuss variance estimation.

Unbiasedness of the Difference-in-Means Estimator

Let us prove this formally. We have n trainers, of whom n_1 are randomly assigned to treatment and $n_0 = n - n_1$ to control. The **difference-in-means estimator** is:

$$\hat{\tau}^{DM} = \bar{Y}_1 - \bar{Y}_0 = \frac{1}{n_1} \sum_{i:D_i=1} Y_i - \frac{1}{n_0} \sum_{i:D_i=0} Y_i$$

Under random assignment:

$$E[\bar{Y}_1] = E\left[\frac{1}{n_1} \sum_{i:D_i=1} Y_i\right] = E[Y_i \mid D_i = 1] = E[Y_i(1) \mid D_i = 1] = E[Y_i(1)]$$

where the last equality uses $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i$. By the same logic:

$$E[\bar{Y}_0] = E[Y_i(0)]$$

Therefore:

$$E[\hat{\tau}^{DM}] = E[\bar{Y}_1 - \bar{Y}_0] = E[Y_i(1)] - E[Y_i(0)] = \tau$$

The difference-in-means estimator is **unbiased** for the ATE under random assignment. This is the foundational result of experimental causal inference, articulated in various forms by Fisher (1935), Neyman (1923/1990), and formalized in the modern potential outcomes framework by Rubin (1974).

The Pewter Protein Balance Table

Nurse Joy runs the trial. Two hundred trainers walk through the doors, each receiving their coin-flip assignment. After enrollment is complete, she compiles a **balance table** — a comparison of pre-treatment characteristics across the two groups — to check whether randomization achieved approximate balance.

Characteristic	Protein ($n_1 = 102$)	Placebo ($n_0 = 98$)	Difference	p-value
Mean trainer level	14.8	14.3	0.5	0.52
% with Water-type Pokemon	28.4%	26.5%	1.9 pp	0.76
% with Grass-type Pokemon	15.7%	17.3%	-1.6 pp	0.74
Mean # of Pokemon on team	3.8	3.9	-0.1	0.67
Mean prior gym badges	0.9	0.8	0.1	0.41
% first-time challengers	61.8%	64.3%	-2.5 pp	0.71
Mean Pokemon avg. level	13.2	12.9	0.3	0.59

Table 2.1: Balance table for the Pewter Protein RCT. No statistically significant differences at $\alpha = 0.05$, consistent with successful randomization. (pp = percentage points.)

None of the p-values are close to significance. The groups look comparable. Of course, even if one or two covariates showed a significant imbalance by chance (which happens roughly 5% of the time per test at $\alpha = 0.05$), this would not invalidate the experiment — it is expected under the null of successful randomization.

Blue's Mistake: Self-Selection is Not Randomization

Right as Nurse Joy is reviewing her balance table, Blue strides into the Pokemon Center, arms full of shopping bags.

> ***Blue's Mistake: "I Don't Need an Experiment"***
>
> *"Hey losers," Blue announces, dumping 50 boxes of Pewter Protein on the counter. "I already KNOW this stuff works. I've been using it for a week and my Squirtle is destroying everything. I don't need some nerdy experiment to tell me what's obvious."*
>
> *Professor Oak sighs. "Blue, you chose to buy Protein because you're wealthy, experienced, and already have a Water-type starter — the perfect counter to Brock's Rock-types. Trainers like you would have won anyway. Your personal experience tells us nothing about whether the Protein itself had any effect."*
>
> *Blue's data illustrates **self-selection bias** perfectly. Among trainers who voluntarily purchase Pewter Protein, the win rate against Brock is 78%. Among those who don't, it's 45%. But this 33 percentage point gap is a mix of the true causal effect of Protein and the fact that Protein buyers are stronger trainers to begin with.*
>
> *Nurse Joy's RCT tells a different story. In the randomized trial, the win rate is 52% in the Protein group and 43% in the Placebo group — a 9 percentage point difference. The true effect of Pewter Protein exists, but it's much smaller than Blue's naive comparison suggests.*

The gap between Blue's estimate (33 pp) and the experimental estimate (9 pp) — a difference of 24 percentage points — is pure selection bias. This is the danger of drawing causal conclusions from observational data without proper adjustment, a theme we will return to throughout this textbook.

2.2 Experimental Design

A randomized experiment does not design itself. The coin flip is the heart of the method, but an enormous amount of thought must go into the architecture surrounding that flip. In this section, we walk through the key design decisions that Professor Oak and Nurse Joy faced when planning the Pewter Protein RCT.

Precise Treatment Definition

A vague treatment yields vague conclusions. "Pewter Protein" could mean many things — different doses, different timings, different formulations. The treatment must be defined with surgical precision.

> ***Definition 2.2 (Treatment Protocol).*** *The treatment in the Pewter Protein RCT is defined as:*
>
> - ***Treatment arm:*** *Exactly one tablet of Pewter Protein (10 mg, Lot #151, manufactured by Silph Co.) administered orally with 200 mL water, exactly 60 minutes before the trainer's scheduled Brock battle.*
> - ***Control arm:*** *Exactly one Placebo Berry tablet, identical in size (8 mm diameter), color (pale blue), weight (0.5 g), and taste (mild Oran Berry flavor), administered under the same conditions.*
> - ***Administration:*** *By Nurse Joy, in a private room at the Pewter City Pokemon Center, following a standardized script.*

This precision serves multiple purposes. First, it ensures replicability — another researcher in Cerulean City could run the same experiment with the same treatment. Second, it anchors the causal estimand: we are estimating the effect of *this specific intervention*, not "supplements in general." Third, it supports blinding: trainers cannot distinguish treatment from control by appearance, taste, or administration context.

Outcome Definition

Brock battles yield multiple potential outcomes, each measuring something different:

Outcome	Type	Definition	Pros	Cons
Beat Brock (yes/no)	Binary	Trainer's team reduces all of Brock's Pokemon to 0 HP	Clear, meaningful	Loses granularity
Total damage dealt	Continuous	Sum of HP damage dealt to Brock's Pokemon across all turns	Granular, high power	May be noisy
Turns to win	Count/Time-to-event	Number of battle turns until victory (censored if trainer loses)	Captures efficiency	Requires handling censoring
Pokemon remaining	Count	Number of trainer's Pokemon still standing at battle's end	Measures dominance	Bounded, discrete

Table 2.2: Candidate outcome measures for the Pewter Protein RCT.

Nurse Joy selects **total damage dealt** (continuous) as the primary outcome and **beat Brock** (binary) as a pre-registered secondary outcome. The continuous measure provides more statistical power (we can detect smaller effects), while the binary measure is the outcome trainers actually care about.

The choice of primary vs. secondary outcomes must be **pre-registered** — declared before the data are analyzed. This prevents the researcher from running multiple analyses and cherry-picking the most favorable result, a practice known as *p-hacking* or *specification searching*.

Power Analysis

Before enrolling a single trainer, Professor Oak insists on a **power analysis** to determine whether 200 trainers is enough to detect a meaningful effect.

Statistical power is the probability of correctly rejecting the null hypothesis when the treatment effect is real. For a two-sample test comparing means, the required sample size per group is approximately:

$$n_{\text{per group}} = \frac{(z_{\alpha/2} + z_{\beta})^2(\sigma_1^2 + \sigma_0^2)}{(\mu_1 - \mu_0)^2}$$

where: - $z_{\alpha/2}$ is the critical value for a two-sided test at significance level α - z_{β} is the critical value corresponding to the desired power $1 - \beta$ - σ_1^2, σ_0^2 are the variances of the outcome in the treatment and control groups - $\mu_1 - \mu_0$ is the **minimum detectable effect** (MDE) — the smallest effect we want to be able to detect

Professor Oak's assumptions, based on pilot data from Viridian City:

- Significance level: $\alpha = 0.05$, so $z_{\alpha/2} = 1.96$
- Power: $1 - \beta = 0.80$, so $z_{\beta} = 0.842$
- Standard deviation of damage dealt in both groups: $\sigma_1 = \sigma_0 = 35$ HP
- Minimum detectable effect: $\mu_1 - \mu_0 = 15$ HP of additional damage

Plugging in:

$$n_{\text{per group}} = \frac{(1.96 + 0.842)^2(35^2 + 35^2)}{15^2} = \frac{(2.802)^2(2450)}{225} = \frac{7.851 \times 2450}{225} = \frac{19{,}235}{225} \approx 85.5$$

Rounding up, we need approximately **86 trainers per group**, or 172 total. Professor Oak rounds up to 200 to provide a buffer for attrition and to increase power slightly. With $n = 200$ (100 per group), the actual power is approximately 87%.

> ***Professor Oak Explains: Effect Sizes and Practical Significance***
> *"A power analysis forces you to think about what effect size would be meaningful. If Pewter Protein only adds 2 HP of damage — less than a single Scratch attack — no trainer would care, even if it were statistically significant. We powered this study to detect a 15 HP difference because that's roughly the damage from one extra Tackle, which could plausibly swing a close battle. Always think about practical significance, not just statistical significance."*

Blocking and Stratified Randomization

Pure coin-flip randomization is simple and valid, but it can be made more efficient. Nurse Joy notices that having a Water-type or Grass-type Pokemon (which are super-effective against Brock's Rock-types) is the single strongest predictor of battle performance. If, by chance, more Water-type trainers ended up in the treatment group, the estimate would be noisier than necessary.

Stratified (block) randomization addresses this. Instead of one big randomization, we randomize *within strata*:

> ***Definition 2.3 (Stratified Randomization).*** *Divide the sample into K strata based on pre-treatment covariates. Within each stratum k, independently randomly assign a fixed number of units to treatment and control.*

For the Pewter Protein RCT, Nurse Joy creates two blocks:

Block	Definition	n	Assigned to Protein	Assigned to Placebo
A: Type advantage	Trainer has ≥ 1 Water or Grass-type Pokemon	78	39	39
B: No type advantage	Trainer has no Water or Grass-type Pokemon	122	61	61
Total		**200**	**100**	**100**

Table 2.3: Stratified randomization scheme for the Pewter Protein RCT.

Within each block, treatment and control are *exactly* balanced on the blocking variable. This eliminates one source of sampling variability, improving precision — especially when the blocking variable is strongly predictive of the outcome. The efficiency gain from blocking is proportional to the R^2 between the blocking variable and the outcome (see Gerber & Green, 2012, Chapter 4).

Importantly, stratified randomization preserves the independence condition $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i \mid X_i$ within each stratum, and the overall ATE can be estimated as a weighted average of within-stratum effects:

$$\hat{\tau}_{\text{stratified}} = \sum_{k=1}^{K} \frac{n_k}{n} \hat{\tau}_k$$

where $\hat{\tau}_k = \bar{Y}_{1k} - \bar{Y}_{0k}$ is the difference in means within stratum k.

Cluster Randomization

What if randomizing individual trainers is logistically impossible? Suppose the Kanto Pokemon League wants to test Pewter Protein across the entire region. It might be easier to randomize at the level of **Pokemon Centers**: all trainers at certain Centers receive Protein, all trainers at others receive Placebo.

This is **cluster randomization**, and it introduces complications:

> ***Definition 2.4 (Cluster Randomized Trial).*** *Units are grouped into clusters (e.g., Pokemon Centers, towns, training schools). Entire clusters, rather than individual units, are randomly assigned to treatment or control.*

The primary cost of cluster randomization is a loss of statistical power. Trainers within the same Pokemon Center are likely more similar to each other (they trained in the same area, faced similar wild Pokemon, may know each other) than trainers across Centers. This **intra-cluster correlation (ICC)** means that adding more trainers within a cluster provides diminishing returns. The effective sample size is:

$$n_{\text{effective}} = \frac{n}{1 + (m - 1)\rho}$$

where n is the total sample size, m is the average cluster size, and ρ is the ICC. If $\rho = 0.05$ and each of 10 Pokemon Centers has 20 trainers ($n = 200, m = 20$), the effective sample size is only $200/(1 + 19 \times 0.05) = 200/1.95 \approx 103$ — nearly half the nominal sample.

For the Pewter Protein RCT, individual randomization is feasible (every trainer visits the same Pokemon Center in Pewter City), so Nurse Joy and Professor Oak wisely avoid the power loss of clustering.

Blinding

Blinding prevents **expectation effects** — changes in behavior caused by knowing one's treatment status rather than by the treatment itself.

Blinding Level	Who is Blinded	Pewter Protein Implementation
Single-blind	Trainers don't know which tablet they received	Protein and Placebo look, taste, and smell identical
Double-blind	Trainers AND Brock don't know treatment status	Brock battles all challengers identically; no access to assignment records
Triple-blind	Trainers, Brock, AND Nurse Joy (analyst)	An independent statistician (Prof. Oak's assistant) analyzes coded data

Table 2.4: Blinding levels in the Pewter Protein RCT.

The Pewter Protein RCT implements **double-blinding**. Trainers cannot tell which tablet they received. Brock, for his part, battles every challenger with the same team (Geodude and Onix) using the same strategy, and has no idea which trainers received Protein. Nurse Joy records outcomes but does not analyze the data until the trial is complete — and even then, treatment codes are not unblinded until after the primary analysis is locked.

Design Summary

Design Element	Decision	Rationale
Population	Trainers registering to challenge Brock at Pewter Gym	Defined, accessible population
Sample size	n = 200 (100 per arm)	87% power to detect 15 HP MDE
Treatment	10 mg Pewter Protein tablet	Standardized dose and formulation
Control	Placebo Berry tablet	Indistinguishable from treatment
Randomization	Stratified by type-advantage (2 blocks)	Improves precision
Primary outcome	Total damage dealt (HP)	Continuous, high power
Secondary outcome	Beat Brock (yes/no)	Binary, practically meaningful
Blinding	Double-blind	Prevents expectation effects
Analysis plan	Pre-registered difference-in-means	Prevents p-hacking
Ethics	Approved by Kanto Pokemon League IRB	Equipoise: genuine uncertainty about Protein efficacy

Table 2.5: Summary of the Pewter Protein RCT design.

2.3 Estimands and Estimators

With the experiment designed and the data collected, we turn to the question of estimation. What exactly are we estimating, and how?

Notation at a Glance: The Symbols You'll See in This Section

The next few pages throw a lot of Greek letters at you. Here is what each one means, in plain English, before we use them in formulas.

Symbol	***Plain-English reading***
n_1, n_0	*How many trainers are in the treatment group (n_1) and the control group (n_0).*
$\bar{Y}_1, \bar{Y}_0$	*The group averages — the mean outcome in the treatment and control groups.*
$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$	*Our point estimate of the treatment effect: just the difference of the two group means. The hat (^) means "estimated from data."*
s_1^2, s_0^2	*The sample variance in each group. Measures how spread out the outcomes are.*
σ_1^2, σ_0^2	*The true (population) variances. Greek letters = truth; Latin letters with hats = estimates.*
$\widehat{SE}$	*The standard error of $\hat{\tau}$ — how much our estimate would wiggle if we re-ran the same experiment.*
α	*The significance level of a test (conventionally 0.05). "How often do we tolerate a false alarm?"*
$1 - \beta$	*The power of a test (conventionally 0.80). "How often do we catch a real effect?"*
$z_{\alpha/2}, z_\beta$	*Just lookup values from the normal curve — 1.96 and 0.842 for $\alpha = 0.05, \beta = 0.20$.*
$\mu_1 - \mu_0$	*The effect size we care about — the smallest difference worth detecting.*
ϱ	*Intra-cluster correlation: "how similar are units in the same cluster?"*
ϵ_i	*The unexplained leftover — the part of trainer i's outcome no variable in the model accounts for.*

The rule of thumb: Greek letters ($\mu, \sigma, \tau, \alpha, \beta$) are usually unknown truths. Latin letters with hats ($\hat{\tau}, \hat{\mu}, s$) are what we compute from data to estimate them.

Estimands: ATE, ATT, and ATC

Recall from Chapter 1 the three core causal estimands:

$$\text{ATE} = E[Y_i(1) - Y_i(0)]$$

$$\text{ATT} = E[Y_i(1) - Y_i(0) \mid D_i = 1]$$

$$\text{ATC} = E[Y_i(1) - Y_i(0) \mid D_i = 0]$$

In an observational study, these can differ — the effect of Protein may be different for trainers who choose to take it (ATT) vs. those who do not (ATC). But under random assignment, treatment status is independent of potential outcomes, so:

$$\text{ATT} = E[Y_i(1) - Y_i(0) \mid D_i = 1] = E[Y_i(1) - Y_i(0)] = \text{ATE}$$

and similarly $\text{ATC} = \text{ATE}$. All three estimands coincide. This is a major simplification: in an RCT, we don't need to worry about which estimand we're targeting. The simple difference in means estimates all of them.

Neyman's Repeated Sampling Framework

Jerzy Neyman, in his foundational 1923 paper (translated to English in 1990), provided the inferential framework that most applied researchers use today. The key idea: treat the observed data as one realization of all possible random assignments, and derive the sampling distribution of the estimator over these hypothetical repetitions.

For a completely randomized experiment with n_1 treated and n_0 control units, the difference-in-means estimator $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$ is unbiased (as shown in Section 2.1), and its variance under repeated sampling is:

$$\text{Var}(\hat{\tau}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0} - \frac{\sigma_\tau^2}{n}$$

where $\sigma_1^2 = \mathrm{Var}(Y_i(1))$, $\sigma_0^2 = \mathrm{Var}(Y_i(0))$, and $\sigma_\tau^2 = \mathrm{Var}(\tau_i) = \mathrm{Var}(Y_i(1) - Y_i(0))$ is the variance of the individual treatment effects.

The last term, σ_τ^2 / n, is non-negative and generally unknown (it depends on the joint distribution of $Y_i(1)$ and $Y_i(0)$, which we never observe). Dropping it gives us a **conservative** variance estimator:

$$\hat{V}[\hat{\tau}] = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}$$

where $s_1^2 = \frac{1}{n_1 - 1} \sum_{i:D_i=1} (Y_i - \bar{Y}_1)^2$ and $s_0^2 = \frac{1}{n_0 - 1} \sum_{i:D_i=0} (Y_i - \bar{Y}_0)^2$ are the sample variances in each group.

> ***Definition 2.5 (Neyman Variance Estimator).*** *The conservative estimator of the variance of $\hat{\tau}$ under complete randomization is:*
>
> $$\hat{V}[\hat{\tau}] = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}$$
>
> *This is conservative in the sense that $E[\hat{V}] \geq \mathrm{Var}(\hat{\tau})$, with equality when $\tau_i = \tau$ for all i (constant treatment effect). The standard error is $\hat{SE} = \sqrt{\hat{V}[\hat{\tau}]}$.*

This is precisely the familiar two-sample t-test variance formula. Neyman's contribution was showing that it is valid for inference about causal effects under the randomization distribution, without requiring any assumptions about a superpopulation or distributional form.

Confidence Intervals and Hypothesis Testing

With the point estimate and standard error in hand, we construct a 95% confidence interval:

$$\hat{\tau} \pm 1.96 \times \hat{SE}$$

and test the null hypothesis $H_0 : \tau = 0$ using the test statistic:

$$t = \frac{\hat{\tau}}{\hat{SE}}$$

Under the null, this is approximately standard normal for large samples (or follows a t-distribution with degrees of freedom given by the Welch-Satterthwaite approximation for finite samples).

Worked Example: The Pewter Protein Results

Nurse Joy collects the data. Here are the summary statistics:

Group	n	Mean damage (HP)	SD (HP)
Protein (treatment)	100	142.7	36.2
Placebo (control)	100	128.4	33.8

Step 1: Point estimate.

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0 = 142.7 - 128.4 = 14.3 \text{ HP}$$

Step 2: Standard error.

$$\hat{SE} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}} = \sqrt{\frac{36.2^2}{100} + \frac{33.8^2}{100}} = \sqrt{\frac{1310.44}{100} + \frac{1142.44}{100}} = \sqrt{13.10 + 11.42} = \sqrt{24.53} = 4.95$$

Step 3: 95% Confidence interval.

$$14.3 \pm 1.96 \times 4.95 = 14.3 \pm 9.7 = [4.6, 24.0]$$

Step 4: Test statistic and p-value.

$$t = \frac{14.3}{4.95} = 2.89$$

The two-sided p-value is $p = 0.004$. At $\alpha = 0.05$, we reject the null hypothesis of no treatment effect.

Interpretation: Pewter Protein increases the total damage dealt to Brock's Pokemon by an estimated 14.3 HP (95% CI: 4.6 to 24.0). This is statistically significant ($p = 0.004$) and practically meaningful — 14.3 HP is roughly equivalent to one additional Tackle attack, which could turn a close loss into a win.

For the secondary binary outcome (beat Brock), the treatment group win rate is 52% vs. 43% in the control group, a difference of 9 percentage points (95% CI: -1.8 to 19.8 pp, $p = 0.10$). The binary outcome is less precisely estimated because binary variables carry less information than continuous ones — a lesson in why the choice of outcome measure affects power.

Regression Adjustment: Lin (2013)

Can we do better than the simple difference in means? **Yes** — without sacrificing the unbiasedness guarantee of randomization.

Lin (2013) proposed the following regression specification for estimating treatment effects in experiments with covariate adjustment:

$$Y_i = \alpha + \tau D_i + \beta X_i + \gamma D_i(X_i - \bar{X}) + \epsilon_i$$

where X_i is a vector of pre-treatment covariates, centered by subtracting the overall mean $\bar{X}$. The OLS estimate of τ in this regression has two important properties:

1. It is **consistent** for the ATE regardless of whether the linear model is correctly specified.
2. It is **at least as precise** as the unadjusted difference-in-means (asymptotically), and strictly more precise when the covariates predict the outcome.

The interaction term $D_i(X_i - \bar{X})$ is critical — it allows the relationship between covariates and outcome to differ across treatment and control groups. Without it, misspecification of the covariate-outcome relationship can actually introduce bias (Freedman, 2008). Lin's specification is robust to this concern.

For the Pewter Protein trial, Nurse Joy includes the covariates from the balance table: trainer level, number of Pokemon, indicator for Water/Grass-type Pokemon, and prior gym badges. The regression-adjusted estimate is:

$$\hat{\tau}_{adj} = 14.8 \text{ HP}, \quad \hat{SE}_{adj} = 3.91$$

The point estimate barely changes (14.8 vs. 14.3 — reassuring, since large changes would suggest imbalance), but the standard error shrinks from 4.95 to 3.91, a 21% reduction. The 95% CI narrows to $[7.1, 22.5]$, and the p-value drops to $p = 0.0002$.

> ***Professor Oak Explains: Why Covariate Adjustment Helps but Doesn't Bias***
>
> *"Here is the key insight: in a randomized experiment, covariate adjustment cannot introduce bias (the covariates are balanced in expectation), but it can reduce variance by explaining outcome variation that is unrelated to treatment. Think of it as noise reduction. The type of Pokemon a trainer has strongly predicts damage dealt, but since randomization balanced this variable across groups, including it in the model just soaks up residual variation, leaving a cleaner estimate of the treatment effect."*

2.4 SUTVA: The Stable Unit Treatment Value Assumption

Every result derived so far — from the unbiasedness of difference-in-means to the validity of confidence intervals — rests on an assumption so fundamental that it often goes unstated. It is called **SUTVA**, and violating it can unravel the entire inferential framework.

> ***Definition 2.6 (Stable Unit Treatment Value Assumption — SUTVA).*** *The potential outcomes for unit i depend only on the treatment assigned to unit i, and not on the treatments assigned to other units. Furthermore, there is only one version of each treatment level. Formally:*
>
> 1. ***No interference:*** $Y_i(D_1, D_2, \ldots, D_n) = Y_i(D_i)$ *for all i. Unit i's outcome depends only on its own treatment, not on the treatment status of any other unit.*
> 2. ***No hidden variations of treatment:*** *For all i, $Y_i(1)$ is a single, well-defined value — there is only one "version" of treatment.*

No Interference

SUTVA's first component requires that one trainer's treatment status does not affect another trainer's outcome. In the Pewter Protein trial, this seems plausible at first glance — each trainer battles Brock independently.

But consider violations:

Sharing. What if trainers in the Pokemon Center lobby share their tablets? "Hey, I got two of these blue things — want one?" If a control trainer receives half a Protein tablet from a treated trainer, both of their potential outcomes are distorted. The control trainer's $Y_i(0)$ is no longer their true untreated outcome.

Information spillover. Suppose trainers wait in a common area before their battles. A trainer who just won on Protein might tell the next trainer, "The Protein works — use Ember on Geodude first, then switch to your Water-type for Onix." Now the control trainer's strategy has been influenced by the treated trainer's experience, even without receiving the treatment.

Brock's adaptation. If Brock notices that Protein trainers hit harder and adjusts his strategy (e.g., using more Potions), then later trainers face a different version of Brock — one whose behavior has been influenced by earlier trainers' treatment status.

To mitigate interference, Nurse Joy schedules trainers at staggered times, prevents them from interacting in the waiting area, and ensures Brock follows a fixed battle script.

No Hidden Variations of Treatment

SUTVA's second component requires that "Pewter Protein" means the same thing for every treated trainer. Violations include:

Storage degradation. Suppose some Protein tablets were stored in a hot stockroom and lost potency, while others were properly refrigerated. Now there are effectively two versions of treatment: full-strength Protein and degraded Protein. The potential outcome $Y_i(1)$ is no longer well-defined — it depends on *which* Protein the trainer received.

Dosing variation. If some trainers accidentally receive 8 mg instead of 10 mg (e.g., tablets from a faulty production batch), treatment is not uniform across the treated group.

Trading. In perhaps the most colorful SUTVA violation imaginable, suppose a treated and control trainer meet in the Pokemon Center lobby, discover their assignments, and *swap tablets*. The treated trainer now has the Placebo, the control trainer has the Protein, and the recorded treatment assignments are wrong for both. This is a form of noncompliance that simultaneously violates both SUTVA components.

Professor Oak addresses these threats through strict standardization: all Protein tablets come from the same lot, stored under controlled conditions, administered by Nurse Joy personally. Trainers are instructed not to share, trade, or discuss their tablets. The protocol isn't perfect — no protocol ever is — but it minimizes SUTVA violations to a manageable level.

> ***Blue's Mistake: Ignoring Interference***
> *Blue, of course, doesn't care about SUTVA. He shares his Protein with three friends, tells everyone in the Pokemon Center about his strategy for beating Brock, and loudly announces which trainers are in the treatment group. "Experiments are for nerds — I'm just helping people out!" Every violation of SUTVA that Nurse Joy carefully prevented, Blue happily introduces.*

2.5 Internal vs. External Validity

An experiment can succeed in answering the question it asks (high internal validity) while failing to answer the question we actually care about (low external validity). These two dimensions of validity are often in tension, and understanding the tradeoff is essential for interpreting experimental results.

Internal Validity

Internal validity asks: did the experiment correctly identify the causal effect of Pewter Protein *in this study population, under these conditions?*

Threats to internal validity include:

Noncompliance. Some trainers assigned to Protein refuse to take the tablet (perhaps they're suspicious of supplements). Some trainers assigned to Placebo obtain Protein elsewhere (perhaps from Blue, who is handing them out in the lobby). If we analyze based on what trainers *actually took* rather than what they were *assigned*, we reintroduce selection bias — the very problem randomization was designed to solve. We address this in Section 2.7.

Attrition. Trainers who lose to Brock badly — their entire team fainted in three turns — may be too embarrassed to report back to Nurse Joy. If attrition is related to treatment status (e.g., Protein trainers who lose feel worse because they "should have" won), the remaining sample is no longer balanced, and our estimate is biased. Formally, attrition means our observed outcome Y_i is missing for some units, and if missingness depends on potential outcomes, we have a problem.

Contamination. Control group trainers learn about Protein from treated trainers and modify their behavior — perhaps they train harder to compensate for not having the supplement. This dilutes the treatment effect toward zero.

Hawthorne effect. Trainers in both groups may battle harder simply because they know they're in a study. If the Hawthorne effect is equal across groups, it doesn't bias the treatment effect estimate (both groups are equally affected). But if being told "you might have received a powerful supplement" motivates Protein trainers more than Placebo trainers — a form of placebo effect — the estimated effect is a composite of the pharmacological effect of Protein and the psychological effect of believing you have an advantage. Double-blinding mitigates this.

External Validity

External validity asks: does the result generalize beyond this specific experiment?

Nurse Joy has convincingly shown that Pewter Protein boosts damage against Brock's Rock-type Pokemon in Pewter City. But consider:

Different gym types. Brock uses Geodude and Onix, both Rock/Ground types with specific weaknesses. Pewter Protein might boost physical Attack stats, which is exactly what you need against Rock-types with high Defense but exploitable weaknesses. Against Misty's Water-types in Cerulean City, the optimal strategy might rely on Special Attack instead. Would Protein help there?

Different populations. Pewter City trainers are typically early in their journey — low-level Pokemon, small teams, limited battle experience. Would Protein have the same effect on experienced trainers with Level 50 Pokemon challenging the Elite Four? There might be ceiling effects (experienced Pokemon already hit the damage cap) or floor effects (novice Pokemon can't absorb the Protein's benefits).

Different contexts. The Pewter Gym is an indoor arena with controlled conditions. Would Protein work the same in a wild Pokemon encounter, where terrain, weather, and surprise play a role? What about double battles, where strategy and team synergy matter more than individual stats?

Dosing and timing. The trial tested exactly 10 mg, 60 minutes before battle. Would 5 mg work? Would 20 mg work better, or cause side effects? What about taking it 2 hours before, or 10 minutes before?

The Tradeoff

Internal and external validity often pull in opposite directions. The tighter you control an experiment — standardized dose, controlled setting, homogeneous population, strict protocol — the higher the internal validity but the narrower the generalizability.

A loose, "pragmatic" trial that enrolled all trainers across Kanto, let them take Protein whenever they wanted, and measured outcomes across all gym types would have higher external validity but would sacrifice internal validity to noncompliance, interference, and treatment variation.

The standard scientific approach is to establish internal validity first — prove the effect exists under controlled conditions — and then investigate generalizability through **replication** across different contexts. Nurse Joy's Pewter City trial is the first step. Future trials in Cerulean City, Vermilion City, and beyond will map the boundaries of the effect.

> ***Professor Oak Explains: Transportability***
> *"The question of whether a causal effect 'transports' from one setting to another is not just a matter of hand-waving. Pearl and Bareinboim (2014) developed a formal theory of* ***transportability****, which uses DAGs to identify exactly which conditions must hold for an experimental result to generalize from a source population to a target population. We'll touch on this in Chapter 8. For now, the lesson is: be precise about what population your experiment represents."*

2.6 Randomization Inference

Everything in Section 2.3 relied on Neyman's framework, which treats the observed data as a draw from a hypothetical superpopulation and derives the sampling distribution over repeated experiments. There is an older, and in some ways more elegant, approach to inference in randomized experiments: **Fisher's randomization inference** (also called **permutation inference**).

Fisher's Sharp Null Hypothesis

> ***Definition 2.7 (Fisher's Sharp Null).*** *The sharp null hypothesis of no treatment effect states:*
>
> $$H_0^F : Y_i(1) = Y_i(0) \quad \text{for all } i = 1, \dots, n$$
>
> *Under this hypothesis, the treatment has absolutely no effect on any unit. This is "sharp" because it specifies both potential outcomes for every unit, not just a population average.*

The sharp null is stronger than Neyman's null ($H_0^N : E[Y_i(1)] - E[Y_i(0)] = 0$), which allows individual effects to be nonzero as long as they average to zero. Under the sharp null, every individual treatment effect is exactly zero: $\tau_i = 0$ for all i.

The Permutation Distribution

Here is the key insight that makes Fisher's approach so powerful: **under the sharp null, all potential outcomes are observed.** If $Y_i(1) = Y_i(0)$ for every trainer, then regardless of whether trainer i was assigned to treatment or control, they would have produced the same outcome. The observed outcome Y_i equals both $Y_i(1)$ and $Y_i(0)$.

This means we can compute what the test statistic *would have been* under every possible random assignment, because the outcomes are fixed — only the assignment changes. The collection of these hypothetical test statistics is the **randomization distribution** (or permutation distribution).

The algorithm is:

1. Compute the observed test statistic T^{obs} (e.g., the difference in means) from the actual data.
2. List all $\binom{n}{n_1}$ possible ways to assign n_1 units to treatment from n total.
3. For each possible assignment $d^{(j)}$, compute the test statistic $T^{(j)}$ using the *same outcomes* but the *reassigned treatment labels*.
4. The **Fisher exact p-value** is the fraction of permutations where the test statistic is at least as extreme as the observed value:

$$p_{\text{Fisher}} = \frac{\#\{j : |T^{(j)}| \geq |T^{obs}|\}}{\binom{n}{n_1}}$$

For large experiments, enumerating all $\binom{n}{n_1}$ permutations is computationally infeasible ($\binom{200}{100} \approx 9 \times 10^{58}$). In practice, we approximate the permutation distribution by drawing a large number of random permutations (e.g., 10,000 or 100,000).

Worked Example: A Small Pewter Protein Trial

To illustrate the full permutation distribution, consider a miniature version of the trial with $n = 10$ trainers, $n_1 = 5$ assigned to Protein and $n_0 = 5$ to Placebo.

Trainer	Treatment (D_i)	Damage Dealt (Y_i)
Ash	1 (Protein)	155
Misty	1 (Protein)	180
Brock Jr.	1 (Protein)	130
Gary	1 (Protein)	145
Leaf	1 (Protein)	160
Jessie	0 (Placebo)	120
James	0 (Placebo)	135
Ritchie	0 (Placebo)	110
Duplica	0 (Placebo)	140
Todd	0 (Placebo)	125

Observed test statistic:

$$T^{obs} = \bar{Y}_1 - \bar{Y}_0 = \frac{155 + 180 + 130 + 145 + 160}{5} - \frac{120 + 135 + 110 + 140 + 125}{5} = 154 - 126 = 28$$

Under the sharp null, every trainer's outcome is fixed regardless of assignment. There are $\binom{10}{5} = 252$ possible assignments. For each, we compute the difference in means between the (hypothetically) treated and control groups.

For instance, one alternative assignment might place {Ash, Misty, Jessie, James, Ritchie} in treatment and {Brock Jr., Gary, Leaf, Duplica, Todd} in control. The test statistic would be:

$$T = \frac{155 + 180 + 120 + 135 + 110}{5} - \frac{130 + 145 + 160 + 140 + 125}{5} = 140 - 140 = 0$$

We compute all 252 such statistics and form the permutation distribution.

> ***Figure 2.1 (Description):*** *A histogram of the 252 values of the difference-in-means test statistic under all possible random assignments of 5 out of 10 trainers to treatment. The distribution is approximately bell-shaped, centered at zero, with most values between -30 and +30. The observed statistic of $T^{obs} = 28$ is marked with a dashed red vertical line in the right tail. Very few permutations yield a test statistic as large or larger than 28 in absolute value, suggesting the result is unlikely under the sharp null.*

Computing the exact Fisher p-value: of the 252 permutations, suppose 18 yield $|T^{(j)}| \geq 28$. Then:

$$p_{\text{Fisher}} = \frac{18}{252} = 0.071$$

At $\alpha = 0.05$, we would not reject the sharp null in this small sample — but the result is suggestive, and the full $n = 200$ trial (Section 2.3) confirms the effect.

Fisher vs. Neyman

The Fisher and Neyman frameworks differ in philosophy and practice:

Feature	Fisher (Randomization Inference)	Neyman (Repeated Sampling)
Null hypothesis	Sharp: $Y_i(1) = Y_i(0)$ for all i	Weak: $E[Y_i(1)] - E[Y_i(0)] = 0$
Inference basis	Permutation distribution over all possible random assignments	Sampling distribution over hypothetical repetitions
Assumption	SUTVA + random assignment	SUTVA + random assignment
Requires	No distributional assumptions	Large-sample normality (for t-test)
Focus	Testing (p-value)	Estimation (point estimate + CI)
Finite sample	Exact p-values	Approximate (conservative)

Table 2.6: Comparison of Fisher and Neyman inferential frameworks.

In practice, both approaches typically yield similar conclusions. Neyman's framework is more useful for estimation (it provides point estimates and confidence intervals), while Fisher's is attractive for exact inference in small samples and for its elegant conceptual simplicity. Modern applied work often uses Neyman's framework for primary analysis and reports randomization inference as a robustness check.

> ***Professor Oak Explains: The Intellectual Rivalry***
> *"Fisher and Neyman famously disagreed about the foundations of statistical inference — their debates in the 1930s were legendary and sometimes acrimonious. Fisher emphasized testing and fiducial probability; Neyman emphasized estimation and confidence intervals. Both contributed indispensable ideas to experimental statistics, and modern practice draws from both traditions. In causal inference, Rubin (1974) and Holland (1986) synthesized these traditions within the potential outcomes framework. For the definitive modern treatment, see Imbens and Rubin (2015)."*

2.7 Practical Complications

The theoretical elegance of randomized experiments meets the messy reality of the Pokemon world. In this section, we discuss the most common practical complications and how to handle them.

Noncompliance

In an ideal experiment, every trainer takes the assigned treatment. In reality, compliance is imperfect.

One-sided noncompliance: Some trainers assigned to Protein refuse to take the tablet. Perhaps they're skeptical, or they feel sick that day, or they're philosophically opposed to supplements. But no control trainer can access Protein (because it's not available outside the trial). This is "one-sided" noncompliance — deviation occurs only in the treatment group.

Two-sided noncompliance: In the Pewter Protein trial, two-sided noncompliance arises because Blue is handing out Protein in the lobby. Some control trainers sneak Protein, and some treatment trainers — perhaps influenced by Blue's skeptics — throw their tablets away. Now compliance is imperfect in both directions.

The temptation is to analyze based on **actual treatment received** rather than *assigned* treatment. This is a grave error — it reintroduces selection bias. Trainers who comply with their assignment may differ systematically from those who don't. If we compare "Protein takers" to "non-takers" regardless of assignment, we are back in the observational world.

Intent-to-Treat Analysis

The standard solution is **Intent-to-Treat (ITT)** analysis: analyze every trainer according to their *assigned* treatment, regardless of compliance.

> ***Definition 2.8 (Intent-to-Treat).*** *The ITT estimand is the causal effect of being assigned to treatment, regardless of whether the treatment was actually received:*
>
> $$\text{ITT} = E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0]$$
>
> *where* Z_i *is the treatment assignment (instrument) and* D_i *is the actual treatment received.*

The ITT preserves the integrity of randomization — since assignment Z_i is random, the ITT comparison is unbiased for the effect of *assignment*. However, it generally underestimates the effect of *treatment* itself, because some assigned trainers didn't actually take Protein. The ITT is sometimes called a "diluted" treatment effect.

In the Pewter Protein trial, suppose 12 of 100 trainers assigned to Protein refused to take it, and 5 of 100 control trainers obtained Protein from Blue. The ITT analysis still compares the 100 assigned-Protein trainers (88 of whom actually took it) against the 100 assigned-Placebo trainers (5 of whom actually took Protein). The ITT estimate will be attenuated relative to the true effect of actually consuming Protein.

Preview: Instrumental Variables and LATE

Can we recover the effect of actually taking Protein, despite noncompliance? **Yes** — using the randomized assignment as an **instrumental variable**. This is the LATE (Local Average Treatment Effect) framework of Imbens and Angrist (1994):

$$\text{LATE} = \frac{\text{ITT}}{\text{Compliance Rate}} = \frac{E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0]}{E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0]}$$

This ratio — known as the **Wald estimator** — identifies the causal effect of Protein for **compliers**, the subpopulation of trainers who take Protein when assigned to it and don't take it when assigned to control. We will develop this machinery fully in Chapter 6 (Fuchsia City & Cinnabar Island: Instrumental Variables).

Attrition

When trainers drop out of the study before outcomes are recorded, the remaining sample may no longer be balanced — even if the original randomization was perfect.

In the Pewter Protein trial, suppose 8 trainers in the Placebo group who lost badly to Brock leave without reporting their results, while only 2 trainers in the Protein group do the same. The remaining data overrepresent trainers who performed well (especially in the control group), biasing the estimated treatment effect downward.

When attrition rates differ across groups, **Lee (2009) bounds** provide a principled approach: trim the group with lower attrition to match the attrition rate of the other group, and report best-case and worst-case effect estimates. If even the worst-case bound excludes zero, the result is robust to attrition.

In practice, the best defense is prevention: Nurse Joy stations an assistant at the gym exit to record outcomes for *every* trainer, win or lose.

Hawthorne Effect

The Hawthorne effect — named after a famous series of experiments at the Hawthorne Works factory in the 1920s — refers to behavior changes caused by the awareness of being observed or studied, rather than by the treatment itself.

In the Pewter Protein trial, trainers who know they're in a study might battle more carefully, use items more strategically, or try harder than they normally would. If this elevated effort is symmetric across treatment and control (both groups try equally hard), it inflates *both* group means but doesn't bias the treatment-control comparison. The estimated effect of Protein remains valid, but it applies to a population of trainers who are "trying their hardest" — which may differ from real-world conditions where trainers battle casually.

If the Hawthorne effect is asymmetric — for example, trainers in the Protein group try harder because they believe they have an advantage — it becomes a confound. Double-blinding is the primary defense.

Ethical Considerations

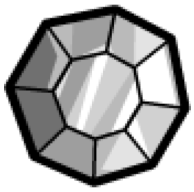
Is it ethical to withhold a potentially beneficial treatment from control trainers? The Pewter Protein trial is justifiable because of **clinical equipoise**: before the trial, there is genuine uncertainty about whether Protein works. If we already knew it worked, randomizing some trainers to Placebo would be denying them a benefit. But we don't know — that's why we're running the experiment.

Additional ethical safeguards in the Pewter Protein RCT:

- **Informed consent:** Trainers are told they will receive *either* Protein or Placebo, and that the goal is to determine whether Protein works. They consent to this uncertainty.
- **No harm:** The Placebo Berry is inert; it doesn't make anyone worse off than they would be without the study. Losing to Brock is disappointing but not dangerous — Nurse Joy heals all Pokemon for free.

- **Data monitoring:** If interim results showed that Protein was causing serious Pokemon health issues, the trial would be stopped immediately.
- **Post-trial access:** After the trial concludes, if Protein is shown to be effective, all control trainers are offered free Protein before their next gym challenge.

Chapter Summary



Boulder Badge earned!

This chapter has covered the theory and practice of randomized controlled trials, the gold standard of causal inference. Here are the key takeaways:

- **Randomization creates independence** between treatment assignment and potential outcomes: $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i$. This eliminates selection bias and balances both observed and unobserved confounders in expectation.
- **The difference-in-means estimator** $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$ is **unbiased** for the ATE under random assignment. No modeling assumptions are required.
- **Experimental design** matters enormously: precise treatment definition, outcome selection, power analysis, stratified randomization, and blinding all improve the quality of the experiment.
- **Under randomization, ATE = ATT = ATC.** The distinction between estimands, which will be crucial in observational settings, collapses in an experiment.
- **Neyman's framework** provides variance estimation ($\hat{V}[\hat{\tau}] = s_1^2/n_1 + s_0^2/n_0$), confidence intervals, and hypothesis tests based on repeated sampling.
- **Regression adjustment** (Lin, 2013) can improve precision by absorbing residual outcome variation, without introducing bias, as long as the specification includes treatment-covariate interactions.
- **SUTVA** — the Stable Unit Treatment Value Assumption — requires no interference between units and no hidden treatment variations. Violations undermine the entire potential outcomes framework.
- **Internal validity** (did the experiment work as designed?) and **external validity** (do the results generalize?) are both important but often trade off against each other.
- **Randomization inference** (Fisher) provides exact p-values by computing the test statistic under all possible random assignments, exploiting the sharp null to treat all potential outcomes as known.
- **Practical complications** — noncompliance, attrition, Hawthorne effects — require careful handling. The ITT analysis preserves the benefits of randomization; IV/LATE (Chapter 6) recovers the effect on compliers.

Professor Oak's Review Questions

1. **Conceptual.** Explain in your own words why randomization balances unobserved confounders. Why can't any observational method achieve this?
2. **Mathematical.** In the Pewter Protein RCT, the difference-in-means estimate is $\hat{\tau} = 14.3$ HP with $\hat{SE} = 4.95$. Suppose we had used a one-sided test ($H_A : \tau > 0$) instead of a two-sided test. What would the p-value be? Would your conclusion change?
3. **Design.** Nurse Joy is planning a follow-up trial to test whether Pewter Protein also works against Misty's Water-type Pokemon. She expects a smaller effect (MDE = 10 HP) because Attack boosts are less relevant against Water-types. How many trainers does she need per group, assuming the same variance ($\sigma = 35$) and 80% power?
4. **SUTVA.** Give a realistic example of a SUTVA violation in the Pewter Protein trial that is NOT mentioned in this chapter. Explain how it would bias the estimated treatment effect and propose a solution.

5. **Fisher vs. Neyman.** In the small-sample worked example (Section 2.6), the Fisher p-value was 0.071 and we did not reject at $\alpha = 0.05$. Using Neyman's framework on the same 10 trainers, compute the difference-in-means, standard error, and 95% confidence interval. Does the confidence interval include zero? Are the two frameworks telling the same story?
6. **External Validity.** A researcher reads about the Pewter Protein trial and concludes: "Pewter Protein works. We should distribute it to all trainers in Kanto before every gym battle." Identify at least three reasons why this conclusion may be premature.

Trainer Challenge Exercises

Exercise 2.1: Balance Check Simulation

Using a programming language of your choice (Python or R recommended), simulate a completely randomized experiment with $n = 200$ trainers. Generate 10 pre-treatment covariates from standard normal distributions (independent of treatment). Assign 100 trainers to treatment and 100 to control. For each covariate, compute the two-sample t-test p-value. Repeat the entire experiment 1,000 times. What fraction of simulations have at least one covariate with $p < 0.05$? What does this tell you about interpreting balance tables?

Exercise 2.2: Power Analysis

A researcher at the Celadon City Game Corner wants to test whether a new TM (Technical Machine) improves Pokemon battle performance. She expects the TM to increase damage by 8 HP, with standard deviations of 25 HP in both groups. Using the formula from Section 2.2, compute the required sample size per group for: (a) 80% power, (b) 90% power, (c) 95% power. Plot the required sample size as a function of power for $1 - \beta \in [0.5, 0.99]$. What happens to sample size requirements as you demand higher power?

Exercise 2.3: Randomization Inference by Hand

Consider the following data from 8 trainers in a mini-experiment:

Trainer	Assignment	Damage
A	Protein	150
B	Protein	130
C	Protein	170
D	Protein	140
E	Placebo	120
F	Placebo	110
G	Placebo	145
H	Placebo	130

a. Compute the observed difference-in-means test statistic.
b. Under the sharp null, enumerate all $\binom{8}{4} = 70$ possible assignments of 4 trainers to treatment. For each, compute the difference-in-means.
c. Plot the permutation distribution and mark the observed statistic.
d. Compute the two-sided Fisher exact p-value. What do you conclude?

Exercise 2.4: Noncompliance and ITT vs. Per-Protocol

In the Pewter Protein trial, suppose the following compliance data are observed:

Assigned Group	Actually Took Protein	Actually Took Placebo	n
Protein	85	15	100
Placebo	8	92	100

The mean outcomes are: Protein group (as assigned) = 142.7 HP, Placebo group (as assigned) = 128.4 HP, Actual Protein takers (regardless of assignment) = 144.2 HP, Actual Placebo takers (regardless of assignment) = 126.8 HP.

a. Compute the ITT estimate and explain what it estimates.
b. Compute the naive "per-protocol" estimate (comparing actual takers vs. non-takers) and explain why it may be biased.

c. Using the Wald estimator, compute the LATE. What population does this estimate apply to?

d. Why is the LATE larger in magnitude than the ITT?

Further Reading

- **Fisher, R.A.** (1935). *The Design of Experiments*. Edinburgh: Oliver and Boyd. The foundational text on randomized experiments and permutation inference.
- **Neyman, J.** (1923/1990). "On the Application of Probability Theory to Agricultural Experiments." Translated and republished in *Statistical Science*, 5(4), 465-472. The original articulation of potential outcomes and the repeated-sampling framework for experimental inference.
- **Rubin, D.B.** (1974). "Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies." *Journal of Educational Psychology*, 66(5), 688-701. The paper that formalized the potential outcomes framework and connected it to randomization.
- **Holland, P.W.** (1986). "Statistics and Causal Inference." *Journal of the American Statistical Association*, 81(396), 945-960. The classic paper stating the "fundamental problem of causal inference" and surveying the Rubin causal model.
- **Lin, W.** (2013). "Agnostic Notes on Regression Adjustments to Experimental Data: Reexamining Freedman's Critique." *Annals of Applied Statistics*, 7(1), 295-318. The definitive treatment of regression adjustment in randomized experiments.
- **Gerber, A.S. & Green, D.P.** (2012). *Field Experiments: Design, Analysis, and Interpretation*. New York: W.W. Norton. An excellent applied textbook on experimental design with extensive coverage of randomization inference, blocking, and noncompliance.
- **Athey, S. & Imbens, G.W.** (2017). "The Econometrics of Randomized Experiments." In *Handbook of Economic Field Experiments*, Vol. 1, 73-140. A modern survey of the econometric theory of randomized experiments, covering design, analysis, and extensions.
- **Imbens, G.W. & Rubin, D.B.** (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences*. Cambridge University Press. The comprehensive graduate-level reference for the potential outcomes approach to causal inference.

Skills to Practice in the Notebook

The notebook `notebooks/ch02_pewter_city.ipynb` is where RCT theory becomes a workable analysis pipeline. Before you claim the Boulder Badge, you should be able to do each of the following end-to-end in code:

1. **Simulate a completely randomized experiment.** Given n trainers and known potential outcomes, flip a coin to assign treatment, compute $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$, and verify empirically that averaging $\hat{\tau}$ over many randomizations recovers the true ATE. This is the core "randomization makes naive estimators unbiased" demo.
2. **Build a balance table.** Using the Pewter Protein dataset (or a simulated one), compute the standardized mean difference (or two-sample t-test) for each pre-treatment covariate across treatment and control. Know what "balanced" looks like and be able to flag a covariate that is not.
3. **Run a power analysis.** Given α, desired power $1 - \beta$, outcome standard deviation σ, and minimum detectable effect (MDE), compute the required sample size per group. Then flip it around: given a fixed n, what MDE can you detect? Both directions matter.
4. **Compute a Neyman standard error and CI by hand (in code).** From raw group means and sample variances, build $\hat{SE}$, a 95% confidence interval, and the two-sided p-value. Do *not* rely on a canned `ttest_ind` call — write the formula yourself at least once. Then cross-check against `scipy.stats.ttest_ind`.
5. **Do a Lin (2013) regression adjustment.** Fit `Y ~ D + (X - mean(X)) + D:(X - mean(X))`. Confirm the treatment coefficient is close to the unadjusted estimate but the standard error shrinks when X predicts Y. Be able to explain in one sentence why this adjustment doesn't bias the estimate.
6. **Run Fisher randomization inference.** For a small dataset, enumerate (or sample) the permutation distribution of $\hat{\tau}$ under the sharp null, compute the exact p-value, and plot the distribution with the observed statistic marked.
7. **Complete the Trainer Challenge Exercises.**
    - **Exercise 2.1:** Balance-check simulation — what fraction of 1,000 random experiments show at least one covariate with $p < 0.05$ by chance?
    - **Exercise 2.2:** Power curve — plot sample size vs. power for $1 - \beta \in [0.5, 0.99]$.
    - **Exercise 2.3:** Randomization inference by hand on the 8-trainer mini-dataset.
    - **Exercise 2.4:** ITT vs. per-protocol vs. LATE on the non-compliance table.

By the end, “run an RCT” should feel like 30 lines of pandas, not a mystery.

Check Your Understanding

Walk through these before challenging Brock. If anything feels shaky, revisit that section — the rest of the book leans hard on Chapter 2’s intuition about randomization.

Questions you should be able to answer out loud, without notes:

- Why does random assignment make the naive difference-in-means an *unbiased* estimate of the ATE? Which term in the selection-bias decomposition does it kill?
- State the independence condition $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i$ and explain what it means in plain English about “the type of trainer who is treated.”
- Why does randomization balance unobserved confounders, not just observed ones? Can any purely observational method do the same?
- Write the Neyman variance estimator $\hat{V}[\hat{\tau}] = s_1^2/n_1 + s_0^2/n_0$ and explain why it is *conservative* (i.e., why the “true” variance is generally smaller).
- What is the difference between σ_1^2, s_1^2, and $\hat{SE}$? When do you use each?
- In an RCT, why do ATE, ATT, and ATC all coincide?
- What is statistical power, and what four inputs determine it? If you double the MDE, what happens to the required sample size?
- Why does stratified (block) randomization reduce variance? When does it *not* help?
- Explain SUTVA in two bullet points (no interference, no hidden variations) and give a concrete Pewter-Protein way each part could fail.
- Compare Fisher’s sharp null and Neyman’s average-effect null. Which test assumes constant effects across units? Which one gives exact p-values in finite samples?
- What does ITT estimate? What does per-protocol estimate? Why is the per-protocol estimate biased under noncompliance?

Tasks you should be able to perform in code:

- Randomly assign treatment to n units and compute $\hat{\tau}$, $\hat{SE}$, 95% CI, and a two-sided p-value — using both your own formula and `scipy`.
- Build a balance table showing the standardized mean difference for every covariate in a dataset.
- Compute required sample size from $(\alpha, 1-\beta, \sigma, \text{MDE})$ and plot the sample-size–vs–power curve.
- Fit a Lin (2013) regression with centered covariates and the treatment-covariate interaction, and compare its SE to the unadjusted SE.
- Run a permutation test: enumerate (or sample) treatment assignments, compute $\hat{\tau}$ under each, and compute the exact Fisher p-value.
- Given a noncompliance table (assigned vs. took), compute the ITT estimate and the Wald/LATE estimate.

If you can explain the answers *and* run the code without a cheat-sheet, you have truly earned the Boulder Badge.

Badge Earned

> ***You’ve earned the Boulder Badge!*** *You now understand how randomization eliminates confounding, why the difference-in-means estimator is unbiased under random assignment, and how to design, analyze, and interpret a randomized controlled trial. You’ve seen both Neyman’s repeated sampling framework and Fisher’s randomization inference, and you know how to handle the practical complications — noncompliance, attrition, and SUTVA violations — that arise in real experiments. Brock would be proud.*

Next: Route 3

The path east from Pewter City winds through Route 3 and the dark tunnels of Mt. Moon. On the other side lies Cerulean City, where the Water-type gym leader Misty has been collecting mountains of observational battle data. Unlike Nurse Joy, Misty has no ability to randomize — she can only observe trainers as they come. How can she draw causal conclusions from this messy, non-experimental data?

The answer involves directed acyclic graphs, the backdoor criterion, and the art of conditioning. Turn the page to Chapter 3: Cerulean City — Observational Studies & DAGs.

Chapter 3: Cerulean City — Observational Studies & Graphical Models

Misty, Cerulean Gym Leader

> *"Water is patient. It doesn't force its way through rock — it finds the path that already exists." — Misty, Cerulean City Gym Leader*

You step into Cerulean City with a fresh Boulder Badge pinned to your trainer card and the confidence of someone who has run a proper randomized experiment. In Pewter City, you learned that randomization is the gold standard — it balances confounders, observed and unobserved, and lets you estimate causal effects with clean authority. But as you walk past the bike shop and toward the city's famous gym, a question begins to nag at you: what happens when you *can't* randomize?

Cerulean City is built around water — the Cerulean Cave to the north, the cape where Bill's lighthouse stands to the east, and the gym where Misty trains her Water-type team at the city's heart. Water, like observational data, flows where it will. You cannot command it into neat experimental channels. You must learn to read its currents.

Misty has a problem. Cerulean Cave, the treacherous cavern north of the city, has long been considered an elite training ground. Trainers who survive its depths seem to emerge stronger — they win more gym badges, perform better in league tournaments, and command higher-level Pokemon. But Misty suspects the relationship isn't so simple. She has years of battle logs from 2,000 trainers who passed through Cerulean City, and she wants to know: **does training in Cerulean Cave actually make trainers better, or do better trainers simply choose to enter the Cave?**

She can't run an experiment — she can't force trainers into the Cave or forbid them from entering. She needs to learn from the data she has. And for that, she needs you to understand observational studies, the biases that haunt them, and the graphical tools that can — sometimes — rescue causal inference from observational chaos.

3.1 Why Observational Studies Are Necessary

The Limits of Experimentation

In Chapter 2, we established that randomized controlled trials (RCTs) are the gold standard for causal inference. If you can randomize treatment assignment, you eliminate confounding by design. The Pewter City Protein experiment was clean: trainers were randomly assigned to receive Protein supplements or placebo, and we measured the effect on battle performance. Beautiful.

But the world — even the Pokemon world — is not a laboratory. Three forces conspire to make experimentation impossible in many settings:

Ethics. Could we randomly assign trainers to enter Cerulean Cave, knowing that some might lose their Pokemon to the powerful wild encounters inside? Could we randomly withhold medical treatment from injured Pokemon to measure its effect? The Kanto Pokemon Ethics Board would never approve such studies. In the human world, we cannot randomly assign people to smoke, to experience poverty, or to receive inferior education. Many of the most important causal questions involve treatments that would be unethical to assign.

Cost and feasibility. Running an RCT requires resources — you need to recruit participants, implement randomization, maintain compliance, and follow up over time. Misty would need to intercept trainers arriving in Cerulean City, randomly assign half to train in the Cave (and somehow enforce this), and then track their subsequent journey across Kanto. The logistical cost would be enormous. Many policy-relevant questions involve treatments that operate at scale (city-level policies, national regulations) where randomization is simply infeasible.

Timeliness. Misty has data *now*. She has five years of battle logs sitting in the Cerulean Gym's Pokedex database. An RCT would take another year to design, implement, and analyze. By then, the League season will be over. In many real-world settings, decisions must be made with the data available, not the data we wish we had.

Misty's Dataset

Misty's observational dataset, `kanto_trainers.csv`, contains records on $n = 2{,}000$ trainers who passed through Cerulean City over the past five years. For each trainer i, she observes:

Variable	Description	Type
`trainer_id`	Unique identifier	ID
`cave_training` (D_i)	Whether trainer trained in Cerulean Cave	Binary (0/1)
`badges_earned` (Y_i)	Total gym badges earned in Kanto	Count (0-8)
`experience_level`	Pre-existing trainer experience (years)	Continuous
`starter_type`	Starter Pokemon type (Fire/Water/Grass)	Categorical
`wealth`	Trainer's family wealth (Pokedollars)	Continuous
`strategy_score`	Measured battle strategy proficiency	Continuous
`natural_talent`	Unobserved innate ability	*Unobserved*

A naive comparison of means yields:

$$\bar{Y}_{\text{cave}} - \bar{Y}_{\text{no cave}} = 5.8 - 3.2 = 2.6 \text{ badges}$$

Trainers who entered the Cave earned, on average, 2.6 more badges than those who did not. Should Misty conclude that Cave training causes a 2.6-badge improvement?

Of course not. You already sense the problem. Trainers who *chose* to enter Cerulean Cave are not a random sample — they tend to be more experienced, wealthier (they can afford supplies for the dangerous cave), and more naturally talented. The comparison is contaminated by **confounding**.

The Promise and Peril of Observational Data

Observational data is abundant and cheap. Every gym in Kanto logs battle outcomes. The Pokemon League maintains records on every registered trainer. Nurse Joy's healing stations track Pokemon injuries across the region. This data exists whether or not anyone designs a study.

The promise: if we can properly account for confounding, observational data can answer causal questions that experiments never could. The peril: if we fail to account for confounding, observational studies will mislead us — and they will mislead us with the false confidence of large sample sizes.

The tools we develop in this chapter — directed acyclic graphs, d-separation, the backdoor criterion, and the do-calculus — are the trainer's guide to navigating observational data. They tell us *when* we can extract causal answers from observational data, *what* we need to condition on, and *when* the data simply cannot answer our question no matter how cleverly we analyze it.

Let us begin with the central threat: confounding.

3.2 Confounding (Formal Treatment)

The Intuition

You're standing outside the Cerulean Gym when Blue strolls up, his Pidgeot preening on his shoulder. "Cave training is the best thing I ever did," he declares. "I trained in Cerulean Cave for two weeks, and look — seven badges already."

You ask: "But Blue, you've been training Pokemon since you were five years old. Your grandfather is Professor Oak. You had a head start on every trainer in Kanto. How do you know it was the Cave and not your experience?"

Blue waves his hand dismissively. "Details."

Blue's mistake is the most fundamental error in causal reasoning: **confounding**. A confounder is a variable that influences both the treatment and the outcome, creating a spurious association that masquerades as a causal effect.

Formal Definition

> ***Definition 3.1 (Confounder — Informal).*** *A variable Z is a confounder of the relationship between treatment D and outcome Y if Z is a common cause of both D and Y. That is, Z causally affects D and Z causally affects Y, creating a non-causal path $D \leftarrow Z \rightarrow Y$.*

We will refine this definition using DAGs in Section 3.5. For now, the intuition suffices: a confounder is a "back door" through which association flows between D and Y without any causal mechanism from D to Y.

In Misty's data, **trainer experience** is a confounder. More experienced trainers are more likely to enter the Cave (they're better prepared for its dangers), and more experienced trainers earn more badges (regardless of whether they enter the Cave). The causal structure is:

$$\text{Experience} \rightarrow \text{Cave Training}$$

$$\text{Experience} \rightarrow \text{Badges Earned}$$

This creates a spurious association between Cave Training and Badges Earned that has nothing to do with the Cave itself.

The Confounding Bias Formula

Let us formalize the bias. Consider the linear model:

$$Y_i = \alpha + \tau D_i + \gamma Z_i + \varepsilon_i \tag{3.1}$$

where Y_i is badges earned, D_i is cave training (binary), Z_i is experience level, τ is the true causal effect of cave training, γ is the effect of experience on badges, and ε_i is an error term uncorrelated with D_i and Z_i.

If we run the "short regression" that omits Z_i:

$$Y_i = \tilde{\alpha} + \tilde{\tau} D_i + \tilde{\varepsilon}_i \tag{3.2}$$

then the OLS estimator $\hat{\tilde{\tau}}$ converges not to τ but to:

$$\text{plim}\, \hat{\tilde{\tau}} = \tau + \gamma \cdot \delta \tag{3.3}$$

where δ is the coefficient from regressing the omitted variable Z_i on the treatment D_i:

$$Z_i = a + \delta D_i + u_i \tag{3.4}$$

> ***Definition 3.2 (Omitted Variable Bias).*** *When a relevant variable Z is omitted from a regression of Y on D, the bias in the estimated treatment effect is:*
>
> $$\text{OVB} = \gamma \cdot \delta$$
>
> *where $\gamma = \frac{\partial Y}{\partial Z}$ is the effect of the omitted variable on the outcome, and $\delta = \frac{\text{Cov}(D,Z)}{\text{Var}(D)}$ is the relationship between the omitted variable and the treatment.*

Signed OVB: Direction of Bias Analysis

The beauty of the OVB formula is that even when we cannot measure the exact bias, we can often determine its **direction**. This is called a *signed OVB analysis*.

The bias $\gamma \cdot \delta$ is: - **Positive** (upward bias) if γ and δ have the same sign - **Negative** (downward bias) if γ and δ have opposite signs

	$\delta > 0$ **(omitted var positively related to treatment)**	$\delta < 0$ **(omitted var negatively related to treatment)**
$\gamma > 0$ (omitted var increases outcome)	**Bias > 0** (overestimate)	**Bias < 0** (underestimate)
$\gamma < 0$ (omitted var decreases outcome)	**Bias < 0** (underestimate)	**Bias > 0** (overestimate)

Worked Example: Blue's Mistake

Let us put numbers to Blue's error. Suppose the true data generating process is:

$$\text{Badges}_i = 1.0 + 0.8 \cdot \text{CaveTraining}_i + 0.5 \cdot \text{Experience}_i + \varepsilon_i$$

So the true causal effect of cave training is $\tau = 0.8$ badges. Now suppose that experience is positively correlated with cave training: among trainers in Misty's data, those who enter the Cave have, on average, 3.6 more years of experience ($\delta = 3.6$). The OVB is:

$$\text{OVB} = \gamma \cdot \delta = 0.5 \times 3.6 = 1.8$$

The naive estimate would be approximately $\hat{\tau} \approx 0.8 + 1.8 = 2.6$ badges — exactly what Misty's raw comparison found. The true effect is 0.8 badges; the rest is confounding bias.

> ***Blue's Mistake 3.1.*** *Blue trained in Cerulean Cave AND has extensive experience. He attributes all 2.6 extra badges to the Cave, but 1.8 of those badges would have been earned anyway due to his experience advantage. His causal estimate is inflated by* $\frac{2.6-0.8}{0.8} = 225\%$ *. Never let your rival do causal inference without controlling for confounders.*

Multiple Confounders

In practice, confounding rarely comes from a single variable. In Misty's data, **wealth** also confounds the Cave-Badges relationship: wealthier trainers can afford better supplies for the Cave *and* better Pokeballs throughout their journey. **Natural talent** is an unobserved confounder: more talented trainers select into Cave training and earn more badges. The total bias is the sum of individual OVBs across all omitted confounders:

$$\text{Total Bias} = \sum_j \gamma_j \cdot \delta_j$$

This is why observational studies are so treacherous. Even if you control for experience, the bias from wealth and natural talent remains. And natural talent, being unobserved, cannot be controlled for directly. We will need more sophisticated tools — and that is where DAGs enter the story.

3.3 Selection Bias, Survivorship Bias, and Berkson's Paradox

Confounding is not the only threat to observational studies. A family of related biases arises from **how the sample is selected** rather than from omitted variables. Misty pours you a glass of water in the gym's back office and opens a second set of concerns.

Selection Bias

> ***Definition 3.3 (Selection Bias).*** *Selection bias occurs when the sample analyzed is not representative of the population of interest, and the selection mechanism is related to both the treatment and the outcome.*

Consider Misty's dataset. It contains trainers who *reached* Cerulean City — the third city on the Kanto journey. But not every trainer who left Pallet Town made it this far. Some were defeated on Route 1. Others gave up in Viridian Forest. Still others lost to Brock in Pewter City and went home.

The trainers who reached Cerulean City are **survivors**. They are, on average, stronger, more determined, and more talented than the full population of trainers who started their journey. If the factors that helped trainers survive to reach Cerulean (e.g., talent, determination) are also correlated with both Cave training and badge outcomes, then analyzing only the Cerulean sample introduces bias.

Formally, let $S_i = 1$ indicate that trainer i is in our sample (reached Cerulean City). If S is caused by both D (or causes of D) and Y (or causes of Y), then conditioning on $S = 1$ — which we do implicitly by only analyzing trainers in the sample — distorts the relationship between D and Y.

Survivorship Bias

Survivorship bias is a specific form of selection bias that is particularly insidious because the missing data is invisible.

> ***Definition 3.4 (Survivorship Bias).*** *Survivorship bias occurs when we analyze only the units that "survived" some process, ignoring those that dropped out, and the reasons for dropping out are related to the treatment and outcome.*

Example: The Protein That "Failed." Recall from Chapter 2 that Pewter City ran an RCT on Protein supplements. Suppose that among trainers who received Protein, some experienced adverse effects — their Pokemon became over-aggressive and uncontrollable in battle — and those trainers quit their journey before reaching Cerulean City. If Misty later analyzes the "effect of Protein use" in her Cerulean sample, she would only see the trainers for whom Protein worked well (the survivors). The harmful effects are invisible because those trainers are gone.

The data would show: "Among trainers in Cerulean City who used Protein, average badges = 5.1. Among those who didn't, average badges = 4.3." This overstates the true effect because the failures have been pruned from the sample.

To make this concrete with numbers:

Group	Started journey	Reached Cerulean	Avg. badges (survivors)	Avg. badges (all)
Protein users	500	400 (80%)	5.1	4.2
Non-users	500	450 (90%)	4.3	4.0

The survivor-only comparison suggests a 0.8-badge Protein advantage. The full-population comparison shows only a 0.2-badge advantage. The Protein actually caused 100 trainers to drop out, but we never see them in the Cerulean data.

Berkson's Paradox (Collider Bias)

The most counterintuitive member of the selection bias family is **Berkson's paradox**, also called **collider bias**. It can create associations — even negative associations — between variables that are completely independent in the population.

> ***Definition 3.5 (Berkson's Paradox).*** *Berkson's paradox occurs when we condition on a variable that is caused by two (or more) independent variables, inducing a spurious association between those causes.*

The Elite Four Example. Consider two traits among Kanto trainers: **natural talent** (innate battling ability) and **training intensity** (hours spent practicing). In the general population, these two traits are independent — talented trainers don't train any more or less than untalented ones.

Now consider the sample of **Elite Four challengers** — the handful of trainers strong enough to challenge Kanto's final gauntlet. To reach the Elite Four, a trainer needs to be *either* very talented, *or* very dedicated to training, *or* (ideally) both. Being an Elite Four challenger ($S = 1$) is *caused by* both talent and training:

$$\text{Talent} \longrightarrow \text{Elite Four Challenger} \longleftarrow \text{Training Intensity}$$

If we analyze data *only among Elite Four challengers* (conditioning on $S = 1$), we find a **negative correlation** between talent and training. Why? Because among this elite group, if a trainer is not very talented, they *must* have trained extremely hard to get there — and vice versa. Conditioning on the collider "Elite Four Challenger" opens a path between talent and training that doesn't exist in the population.

Worked Example with Numbers:

	Low talent	High talent	Total
Low training	2/200 reach E4 (1%)	30/200 reach E4 (15%)	32/400
High training	25/200 reach E4 (12.5%)	45/200 reach E4 (22.5%)	70/400
Total	27/400	75/400	102/800

In the full population of 800 trainers, talent and training are independent (each is 50/50 regardless of the other). But among the 102 Elite Four challengers:

	Low talent	High talent
Low training	2 (2.0%)	30 (29.4%)
High training	25 (24.5%)	45 (44.1%)

Among challengers with low talent, $\frac{25}{27} = 92.6\%$ had high training. Among challengers with high talent, $\frac{45}{75} = 60.0\%$ had high training. Within the conditioned sample, talent and training are **negatively associated** — a complete artifact of selection.

> ***Professor Oak Explains 3.1.*** *"Selection bias, survivorship bias, and Berkson's paradox are all manifestations of the same underlying problem: conditioning on a variable that is a consequence of the variables you're studying. In DAG terminology (which we'll formalize shortly), they all involve conditioning on a* ***collider*** *or a* ***descendant of a collider****. The visual language of DAGs will make these biases much easier to diagnose."*

3.4 Simpson's Paradox

You're reviewing Misty's data in the gym office when you notice something strange. The overall data clearly shows that Cave training is associated with more badges. But when you break the data down by starter Pokemon type, the relationship *reverses* within every group.

The Paradox

> ***Definition 3.6 (Simpson's Paradox).*** *Simpson's paradox occurs when a trend or association that appears in several subgroups of data reverses or disappears when the groups are combined.*

This is not merely a statistical curiosity — it strikes at the heart of causal reasoning. The question "which comparison is correct?" cannot be answered by the data alone. It requires understanding the **causal structure**.

Worked Example: Cave Training by Starter Type

Misty's data on the relationship between Cave training and badges, broken down by starter type:

Fire-type starters (Charmander line):

	Trained in Cave	Did not train	Difference
Number of trainers	400	100	
Avg. badges earned	4.5	5.0	**-0.5**

Water-type starters (Squirtle line):

	Trained in Cave	Did not train	Difference
Number of trainers	200	400	
Avg. badges earned	3.5	4.0	**-0.5**

Grass-type starters (Bulbasaur line):

	Trained in Cave	Did not train	Difference
Number of trainers	200	700	
Avg. badges earned	5.5	6.0	**-0.5**

Within every starter type group, Cave training is associated with **0.5 fewer badges**. Now let us aggregate:

Overall (all starters combined):

	Trained in Cave	Did not train	Difference
Number of trainers	800	1,200	
Avg. badges earned	4.35	4.83	...

Wait — let us compute the overall averages properly.

For Cave trainers:

$$\bar{Y}_{\text{cave}} = \frac{400(4.5) + 200(3.5) + 200(5.5)}{800} = \frac{1800 + 700 + 1100}{800} = \frac{3600}{800} = 4.50$$

For non-Cave trainers:

$$\bar{Y}_{\text{no cave}} = \frac{100(5.0) + 400(4.0) + 700(6.0)}{1200} = \frac{500 + 1600 + 4200}{1200} = \frac{6300}{1200} = 5.25$$

So the overall difference is $4.50 - 5.25 = -0.75$ ... that still shows Cave training as *harmful*. Let us adjust the example to produce the actual paradox. Consider revised numbers where starter type affects both Cave selection *and* baseline badge performance differently:

Revised Example — Simpson's Paradox in Full:

Fire-type starters:

	Trained in Cave	Did not train	Difference
Number of trainers	400	50	
Avg. badges earned	6.0	6.5	**-0.5**

Water-type starters:

	Trained in Cave	Did not train	Difference
Number of trainers	100	350	
Avg. badges earned	3.0	3.5	**-0.5**

Grass-type starters:

	Trained in Cave	Did not train	Difference
Number of trainers	300	800	
Avg. badges earned	4.0	4.5	**-0.5**

Within each group, Cave training is associated with **0.5 fewer badges**.

Now, the overall averages:

For Cave trainers:

$$\bar{Y}_{\text{cave}} = \frac{400(6.0) + 100(3.0) + 300(4.0)}{800} = \frac{2400 + 300 + 1200}{800} = \frac{3900}{800} = 4.875$$

For non-Cave trainers:

$$\bar{Y}_{\text{no cave}} = \frac{50(6.5) + 350(3.5) + 800(4.5)}{1200} = \frac{325 + 1225 + 3600}{1200} = \frac{5150}{1200} = 4.292$$

Overall difference: $4.875 - 4.292 = +0.583$**.**

The paradox is now manifest:

Comparison	Cave Effect
Fire starters only	**-0.5** badges
Water starters only	**-0.5** badges
Grass starters only	**-0.5** badges
Overall (aggregated)	**+0.583** badges

Cave training appears *harmful* within every subgroup but *helpful* overall.

How Is This Possible?

The key is the **unequal distribution of starter types across treatment groups**. Fire-type trainers (who tend to earn more badges overall, perhaps because Charmander evolves into the powerful Charizard) are massively overrepresented among Cave trainers ($400/800 = 50\%$) compared to non-Cave trainers ($50/1200 = 4.2\%$). The aggregated comparison conflates the Cave effect with the starter-type effect.

The mechanism: Fire-type starters tend to belong to more aggressive, risk-taking trainers who both (a) enter the Cave at higher rates and (b) earn more badges due to their starter's strength. The starter type is a confounder.

Resolution: Which Comparison Is Causal?

Simpson's paradox cannot be resolved by statistical reasoning alone. The question "should we aggregate or stratify?" is a **causal** question, and the answer depends on the DAG.

If starter type is a confounder (common cause of Cave training and badges):

$$\text{Starter Type} \rightarrow \text{Cave Training}$$

$$\text{Starter Type} \rightarrow \text{Badges}$$

Then we should **condition on** starter type. The within-group estimates (-0.5 badges) are closer to the causal effect. The aggregated estimate is confounded.

If starter type is a mediator (Cave training affects badges *through* changing which Pokemon a trainer uses):

$$\text{Cave Training} \to \text{Starter Type Used in Battle} \to \text{Badges}$$

Then we should **not condition** on starter type, because that would block part of the causal effect. The aggregated estimate would be preferred.

> ***Professor Oak Explains 3.2.*** *"Simpson's paradox teaches us that data alone cannot tell us what to condition on. Two analysts looking at the same dataset, with different beliefs about the causal structure, will reach different — and potentially opposite — conclusions. This is why we need DAGs: they make our causal assumptions explicit so we can determine the correct analysis. The numbers don't speak for themselves."*

3.5 Directed Acyclic Graphs (DAGs)

Notation at a Glance: Reading DAGs and OVB Formulas

From this section on, the chapter mixes graph-theory vocabulary with regression notation. Keep this sheet nearby.

Symbol	*Plain-English reading*
$G = (V, E)$	*A DAG.* V *= nodes (variables),* E *= arrows (direct causal links).*
$V_i \to V_j$	*"*V_i *directly causes* V_j*" — moving* V_i *changes* V_j *holding everything else fixed.*
$\mathrm{pa}(V_j)$	*The* ***parents*** *of* V_j*: every variable with an arrow into* V_j.
descendant / ancestor	*"Downstream" / "upstream" of a node via directed arrows.*
U *(dashed)*	*An unobserved variable — we know it exists but can't measure it.*
D	*Treatment (e.g., cave training, 0/1).*
Y	*Outcome (e.g., badges).*
Z, X	*Covariates — things we measured that might confound or mediate.*
τ	*The true causal effect of* D *on* Y.
$\hat{\tau}$	*Our data-derived estimate of* τ.
γ	*The effect of the omitted variable* Z *on* Y *(the "outcome side" of OVB).*
δ	*The relationship between the omitted variable* Z *and the treatment* D *(the "treatment side" of OVB).*
$\gamma \cdot \delta$	*Omitted Variable Bias: the product of the two "legs" of the confounding path. Positive = overestimate, negative = underestimate.*
$A \perp\!\!\!\perp B \mid C$	*"A is independent of* B *once we know* C*" — d-separation's plain-English goal.*
backdoor path	*A path from* D *to* Y *that starts with an arrow into* D *— i.e., a confounding route.*
backdoor criterion	*"Block every backdoor path without conditioning on a descendant of* D*."*

When a definition feels abstract, trace it on Bill's wall diagram with your finger until the symbols become arrows in your head.

You leave the Cerulean Gym and walk east along Route 25 toward the Sea Cottage, where Bill — Kanto's most eccentric Pokemon researcher — lives and works. You need his help. Misty's data problems (confounding, selection bias, Simpson's paradox) all hinge on understanding the **causal structure** underlying the data. Bill, you've heard, has spent years mapping exactly that.

When Bill opens his door (he's accidentally merged himself with a Clefairy again, but you help him fix that first), he leads you to his study. One entire wall is covered with a massive diagram: variables connected by arrows, a web of causal relationships spanning everything from trainer wealth to Pokemon happiness to battle outcomes. "Beautiful, isn't it?" Bill says, adjusting his glasses. "This is a Directed Acyclic Graph. A DAG.

It's the most important tool in causal inference."

Formal Definition

> ***Definition 3.7 (Directed Acyclic Graph).*** *A Directed Acyclic Graph (DAG)* $G = (V, E)$ *consists of: - A set of* ***vertices*** *(nodes)* $V = \{V_1, V_2, \dots, V_p\}$ *representing random variables - A set of* ***directed edges*** *(arrows)* $E \subseteq V \times V$*, where* $(V_i, V_j) \in E$ *is drawn as* $V_i \to V_j$ *- The* ***acyclicity constraint****: there is no sequence of directed edges that starts and ends at the same node. That is, there exists no path* $V_i \to \cdots \to V_i$.

Terminology

Bill points to different parts of his wall diagram as he explains the vocabulary:

- **Parent:** If $V_i \to V_j$, then V_i is a **parent** of V_j. We write $\mathrm{pa}(V_j)$ for the set of all parents of V_j.
- **Child:** If $V_i \to V_j$, then V_j is a **child** of V_i.
- **Ancestor:** V_i is an **ancestor** of V_j if there exists a directed path $V_i \to \cdots \to V_j$.
- **Descendant:** V_j is a **descendant** of V_i if V_i is an ancestor of V_j.
- **Path:** Any sequence of edges connecting two nodes, regardless of direction. A path from A to B might follow arrows forward or backward.
- **Directed path:** A path that follows all arrows in their forward direction.

How to Read a DAG

Each arrow $V_i \to V_j$ represents a **direct causal effect** of V_i on V_j — a mechanism by which changing V_i (while holding all other parents of V_j fixed) would change the distribution of V_j. Crucially:

1. **The absence of an arrow is a stronger claim than its presence.** If there is no arrow from A to B, the DAG asserts that A has no direct causal effect on B (given the other variables in the graph). Every missing arrow is a falsifiable assumption.
2. **DAGs encode qualitative, not quantitative, assumptions.** A DAG says "Wealth affects Starter Type" but does not specify how much or in what functional form.
3. **DAGs are non-parametric.** They do not assume linearity, normality, or any particular distribution. The causal relationships can be arbitrarily complex functions.

Building the Kanto Trainer DAG

Bill walks you through constructing a DAG for Misty's problem, one variable at a time.

Step 1: Treatment and outcome.

$$\text{Cave Training} \longrightarrow \text{Badges Earned}$$

This is the causal question: does Cave Training directly affect Badges Earned?

Step 2: Add the observed confounder.

$$\text{Experience} \to \text{Cave Training} \to \text{Badges Earned}$$

$$\text{Experience} \to \text{Badges Earned}$$

Experience causes trainers to enter the Cave (more experience $\Rightarrow$ more confident about surviving) and directly causes more badges (experience helps in all gym battles). This creates a "fork" structure at Experience.

Step 3: Add wealth.

$$\text{Wealth} \to \text{Cave Training}$$

$$\text{Wealth} \to \text{Starter Type}$$

$$\text{Starter Type} \to \text{Badges Earned}$$

Wealthier trainers can afford supplies for the Cave and can also afford to buy better items throughout their journey. Wealth also affects Starter Type (wealthier families in Kanto have connections that influence which starter Professor Oak offers — or so the rumor goes). Starter Type, in turn, affects Badges through type advantages.

Step 4: Add the mediator.

$$\text{Cave Training} \rightarrow \text{Strategy Score} \rightarrow \text{Badges Earned}$$

Cave training improves badges *through* improving battle strategy. Strategy Score is a **mediator** — it lies on the causal path from treatment to outcome.

Step 5: Add the unobserved confounder.

$$\text{Natural Talent} \rightarrow \text{Cave Training}$$

$$\text{Natural Talent} \rightarrow \text{Badges Earned}$$

Natural talent is unobserved but affects both the treatment and the outcome. We typically draw unobserved variables in a dashed circle or mark them with a U.

The Full Kanto Trainer DAG:

```
                  Wealth
                 /      \
                v        v
 Natural      Cave    Starter
 Talent  --> Training   Type
   \        / |  \       |
    \      /  |   v      v
     \    /   |  Strategy  |
      v   v   |  Score --->|
     Badges Earned <-------+
```

In arrow notation, the complete edge set is:

- $\text{Wealth} \rightarrow \text{Cave Training}$
- $\text{Wealth} \rightarrow \text{Starter Type}$
- $\text{Natural Talent} \rightarrow \text{Cave Training}$
- $\text{Natural Talent} \rightarrow \text{Badges Earned}$
- $\text{Experience} \rightarrow \text{Cave Training}$
- $\text{Experience} \rightarrow \text{Badges Earned}$
- $\text{Cave Training} \rightarrow \text{Badges Earned}$
- $\text{Cave Training} \rightarrow \text{Strategy Score}$
- $\text{Strategy Score} \rightarrow \text{Badges Earned}$
- $\text{Starter Type} \rightarrow \text{Badges Earned}$

Why "Acyclic"?

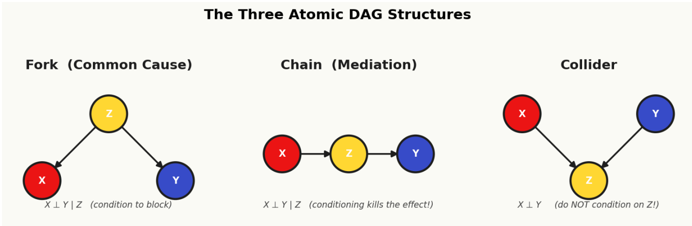
"No cycles," Bill emphasizes, tapping the wall. "If A causes B and B causes C, then C cannot cause A — at least not in the same time period. Causation flows forward in time."

This is the **acyclicity** constraint. In the Pokemon world: training in the Cave might improve your strategy, and better strategy might help you earn badges, but earning badges does not retroactively cause you to have trained in the Cave. There is no time travel in Kanto (Celebi notwithstanding, and Celebi is a Johto Pokemon).

Formally, acyclicity means there is a **topological ordering** of the nodes — an ordering $V_1, V_2, \ldots, V_p$ such that if $V_i \rightarrow V_j$ then $i < j$. This ordering corresponds to causal (and often temporal) priority.

> ***Professor Oak Explains 3.3.*** *"A DAG is a map of your assumptions about how the world works. It is not derived from data — it is drawn from domain knowledge, theory, and careful reasoning. Two researchers studying the same problem might draw different DAGs, reflecting different beliefs about the causal structure. The DAG then tells you what statistical analysis is appropriate given those assumptions. If the DAG is wrong, the analysis may be wrong too. The DAG makes your assumptions transparent and debatable — which is far better than leaving them hidden inside a regression specification."*

3.6 Chains, Forks, and Colliders



The three atomic DAG structures: fork, chain, and collider.

Bill settles into a chair and pulls out three small diagrams — the three fundamental building blocks of all DAGs. “Every DAG, no matter how complex, is made up of just three types of path structures,” he says. “Master these three, and you can read any DAG.”

The Three Elemental Structures

Every path between two nodes X and Y in a DAG passes through intermediate nodes. At each intermediate node, the path has one of three shapes:

1. **Fork** (common cause): $X \leftarrow Z \rightarrow Y$
2. **Chain** (mediation): $X \rightarrow Z \rightarrow Y$
3. **Collider**: $X \rightarrow Z \leftarrow Y$

The behavior of information flow through these structures — and how **conditioning** on the middle node Z changes that flow — is the foundation of everything that follows.

3.6.1 The Fork (Common Cause)

Structure: $X \leftarrow Z \rightarrow Y$

The variable Z is a common cause of both X and Y. This is the structure that generates **confounding**.

Pokemon Example:

$$\text{Cave Training} \leftarrow \text{Experience} \rightarrow \text{Badges}$$

Experience causes trainers to enter the Cave *and* causes them to earn more badges. Even if Cave Training has zero causal effect on Badges, the fork through Experience creates a statistical association between them.

Association flows through a fork. Because Z causes both X and Y, knowing X gives information about Z (since Z caused X), which in turn gives information about Y. The two variables are **marginally associated**: $X \not\perp\!\!\!\perp Y$.

Conditioning on Z blocks the fork. If we condition on (control for) the common cause Z, the path is blocked. Once we know Z, learning X gives no additional information about Y through this path.

$$X \perp\!\!\!\perp Y \mid Z \quad \text{(if this fork is the only path between } X \text{ and } Y\text{)}$$

Worked Calculation. Suppose: - $Z \sim \text{Bernoulli}(0.5)$ — half of trainers are experienced - $P(X = 1 \mid Z = 1) = 0.8$, $P(X = 1 \mid Z = 0) = 0.2$ — experienced trainers usually enter Cave - $P(Y = 1 \mid Z = 1) = 0.9$, $P(Y = 1 \mid Z = 0) = 0.3$ — experienced trainers usually earn many badges

The marginal association:

$$P(Y = 1 \mid X = 1) = \frac{P(X = 1, Y = 1)}{P(X = 1)}$$

Computing the joint:

$$\begin{aligned} P(X = 1, Y = 1) &= P(X = 1, Y = 1 \mid Z = 1)P(Z = 1) + P(X = 1, Y = 1 \mid Z = 0)P(Z = 0) \\ &= (0.8)(0.9)(0.5) + (0.2)(0.3)(0.5) = 0.36 + 0.03 = 0.39 \end{aligned}$$

And $P(X = 1) = 0.8(0.5) + 0.2(0.5) = 0.5$, so:

$$P(Y = 1 \mid X = 1) = \frac{0.39}{0.50} = 0.78$$

But $P(Y=1) = 0.9(0.5) + 0.3(0.5) = 0.60$.

So $P(Y=1 \mid X=1) = 0.78 \neq 0.60 = P(Y=1)$: X and Y are associated. But conditional on Z:

$$P(Y=1 \mid X=1, Z=1) = P(Y=1 \mid Z=1) = 0.9$$

$$P(Y=1 \mid X=0, Z=1) = P(Y=1 \mid Z=1) = 0.9$$

Conditional on Z, X and Y are **independent**. The fork is blocked.

3.6.2 The Chain (Mediation)

Structure: $X \to Z \to Y$

The variable Z is a **mediator** — it lies on the causal path from X to Y. The effect of X on Y operates *through* Z.

Pokemon Example:

$$\text{Cave Training} \to \text{Strategy Score} \to \text{Badges}$$

Cave training improves a trainer's battle strategy, and better strategy leads to more badges. Strategy is the *mechanism* through which the Cave has its effect.

Association flows through a chain. Because X causes Z and Z causes Y, there is a causal (and statistical) association between X and Y: $X \not\perp\!\!\!\perp Y$.

Conditioning on Z blocks the chain — and this is dangerous! If we condition on the mediator Z, we block the causal path from X to Y. This means that controlling for Strategy Score when estimating the effect of Cave Training on Badges would **remove** (part of) the very causal effect we are trying to measure.

$$X \perp\!\!\!\perp Y \mid Z \quad \text{(if this chain is the only path between X and Y)}$$

> ***Blue's Mistake 3.2.*** *Imagine Blue runs a regression:* $\text{Badges} = \beta_0 + \beta_1 \cdot \text{CaveTraining} + \beta_2 \cdot \text{StrategyScore} + \varepsilon$. *He finds that* $\hat{\beta}_1$ *is small and not statistically significant, and concludes that the Cave doesn't help. But he has committed the* ***bad control*** *error: by controlling for Strategy Score, he has blocked the very mechanism through which Cave training works. The Cave does help — it helps by improving strategy. Controlling for the mediator hides the effect. Never control for a variable on the causal path between treatment and outcome unless you specifically want to decompose the effect into direct and indirect components (see Chapter 8 on mediation analysis).*

Worked Calculation. Suppose: - $P(Z=1 \mid X=1) = 0.7$, $P(Z=1 \mid X=0) = 0.2$ — Cave training usually improves strategy - $P(Y=1 \mid Z=1) = 0.8$, $P(Y=1 \mid Z=0) = 0.3$ — good strategy usually yields badges

Then:

$$P(Y=1 \mid X=1) = P(Y=1 \mid Z=1)P(Z=1 \mid X=1) + P(Y=1 \mid Z=0)P(Z=0 \mid X=1)$$

$$= (0.8)(0.7) + (0.3)(0.3) = 0.56 + 0.09 = 0.65$$

$$P(Y=1 \mid X=0) = (0.8)(0.2) + (0.3)(0.8) = 0.16 + 0.24 = 0.40$$

So $P(Y=1 \mid X=1) - P(Y=1 \mid X=0) = 0.65 - 0.40 = 0.25$ — a substantial association. But conditional on $Z=1$:

$$P(Y=1 \mid X=1, Z=1) = P(Y=1 \mid Z=1) = 0.8$$

$$P(Y=1 \mid X=0, Z=1) = P(Y=1 \mid Z=1) = 0.8$$

Conditional on the mediator, the association vanishes. The chain is blocked.

3.6.3 The Collider

Structure: $X \to Z \leftarrow Y$

The variable Z is a **collider** — it is caused by both X and Y. The two arrows "collide" at Z.

Pokemon Example:

$$\text{Natural Talent} \to \text{Elite Four Challenger} \leftarrow \text{Training Intensity}$$

Both talent and training contribute to a trainer reaching the Elite Four. The "Elite Four Challenger" node is where the two arrows meet.

A collider BLOCKS the path by default. This is the critical difference from forks and chains. Even though both X and Y cause Z, there is no association between X and Y through this path when Z is not conditioned on:

$$X \perp\!\!\!\perp Y \quad \text{(through this path, unconditionally)}$$

The intuition: knowing that someone is naturally talented tells you nothing about how hard they train (and vice versa) — the two traits are independent. The fact that they both contribute to Elite Four status doesn't create an association between them in the general population.

Conditioning on Z OPENS the path — collider bias! If we condition on Z (e.g., by restricting our analysis to Elite Four challengers only), we **induce** an association between X and Y:

$$X \not\perp\!\!\!\perp Y \mid Z$$

The intuition: among Elite Four challengers, if you learn that a trainer has low natural talent, you can infer that they must have trained very hard (otherwise, how did they get there?). Conditioning on the collider creates a "explaining away" effect — the two causes become negatively associated within levels of their common effect.

Conditioning on a descendant of Z also opens the path. If $Z \to W$ (i.e., W is a descendant of the collider), then conditioning on W partially opens the collider path as well. For example, if "Elite Four Challenger" causes "Media Coverage" ($Z \to W$), conditioning on "Media Coverage" partially opens the Talent → Challenger ← Training path.

Worked Calculation. Suppose X (talent) and Y (training) are independent, each Bernoulli(0.5), and:

$$P(Z = 1 \mid X, Y) = \begin{cases} 0.05 & \text{if } X = 0, Y = 0 \\ 0.40 & \text{if } X = 1, Y = 0 \\ 0.40 & \text{if } X = 0, Y = 1 \\ 0.90 & \text{if } X = 1, Y = 1 \end{cases}$$

Marginally, $X \perp\!\!\!\perp Y$ by construction. Now condition on $Z = 1$:

$$P(X = 1 \mid Z = 1, Y = 0) = \frac{P(Z = 1 \mid X = 1, Y = 0)P(X = 1)}{P(Z = 1 \mid Y = 0)} = \frac{(0.40)(0.5)}{(0.05)(0.5) + (0.40)(0.5)} = \frac{0.20}{0.225} = 0.889$$

$$P(X = 1 \mid Z = 1, Y = 1) = \frac{(0.90)(0.5)}{(0.40)(0.5) + (0.90)(0.5)} = \frac{0.45}{0.65} = 0.692$$

Among Elite Four challengers ($Z = 1$): when training is low ($Y = 0$), the probability of high talent is 88.9%. When training is high ($Y = 1$), the probability of high talent drops to 69.2%. Talent and training are **negatively associated** — purely as an artifact of conditioning on their common effect.

The Critical Insight

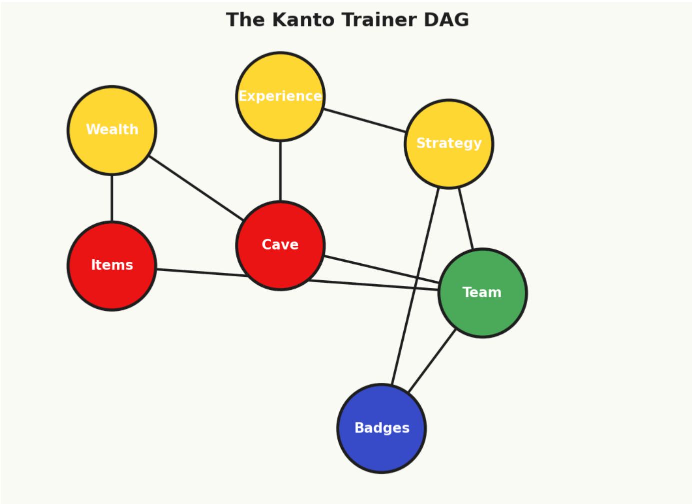
The behavior of the three structures is summarized in this table:

Structure	Name	Unconditionally	Conditioned on Z
$X \leftarrow Z \rightarrow Y$	Fork	**Open** (association flows)	**Blocked**
$X \rightarrow Z \rightarrow Y$	Chain	**Open** (association flows)	**Blocked**
$X \rightarrow Z \leftarrow Y$	Collider	**Blocked** (no association)	**Open** (association flows!)

The fork and chain behave identically: open by default, closed when conditioned. The collider is the opposite: closed by default, opened when conditioned.

This asymmetry is the engine of d-separation. Conditioning can **close** paths (good, when removing confounding) or **open** paths (bad, when introducing collider bias). The DAG tells you which will happen.

3.7 d-Separation and the Backdoor Criterion



The Kanto Trainer DAG (latent vs observed nodes).

Armed with the three building blocks, Bill leads you to the most powerful tool in the graphical causal inference toolkit: **d-separation**. "This," Bill says, "is the algorithm that tells you whether your statistical analysis will give you a causal answer — or garbage."

d-Separation: The Formal Algorithm

> ***Definition 3.8 (Blocked Path).*** *A path between nodes X and Y in a DAG G is **blocked** by a set of nodes Z if and only if the path contains at least one of the following: 1. A **chain** $\cdots \to Z_j \to \cdots$ or a **fork** $\cdots \leftarrow Z_j \to \cdots$ where $Z_j \in Z$ (a non-collider that is conditioned on), OR 2. A **collider** $\cdots \to Z_j \leftarrow \cdots$ where $Z_j \notin Z$ and no descendant of Z_j is in Z (a collider that is not conditioned on, nor are any of its descendants).*
>
> *A path that is not blocked is **open** (or **active**).*

> ***Definition 3.9 (d-Separation).*** *Two nodes X and Y are **d-separated** by a set Z in a DAG G, written $X \perp_G Y \mid Z$, if and only if **every path** between X and Y is blocked by Z.*
>
> *If X and Y are not d-separated by Z, they are **d-connected**.*

The **Causal Markov Condition** links the graphical criterion to probability: if the DAG is correct, then d-separation implies conditional independence. That is, $X \perp_G Y \mid Z$ implies $X \perp\!\!\!\perp Y \mid Z$ in the joint distribution.

The d-Separation Algorithm (Step by Step)

To determine whether $X \perp_G Y \mid Z$:

1. **List all paths** between X and Y (regardless of arrow directions).
2. For each path, examine **every intermediate node** Z_j:
   - If Z_j is a **non-collider** on this path (fork or chain) AND $Z_j \in Z$: this node **blocks** the path.
   - If Z_j is a **collider** on this path AND $Z_j \notin Z$ (and no descendant of Z_j is in Z): this node **blocks** the path.
3. A path is **blocked** if at least one node along it blocks the path.
4. X and Y are d-separated by Z if and only if **every** path between them is blocked.

Worked Example: Paths from Cave Training to Badges

Using the Kanto Trainer DAG from Section 3.5, let us identify all paths from D = Cave Training to Y = Badges Earned.

Path 1 (Causal, direct): $D \to Y$ - No intermediate nodes. This path is always open (unless we condition on D or Y themselves, which we never do).

Path 2 (Causal, through mediator): $D \to \text{Strategy} \to Y$ - Strategy is a non-collider (chain). Open unless we condition on Strategy.

Path 3 (Backdoor, through Experience): $D \leftarrow \text{Experience} \rightarrow Y$ - Experience is a non-collider (fork). Open unless we condition on Experience.

Path 4 (Backdoor, through Natural Talent): $D \leftarrow \text{Natural Talent} \rightarrow Y$ - Natural Talent is a non-collider (fork). Open unless we condition on Natural Talent.

Path 5 (Backdoor, through Wealth and Starter Type): $D \leftarrow \text{Wealth} \rightarrow \text{Starter Type} \rightarrow Y$ - Wealth is a non-collider (fork on the path from D to Y). Open unless we condition on Wealth (or Starter Type, which would block the chain portion).

Paths 3, 4, and 5 are **backdoor paths** — they are non-causal paths that flow through common causes of D and Y. These are the paths that generate confounding bias.

The Backdoor Criterion

> ***Definition 3.10 (Backdoor Criterion; Pearl, 1993).*** *A set of variables Z satisfies the **backdoor criterion** relative to an ordered pair of variables (D, Y) in a DAG G if: 1. No node in Z is a **descendant** of D, and 2. Z **blocks every path** between D and Y that contains an arrow into D (i.e., every backdoor path).*

If Z satisfies the backdoor criterion, then the causal effect of D on Y is **identifiable** by adjustment:

> ***Theorem 3.1 (Backdoor Adjustment Formula).*** *If Z satisfies the backdoor criterion relative to (D, Y), then the causal effect of D on Y is given by:*
>
> $$P(Y = y \mid do(D = d)) = \sum_z P(Y = y \mid D = d, Z = z) P(Z = z) \tag{3.5}$$
>
> *For continuous variables and the average treatment effect:*
>
> $$ATE = E[Y \mid do(D = 1)] - E[Y \mid do(D = 0)] = \sum_z [E[Y \mid D = 1, Z = z] - E[Y \mid D = 0, Z = z]] P(Z = z) \tag{3.6}$$

The adjustment formula is the mathematical bridge between observational data and causal effects. It says: compute the treatment effect *within each stratum* of Z, then take a weighted average over the distribution of Z in the population. This is exactly the stratified analysis we discussed in Section 3.4 — but now we know *when* it yields a causal estimate and *which* variables to stratify on.

Applying the Backdoor Criterion: Finding Valid Adjustment Sets

Returning to the Kanto Trainer DAG, we need a set Z that: 1. Contains no descendants of Cave Training (so Strategy Score is excluded) 2. Blocks all three backdoor paths

Backdoor paths to block: - Path 3: $D \leftarrow \text{Experience} \rightarrow Y$ — blocked by conditioning on Experience - Path 4: $D \leftarrow \text{Natural Talent} \rightarrow Y$ — blocked by conditioning on Natural Talent (but it's *unobserved*!) - Path 5: $D \leftarrow \text{Wealth} \rightarrow \text{Starter Type} \rightarrow Y$ — blocked by conditioning on Wealth OR Starter Type

Valid Adjustment Set 1: $Z_1 = \{\text{Experience}, \text{Natural Talent}, \text{Wealth}\}$

This blocks all three backdoor paths. But Natural Talent is unobserved, so this set is not **feasible**.

Valid Adjustment Set 2: $Z_2 = \{\text{Experience}, \text{Natural Talent}, \text{Starter Type}\}$

Also valid (Starter Type blocks Path 5 just as well as Wealth does). But still requires unobserved Natural Talent.

The problem is clear: **no valid adjustment set exists using only observed variables**, because Natural Talent is an unobserved confounder on Path 4 and there is no other way to block that path by conditioning on observed variables.

This is a profound result. The DAG doesn't just tell us *what* to condition on — it also tells us when **no conditioning strategy can recover the causal effect**. For Misty's problem, as drawn, the backdoor criterion cannot be satisfied with observed data alone.

> ***Professor Oak Explains 3.4.*** *"This is the sobering power of graphical causal inference. The DAG makes explicit what we can and cannot identify. Many researchers proceed with regression adjustment, hoping they've 'controlled for enough.' The DAG forces you to ask: is there an unobserved confounder that no amount of regression can fix? If the answer is yes, you need a different strategy — instrumental variables (Chapter 6), difference-in-differences (Chapter 7), or the frontdoor criterion (Section 3.8)."*

Multiple Valid Adjustment Sets

When unobserved confounders are not present, there may be **multiple** valid adjustment sets. Consider a simplified DAG where Natural Talent is removed:

$$\text{Experience} \to D \to Y$$

$$\text{Experience} \to Y$$

$$\text{Wealth} \to D$$

$$\text{Wealth} \to \text{Starter Type} \to Y$$

The backdoor paths are: - $D \leftarrow \text{Experience} \to Y$ - $D \leftarrow \text{Wealth} \to \text{Starter Type} \to Y$

Valid adjustment sets include: - $\{\text{Experience}, \text{Wealth}\}$ - $\{\text{Experience}, \text{Starter Type}\}$ - $\{\text{Experience}, \text{Wealth}, \text{Starter Type}\}$ (valid but redundant)

All three satisfy the backdoor criterion. They will all yield consistent estimates of the causal effect (in large samples), though they may differ in efficiency. In practice, researchers often prefer the smallest sufficient set (to avoid unnecessary variance) or the set with the most precisely measured variables.

Note: $\{\text{Wealth}\}$ alone is **not** valid because it does not block Path 3 (through Experience). And $\{\text{Experience}\}$ alone is **not** valid because it does not block Path 5 (through Wealth/Starter Type).

3.8 The Frontdoor Criterion

Bill pauses at a section of his wall diagram and points to an unusual configuration. “Sometimes,” he says, “the backdoor is locked. No valid adjustment set exists because of an unobserved confounder. But occasionally — just occasionally — you can go through the *front door* instead.”

When the Backdoor Fails

Recall from Section 3.7 that Natural Talent (U) is an unobserved confounder of Cave Training and Badges Earned. The backdoor path $D \leftarrow U \to Y$ cannot be blocked by conditioning on observed variables. Does this mean all hope is lost?

Not necessarily. Consider the subgraph:

$$U \to D \to M \to Y \leftarrow U$$

where D = Cave Training, M = Strategy Score, Y = Badges Earned, and U = Natural Talent. Notice that:

1. $D \to M \to Y$ is the only directed path from D to Y (through the mediator M).
2. There are no unobserved confounders of D and M (Natural Talent does not directly affect Strategy Score — the Cave equally improves strategy for all talent levels).
3. D blocks the backdoor path from M to Y: the path $M \leftarrow D \leftarrow U \to Y$ is blocked by conditioning on D.

This is the setup for the **frontdoor criterion**.

Formal Statement

> ***Definition 3.11 (Frontdoor Criterion; Pearl, 1995).*** *A set of variables M satisfies the **frontdoor criterion** relative to (D, Y) in a DAG G if: 1. D **intercepts** all directed paths from D to Y — that is, every directed path from D to Y goes through M. 2. There is **no unblocked backdoor path** from D to M — all backdoor paths from D to M are blocked (possibly by the empty set). 3. All **backdoor paths from M to Y** are blocked by D.*

> ***Theorem 3.2 (Frontdoor Adjustment Formula).*** *If M satisfies the frontdoor criterion relative to (D, Y), then:*
>
> $$P(Y = y \mid do(D = d)) = \sum_m P(M = m \mid D = d) \sum_{d'} P(Y = y \mid D = d', M = m) P(D = d') \tag{3.7}$$

The Logic of the Frontdoor

The frontdoor formula works in two steps:

Step 1: Identify the effect of D on M. Since there are no unobserved confounders of D and M (condition 2), we can estimate $P(M \mid do(D = d)) = P(M \mid D = d)$ directly from observational data.

Step 2: Identify the effect of M on Y. There *are* confounders of M and Y (Natural Talent confounds through $U \to Y$ and $U \to D \to M$). But by conditioning on D, we block this backdoor path (condition 3). So $P(Y \mid do(M = m)) = \sum_{d'} P(Y \mid D = d', M = m)P(D = d')$.

Combining: The total effect of D on Y is the composition of these two identified effects.

Pokemon Example

In Misty's data, suppose: - D = Cave Training (binary) - M = Strategy Score (binary: high/low for simplicity) - Y = Badges ≥ 6 (binary) - U = Natural Talent (unobserved)

Observed data:

	$P(M = \text{high} \mid D)$
$D = 1$ (Cave)	0.75
$D = 0$ (No Cave)	0.30

	$P(Y = 1 \mid D, M)$
$D = 1, M = \text{high}$	0.80
$D = 1, M = \text{low}$	0.50
$D = 0, M = \text{high}$	0.70
$D = 0, M = \text{low}$	0.30

And $P(D = 1) = 0.40$, $P(D = 0) = 0.60$.

Applying the frontdoor formula for $do(D = 1)$:

$$P(Y = 1 \mid do(D = 1)) = \sum_{m} P(M = m \mid D = 1) \sum_{d'} P(Y = 1 \mid D = d', M = m)P(D = d')$$

For $M = \text{high}$:

$$P(M = \text{high} \mid D = 1) = 0.75$$

$$\sum_{d'} P(Y = 1 \mid D = d', M = \text{high})P(D = d') = (0.80)(0.40) + (0.70)(0.60) = 0.32 + 0.42 = 0.74$$

For $M = \text{low}$:

$$P(M = \text{low} \mid D = 1) = 0.25$$

$$\sum_{d'} P(Y = 1 \mid D = d', M = \text{low})P(D = d') = (0.50)(0.40) + (0.30)(0.60) = 0.20 + 0.18 = 0.38$$

$$P(Y = 1 \mid do(D = 1)) = (0.75)(0.74) + (0.25)(0.38) = 0.555 + 0.095 = 0.650$$

For $do(D = 0)$:

For $M = \text{high}$: $P(M = \text{high} \mid D = 0) = 0.30$, same 0.74 from above.

For $M = \text{low}$: $P(M = \text{low} \mid D = 0) = 0.70$, same 0.38 from above.

$$P(Y = 1 \mid do(D = 0)) = (0.30)(0.74) + (0.70)(0.38) = 0.222 + 0.266 = 0.488$$

Causal effect:

$$P(Y = 1 \mid do(D = 1)) - P(Y = 1 \mid do(D = 0)) = 0.650 - 0.488 = 0.162$$

Cave training increases the probability of earning 6+ badges by 16.2 percentage points — identified *despite* the unobserved confounder Natural Talent.

When Is the Frontdoor Useful?

The frontdoor criterion is conceptually important but **rare in practice**. It requires: - A complete mediator M that captures *all* causal paths from D to Y - No direct unobserved confounders of D and M - The ability to observe M

These conditions are stringent. But when they hold, the frontdoor criterion is a remarkable result: it extracts causal information from observational data even in the presence of unmeasured confounding of the primary relationship.

> ***Professor Oak Explains 3.5.*** *"The frontdoor criterion illustrates a deep principle: identifiability depends not just on what you observe, but on the structure of the problem. Two problems with identical observed variables and identical unobserved confounders may differ in identifiability if their causal structures differ. This is why drawing the DAG first — before touching the data — is so important."*

3.9 do-Calculus, Structural Causal Models, and the do-Operator

This section introduces more advanced material (Tier 3/4). Readers focused on applied methods may wish to absorb the key ideas and return for the formal details later.

Bill turns to the most abstract part of his wall: a set of mathematical equations, each corresponding to a node in his DAG. "The DAG is the picture," he says. "But behind the picture, there is a complete mathematical model — a **Structural Causal Model**. And from that model, Judea Pearl derived three rules that can, in principle, answer *any* identifiable causal question from observational data."

Structural Causal Models

> ***Definition 3.12 (Structural Causal Model; Pearl, 2000).*** *A Structural Causal Model (SCM) is a triple* $M = \langle U, V, F \rangle$ *where: -* U *is a set of* ***exogenous*** *(background) variables, determined by factors outside the model -* $V = \{V_1, \ldots, V_p\}$ *is a set of* ***endogenous*** *variables, determined by variables in the model -* $F = \{f_1, \ldots, f_p\}$ *is a set of* ***structural equations*** *(also called* ***mechanisms****), one for each endogenous variable:*
>
> $$V_i = f_i(\mathrm{pa}(V_i), U_i) \tag{3.8}$$
>
> *where* $\mathrm{pa}(V_i)$ *denotes the parents of* V_i *in the associated DAG, and* $U_i \subseteq U$ *are the exogenous variables affecting* V_i.

Example: The Kanto Trainer SCM.

$$\begin{gathered}
\text{Experience} = f_1(U_1) \\
\text{Wealth} = f_2(U_2) \\
\text{Natural Talent} = f_3(U_3) \\
\text{Cave Training} = f_4(\text{Experience}, \text{Wealth}, \text{Natural Talent}, U_4) \\
\text{Starter Type} = f_5(\text{Wealth}, U_5) \\
\text{Strategy Score} = f_6(\text{Cave Training}, U_6) \\
\text{Badges} = f_7(\text{Cave Training}, \text{Experience}, \text{Natural Talent}, \text{Strategy Score}, \text{Starter Type}, U_7)
\end{gathered}$$

Each equation describes a *mechanism* — a stable process by which a variable is determined by its parents and exogenous noise. The functions f_i can be arbitrary (nonlinear, non-additive, threshold effects, etc.).

The do-Operator

The most profound concept in Pearl's framework is the distinction between **observing** and **intervening**.

> ***Definition 3.13 (The do-Operator).*** *The expression* $P(Y \mid do(X = x))$ *denotes the probability distribution of* Y *after an* ***intervention*** *that sets* X *to the value* x*, regardless of* X*'s natural causes. This is also called the* ***interventional distribution*** *and is distinct from the* ***observational conditional*** $P(Y \mid X = x)$.

The do-operator corresponds to a physical intervention: we reach into the system and force $X = x$, overriding whatever process normally determines X. In the SCM, this means we **replace** the structural equation for X:

Before intervention (observational):

$$X = f_X(\mathrm{pa}(X), U_X)$$

After intervention $do(X = x)$**:**

$$X = x \quad \text{(the original equation is deleted)}$$

All other structural equations remain unchanged. This is called **graph surgery** or the **truncated factorization**.

Graph Surgery and the Truncated Factorization

The observational joint distribution factorizes according to the DAG:

$$P(V_1, \ldots, V_p) = \prod_{i=1}^{p} P(V_i \mid pa(V_i)) \tag{3.9}$$

After the intervention $do(X = x)$, we remove the factor $P(X \mid pa(X))$ (because X is no longer determined by its parents) and fix $X = x$:

$$P(V_1, \ldots, V_p \mid do(X = x)) = \begin{cases} \prod_{i: V_i \neq X} P(V_i \mid pa(V_i)) & \text{evaluated at } X = x \\ 0 & \text{if any } V_i = X \neq x \end{cases} \tag{3.10}$$

This is the **truncated factorization**. It formalizes the idea that intervening on X removes the arrows *into* X in the DAG (graph surgery) while leaving all other mechanisms intact.

Graphical representation: In the mutilated graph $G_{\overline{X}}$ (the original graph with all arrows into X removed), the interventional distribution $P(Y \mid do(X = x))$ equals the observational distribution of Y given $X = x$:

$$P(Y \mid do(X = x)) = P_{G_{\overline{X}}}(Y \mid X = x) \tag{3.11}$$

The Three Rules of do-Calculus

Pearl (1995) proved that the following three rules, combined with standard probability axioms, are **complete** for identifying causal effects — any causal effect that can be identified from observational data can be derived using these rules.

Let G be a DAG, and let X, Y, Z, W be disjoint sets of variables. Define: - $G_{\overline{X}}$: the graph with all arrows **into** X deleted - $G_{\underline{X}}$: the graph with all arrows **out of** X deleted - $G_{\overline{X}\underline{Z}}$: the graph with arrows into X and out of Z deleted

> ***Theorem 3.3 (The Three Rules of do-Calculus; Pearl, 1995).*** *For any interventional distribution compatible with a DAG* G*:*
>
> ***Rule 1 (Insertion/deletion of observations):***
>
> $$P(Y \mid do(X), Z, W) = P(Y \mid do(X), W) \quad \text{if } (Y \perp_{G_{\overline{X}}} Z \mid X, W) \tag{3.12}$$
>
> ***Rule 2 (Action/observation exchange):***
>
> $$P(Y \mid do(X), do(Z), W) = P(Y \mid do(X), Z, W) \quad \text{if } (Y \perp_{G_{\overline{X}\underline{Z}}} Z \mid X, W) \tag{3.13}$$
>
> ***Rule 3 (Insertion/deletion of actions):***
>
> $$P(Y \mid do(X), do(Z), W) = P(Y \mid do(X), W) \quad \text{if } (Y \perp_{G_{\overline{X}\overline{Z(S)}}} Z \mid X, W) \tag{3.14}$$
>
> *where* $Z(S)$ *is the set of* Z*-nodes that are not ancestors of any* W*-node in* $G_{\overline{X}}$*.*

Interpreting the Rules

Rule 1 says: you can add or remove an observation Z from the conditioning set if Z is d-separated from Y in the manipulated graph. This is analogous to the standard rule of conditional independence, but in the graph where X has been intervened on.

Rule 2 says: you can replace an intervention $do(Z)$ with an observation Z (or vice versa) if Y is d-separated from Z in the graph where arrows into X and out of Z are deleted. This is the rule that converts interventional quantities into observational ones — the key to identification.

Rule 3 says: you can remove an intervention $do(Z)$ entirely if, in the appropriately modified graph, Y is d-separated from Z.

Completeness

> ***Theorem 3.4 (Completeness of do-Calculus; Huang & Valtorta, 2006; Shpitser & Pearl, 2006).*** *The three rules of do-calculus, together with standard probability axioms, are* ***complete*** *for identifying causal effects. That is: if a causal effect* $P(Y \mid do(X))$ *can be expressed as a function of observational distributions, then it can be derived by a finite sequence of applications of Rules 1-3.*

This is a landmark result. It means that: 1. The do-calculus provides a **complete algorithmic procedure** for causal identification. 2. If the do-calculus cannot identify an effect, then the effect is **genuinely non-identifiable** from the given DAG and observational data — no clever statistical trick can recover it. 3. The DAG encodes **all** the information needed to determine what can and cannot be learned from observational data.

Why This Matters

The practical implication is both empowering and humbling:

Empowering: Given a correct DAG, we have a systematic, mechanical procedure for determining whether a causal question is answerable from observational data, and if so, what formula to use. The backdoor formula (Equation 3.5) and the frontdoor formula (Equation 3.7) are both special cases derivable from the do-calculus.

Humbling: When the do-calculus says "non-identifiable," there is no escape. No amount of data, no sophisticated machine learning algorithm, no clever regression specification can identify the effect. The only options are: (a) collect additional data (new variables that change the DAG), (b) make additional assumptions (parametric restrictions), or (c) use a different identification strategy (IV, DiD, RDD — the subjects of future chapters).

> ***Blue's Mistake 3.3.*** *Blue, having learned about regression in school, runs a kitchen-sink regression with every variable he can find and declares the causal effect identified. But regression is just a tool for implementing the adjustment formula — it doesn't justify which variables to adjust for. Without a DAG, Blue has no way to know whether his variables satisfy the backdoor criterion, whether he's controlling for a mediator (biasing his estimate), or whether he's conditioning on a collider (introducing new bias). "I controlled for everything" is not a causal argument. A DAG is.*

Chapter Summary

Cascade Badge earned!

You leave Bill's cottage as the sun sets over the Cape, the orange light glinting off the water. You came to Cerulean City with a naive comparison of means. You leave with a toolkit for reading the causal structure of the world.

Key concepts from this chapter:

1. **Observational studies** are necessary when experiments are unethical, infeasible, or impractical. They are abundant but vulnerable to bias.
2. **Confounding** occurs when a common cause of treatment and outcome creates a spurious association. The Omitted Variable Bias formula $OVB = \gamma \cdot \delta$ quantifies the bias from omitting a confounder, and signed OVB analysis determines its direction.
3. **Selection bias, survivorship bias, and Berkson's paradox** arise from conditioning on variables caused by treatment or outcome. They can create, remove, or reverse associations.
4. **Simpson's paradox** demonstrates that aggregate and subgroup associations can point in opposite directions. The resolution requires causal reasoning, not statistical reasoning.
5. **Directed Acyclic Graphs (DAGs)** are the formal language for encoding causal assumptions. Each node is a variable; each arrow represents a direct causal effect; the absence of an arrow is a testable assumption.

6. **Three elemental structures** determine information flow: forks (common causes, creating confounding), chains (mediators, transmitting causal effects), and colliders (common effects, blocking association by default but opening it when conditioned on).
7. **d-Separation** is the algorithmic criterion for reading conditional independence from a DAG. It determines which associations are causal and which are spurious.
8. **The Backdoor Criterion** identifies valid adjustment sets for estimating causal effects. The **Adjustment Formula** converts observational conditionals into interventional quantities.
9. **The Frontdoor Criterion** enables causal identification through a complete mediator, even in the presence of unobserved confounders.
10. **Structural Causal Models** and the **do-calculus** provide the complete mathematical foundation for causal inference from observational data. The three rules of do-calculus are provably complete: they can derive any identifiable causal effect.

Professor Oak's Review Questions

> *Test your understanding of this chapter's core concepts.*

1. **Confounding.** Explain, in your own words, why a naive comparison of outcomes between trainers who entered Cerulean Cave and those who did not is likely biased. Identify at least two confounders and sign the direction of bias for each.
2. **Selection bias vs. confounding.** What is the difference between confounding bias and selection bias? Give an example of each using the Cerulean City setting. Can the same variable produce both types of bias in different analyses?
3. **Simpson's paradox.** Construct your own numerical example (different from the one in Section 3.4) where an association reverses upon stratification. Explain why the reversal occurs and which comparison is causal under your assumed DAG.
4. **The three structures.** For each of the following, state whether conditioning on Z opens or closes the path, and whether this is desirable or harmful for estimating the effect of X on Y:
   - a. $X \leftarrow Z \rightarrow Y$
   - b. $X \rightarrow Z \rightarrow Y$
   - c. $X \rightarrow Z \leftarrow Y$
5. **d-Separation.** In the Kanto Trainer DAG (Section 3.5), determine whether Cave Training and Starter Type are d-separated conditional on: (a) the empty set, (b) {Wealth}, (c) {Wealth, Experience}. Show your work by listing all paths and checking each.
6. **Backdoor criterion.** Consider the simplified DAG (without Natural Talent) in Section 3.7. List all minimal sufficient adjustment sets. For each, verify that it satisfies both conditions of the backdoor criterion.

Trainer Challenge Exercises

> *These exercises require deeper analysis and creative application of the chapter's tools.*

Exercise 3.1: Draw the DAG. Misty tells you that in addition to the variables in the Kanto Trainer DAG, she has data on `pokemon_center_visits` (how often a trainer visits a Pokemon Center) and `team_size` (number of Pokemon in the trainer's party). She believes: - Experience affects team size (experienced trainers catch more Pokemon) - Team size affects badges earned (more Pokemon = more strategic options) - Cave training affects Pokemon Center visits (the Cave is dangerous) - Pokemon Center visits affect badges earned (healed Pokemon perform better) - Wealth affects Pokemon Center visits (healing items reduce the need for Centers)

Draw the extended DAG incorporating these new variables. Identify all paths from Cave Training to Badges Earned. Which are causal? Which are backdoor paths? Find a valid adjustment set using only observed variables (assume Natural Talent is still unobserved).

Exercise 3.2: Collider detection. A sports journalist in Saffron City reports: "Among Pokemon League champions, there is a strong negative correlation between natural talent and training hours. This proves that talented trainers are lazy!" Explain, using the concept of collider bias, why this conclusion is flawed. Draw the relevant DAG and show how conditioning on "champion" induces the spurious correlation.

Exercise 3.3: Frontdoor in practice. Suppose a researcher wants to estimate the causal effect of a new Pokemon training app (D) on tournament wins (Y). The app works entirely by improving trainers' knowledge of type matchups (M). There is an unobserved confounder: trainers who download the app tend to be more motivated (U), and motivation independently affects tournament wins. Draw the DAG, verify the frontdoor criterion, and write out the frontdoor adjustment formula for this specific problem.

Exercise 3.4: do-Calculus application. Consider the DAG: $Z \to X \to Y, Z \to Y, U \to X, U \to Y$ where U is unobserved. Show that the backdoor criterion fails (no valid adjustment set exists using observed variables). Then use Rule 2 of the do-calculus to show that $P(Y \mid do(X = x))$ is nonetheless identifiable by adjusting for Z. (Hint: consider the graph $G_{\underline{X}}$ and check d-separation.)

Further Reading

- **Pearl, J. (2009).** *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press. — The definitive treatment of DAGs, do-calculus, and structural causal models. Essential for anyone serious about graphical causal inference.
- **Pearl, J., Glymour, M., & Jewell, N. P. (2016).** *Causal Inference in Statistics: A Primer*. Wiley. — An accessible introduction to Pearl's framework, with worked examples and exercises. Excellent companion to this chapter.
- **Hernan, M. A., & Robins, J. M. (2020).** *Causal Inference: What If*. Chapman & Hall/CRC. — A masterful treatment of causal inference from an epidemiological perspective, covering DAGs, confounding, selection bias, and identification strategies. Freely available online.
- **Spirtes, P., Glymour, C., & Scheines, R. (2000).** *Causation, Prediction, and Search* (2nd ed.). MIT Press. — The foundational text on causal discovery — learning DAGs from data. Introduces the PC algorithm and the faithfulness condition.
- **Greenland, S., Pearl, J., & Robins, J. M. (1999).** Causal diagrams for epidemiologic research. *Epidemiology*, 10(1), 37-48. — The landmark paper that introduced DAGs to epidemiology. Clear, practical, and still highly relevant.

Skills to Practice in the Notebook

The `notebooks/ch03_cerulean_city.ipynb` notebook turns DAGs from wall art into running code. Before you claim the Cascade Badge, you should be able to do the following fluently:

1. **Build a DAG in code.** Using `networkx` (or the provided `kanto_dag` helper), represent a DAG as nodes + directed edges. Add, remove, and query parents/children/ancestors/descendants programmatically. If you can't construct a DAG in five lines, go back to Section 3.5.
2. **Simulate data from a known DAG.** Write a function that takes a DAG specification (with linear coefficients) and returns a dataframe of n observations consistent with it. This is how you will stress-test every estimator in the rest of the book — you *must* be able to generate data where the truth is known.
3. **Demonstrate OVB numerically.** Simulate data from the Kanto Trainer DAG. Fit the "short" regression (Badges ~ CaveTraining) and the "long" regression (Badges ~ CaveTraining + Experience). Show that the short regression's coefficient equals $\tau + \gamma \cdot \delta$, matching the OVB formula.
4. **Reproduce Simpson's paradox.** Generate (or use the provided) cave-training dataset where the overall effect flips sign when you split by starter type. Compute the aggregated and stratum-specific means and explain which comparison is causal.
5. **Verify d-separation on small DAGs.** For a fork, chain, and collider, compute sample correlations in a simulated dataset and verify that conditioning on the middle node (1) removes the association for forks and chains and (2) *creates* association for colliders. This is the "collider bias experiment" — everyone should see it with their own eyes once.
6. **Apply the backdoor criterion.** Given a DAG, enumerate all paths from D to Y, classify each as causal or backdoor, and find a minimal adjustment set that blocks every backdoor path without conditioning on a descendant of D. Use `dagitty` (via `pydagitty`) or the provided utility to cross-check your answer.
7. **Complete the Trainer Challenge Exercises.** The notebook's challenges walk you through: (a) identifying confounders vs. mediators vs. colliders in a mystery DAG, (b) computing adjustment sets for multiple candidate estimands, and (c) demonstrating Berkson's paradox in simulation.

Check Your Understanding

Before boarding the S.S. Anne, run this gauntlet. Every concept here will be assumed when Chapter 4 builds matching estimators.

Questions you should be able to answer out loud, without notes:

- What is a DAG? What do the arrows mean, and what does a *missing* arrow assert?

- Define confounder, mediator, and collider in terms of arrows. Why is conditioning on a confounder *required* but conditioning on a mediator *forbidden*?
- State the OVB formula $\text{Bias} = \gamma \cdot \delta$ and explain what each factor represents in words.
- What is a backdoor path? Give a plain-English explanation of *why* backdoor paths create spurious associations.
- State the backdoor criterion. Why is the "no descendants of D" clause in there?
- Describe the three elemental structures (fork, chain, collider). For each, say whether conditioning on the middle node opens or closes the path.
- Explain Simpson's paradox in terms of a DAG. Which comparison (aggregated or stratified) is the causal one, and *why*?
- Explain Berkson's paradox using a selection-into-sample DAG. What node is the collider, and what spurious correlation does it create?
- State d-separation informally. When does d-separation imply conditional independence in the data?
- Why is a *missing* arrow a stronger assumption than a present arrow?
- What is the frontdoor criterion, and when would you use it instead of the backdoor criterion?

Tasks you should be able to perform in code:

- Represent a DAG, query its parents/children/ancestors/descendants, and list all paths between two nodes.
- Simulate data from a user-specified DAG with linear structural equations.
- Fit short and long regressions on simulated data and verify that their difference matches $\gamma \cdot \delta$ to within Monte Carlo noise.
- Reproduce Simpson's paradox — both the aggregated and the stratified comparisons — and explain which is causal given the DAG.
- Check a proposed adjustment set against the backdoor criterion programmatically (by enumerating paths).
- Run the collider-bias experiment: generate two independent variables, create a third as their sum (the collider), subset on the collider, and observe the induced (spurious) correlation.

If you can do every one of these, the Cascade Badge is yours.

> *The S.S. Anne awaits in Vermilion Harbor. Among 400 passengers, you'll search for your causal "twin" — a trainer just like you who made a different choice. In Chapter 4, we learn* ***matching and propensity scores****: how to construct counterfactuals from observational data by finding your closest match in the data. Pack your bags — the ship departs at dawn.*

Chapter 4: Vermilion City — Matching & Subclassification

#025 Pikachu

#026 Raichu

#100 Voltorb

Vermilion's Electric-type roster — treatment groups we'll try to match.

Electric-type — Lt. Surge's specialty

Lt. Surge, Vermilion Gym Leader

> *"The best trainers don't just fight hard — they find the right opponent to learn from." — Lt. Surge, Vermilion City Gym Leader*

The S.S. Anne sits in Vermilion Harbor, its hull gleaming under the Kanto sun. Four hundred trainers have disembarked after a weeklong cruise, and among them a fierce debate has broken out: does Lt. Surge's "Thunder Training" program actually make trainers better at the Vermilion Gym? Some passengers completed the program during the cruise; others spent their time at the buffet, the pool, or the onboard Pokemon Center. Now, with the Gym battle looming, we have the perfect observational study.

In Chapter 3, we learned to draw DAGs and identify which variables confound the relationship between a treatment and an outcome. But identifying confounders is only half the battle. We now need a *strategy* for adjusting for those confounders so that we can isolate the causal effect of Thunder Training on Gym performance. This chapter introduces the oldest and most intuitive family of such strategies: **matching**.

The logic is beautifully simple. If we want to know whether Thunder Training helps, we should compare a trained trainer to an untrained trainer who is *otherwise identical* — the same number of badges, the same team level, the same battle experience. If the trained trainer still performs better, we can attribute the difference to the training itself. The challenge, as we will see, is that "otherwise identical" is far harder to achieve than it sounds.

4.1 The Intuition Behind Matching

The S.S. Anne Passenger Manifest

Let us formalize the setting. We observe $n = 400$ trainers who disembarked from the S.S. Anne. For each trainer i, we record the following:

- $D_i \in \{0, 1\}$: whether trainer i completed Thunder Training ($D_i = 1$) or not ($D_i = 0$).
- Y_i: performance score at the Vermilion Gym (0-100 scale, combining damage dealt, Pokemon remaining, and time to victory or defeat).
- X_i: a vector of pre-treatment covariates, including:

Variable	Description	Range
`badges`	Number of Gym badges earned before Vermilion	0-7
`team_level`	Average level of the trainer's six Pokemon	10-55
`battle_exp`	Total battles fought (career)	5-500
`strategy_score`	Score on a tactical reasoning assessment	0-100
`electric_knowledge`	Prior knowledge of Electric-type matchups	0-10
`pokedex_count`	Number of species registered in Pokedex	5-151
`has_ground_type`	Whether the team includes a Ground-type Pokemon	0 or 1
`items_carried`	Number of healing/battle items in the bag	0-20
`hours_training`	Hours spent training Pokemon in the past month	0-200
`cruise_day_joined`	Day of the cruise when training began (1-7, or 0 if untreated)	0-7

Of the 400 trainers, 160 completed Thunder Training ($D = 1$) and 240 did not ($D = 0$). A naive comparison of group means yields:

$$\bar{Y}_1 - \bar{Y}_0 = 72.4 - 58.1 = 14.3 \text{ points}$$

Thunder Training graduates scored, on average, 14.3 points higher at the Vermilion Gym. Case closed?

Why Simple Comparison Fails

Not remotely. Consider the covariate means by treatment group:

Covariate	Thunder Training ($D = 1$)	No Training ($D = 0$)	Difference
`badges`	3.8	2.1	+1.7
`team_level`	34.2	25.6	+8.6
`battle_exp`	187.3	98.4	+88.9
`strategy_score`	68.5	52.3	+16.2
`electric_knowledge`	6.2	3.8	+2.4
`has_ground_type`	0.71	0.43	+0.28

The trainers who chose Thunder Training were *already better*. They had more badges, higher-level Pokemon, more battle experience, and greater strategic sophistication. The 14.3-point raw difference confounds the causal effect of Thunder Training with these pre-existing advantages.

In the language of potential outcomes from Chapter 1, the problem is *selection bias*:

$$\bar{Y}_1 - \bar{Y}_0 = \underbrace{E[Y_i(1) - Y_i(0) \mid D_i = 1]}_{\text{ATT}} + \underbrace{E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]}_{\text{Selection bias}}$$

The second term — selection bias — reflects the fact that Thunder Training participants would have performed better *even without the training*, simply because they are more skilled. Our goal is to eliminate this term.

The Matching Idea

Matching attacks selection bias with a beautifully direct strategy: for each treated trainer, find an untreated trainer with the same covariates. If trainer i completed Thunder Training, has 4 badges, a team level of 32, and 150 career battles, we search the control group for trainer j who also has 4 badges, a team level of 32, and 150 career battles — but who skipped the training. The difference $Y_i - Y_j$ estimates the individual treatment effect for someone with those characteristics, and the average across all matched pairs estimates the **Average Treatment Effect on the Treated (ATT)**:

$$\hat{\text{ATT}} = \frac{1}{n_1} \sum_{i:D_i=1} (Y_i - Y_{j(i)})$$

where $j(i)$ denotes the matched control unit for treated unit i, and n_1 is the number of treated units.

The logic is identical to the ideal experiment we cannot run. In an RCT, randomization ensures that treated and control groups are comparable on all covariates, both observed and unobserved. Matching tries to *construct* a comparison group that resembles the treated group on observed covariates. If we succeed, and if there are no unobserved confounders (the "conditional ignorability" assumption from Chapter 3), then the matched comparison is as good as a randomized experiment.

> ***Blue's Mistake: The Lazy Comparison***
>
> *Blue bursts into the Vermilion Pokemon Center waving a printout. "I computed the average Gym score for Thunder Training graduates and non-participants," he announces. "The difference is 14.3 points! Thunder Training is incredible!"*
>
> *Red shakes his head. "Blue, the graduates already had more badges and higher-level Pokemon. You're comparing Dragonite trainers to Rattata trainers and crediting the difference to a one-week course."*
>
> *Blue's mistake is the most fundamental error in observational causal inference: treating an associational comparison as a causal one. Without adjusting for confounders, the raw mean difference is a biased estimate of the causal effect. Blue is, quite literally, comparing apples to oranges — and then concluding that apples taste better because they're apples.*
>
> *To his credit, Blue did collect the right data. His mistake is in the analysis, not the measurement. As we will see, matching provides a principled way to use that data correctly.*

What Matching Requires

For matching to yield a valid causal estimate, we need two key assumptions (which we formalized as DAG criteria in Chapter 3):

1. **Conditional Ignorability (Unconfoundedness):** $Y(0), Y(1) \perp D \mid X$. Given the observed covariates, treatment assignment is as good as random. There are no unobserved confounders.
2. **Overlap (Common Support):** $0 < P(D = 1 \mid X = x) < 1$ for all x in the support of X. For every combination of covariate values observed among the treated, there exists at least some probability of finding a control unit with the same values.

If both assumptions hold, matching on X eliminates selection bias completely. The question is *how* to match. The next several sections explore increasingly sophisticated answers.

4.2 Exact Matching and Coarsened Exact Matching (CEM)

Exact Matching: The Gold Standard in Theory

The most straightforward approach to matching is **exact matching**: for each treated unit, find a control unit with *exactly the same values* on every covariate. If treated trainer i has $X_i = (4\ \text{badges}, 32\ \text{team level}, 150\ \text{battles}, 68\ \text{strategy}, \dots)$, we require control trainer j to have $X_j = (4, 32, 150, 68, \dots)$ identically.

> ***Definition 4.1 (Exact Matching).*** *A matched pair (i, j) is an exact match if $X_i = X_j$, where i is a treated unit and j is a control unit. The exact matching estimator of the ATT is:*
>
> $$\hat{ATT}_{\text{exact}} = \frac{1}{n_1} \sum_{i:D_i=1} (Y_i - \bar{Y}_{0,M(i)})$$
>
> *where $M(i) = \{j : D_j = 0, X_j = X_i\}$ is the set of exact matches for treated unit i, and $\bar{Y}_{0,M(i)}$ is the average outcome among those matches.*

Exact matching has an obvious appeal: if we find exact matches, there is zero covariate imbalance within matched sets, and the estimator is unbiased under conditional ignorability. But there is a fatal practical problem.

The Curse of Dimensionality

Consider our S.S. Anne data. We have 10 covariates, many of which are continuous or take many discrete values. Even if we discretize `team_level` into just 10 bins, `battle_exp` into 10 bins, `strategy_score` into 10 bins, and so on, the covariate space contains:

$$8 \times 10 \times 10 \times 10 \times 11 \times 147 \times 2 \times 21 \times 10 \times 8 \approx 2.7 \times 10^{10} \text{ cells}$$

With only 400 trainers, the vast majority of these cells are empty. Finding a control trainer who matches a treated trainer on *all* covariates simultaneously is virtually impossible. In practice, exact matching with more than 3-4 discrete covariates produces very few matches, leaving most treated units unmatched and the analysis severely underpowered.

This is the **curse of dimensionality**: as the number of covariates grows, the volume of the covariate space explodes exponentially, and the data become extremely sparse.

Coarsened Exact Matching (CEM)

Iacus, King, and Porro (2012) proposed an elegant compromise: **Coarsened Exact Matching (CEM)**. The idea is to *coarsen* each covariate into a small number of bins, then perform exact matching on the coarsened values.

> ***Definition 4.2 (Coarsened Exact Matching).*** *Let $\tilde{X}_k$ denote the coarsened version of covariate X_k, obtained by mapping continuous values into discrete bins. CEM proceeds as follows:*
>
> 1. ***Coarsen*** *each covariate: define bins for each X_k.*
> 2. ***Exact match*** *on the coarsened covariates: place all units into strata defined by unique combinations of $(\tilde{X}_1, \tilde{X}_2, \ldots, \tilde{X}_K)$.*
> 3. ***Prune*** *any stratum that does not contain at least one treated and one control unit.*
> 4. ***Weight*** *remaining units within each stratum to account for different numbers of treated and control units.*

Worked Example. Suppose we coarsen three key covariates as follows:

Covariate	Original Range	Coarsened Bins
`badges`	0-7	{0-2: Low, 3-5: Mid, 6-7: High}
`team_level`	10-55	{10-25: Low, 26-40: Mid, 41-55: High}
`has_ground_type`	0, 1	{0: No, 1: Yes}

This creates $3 \times 3 \times 2 = 18$ strata. Consider four of them:

Stratum	Badges	Team Level	Ground?	Treated (n_1)	Control (n_0)	Status
A	Low	Low	No	3	28	Matched
B	Mid	Mid	Yes	22	15	Matched
C	High	High	Yes	18	2	Matched
D	High	High	No	8	0	**Pruned**

Stratum D is pruned because it contains no control units — these 8 treated trainers cannot be matched and are excluded from the analysis. Within surviving strata, we assign weights to ensure that the treated-to-control ratio is balanced.

For stratum s with n_{1s} treated units and n_{0s} control units, each control unit receives weight:

$$w_s = \frac{n_{1s}/n_1}{n_{0s}/n_0}$$

where n_1 and n_0 are the total treated and control counts across all matched strata.

The CEM estimate of the ATT within stratum B, for example, would be:

$$\hat{ATT}_B = \bar{Y}_{1,B} - \bar{Y}_{0,B}$$

and the overall ATT is the weighted average across strata:

$$\hat{ATT}_{CEM} = \sum_s \frac{n_{1s}}{n_1^{\text{matched}}} \hat{ATT}_s$$

The Coarsening Tradeoff

CEM introduces a fundamental tradeoff:

- **More coarsening** (fewer, wider bins): more matches are found, but units within a stratum may differ substantially on the original covariates. The matching is less precise, introducing potential bias.
- **Less coarsening** (more, narrower bins): matches are more precise, but fewer strata survive pruning, and more treated units are discarded. The analysis may become underpowered.

The researcher must navigate this tradeoff thoughtfully. CEM's key advantage is its *transparency*: the researcher chooses the coarsening explicitly and can evaluate the resulting covariate balance directly. There is no hidden model dependence — the tradeoffs are visible.

Professor Oak Explains: Why CEM Is a "Method of Monotone Imbalance Bounding"

"CEM has a remarkable theoretical property," Professor Oak notes, adjusting his glasses. "The maximum imbalance between treated and control groups is bounded by the coarsening the researcher chooses. If you coarsen `team_level` *into bins of width 10, you know that any matched treated-control pair differs by at most 10 levels on that variable. This bound holds regardless of the data — it is set ex ante by the researcher."*

"Compare this to other matching methods where the degree of imbalance is discovered only after matching and depends on the data in complex ways. CEM lets you choose your acceptable imbalance first and then find matches that satisfy it. That is a powerful guarantee."

4.3 Distance-Based Matching

Notation at a Glance: Matching and Propensity Scores

The matching literature has a lot of decorated X's and scripty $\mathcal{M}$'s. Here is the plain-English decoder.

Symbol	*Plain-English reading*
X_i	*The vector of covariates for trainer i — e.g.,* `[badges, team_level, experience]`.
$X_i - X_j$	*How far apart two trainers are on each covariate, coordinate by coordinate.*
Σ	*The covariance matrix of X in the full sample — encodes "how much each covariate varies" and "which pairs move together."*
Σ^{-1}	*Its inverse — used to rescale differences so every covariate matters on a comparable footing.*
$d_M(X_i, X_j)$	*Mahalanobis distance — how "similar" trainers i and j are, after rescaling. Small = good match.*
$e(X_i) = P(D_i = 1 \mid X_i)$	*The **propensity score** — the probability that a trainer with covariates X_i ends up treated. A single number summarizing X_i's effect on assignment.*
$e(X_i) = 0.5$	*"Given this trainer's covariates, treatment is a coin flip." The sweet spot for matching.*
$e(X_i) \approx 0$ *or* 1	*"Essentially never/always treated" — these units hurt positivity and should be flagged.*
$\mathcal{M}(i)$	*The set of control units that trainer i can be matched to (after applying a caliper).*
caliper c	*Maximum distance allowed for a match. Tighter $c \to$ better matches, smaller sample.*
w_i	*Weight assigned to observation i (e.g., 1 for matched, 0 for discarded; or a subclass weight).*
ATT	*Average Treatment effect on the Treated — the estimand matching typically targets.*
SMD	*Standardized Mean Difference — the balance diagnostic: $\lvert\bar{X}_T - \bar{X}_C\rvert / s_{\text{pooled}}$. Target < 0.1.*

*Keep in mind: the propensity score $e(X)$ is not a prediction we care about. It is a **balancing score** — a tool for making treated and control groups look alike. Its accuracy matters less than the balance it produces.*

Exact matching (even coarsened) becomes unwieldy as the number of covariates grows. An alternative is to define a *distance metric* between units and match each treated unit to its nearest control neighbor in the covariate space.

Mahalanobis Distance

The most common distance metric for matching is the **Mahalanobis distance**, which accounts for the variance and correlation structure of the covariates:

> ***Definition 4.3 (Mahalanobis Distance).*** *The Mahalanobis distance between two covariate vectors X_i and X_j is:*
>
> $$d_M(X_i, X_j) = \sqrt{(X_i - X_j)^T \Sigma^{-1} (X_i - X_j)}$$
>
> *where Σ is the sample covariance matrix of X (typically computed from the full sample or the control group).*

The Mahalanobis distance generalizes Euclidean distance by "standardizing" the covariates. If two covariates are measured on very different scales — `battle_exp` ranges from 5 to 500, while `badges` ranges from 0 to 7 — Euclidean distance would be dominated by `battle_exp`. The Mahalanobis distance rescales by the inverse covariance matrix, giving each covariate appropriate weight and accounting for correlations between covariates.

Worked Example. Consider two treated trainers and three potential control matches, using only two covariates for illustration: `badges` (X_1) and `team_level` (X_2).

Trainer	Type	Badges (X_1)	Team Level (X_2)
Alice	Treated	4	32
Bob	Control	4	30
Carol	Control	3	35
Dan	Control	5	28

Suppose the sample covariance matrix (from all 400 trainers) is:

$$\Sigma = \begin{pmatrix} 3.2 & 4.5 \\ 4.5 & 64.0 \end{pmatrix}, \quad \Sigma^{-1} = \begin{pmatrix} 0.366 & -0.026 \\ -0.026 & 0.018 \end{pmatrix}$$

The Mahalanobis distance from Alice to each control is:

Alice to Bob: $\Delta X = (0, 2)$

$$d_M = \sqrt{(0,2)\begin{pmatrix} 0.366 & -0.026 \\ -0.026 & 0.018 \end{pmatrix}\begin{pmatrix} 0 \\ 2 \end{pmatrix}} = \sqrt{(0,2)\begin{pmatrix} -0.052 \\ 0.036 \end{pmatrix}} = \sqrt{0.072} = 0.268$$

Alice to Carol: $\Delta X = (-1, 3)$

$$d_M = \sqrt{(-1,3)\begin{pmatrix} 0.366 & -0.026 \\ -0.026 & 0.018 \end{pmatrix}\begin{pmatrix} -1 \\ 3 \end{pmatrix}} = \sqrt{(-1,3)\begin{pmatrix} -0.444 \\ 0.080 \end{pmatrix}} = \sqrt{0.444 + 0.240} = \sqrt{0.684} = 0.827$$

Alice to Dan: $\Delta X = (1, -4)$

$$d_M = \sqrt{(1,-4)\begin{pmatrix} 0.366 & -0.026 \\ -0.026 & 0.018 \end{pmatrix}\begin{pmatrix} 1 \\ -4 \end{pmatrix}} = \sqrt{(1,-4)\begin{pmatrix} 0.470 \\ -0.098 \end{pmatrix}} = \sqrt{0.470 + 0.392} = \sqrt{0.862} = 0.929$$

Bob is the nearest neighbor ($d_M = 0.268$), which makes intuitive sense: he matches Alice on badges exactly and is very close on team level.

Nearest-Neighbor Matching

1:1 matching pairs each treated unit with its single closest control unit. **k:1 matching** pairs each treated unit with its k nearest controls, using the average of their outcomes as the imputed counterfactual.

A key choice is **with or without replacement**:

- **Without replacement:** Each control unit can be matched to at most one treated unit. This avoids using the same control observation multiple times, but may force poor matches if the pool of unmatched controls shrinks. The order in which treated units are matched matters, introducing some arbitrariness.
- **With replacement:** Each control unit can serve as a match for multiple treated units. This generally produces better matches (the best control is always available), but reduces the effective sample size because some control observations receive heavy weight.

Caliper Matching

A **caliper** imposes a maximum acceptable distance: if no control unit is within the caliper radius of a treated unit, that treated unit goes unmatched.

> ***Definition 4.4 (Caliper Matching).*** *Given a caliper width $c > 0$, the match set for treated unit i is:*
>
> $$M(i) = \{j : D_j = 0,\ d(X_i, X_j) \leq c\}$$
>
> *If $M(i) = \varnothing$, unit i is discarded.*

The caliper introduces a **bias-variance tradeoff**:

- **Tight caliper** (c small): matches are excellent (low bias), but many treated units are unmatched, reducing sample size and increasing variance. The estimand shifts toward the ATT among *matchable* treated units, which may differ from the overall ATT.
- **Loose caliper** (c large): most treated units are matched (low variance), but some matches are poor (higher bias).

A common rule of thumb, due to Austin (2011), is to set the caliper at 0.2 standard deviations of the logit of the propensity score.

> ***Professor Oak Explains: Which Distance Metric?***
>
> *"When you have a small number of continuous covariates — say, five or fewer — Mahalanobis distance matching works very well," Professor Oak advises. "It handles scale differences and correlations naturally. However, as the dimension grows, Mahalanobis distance can degrade because the inverse covariance matrix becomes unstable, and the notion of 'nearest neighbor' becomes less meaningful in high-dimensional space."*
>
> *"For higher-dimensional problems, the propensity score — which we will encounter in the next section — collapses all covariates into a single dimension. This is its great power, and also, as we shall see, the source of some controversy."*

4.4 The Propensity Score

Motivation: The Dimensionality Problem Revisited

Matching on many covariates simultaneously is difficult. With K covariates, we are searching for neighbors in K-dimensional space, and the curse of dimensionality makes close neighbors increasingly rare. What if we could reduce the problem to matching on a *single number* that summarizes all the relevant information in X?

This is precisely what the **propensity score** achieves.

> ***Definition 4.5 (Propensity Score).*** *The propensity score is the conditional probability of receiving treatment given the observed covariates:*
>
> $$e(X_i) = P(D_i = 1 \mid X_i)$$
>
> *For the S.S. Anne study, $e(X_i)$ is the probability that trainer i completed Thunder Training, given their badges, team level, battle experience, and other observed characteristics.*

The Rosenbaum-Rubin Theorem

The propensity score's power comes from a remarkable theorem proved by Rosenbaum and Rubin (1983):

> ***Theorem 4.1 (Propensity Score Theorem, Rosenbaum & Rubin, 1983).*** *If treatment assignment is strongly ignorable given X — that is, $Y(0), Y(1) \perp D \mid X$ and $0 < e(X) < 1$ — then treatment assignment is also strongly ignorable given the propensity score alone:*
>
> $$Y(0), Y(1) \perp D \mid e(X)$$

This theorem says that if conditioning on the full covariate vector X is sufficient to eliminate confounding, then conditioning on the scalar $e(X)$ is *also* sufficient. We can collapse ten covariates into one number without losing any information relevant to confounding.

Proof Sketch: The Balancing Property

The theorem rests on a key property: the propensity score *balances* the covariates between treated and control groups.

> ***Lemma 4.1 (Balancing Property).*** $D \perp X \mid e(X)$. *That is, within strata defined by the propensity score, the distribution of covariates is the same for treated and control units.*

Proof sketch. For any measurable set A in the covariate space:

$$P(X \in A \mid e(X) = p, D = 1) = P(X \in A \mid e(X) = p, D = 0)$$

To see why, note that:

$$P(D = 1 \mid X, e(X)) = P(D = 1 \mid X) = e(X)$$

The first equality holds because $e(X)$ is a function of X (so conditioning on both X and $e(X)$ is the same as conditioning on X). Therefore, within a group of units that all share the same propensity score $e(X) = p$, the treatment probability is p for every unit regardless of their specific covariate values. This means treatment assignment within a propensity score stratum is effectively *random* with respect to X — exactly the balancing property.

Now, if $Y(0), Y(1) \perp D \mid X$ and $D \perp X \mid e(X)$, it follows that $Y(0), Y(1) \perp D \mid e(X)$. Intuitively: conditioning on $e(X)$ makes D independent of X, and conditioning on X makes D independent of potential outcomes, so conditioning on $e(X)$ suffices to make D independent of potential outcomes.

Estimating the Propensity Score

In practice, $e(X)$ is unknown and must be estimated from the data. The most common approach is **logistic regression**:

$$\log \frac{e(X_i)}{1 - e(X_i)} = \beta_0 + \beta_1 \cdot \text{badges}_i + \beta_2 \cdot \text{team\_level}_i + \beta_3 \cdot \text{battle\_exp}_i + \beta_4 \cdot \text{strategy\_score}_i + \cdots$$

For the S.S. Anne data, suppose we estimate the following model:

Covariate	Coefficient ($\hat{\beta}$)	Std. Error	z-value	p-value
Intercept	-4.821	0.612	-7.88	< 0.001
`badges`	0.342	0.078	4.38	< 0.001
`team_level`	0.058	0.014	4.14	< 0.001
`battle_exp`	0.004	0.001	3.12	0.002
`strategy_score`	0.031	0.008	3.88	< 0.001
`electric_knowledge`	0.189	0.052	3.63	< 0.001
`has_ground_type`	0.614	0.198	3.10	0.002
`items_carried`	0.023	0.031	0.74	0.458
`hours_training`	0.007	0.003	2.33	0.020
`pokedex_count`	0.005	0.003	1.67	0.095

Interpretation. Each additional badge increases the log-odds of participating in Thunder Training by 0.342 (equivalently, the odds ratio is $e^{0.342} \approx 1.41$). Having a Ground-type Pokemon increases the odds by a factor of $e^{0.614} \approx 1.85$, which makes sense: trainers who already have a type advantage against Electric types may be particularly interested in learning to exploit it.

For a specific trainer — say, Alice with 4 badges, team level 32, 150 battles, strategy score 68, electric knowledge 7, Ground-type present, 8 items, 45 hours training, and 62 Pokedex entries — the estimated propensity score is:

$$\begin{aligned}
\hat{e}(X_{\text{Alice}}) &= \text{logit}^{-1}(-4.821 + 0.342(4) + 0.058(32) + 0.004(150) + 0.031(68) + 0.189(7) + 0.614(1) + 0.023(8) + 0.007(45) + 0.005(62)) \\
&= \text{logit}^{-1}(-4.821 + 1.368 + 1.856 + 0.600 + 2.108 + 1.323 + 0.614 + 0.184 + 0.315 + 0.310) \\
&= \text{logit}^{-1}(3.857) = \frac{1}{1 + e^{-3.857}} = 0.979
\end{aligned}$$

Alice is almost certainly a Thunder Training participant, which is consistent with her strong profile.

Checking Overlap

Before using propensity scores for matching, we must verify the **overlap** (common support) assumption: the propensity score distributions for treated and control groups must overlap substantially. If there are regions where the propensity score is near 0 or 1, there are treated units with no comparable controls (or vice versa), and matching in those regions is unreliable.

A summary of the propensity score distribution in our S.S. Anne data:

Statistic	Treated ($D = 1$)	Control ($D = 0$)
Min	0.08	0.02
Q1	0.35	0.12
Median	0.58	0.25
Q3	0.78	0.42
Max	0.98	0.89

The distributions overlap in the range $[0.08, 0.89]$. Treated trainers with propensity scores above 0.89 have no comparable controls; these units are in the region of *non-overlap* and should be handled carefully (trimmed, or given a sensitivity analysis).

What the Propensity Score Does and Does Not Do

The propensity score is a powerful tool, but it is essential to understand its limitations:

What it does: - Balances *observed* covariates between treated and control groups. - Reduces a multi-dimensional matching problem to a one-dimensional one. - Enables consistent estimation of causal effects under the assumption that all confounders are observed.

What it does NOT do: - Balance *unobserved* covariates. If a confounder is missing from the propensity score model, matching on the score does not adjust for it. - Guarantee that the propensity score model is correctly specified. If the true model involves interactions or nonlinearities that the logistic regression omits, the estimated score may fail to balance covariates. - Eliminate all sources of bias. The propensity score is only as good as the conditional ignorability assumption. If that assumption fails, no propensity score method can save us.

> ***Blue's Mistake: "The Score Handles Everything"***
>
> *Blue has heard about propensity scores and is excited. "So I just run a logistic regression, compute the score, and match on it? That handles all confounding?"*
>
> *"Only confounding from observed variables," Red corrects. "If there's a confounder you didn't include in your model — like natural talent or patience — the propensity score won't help."*
>
> *"But my model has an R^2 of 0.35! That's pretty good!"*
>
> *"The R^2 of your propensity score model is irrelevant to whether you've captured all confounders. A model that perfectly predicts treatment based on five variables still misses the sixth if you didn't include it. Propensity scores balance what you put in. They have no magical power over what you leave out."*

4.5 Propensity Score Matching

The Procedure

With estimated propensity scores in hand, we can match treated and control units on this single dimension:

> ***Definition 4.6 (Propensity Score Matching).*** *For each treated unit i, find the control unit j that minimizes $|\hat{e}(X_i) - \hat{e}(X_j)|$. The PSM estimator of the ATT is:*
>
> $$\hat{ATT}_{PSM} = \frac{1}{n_1} \sum_{i:D_i=1} (Y_i - Y_{j(i)})$$

As with Mahalanobis matching, we must choose:

1. **1:1 or k:1 matching**: More matches per treated unit reduces variance but may introduce worse matches.
2. **With or without replacement**: With replacement generally yields better matches but complicates variance estimation.
3. **Caliper**: A maximum allowable distance on the propensity score scale (commonly 0.2 SD of the logit propensity score).

Greedy vs. Optimal Matching

Greedy (nearest-neighbor) matching proceeds sequentially: for each treated unit (in some order), find the closest available control. This is fast but can produce suboptimal pairings — the first treated unit may "steal" a control that would have been a better match for a later treated unit.

Optimal matching minimizes the total distance across all matched pairs simultaneously, solving a combinatorial optimization problem (typically a minimum-cost flow problem). This guarantees the globally best set of pairings but is computationally more expensive. For most datasets of moderate size, optimal matching is feasible and preferred.

Worked Example

Let us trace through propensity score matching for a small subset of S.S. Anne passengers. Consider five treated and eight control trainers:

Trainer	D	$\hat{e}(X)$	Gym Score (Y)
T1	1	0.72	78
T2	1	0.45	65
T3	1	0.61	71
T4	1	0.83	82
T5	1	0.38	60
C1	0	0.40	58
C2	0	0.69	70
C3	0	0.55	63
C4	0	0.22	48
C5	0	0.47	62
C6	0	0.75	72
C7	0	0.60	64
C8	0	0.34	55

1:1 nearest-neighbor matching without replacement:

Treated	$\hat{e}$	Best Match	$\hat{e}$ (match)	$\lvert\Delta\hat{e}\rvert$	Y_T	Y_C	$Y_T - Y_C$
T1	0.72	C6	0.75	0.03	78	72	+6
T2	0.45	C5	0.47	0.02	65	62	+3
T3	0.61	C7	0.60	0.01	71	64	+7
T4	0.83	C2	0.69	0.14	82	70	+12
T5	0.38	C1	0.40	0.02	60	58	+2

$$\hat{ATT} = \frac{6 + 3 + 7 + 12 + 2}{5} = \frac{30}{5} = 6.0$$

Note that T4's match (C2, $\hat{e} = 0.69$) is the weakest — the distance is 0.14. If we had used a caliper of 0.10, T4 would have gone unmatched, and the ATT would have been estimated from only 4 pairs:

$$\hat{ATT}_{\text{caliper}} = \frac{6 + 3 + 7 + 2}{4} = 4.5$$

The choice of caliper thus affects both the estimate and the estimand (the population for which the ATT is estimated).

The Abadie and Imbens Warning

Abadie and Imbens (2016) proved a cautionary result: propensity score matching can *increase* the distance between matched units on the original covariates, even when the propensity scores are close. Two trainers with $\hat{e} = 0.50$ might have arrived at that score via very different covariate profiles — one with many badges but low strategy, another with few badges but high strategy.

> ***Definition 4.7 (The PSM Imbalance Risk, Abadie & Imbens, 2016).*** *Matching on the estimated propensity score does not generally minimize covariate imbalance. If the propensity score model is misspecified, PSM can increase the mean squared difference in covariates between matched pairs relative to the unmatched sample.*

This result has led some methodologists — most notably King and Nielsen (2019) — to argue that propensity score matching should be used with caution, and that direct covariate matching (e.g., Mahalanobis distance, CEM) may be preferable when the number of covariates is manageable.

The practical recommendation: after propensity score matching, *always* check covariate balance (Section 4.7). If balance is poor, the propensity score model may need re-specification.

Standard Error Estimation

Standard errors for matching estimators are subtle. The naive formula — treating matched pairs as independent — ignores two sources of uncertainty:

1. **Estimation of the propensity score.** The score is estimated, not known, and this estimation uncertainty should propagate into the standard error of the ATT.
2. **Matching with replacement.** When a control unit is used multiple times, the effective sample size is reduced, and observations are no longer independent.

Abadie and Imbens (2006) derived analytical standard errors for nearest-neighbor matching estimators that account for matching uncertainty. In practice, many analysts use **bootstrap** standard errors, though Abadie and Imbens (2008) showed that the standard bootstrap is not valid for nearest-neighbor matching without modification. The subsampling bootstrap or the analytical formula should be used instead.

4.6 Subclassification / Stratification

The Idea

Instead of finding a single matched partner for each treated unit, **subclassification** (also called **stratification**) divides the propensity score into strata and computes the treatment effect *within* each stratum.

> ***Definition 4.8 (Subclassification on the Propensity Score).*** *Divide the estimated propensity score into* S *strata defined by cutpoints* $c_0 < c_1 < \cdots < c_S$. *Within stratum* s, *estimate the treatment effect as:*
>
> $$\hat{\tau}_s = \bar{Y}_{1s} - \bar{Y}_{0s}$$
>
> *The ATT is estimated as a weighted average:*
>
> $$\widehat{ATT}_{sub} = \sum_{s=1}^{S} \frac{n_{1s}}{n_1} \hat{\tau}_s$$
>
> *where* n_{1s} *is the number of treated units in stratum* s *and* $n_1 = \sum_s n_{1s}$.

Cochran's Rule of Five

A classical result due to Cochran (1968) provides practical guidance: subclassification into **five strata** based on a single confounding covariate removes approximately 90% of the bias due to that covariate. With the propensity score summarizing multiple covariates, five to ten strata are typically sufficient for substantial bias reduction.

Worked Example

We divide the estimated propensity scores from our S.S. Anne data into five quintile-based strata:

Stratum	$\hat{e}$ Range	Treated (n_{1s})	Control (n_{0s})	$\bar{Y}_{1s}$	$\bar{Y}_{0s}$	$\hat{\tau}_s$
1	[0.02, 0.20)	8	72	52.3	47.1	5.2
2	[0.20, 0.35)	22	68	59.8	53.6	6.2
3	[0.35, 0.55)	38	52	66.4	60.1	6.3
4	[0.55, 0.75)	48	36	73.2	66.8	6.4
5	[0.75, 0.98]	44	12	80.5	73.3	7.2

The ATT estimate via subclassification:

$$\hat{ATT}_{sub} = \frac{8}{160}(5.2) + \frac{22}{160}(6.2) + \frac{38}{160}(6.3) + \frac{48}{160}(6.4) + \frac{44}{160}(7.2)$$

$$= 0.050(5.2) + 0.138(6.2) + 0.238(6.3) + 0.300(6.4) + 0.275(7.2)$$

$$= 0.260 + 0.856 + 1.499 + 1.920 + 1.980 = 6.52$$

This estimate of 6.52 points is substantially smaller than the naive difference of 14.3 points, confirming that much of the raw gap was due to selection bias. The matching estimate of 6.0 from Section 4.5 is similar, providing reassuring convergence across methods.

Advantages Over Matching

Subclassification has several practical advantages:

1. **No units are discarded.** Every treated and control unit appears in exactly one stratum (assuming overlap). This preserves the full sample, unlike matching with calipers which can discard treated units.
2. **Computational simplicity.** No optimization problem is required — just sorting and stratifying.
3. **Transparency.** The within-stratum comparisons are easy to inspect. If one stratum shows a wildly different treatment effect, it may signal model misspecification or effect heterogeneity.
4. **Variance estimation is straightforward.** Standard errors can be computed within each stratum using conventional formulas and combined using the delta method.

The main disadvantage is *residual confounding*: within each stratum, the propensity score is not perfectly constant, so some covariate imbalance may remain. Using more strata mitigates this but reduces the sample size per stratum.

4.7 Covariate Balance Assessment

Matching and subclassification aim to eliminate covariate imbalance between treated and control groups. The most important diagnostic step is to verify that they succeeded. Unlike regression, where model fit is assessed via residuals and R^2, matching quality is assessed via **covariate balance**.

Standardized Mean Difference

The primary balance metric is the **Standardized Mean Difference (SMD)**:

Definition 4.9 (Standardized Mean Difference). *For covariate X_k, the SMD is:*

$$SMD_k = \frac{\bar{X}_{k,1} - \bar{X}_{k,0}}{\sqrt{(s^2_{k,1} + s^2_{k,0})/2}}$$

where $\bar{X}_{k,1}$ and $\bar{X}_{k,0}$ are the covariate means in the treated and matched control groups, and $s^2_{k,1}$ and $s^2_{k,0}$ are the corresponding variances.

The SMD expresses the group difference in standard deviation units, making it comparable across covariates measured on different scales. A widely used rule of thumb (Rosenbaum & Rubin, 1985; Austin, 2009):

- $|SMD| < 0.1$: **Good balance**. Negligible difference.
- $0.1 \le |SMD| < 0.25$: **Moderate imbalance**. May warrant concern.
- $|SMD| \ge 0.25$: **Substantial imbalance**. The matching has failed to adequately balance this covariate.

Variance Ratios

In addition to means, we should check that the *variances* of covariates are similar across groups. The **variance ratio** is:

$$\mathrm{VR}_k = \frac{s_{k,1}^2}{s_{k,0}^2}$$

A ratio near 1.0 indicates balanced variances. Rubin (2001) recommends that variance ratios fall between 0.5 and 2.0. A variance ratio far from 1 suggests that matching has created groups with different distributional shapes, even if the means are balanced.

Balance Table: Before and After Matching

The following table summarizes covariate balance for the S.S. Anne data before matching and after 1:1 propensity score matching:

Covariate	Before Matching		After Matching	
	SMD	VR	SMD	VR
`badges`	0.85	1.12	0.04	0.98
`team_level`	0.91	0.87	0.06	1.03
`battle_exp`	0.78	1.45	0.08	1.11
`strategy_score`	0.82	0.93	0.05	0.97
`electric_knowledge`	0.72	1.08	0.03	1.01
`has_ground_type`	0.58	—	0.07	—
`items_carried`	0.21	1.22	0.02	1.04
`hours_training`	0.64	1.31	0.09	1.08
`pokedex_count`	0.43	0.91	0.06	0.99

Before matching, every covariate shows substantial imbalance (SMDs ranging from 0.21 to 0.91). After matching, all SMDs fall below 0.10 and all variance ratios are between 0.5 and 2.0. This is excellent balance.

Love Plots

A **Love plot** (named after Thomas Love) is a horizontal dot plot that displays the SMD for each covariate, with markers for "before matching" and "after matching." Vertical reference lines are drawn at ± 0.1 (or ± 0.25) to indicate the balance threshold.

```
                     Love Plot: Covariate Balance
             −0.25          0.0          +0.25         +0.50         +0.75         +1.0
               |              |              |              |              |              |
badges         |          ●   |              |              |              |    ○         |
team_level     |          ●   |              |              |              |     ○        |
battle_exp     |           ●  |              |              |          ○   |              |
strategy_score |          ●   |              |              |              |    ○         |
elec_knowledge |         ●    |              |              |         ○    |              |
has_ground     |           ●  |              |              |    ○         |              |
items_carried  |          ●   |              |              |              |              |
               |              |         ○    |              |              |              |
hours_training |           ●  |              |              |       ○      |              |
pokedex_count  |           ●  |              |         ○    |              |              |
               |              |              |              |              |              |
               └──────────────┴──────────────┘
                        ● After     ○ Before
                        Dashed lines at ±0.1
```

In a proper graphical implementation (see the companion Jupyter notebook), the Love plot provides an immediate visual summary: all "after" markers should cluster near zero, while the "before" markers may be scattered far from zero.

What to Do When Balance Is Poor

If matching fails to achieve adequate balance on one or more covariates, the analyst should:

1. **Re-specify the propensity score model.** Add interaction terms (e.g., `badges` × `team_level`), polynomial terms (e.g., `battle_exp`^{2}), or additional covariates that were initially excluded.
2. **Try a different matching method.** If propensity score matching produces poor balance, consider Mahalanobis distance matching, CEM, or a combination (e.g., Mahalanobis matching within propensity score calipers).
3. **Use a tighter caliper.** Reducing the caliper width forces closer matches but at the cost of losing some treated units.

4. **Increase the matching ratio.** Moving from 1:1 to k:1 matching can improve balance by averaging over multiple controls.
5. **Consider alternative estimators.** If matching proves difficult, inverse probability weighting (Chapter 5) or doubly robust estimation may be more appropriate.

> ***Professor Oak Explains: Balance, Not Prediction***
>
> *"A common misunderstanding is that the propensity score model should be evaluated by how well it predicts treatment assignment," Professor Oak cautions. "In fact, the goal is not prediction but balance. A model with a modest pseudo-R^2 but excellent covariate balance is preferable to one with a high pseudo-R^2 but residual imbalance."*
>
> *"This is why we assess matching quality through balance diagnostics — SMDs, variance ratios, Love plots — rather than model fit statistics. The propensity score is a tool for balance, not a prediction model. Its quality is judged entirely by whether it achieves balance."*

4.8 Limitations of Matching

Matching is intuitive, transparent, and powerful. But like every method in causal inference, it rests on assumptions that may be violated. Understanding these limitations is essential for responsible application — and for knowing when to reach for other tools.

The Critical Assumption: No Unobserved Confounders

Matching adjusts for *observed* covariates. The validity of matching-based causal estimates depends entirely on the **conditional ignorability** assumption:

$$Y(0), Y(1) \perp D \mid X$$

If there exists a confounder U that affects both treatment and outcome but is not included in X, matching on X alone does not eliminate the bias.

The S.S. Anne Unmasking. Suppose that two variables we *did not observe* also influence both Thunder Training participation and Gym performance:

- **Patience** (U_1): patient trainers are more likely to complete a week-long training program *and* more likely to persevere through a difficult Gym battle.
- **Natural Talent** (U_2): naturally talented trainers are drawn to intensive training *and* perform better in any battle, regardless of training.

The true causal DAG includes:

```
            badges ──────→ Thunder Training ──────→ Gym Score
              ↑                  ↑                      ↑
          team_level        patience (U₁)          patience (U₁)
              ↑                  ↑                      ↑
        battle_exp        nat. talent (U₂)      nat. talent (U₂)
              ↑
      strategy_score
              ↑
    electric_knowledge
```

Because U_1 and U_2 are unobserved, they do not appear in our propensity score model. Matching on observed covariates does not block the backdoor paths through U_1 and U_2. Our estimated ATT of approximately 6.0-6.5 points is *biased upward*: part of it reflects the advantage that patient, naturally talented trainers have even without Thunder Training.

If the true causal effect is, say, 4.0 points, then roughly 2.0-2.5 points of our estimate is residual confounding bias due to the unobserved variables. No amount of re-specifying the propensity score model or trying different matching algorithms can fix this — the problem is in the data, not the method.

Positivity Violations

The **overlap** assumption requires that for every covariate pattern observed among the treated, there is a non-zero probability of being a control. In practice, this fails in predictable ways:

- **Structural violations:** Some trainers may *always* or *never* participate in Thunder Training given their characteristics. For example, trainers with zero badges and team level below 15 may never have been offered the program (selection by the cruise director). These trainers have $e(X) = 0$, and no matching can create comparisons for treated units in this region.

- **Random violations:** Even without structural reasons, small sample sizes can produce regions of covariate space where only treated or only control units are observed.

Positivity violations manifest as extreme propensity scores (near 0 or 1) and require the analyst to either trim the sample (restricting inference to the region of overlap) or use extrapolation-based methods (regression), which introduce model dependence.

Model Dependence

Ho, Imai, King, and Stuart (2007) demonstrated that when matching is imperfect — when matched units still differ on covariates — the treatment effect estimate can be sensitive to how the researcher handles this residual imbalance. Different matching specifications (1:1 vs. 5:1, with vs. without replacement, different calipers) can yield different results.

This **model dependence** is not unique to matching — regression suffers from it too — but it means that analysts should report results across multiple reasonable specifications. If the conclusions are robust across specifications, we can be more confident. If they vary substantially, the evidence is weaker.

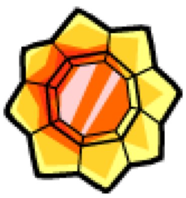
From Matching to Stronger Designs

These limitations of matching are not reasons to abandon it. Matching remains one of the most transparent and interpretable methods for observational causal inference. But they motivate the tools we will develop in subsequent chapters:

- **Chapter 5 (Celadon City)** introduces **regression** and **inverse probability weighting** as alternatives to matching, and shows how combining them yields **doubly robust** estimators that are consistent if *either* the outcome model or the propensity score model is correctly specified.
- **Chapter 6 (Fuchsia & Cinnabar)** introduces **instrumental variables**, which can identify causal effects even in the presence of unobserved confounders — precisely the scenario where matching fails. The key is finding a variable that affects treatment but has no direct effect on the outcome.
- **Sensitivity analysis** techniques (which we will encounter along the way) allow us to ask: "How strong would an unobserved confounder have to be to overturn our conclusions?" If the answer is "implausibly strong," we can be more confident despite the theoretical vulnerability.

> ***Professor Oak Explains: Matching Is a Design, Not Just an Estimator***
>
> *"I want to emphasize something that many researchers overlook," Professor Oak says. "Matching is not just a statistical technique — it is a research design. When you match, you are constructing a comparison group that approximates what a randomized experiment would have produced. This is a design step, just as choosing your sample or your treatment is a design step."*
>
> *"The practical implication is that matching should be done before looking at outcomes. Assess covariate balance, iterate on the propensity score model, try different matching specifications — all without peeking at* Y*. Only after you are satisfied with the matched sample should you estimate the treatment effect. This separation of design and analysis protects against the temptation to choose the matching specification that gives the 'best' result."*

Chapter Summary



Thunder Badge earned!

In this chapter, we explored the family of matching methods for causal inference in observational studies, using the S.S. Anne Thunder Training study as our running example.

Key concepts:

1. **Matching intuition.** Compare treated units to untreated units that are similar on observed covariates. This approximates the comparison that a randomized experiment would provide.
2. **Exact and Coarsened Exact Matching.** Exact matching requires identical covariate values, which is infeasible in high dimensions. CEM coarsens covariates into bins and matches exactly on bins, offering a transparent bias-variance tradeoff.

3. **Distance-based matching.** The Mahalanobis distance accounts for covariate scaling and correlations. Nearest-neighbor matching (with or without replacement, with or without calipers) finds close matches in covariate space.
4. **The propensity score.** The probability of treatment given covariates, $e(X) = P(D = 1|X)$. The Rosenbaum-Rubin theorem shows that conditioning on this scalar is sufficient to eliminate confounding from observed variables. Typically estimated via logistic regression.
5. **Propensity score matching.** Match treated and control units on estimated propensity scores. Computationally simple but may increase covariate imbalance if the score model is misspecified (Abadie & Imbens, 2016).
6. **Subclassification.** Divide the propensity score into strata, estimate effects within strata, and combine. Uses all observations and avoids discarding data.
7. **Covariate balance assessment.** Standardized mean differences, variance ratios, and Love plots are essential diagnostics. The goal of matching is balance, not prediction.
8. **Limitations.** Matching adjusts only for observed confounders. Unobserved confounders, positivity violations, and model dependence remain threats. These motivate the methods of subsequent chapters.

The journey so far. In Pallet Town, we learned the language of potential outcomes. In Pewter City, we saw that randomization solves the causal inference problem by design. In Cerulean City, we learned to identify confounders using DAGs. Here in Vermilion City, we took our first step toward *adjusting* for confounders in observational data — finding each trainer's twin. But the twin we found may not be as identical as we hoped. Hidden variables lurk beneath the surface, and our next challenges will equip us with more powerful tools to confront them.

Professor Oak's Review Questions

1. **Conceptual.** Explain, in your own words, why the raw difference in Gym scores between Thunder Training graduates and non-participants ($\bar{Y}_1 - \bar{Y}_0 = 14.3$) is not a valid estimate of the causal effect of Thunder Training. Decompose this difference into the ATT and the selection bias term.
2. **Matching mechanics.** Suppose you have three covariates, each taking 5 possible values. How many cells does the exact matching table have? If you have 200 treated units and 300 control units, what fraction of cells do you expect to contain at least one treated *and* one control unit? What does this imply for the feasibility of exact matching?
3. **Propensity score theory.** State the balancing property of the propensity score. Why does this property imply that conditioning on $e(X)$ is sufficient for causal identification, even though $e(X)$ is a scalar summary of a vector X?
4. **Critical thinking.** Blue matches treated and control trainers on their propensity scores and finds that balance on `strategy_score` is poor (SMD = 0.32). He declares that the propensity score "doesn't work." What are three concrete steps he should take before abandoning the method?
5. **Assumptions.** Suppose an unobserved variable, `motivation`, influences both the decision to undertake Thunder Training and performance at the Vermilion Gym. Draw the DAG including this variable. Does matching on observed covariates block the backdoor path through `motivation`? What methods from later chapters might address this problem?
6. **Subclassification.** Cochran (1968) showed that five strata remove approximately 90% of bias from a single covariate. Does this guarantee vanish if the propensity score model is misspecified? Why or why not?

Trainer Challenge Exercises

Exercise 4.1: Computing Mahalanobis Distances

Consider the following four trainers and two covariates:

Trainer	Treatment	Badges (X_1)	Strategy (X_2)
Ash	1	3	72
Misty	0	5	65
Brock	0	4	70
Erika	0	2	78

The covariance matrix is:

$$\Sigma = \begin{pmatrix} 1.5 & -3.0 \\ -3.0 & 28.0 \end{pmatrix}$$

a. Compute Σ^{-1}.

b. Compute the Mahalanobis distance from Ash to each control trainer (Misty, Brock, Erika).

c. Which trainer is Ash's nearest neighbor? Is this the same trainer who would be selected by Euclidean distance?

d. Explain intuitively why the Mahalanobis distance might differ from the Euclidean distance in this case.

Exercise 4.2: Propensity Score Estimation and Matching

A logistic regression for Thunder Training participation yields the following model:

$$\log \frac{e(X)}{1 - e(X)} = -3.5 + 0.40 \cdot \text{badges} + 0.05 \cdot \text{team\_level} + 0.25 \cdot \text{has\_ground}$$

Consider the following six trainers:

Trainer	D	Badges	Team Level	Has Ground	Y
A	1	4	30	1	75
B	1	3	35	0	68
C	1	5	28	1	80
D	0	3	32	1	62
E	0	4	25	0	58
F	0	2	38	1	55

a. Compute the estimated propensity score $\hat{e}(X)$ for each trainer.

b. Perform 1:1 nearest-neighbor matching (without replacement) on the propensity score, matching each treated trainer to a control.

c. Compute the matched ATT estimate.

d. Assess covariate balance before and after matching by computing the SMD for each covariate. Does matching improve balance?

Exercise 4.3: Subclassification

Using the estimated propensity scores from the full S.S. Anne dataset (provided in the companion notebook), divide the sample into 5 strata based on quintiles of the propensity score distribution.

a. Compute the number of treated and control units in each stratum. Are there any strata with very few treated or very few control units?

b. Compute the within-stratum treatment effect $\hat{\tau}_s$ for each stratum.

c. Compute the overall ATT using the subclassification formula. Compare this to the matching estimate from Section 4.5.

d. Now divide the sample into 10 strata instead of 5. How does the ATT estimate change? What happens to the standard error?

Exercise 4.4: Balance Table Construction

Using the matched sample from Exercise 4.2 (or the full matched S.S. Anne sample from the companion notebook):

a. Construct a complete balance table showing, for each covariate: the treated mean, the control mean, the SMD, and the variance ratio, both before and after matching.

b. Create a Love plot visualizing the SMDs before and after matching.

c. Identify any covariates with $|\text{SMD}| > 0.1$ after matching. For each such covariate, propose a modification to the propensity score model that might improve balance (e.g., adding an interaction term, a polynomial, or re-specifying the functional form).

Further Reading

- **Rosenbaum, P. R. & Rubin, D. B. (1983).** "The Central Role of the Propensity Score in Observational Studies for Causal Effects." *Biometrika*, 70(1), 41-55. The foundational paper introducing the propensity score and proving the balancing theorem.
- **Iacus, S. M., King, G., & Porro, G. (2012).** "Causal Inference without Balance Checking: Coarsened Exact Matching." *Political Analysis*, 20(1), 1-24. Introduces CEM and the concept of monotone imbalance bounding.
- **Stuart, E. A. (2010).** "Matching Methods for Causal Inference: A Review and a Look Forward." *Statistical Science*, 25(1), 1-21. An excellent comprehensive review of matching methods.
- **Abadie, A. & Imbens, G. W. (2006).** "Large Sample Properties of Matching Estimators for Average Treatment Effects." *Econometrica*, 74(1), 235-267. Derives the asymptotic properties and standard errors of matching estimators.
- **Abadie, A. & Imbens, G. W. (2016).** "Matching on the Estimated Propensity Score." *Econometrica*, 84(2), 781-807. Shows that propensity score matching can increase covariate imbalance under misspecification.
- **Ho, D. E., Imai, K., King, G., & Stuart, E. A. (2007).** "Matching as Nonparametric Preprocessing for Reducing Model Dependence in Parametric Causal Inference." *Political Analysis*, 15(3), 199-236. Argues for matching as a preprocessing step to reduce model dependence in subsequent regression.
- **King, G. & Nielsen, R. (2019).** "Why Propensity Scores Should Not Be Used for Matching." *Political Analysis*, 27(4), 435-454. A provocative critique arguing that PSM can increase imbalance, approximates a random experiment poorly, and should be replaced by other methods.
- **Cochran, W. G. (1968).** "The Effectiveness of Adjustment by Subclassification in Removing Bias in Observational Studies." *Biometrics*, 24(2), 295-313. The classic result that five strata remove approximately 90% of bias.
- **Austin, P. C. (2011).** "An Introduction to Propensity Score Methods for Reducing the Effects of Confounding in Observational Studies." *Multivariate Behavioral Research*, 46(3), 399-424. An accessible introduction to propensity score methods for applied researchers.

Skills to Practice in the Notebook

The notebook `notebooks/ch04_vermilion_city.ipynb` is where matching stops being a diagram and becomes an estimator. Complete each of the following to claim the Thunder Badge:

1. **Implement exact matching from scratch.** Given a dataset with a discrete covariate (e.g., hometown), pair each treated unit with controls that share the same value. Compute the matched ATT as the mean of within-pair differences. No library — do it in pandas first.
2. **Compute Mahalanobis distances and do 1:1 nearest-neighbor matching.** Use `scipy.spatial.distance.mahalanobis` (or your own implementation) to find the closest control for each treated unit on a small set of covariates. Then pair outcomes and compute $\hat{\tau}_{\mathrm{ATT}}$. Do this both *with* and *without* replacement and compare.
3. **Estimate a propensity score.** Fit a logistic regression (or a gradient boosted classifier) for $P(D = 1 \mid X)$. Plot overlap: histograms of $\hat{e}(X)$ for treated and control. Be able to *spot* positivity violations visually.
4. **Do propensity score matching with a caliper.** Match each treated unit to its nearest control within a caliper of $0.2 \times \mathrm{SD}(\mathrm{logit}(\hat{e}))$. Count how many treated units are dropped. Compute the caliper-matched ATT.
5. **Check balance before and after matching.** For every covariate, compute the standardized mean difference (SMD) in the raw data and in the matched sample. Plot a "love plot" (SMD before vs. after) and know what "success" looks like (all SMDs below 0.1).
6. **Run subclassification / stratification.** Split the sample into 5 propensity-score bins. Within each bin, compute a treated-vs-control difference. Combine bin-level estimates into an overall ATE and ATT using the appropriate weights. Compare against the matched estimate.
7. **Complete the Trainer Challenge Exercises.** The notebook walks you through: (a) rebuilding the Rosenbaum-Rubin balancing result in simulation, (b) showing how caliper width trades off bias vs. variance empirically, and (c) diagnosing a dataset where matching *fails* because positivity is violated.

Check Your Understanding

Before challenging Lt. Surge, work through these. Matching is the foundation for Chapter 5's regression adjustment and the weighting methods in Chapter 6 — do not move forward with gaps.

Questions you should be able to answer out loud, without notes:

- Why do we match? What is the "counterfactual twin" intuition in one sentence?
- State the ignorability assumption $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i \mid X_i$ and say precisely what matching assumes and does not assume.
- Why does the curse of dimensionality make exact matching on many covariates impractical?
- Define the Mahalanobis distance and explain in plain English what the Σ^{-1} factor *does* (why not just Euclidean distance?).
- Define the propensity score $e(X) = P(D = 1 \mid X)$. Why is it a *balancing* score rather than a *prediction* score?
- State the Rosenbaum-Rubin theorem in one sentence. What's the practical payoff?
- What is positivity? Why do $e(X)$ values near 0 or 1 break matching?
- Compare matching with replacement and without replacement. When does each win?
- What does a caliper do, and what is the bias-variance tradeoff it controls?
- What is the standardized mean difference (SMD), and what threshold flags "acceptable balance"?
- Why does matching target the ATT rather than the ATE, and how would you adapt the method to estimate the ATE instead?
- Name three ways matching can fail even when the code runs cleanly. (Hidden bias, poor overlap, model misspecification of $e(X)$, etc.)

Tasks you should be able to perform in code:

- Fit a propensity score model, plot the overlap histogram, and flag units with $\hat{e}(X) < 0.05$ or > 0.95.
- Do 1:1 and k:1 nearest-neighbor matching (with and without replacement) using a Mahalanobis or propensity-score distance.
- Compute the ATT from a matched sample and get a valid (Abadie-Imbens) standard error — or at least a bootstrap one.
- Compute SMDs for every covariate, pre- and post-match, and draw the love plot.
- Run propensity-score subclassification with K bins and aggregate to ATT and ATE.
- Spot and *report* a positivity violation in the data rather than silently ignoring it.

Do all this and the Thunder Badge is yours.

Badge Earned: Thunder Badge

```
╔═══════════════════════════════════════════════╗
║                                               ║
║            ⚡ THUNDER BADGE  ⚡               ║
║                                               ║
║   Awarded for mastering matching methods,     ║
║   propensity scores, and the art of finding   ║
║   each trainer's counterfactual twin.         ║
║                                               ║
║   Bearer understands:                         ║
║     • Exact, coarsened, and distance matching ║
║     • The Rosenbaum–Rubin balancing theorem   ║
║     • Propensity score estimation & matching  ║
║     • Subclassification / stratification      ║
║     • Covariate balance diagnostics           ║
║     • Limitations of selection on observables ║
║                                               ║
║          Lt. Surge, Vermilion City            ║
╚═══════════════════════════════════════════════╝
```

Next Stop: Celadon City

Celadon City's Department Store has everything a trainer could want — potions, TMs, evolution stones, and an overwhelming number of choices. But who buys what, and why? When we observe trainers purchasing Rare Candies and later winning battles, how do we separate the effect of the candy from the wealth and experience that let them afford it in the first place? The answer will teach you about regression, inverse probability weighting, and the art of being doubly robust — building an estimator that works even when you get one of your models wrong. Grab your shopping list and your covariance matrix. The Celadon Department Store awaits…

Chapter 5: Celadon City — Regression, Weighting, & Doubly Robust Methods

#043 Oddish

#044 Gloom

#045 Vileplume

Erika's Grass-type lineage — a reminder that regression can adjust for anything, if the DAG says so.

Grass-type — Erika's specialty

Erika, Celadon Gym Leader

> *"Celadon City — the city of rainbow dreams."*

Celadon City is the commercial heart of Kanto. The Department Store towers above the skyline, its six floors stocked with everything a trainer could desire: TMs for new battle moves, stat-boosting vitamins like Calcium and Iron, evolution stones, and the coveted Rare Candy. Trainers flock here from across the region, wallets open, convinced that spending more on premium items will transform them into Pokemon League champions.

But does spending at the Celadon Department Store actually *cause* better battle outcomes? Or do wealthier trainers — who can afford to spend lavishly — also happen to train harder, travel to more routes, and accumulate more experience? If so, naive comparisons between big spenders and frugal trainers will confuse the effect of spending with the effect of wealth and dedication.

This is the problem of confounding, and in this chapter, we develop the three most important tools in the applied researcher's arsenal for addressing it: **regression adjustment**, **inverse probability weighting**, and **doubly robust estimation**. Along the way, we will navigate two dangerous pitfalls — omitted variable bias and post-treatment bias — that can silently undermine even well-intentioned analyses.

And then there is the Game Corner. Behind the cheerful facade of slot machines and prize exchanges, Team Rocket runs an elaborate scheme: subsidized items funneled to trainers they want to recruit. These subsidies are not handed out at random. Team Rocket targets trainers with specific characteristics — making the "treated" group deeply unrepresentative of the broader trainer population. This is precisely the setting where inverse probability weighting earns its keep.

Gym Leader Erika, the serene master of Grass-type Pokemon, will not be swayed by hand-waving or casual correlations. She demands rigorous statistical proof — a proper identification strategy — before she will acknowledge any causal claim. Earn her respect, and you earn the Rainbow Badge.

5.1 Regression for Causal Inference

5.1.1 OLS: Not Just for Prediction

In Chapter 3, we introduced DAGs and the backdoor criterion as tools for reasoning about which variables to condition on. In Chapter 4, we used matching and propensity scores to create comparable treatment and control groups. Now we turn to the workhorse of empirical social science: **ordinary least squares (OLS) regression**.

Most students first encounter OLS in a statistics or econometrics course focused on *prediction* — finding the linear function that best fits the data. But regression can also serve as a tool for *causal inference*, provided we are willing to make specific assumptions about the data-generating process. The key distinction is this: in predictive modeling, we care about $\hat{Y}$; in causal inference, we care about $\hat{\tau}$, the coefficient on the treatment variable.

Consider our Celadon City question. We observe n trainers, each with:

- Y_i: battle outcome (e.g., gym badges earned, tournament wins)
- D_i: a binary indicator for whether the trainer spent heavily at the Department Store ($D_i = 1$) or not ($D_i = 0$)
- X_i: a vector of pre-treatment covariates (wealth, prior experience, number of routes traveled, starter Pokemon type, etc.)

We write the **causal regression model**:

$$Y_i = \alpha + \tau D_i + X_i' \beta + \epsilon_i$$

where τ is the parameter we hope captures the causal effect of Department Store spending on battle outcomes. But when does the OLS estimator $\hat{\tau}_{\text{OLS}}$ actually recover a causal effect?

5.1.2 The Conditional Mean Independence Assumption

The answer hinges on a critical assumption. Recall from Chapter 1 that each trainer has two potential outcomes: $Y_i(1)$ (outcome if they spend heavily) and $Y_i(0)$ (outcome if they do not). The average treatment effect is $\tau_{\text{ATE}} = E[Y(1) - Y(0)]$.

> ***Definition 5.1 (Conditional Mean Independence).*** *Treatment D satisfies conditional mean independence given covariates X if:*
>
> $$E[Y(0) \mid X, D] = E[Y(0) \mid X]$$
>
> *and*
>
> $$E[Y(1) \mid X, D] = E[Y(1) \mid X]$$
>
> *That is, once we condition on X, knowing a trainer's treatment status provides no additional information about their potential outcomes.*

This is a weaker requirement than the full conditional independence assumption $\{Y(0), Y(1)\} \perp\!\!\!\perp D \mid X$ that we used in Chapter 4. We only need *mean* independence, not independence of the entire distribution. In DAG language, conditional mean independence holds when X blocks all backdoor paths between D and Y — at least with respect to the conditional expectation function.

5.1.3 When Does OLS Estimate the ATE?

Under conditional mean independence, OLS recovers the ATE under additional conditions:

1. **Linearity of the conditional expectation.** The true conditional expectation $E[Y(0) \mid X]$ is linear in X: that is, $E[Y(0) \mid X] = \alpha + X'\beta$. This is a functional form restriction.
2. **Constant (homogeneous) treatment effects.** The causal effect $\tau_i = Y_i(1) - Y_i(0) = \tau$ is the same for all trainers i. Under heterogeneous effects, OLS recovers a variance-weighted average of conditional treatment effects, which generally differs from the ATE.
3. **Correct specification.** The control variables X must include all confounders — variables that jointly affect D and Y. Omitting a confounder leads to omitted variable bias (Section 5.2), while including post-treatment variables leads to bad control bias (Section 5.3).

> ***Professor Oak Explains: The Regression Anatomy***
>
> *"Think of regression as doing two things simultaneously, young trainer. First, it uses the covariates X to predict treatment D and outcome Y. Then it correlates the residuals — the parts of D and Y that X cannot explain. This is exactly the Frisch-Waugh-Lovell (FWL) theorem, and it gives you powerful intuition for what regression actually does."*

5.1.4 The Frisch-Waugh-Lovell Theorem

The FWL theorem provides the deepest intuition for how regression achieves causal adjustment. It states that the coefficient $\hat{\tau}$ from the full regression $Y_i = \alpha + \tau D_i + X_i'\beta + \epsilon_i$ is numerically identical to the coefficient from a bivariate regression of residualized Y on residualized D:

> ***Theorem 5.1 (Frisch-Waugh-Lovell).*** *Let $\tilde{D}_i$ be the residual from regressing D_i on X_i, and let $\tilde{Y}_i$ be the residual from regressing Y_i on X_i. Then:*
>
> $$\hat{\tau}_{OLS} = \frac{\sum_i \tilde{D}_i \tilde{Y}_i}{\sum_i \tilde{D}_i^2}$$
>
> *This is the slope from the simple regression $\tilde{Y}_i = \tau \tilde{D}_i + u_i$.*

The intuition is "partialling out." Step 1: Regress D on X and save the residuals $\tilde{D}$. These residuals represent the variation in treatment status that is *not* predicted by the covariates — the "as-good-as-random" part of treatment, if our model is correct. Step 2: Regress Y on X and save the residuals $\tilde{Y}$. Step 3: Regress $\tilde{Y}$ on $\tilde{D}$. This final regression uses only the variation in treatment that cannot be explained by confounders to estimate the effect on the part of the outcome that is similarly unexplained.

In our Celadon City context, FWL tells us: first strip out the predictable relationship between trainer wealth/experience and Department Store spending. Then strip out the predictable relationship between wealth/experience and battle outcomes. Finally, ask: among trainers with similar wealth and experience, does the *residual* variation in spending predict *residual* variation in outcomes?

5.1.5 Saturated Models and Nonparametric Regression

An important special case arises when we fully saturate the model with interactions. If X is discrete — say, a set of binary indicators for wealth level (low, medium, high) and experience tier (novice, intermediate, expert) — then a fully saturated regression includes a dummy for every cell in the cross-tabulation plus interactions with treatment. In this case, the regression coefficient is a weighted average of within-cell treatment-control differences, and the linearity assumption is automatically satisfied.

Formally, with K discrete cells, the saturated model is:

$$Y_i = \sum_{k=1}^{K} \alpha_k \cdot 1(X_i \in \text{cell } k) + \sum_{k=1}^{K} \tau_k \cdot D_i \cdot 1(X_i \in \text{cell } k) + \epsilon_i$$

The OLS estimate of τ_k equals the difference in sample means between treated and control units within cell k. The overall $\hat{\tau}$ is a *precision-weighted* average of these cell-specific effects — cells with more observations and less variance receive more weight. Note carefully: this is not the same as the ATE, which would weight by cell *population shares*. This distinction matters when treatment effects are heterogeneous and the distribution of covariates differs between treatment and control groups.

5.1.6 Regression as Matching

Regression and matching are closely related. Both condition on covariates X to compare like with like. But they differ in how they weight observations:

Feature	Matching	Regression
Functional form	Nonparametric	Assumes linearity (unless saturated)
Weighting	Equal weight to each matched pair/stratum	Precision-weighted (more weight to higher-variance cells)
Extrapolation	Refuses to compare if no match exists	Extrapolates linearly into regions with no data
Curse of dimensionality	Severe with many covariates	Handles high-dimensional X via parametric structure

Regression's willingness to extrapolate is both its greatest strength and its greatest vulnerability. When treatment and control groups have good overlap (common support), regression and matching tend to agree. When overlap is poor, regression cheerfully extrapolates — a property we should regard with suspicion.

5.1.7 Worked Example: Badges on Department Store Spending

Suppose we observe the following data for six Celadon City trainers:

Trainer	Badges (Y)	Dept Store (D)	Wealth (W)	Experience (E)
Ash	5	0	2	3
Misty	7	1	4	5
Brock	6	1	3	4
Gary	8	1	5	6
Jessie	3	0	1	2
James	4	0	2	1

Naive comparison (ignoring confounders):

$$\bar{Y}_{D=1} - \bar{Y}_{D=0} = \frac{7+6+8}{3} - \frac{5+3+4}{3} = 7.0 - 4.0 = 3.0$$

This suggests spending at the Department Store increases badges by 3. But treated trainers are wealthier and more experienced.

Short regression (no controls): $Y_i = \alpha + \tau D_i + \epsilon_i$ yields $\hat{\tau}_{short} = 3.0$ — identical to the naive comparison.

Long regression (with controls): $Y_i = \alpha + \tau D_i + \beta_1 W_i + \beta_2 E_i + \epsilon_i$. Running this regression (which we encourage the reader to verify in the companion notebook), we obtain approximately:

$$\hat{\tau}_{long} \approx 0.8$$

The coefficient drops from 3.0 to 0.8 once we control for wealth and experience. Most of the naive "effect" was driven by confounding: wealthier, more experienced trainers both spend more and earn more badges. The residual effect of 0.8 — if our model is correctly specified — represents the causal contribution of Department Store spending, holding wealth and experience fixed.

> ***Key Insight.*** *The dramatic decline from $\hat{\tau}_{short} = 3.0$ to $\hat{\tau}_{long} = 0.8$ is driven by omitted variable bias in the short regression. We formalize this in Section 5.2.*

5.2 Omitted Variable Bias

5.2.1 The OVB Formula

Omitted variable bias (OVB) is the single most important concept in applied causal inference with observational data. It formalizes what happens when a relevant confounder is left out of a regression.

Consider two regressions:

- **Short regression**: $Y_i = \alpha_s + \tau_s D_i + \epsilon_i^s$
- **Long regression**: $Y_i = \alpha_\ell + \tau_\ell D_i + \gamma W_i + \epsilon_i^\ell$

where W is the omitted variable (e.g., wealth). The OVB formula relates the two coefficients:

> ***Theorem 5.2 (Omitted Variable Bias Formula).***
>
> $$\hat{\tau}_{short} = \hat{\tau}_{long} + \hat{\delta} \cdot \hat{\gamma}$$
>
> *where: - $\hat{\delta}$ is the coefficient from regressing W on D: the relationship between the omitted variable and treatment - $\hat{\gamma}$ is the coefficient on W in the long regression: the relationship between the omitted variable and the outcome, controlling for treatment*
>
> *Equivalently, the **bias** is:*
>
> $$OVB = \hat{\tau}_{short} - \hat{\tau}_{long} = \hat{\delta} \cdot \hat{\gamma}$$

The formula decomposes the bias into two intuitive pieces. The omitted variable biases the treatment effect estimate only if it is correlated with *both* the treatment ($\hat{\delta} \neq 0$) and the outcome ($\hat{\gamma} \neq 0$). If the omitted variable is unrelated to treatment or unrelated to the outcome, there is no bias.

5.2.2 Signing the Bias

Even when we cannot estimate the bias precisely (because we do not observe W), we can often *sign* it using substantive reasoning. This is one of the most valuable skills in applied work.

$\hat{\delta}$ (Omitted var ~ Treatment)	$\hat{\gamma}$ (Omitted var ~ Outcome)	Bias Direction
+	+	**Positive** (upward)
+	−	**Negative** (downward)
−	+	**Negative** (downward)
−	−	**Positive** (upward)

In our Department Store example:

- **Omitted variable**: trainer "grit" (unobserved dedication and work ethic)
- **$\hat{\delta}$ (grit ~ spending)**: Grittier trainers probably spend more at the Department Store because they invest in every possible advantage. So $\hat{\delta} > 0$.
- **$\hat{\gamma}$ (grit ~ badges)**: Grittier trainers earn more badges because they train harder, regardless of spending. So $\hat{\gamma} > 0$.
- **Bias direction**: $\hat{\delta} \cdot \hat{\gamma} > 0$. The bias is **positive**: omitting grit makes the Department Store spending effect appear *larger* than it truly is.

This matches our worked example: $\hat{\tau}_{\text{short}} = 3.0$ overstated the controlled estimate of $\hat{\tau}_{\text{long}} = 0.8$.

5.2.3 Worked Example: Computing OVB

Returning to our six-trainer dataset, let us verify the OVB formula for the case where wealth is omitted.

Short regression: $Y_i = \alpha_s + \tau_s D_i + \epsilon_i^s$, yielding $\hat{\tau}_{\text{short}} = 3.0$.

Long regression: $Y_i = \alpha_\ell + \tau_\ell D_i + \gamma W_i + \epsilon_i^\ell$. Suppose this yields $\hat{\tau}_{\text{long}} = 1.2$ and $\hat{\gamma} = 0.9$.

Auxiliary regression: $W_i = a + \delta D_i + u_i$. The mean wealth among treated trainers is $(4 + 3 + 5)/3 = 4.0$; among control trainers, $(2 + 1 + 2)/3 = 1.67$. So $\hat{\delta} = 4.0 - 1.67 = 2.33$.

Verification:

$$\hat{\tau}_{\text{short}} = \hat{\tau}_{\text{long}} + \hat{\delta} \cdot \hat{\gamma}$$

$$3.0 \approx 1.2 + 2.33 \times 0.9$$

$$3.0 \approx 1.2 + 2.1 = 3.3$$

The approximation is not exact here because we are using a simplified example with few observations and have omitted the second covariate (experience), but the direction and magnitude of the bias are clear. The OVB of approximately $+2.1$ accounts for the bulk of the difference between the naive and controlled estimates.

5.2.4 Oster (2019): Coefficient Stability and Bounding the Bias

A critical question in any observational study is: *Could unobserved confounders explain away the entire estimated effect?* Emily Oster (2019) formalized a method for assessing this, building on the ideas of Altonji, Elder, and Taber (2005).

The key insight is to examine how the treatment coefficient *changes* when we add observed controls, and then extrapolate to ask how much it would change if we could also add unobserved controls.

> ***Definition 5.2 (Oster's δ).*** *Define the proportional selection parameter:*
>
> $$\delta = \frac{\hat{\tau}_{\text{long}} \cdot (\tilde{R}^2_{\text{max}} - \tilde{R}^2_{\text{long}})}{\hat{\tau}_{\text{long}} - \hat{\tau}_{\text{short}}} \cdot \frac{\tilde{R}^2_{\text{long}} - \tilde{R}^2_{\text{short}}}{\tilde{R}^2_{\text{max}} - \tilde{R}^2_{\text{long}}}$$
>
> *More precisely, the bias-adjusted estimate sets the treatment effect to zero and solves for the δ that rationalizes $\hat{\tau} = 0$. A common formulation is:*
>
> $$\delta = \frac{(\hat{\tau}_{\text{long}})(\tilde{R}^2_{\text{max}} - \tilde{R}^2_{\text{long}})}{(\tilde{R}^2_{\text{long}} - \tilde{R}^2_{\text{short}})(\hat{\tau}_{\text{short}} - \hat{\tau}_{\text{long}})}$$
>
> *When $\delta > 1$, unobservables would need to be more important than observables to fully explain away the effect. When $\delta < 1$, even moderate unobserved confounding could eliminate the estimated effect.*

The mechanics are:

1. Run the **short regression** (no controls). Record $\hat{\tau}_{\text{short}}$ and R^2_{short}.
2. Run the **long regression** (with all observable controls). Record $\hat{\tau}_{\text{long}}$ and R^2_{long}.
3. Choose R^2_{max}, the hypothetical R^2 if we could observe *everything*. A common choice is $R^2_{\text{max}} = \min(1, 1.3 \times R^2_{\text{long}})$.
4. Compute δ. If $\delta \gg 1$, the result is robust to selection on unobservables.

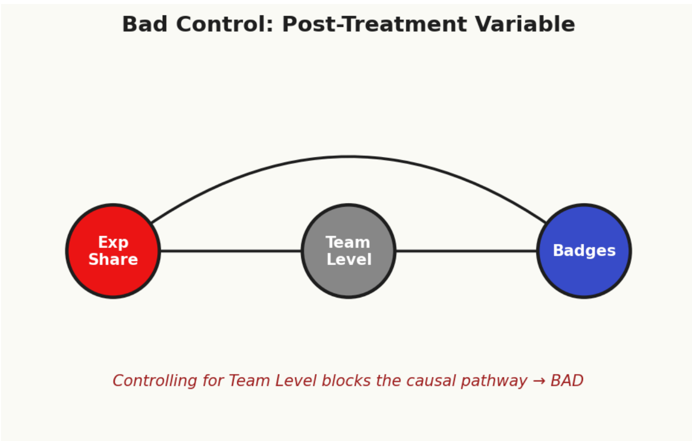
In our Department Store example, suppose $R^2_{\text{short}} = 0.40$ and $R^2_{\text{long}} = 0.82$, with $R^2_{\text{max}} = 1.0$. Then:

$$\delta = \frac{0.8 \times (1.0 - 0.82)}{(0.82 - 0.40) \times (3.0 - 0.8)} = \frac{0.8 \times 0.18}{0.42 \times 2.2} = \frac{0.144}{0.924} \approx 0.156$$

Since $\delta \approx 0.156 < 1$, this tells us that even unobservables that are only 15.6% as important as our observed controls (wealth and experience) could explain away the remaining effect. This is concerning — the estimated effect of Department Store spending is fragile. Erika would raise an eyebrow.

> ***Professor Oak Explains: Interpreting Oster's δ***
>
> *"Think of δ as a stress test, young trainer. If $\delta = 3$, then unobservables would need to be three times as important as your observed controls to eliminate the effect. That is reassuring. If $\delta = 0.5$, even modest unobserved confounding could do the job. When you publish your research, always report δ alongside your main estimates. Referees — and Gym Leaders — will respect the honesty."*

5.3 Bad Controls and Post-Treatment Bias



Controlling for a post-treatment variable blocks the causal pathway.

5.3.1 The Most Common Mistake

If omitted variable bias is the *most important* concept in applied causal inference, **bad control bias** (also called **post-treatment bias** or **overcontrol bias**) is the *most common mistake*. It arises when researchers include variables in their regressions that are affected by the treatment — descendants of D on the DAG — in an effort to "control for everything."

The logic seems reasonable: "I want to isolate the effect of D on Y, so I should control for as many variables as possible." This is the **kitchen sink** approach, and it is wrong.

> ***Definition 5.3 (Bad Control).*** *A variable is a bad control if it is a descendant of the treatment variable D on the causal DAG. Conditioning on a bad control opens spurious paths, introduces collider bias, or blocks causal pathways — any of which distorts the estimated treatment effect.*

5.3.2 The Exp Share Example

Consider the effect of the Exp Share item on gym badges earned. The Exp Share distributes experience points to all Pokemon in a trainer's party, not just the one that battles. We want to estimate:

$$\text{Treatment }(D) : \text{Exp Share usage} \longrightarrow \quad \text{Outcome }(Y) : \text{Gym badges earned}$$

A natural confounder is the trainer's *initial experience* — how many battles they had fought before obtaining the Exp Share. We should control for this. But a tempting (and dangerous) "control" is the trainer's **average team level** (L) at the time badges are measured.

The DAG looks like this:

```
   Initial Experience
      /          \
     v            v
 Exp Share (D) --> Avg Team Level (L) --> Badges (Y)
     \                                    ^
      \----------------------------------/
```

Average team level L is a **descendant** of treatment D: using the Exp Share causes team levels to rise. Controlling for L blocks part of the causal pathway from D to Y and absorbs the very effect we are trying to measure. The regression coefficient on D, after controlling for L, captures only the *direct* effect of Exp Share on badges that does *not* operate through team level — which may be nearly zero, even if the total causal effect is large.

Formally, the total effect is:

$$\tau_{\text{total}} = \tau_{\text{direct}} + \tau_{\text{indirect}}$$

where τ_{indirect} flows through L. Controlling for L eliminates τ_{indirect}, leaving only τ_{direct}. If the Exp Share helps you earn badges *precisely because* it raises your team level, then $\tau_{\text{direct}} \approx 0$ and you would erroneously conclude the Exp Share has no effect.

5.3.3 Collider Bias from Bad Controls

The situation can be even worse than merely blocking causal channels. When a post-treatment variable is a *collider* — a variable caused by both the treatment and the outcome (or by the treatment and an unobserved confounder of the outcome) — conditioning on it opens a spurious path and can introduce bias where none existed.

Consider this DAG:

```
    D ---------> Y
     \          ^
      v        /
       M <----U
```

Here M is affected by both D and an unobserved variable U that also affects Y. Controlling for M opens the path $D \to M \leftarrow U \to Y$, creating a spurious association between D and Y through U.

5.3.4 M-Bias (Butterfly Bias)

A subtler case is M-bias, named for the shape of its DAG:

```
   U1        U2
  /  \      /  \
 v    v    v    v
 D    M    Y
      ^
      |
```

More precisely:

```
U1 --> D
U1 --> M
U2 --> M
U2 --> Y
```

Here M is a collider on the path $D \leftarrow U_1 \rightarrow M \leftarrow U_2 \rightarrow Y$. Without conditioning on M, this path is blocked (because M is a collider). But conditioning on M *opens* it, creating a spurious association between D and Y even though there was no confounding to begin with.

M-bias is controversial because it requires a very specific (and arguably uncommon) causal structure. But it illustrates a crucial principle: **adding controls is not always innocent**.

5.3.5 DAG-Based Variable Selection

The correct approach to variable selection for causal inference is to use the **backdoor criterion** from Chapter 3:

> ***Rule 5.1 (Variable Selection for Regression).*** *Include a variable in your regression if and only if:*
>
> 1. *It is needed to **block a backdoor path** between D and Y, AND*
> 2. *It is **not a descendant** of the treatment D.*
>
> *If both conditions are met, the variable is a "good control." If condition 2 is violated, the variable is a "bad control" regardless of whether it satisfies condition 1.*

In practice, this means:

- **Good controls**: pre-treatment covariates that are causes (or proxies for causes) of both D and Y. Examples: trainer's initial wealth, starter Pokemon type, hometown.
- **Bad controls**: post-treatment variables that are consequences of D. Examples: average team level after using Exp Share, number of TMs purchased after deciding to invest in the Department Store, Pokemon friendship level after treatment.
- **Neutral variables**: pre-treatment variables that predict Y but not D (or vice versa). Including these does not bias $\hat{\tau}$ but may improve precision (if they predict Y) or waste degrees of freedom (if they predict neither).

> ***Blue's Mistake: The Kitchen Sink Regression***
>
> *Blue, the player's rival, runs the following regression to estimate the effect of Exp Share on badges:*
>
> $$\text{Badges}_i = \alpha + \tau \cdot \text{ExpShare}_i + \beta_1 \cdot \text{TeamLevel}_i + \beta_2 \cdot \text{MovesLearned}_i + \beta_3 \cdot \text{ItemsUsed}_i + \epsilon_i$$
>
> *He proudly reports: "See? The Exp Share coefficient is basically zero. Exp Shares don't help at all!"*
>
> *But TeamLevel, MovesLearned, and ItemsUsed are all consequences of using the Exp Share. By controlling for the mechanisms through which the treatment operates, Blue has surgically removed the very effect he was trying to measure.*
>
> *When confronted, Blue says: "But I controlled for everything!" This is exactly the problem. In causal inference, controlling for everything is not a virtue — it is a recipe for bias. The kitchen sink approach — throwing every available variable into the regression — is one of the most common errors in applied research.*
>
> ***The lesson****: More controls are not always better. Let the DAG guide your choices.*

5.3.6 A Decision Framework

When you encounter a candidate control variable Z, ask these questions in order:

1. **Is Z affected by the treatment D?** If yes, it is a bad control. Do not include it.
2. **Is Z a confounder (causes both D and Y)?** If yes, it is a good control. Include it.
3. **Is Z an instrumental variable (causes D but not Y directly)?** Including it does not bias $\hat{\tau}$ but may reduce precision in finite samples.
4. **Is Z a pure predictor of Y (affects Y but not D)?** Including it reduces residual variance and improves precision without affecting consistency.

When in doubt, draw the DAG first and apply the backdoor criterion.

5.4 Inverse Probability Weighting (IPW)

Notation at a Glance: IPW and Doubly Robust Estimators

The rest of this chapter leans heavily on propensity weights and doubly robust corrections. Keep this next to you.

Symbol	***Plain-English reading***
$e(X_i) = P(D_i = 1 \mid X_i)$	*The **propensity score** — each trainer's probability of getting treated, given their observed covariates.*
$\hat{e}(X_i)$	*What we estimate for $e(X_i)$ from a model (usually logistic regression).*
$1 - e(X_i)$	*The matching "control side" probability — how likely this trainer was to end up in the control group.*
$w_i = D_i / e(X_i) + (1 - D_i)/(1 - e(X_i))$	*The IPW weight — "how much should we upweight trainer i to pretend treatment was random?"*
$\hat{\tau}_{HT}$	*The Horvitz-Thompson estimator — unbiased but unstable when weights get huge.*
$\hat{\tau}_{H}$	*The Hajek (normalized) estimator — slightly biased in finite samples but much stabler.*
$m_d(X_i) = E[Y \mid X_i, D_i = d]$	*The **outcome model** — what we'd predict for trainer i if treated ($d = 1$) or control ($d = 0$).*
$\hat{m}_d(X_i)$	*The estimated outcome model (e.g., from OLS or a random forest).*
$\hat{\tau}_{AIPW}$	*The augmented / doubly robust estimator — combines $\hat{e}$ and $\hat{m}$; works if either is correct.*
ϵ_i	*Regression residual: the part of Y_i the model couldn't explain.*
ATO	*Average Treatment effect on the Overlap population — what overlap weights target.*
trimming at α	*"Drop anyone with $\hat{e}(X) \notin [\alpha, 1 - \alpha]$." Changes the estimand to the trimmed population.*
"doubly robust"	*If either the outcome model $\hat{m}_d$ or the propensity model $\hat{e}$ is right, the estimator is still consistent. Not "double the guarantees" — more like "a safety net for whichever model you trust less."*

The most important single insight: the propensity score is not a prediction you care about. It is a reweighting device. Its job is to make the treated and control populations look exchangeable, not to predict treatment assignment accurately.

5.4.1 The Game Corner Problem

Behind the neon lights of Celadon's Game Corner, Team Rocket runs a clandestine operation. Among their schemes: subsidizing premium battle items (Rare Candies, PP Ups, stat vitamins) for trainers they hope to recruit. But Team Rocket does not hand out subsidies randomly. They target trainers who are *already strong* — those with more badges, higher-level Pokemon, and demonstrated battle prowess. After all, Team Rocket wants useful recruits, not novices.

This creates a selection problem. If we compare trainers who received subsidized items (treated) to those who did not (control), the treated group is systematically different: stronger, more experienced, and more battle-hardened. The "treatment" group is **unrepresentative** of the broader population. A naive comparison would confuse the effect of the items with the pre-existing superiority of the trainers who received them.

Regression adjustment, as in Section 5.1, is one solution. But regression relies on correctly specifying the functional form of $E[Y \mid X, D]$. **Inverse probability weighting (IPW)** offers an alternative that models the *treatment assignment mechanism* rather than the outcome. It works by reweighting observations to create a **pseudo-population** in which treatment assignment is independent of covariates.

5.4.2 The Propensity Score, Revisited

In Chapter 4, we defined the propensity score:

$$e(X_i) = P(D_i = 1 \mid X_i)$$

the probability that trainer i receives treatment, given their observed characteristics. For the Game Corner scheme, $e(X_i)$ represents Team Rocket's "targeting function" — how likely they are to subsidize a trainer with covariates X_i.

Under the assumptions of **unconfoundedness** ($\{Y(0), Y(1)\} \perp\!\!\!\perp D \mid X$) and **overlap** ($0 < e(X) < 1$ for all X), the propensity score is sufficient for identifying causal effects (Rosenbaum and Rubin, 1983).

5.4.3 The Horvitz-Thompson Estimator

The foundational IPW estimator is the **Horvitz-Thompson (HT) estimator**, originally developed for survey sampling (Horvitz and Thompson, 1952):

Definition 5.4 (Horvitz-Thompson Estimator). *The IPW estimator of the ATE is:*

$$\hat{\tau}_{HT} = \frac{1}{n}\sum_{i=1}^{n}\frac{D_i Y_i}{e(X_i)} - \frac{1}{n}\sum_{i=1}^{n}\frac{(1-D_i)Y_i}{1-e(X_i)}$$

The logic is beautifully intuitive. Consider a treated trainer with propensity score $e(X_i) = 0.8$. This trainer had an 80% probability of being treated, meaning they are "overrepresented" among the treated: for every trainer like them who was treated, about 0.25 trainers like them were not. To make the treated group representative of the full population, we downweight this trainer by $1/e(X_i) = 1/0.8 = 1.25$.

Conversely, a treated trainer with $e(X_i) = 0.2$ is "underrepresented" — they had only a 20% chance of treatment but happened to be treated anyway. We upweight them by $1/0.2 = 5$ to compensate.

For the control group, the same logic applies with $1 - e(X_i)$: a control trainer with $e(X_i) = 0.8$ had only a 20% chance of being a control unit, so they are underrepresented in the control group and receive weight $1/(1 - 0.8) = 5$.

Professor Oak Explains: The Pseudo-Population

"Here is the key insight, trainer. In the observed data, treated and control groups look different because treatment was not assigned randomly. IPW creates a pseudo-population — a weighted version of the data — where treatment is independent of covariates. In this pseudo-population, it is as if treatment were randomly assigned. Each trainer is weighted by the inverse of their probability of receiving the treatment they actually received. The resulting weighted sample represents the population we would have observed under random assignment."

Formally, the weight for unit i is:

$$w_i = \frac{D_i}{e(X_i)} + \frac{1-D_i}{1-e(X_i)}$$

Under correct specification of the propensity score, these weights satisfy $E[w_i \mid X_i] = 1$ for all X_i, and the weighted distribution of X is balanced between treated and control groups.

5.4.4 The Hajek (Normalized) Estimator

In practice, the Horvitz-Thompson estimator can be unstable because the weights $1/e(X_i)$ and $1/(1 - e(X_i))$ do not necessarily sum to $n/2$ in finite samples. The **Hajek estimator** normalizes the weights:

Definition 5.5 (Hajek Estimator). *The normalized IPW estimator is:*

$$\hat{\tau}_{H} = \frac{\sum_{i=1}^{n}\frac{D_i Y_i}{e(X_i)}}{\sum_{i=1}^{n}\frac{D_i}{e(X_i)}} - \frac{\sum_{i=1}^{n}\frac{(1-D_i)Y_i}{1-e(X_i)}}{\sum_{i=1}^{n}\frac{(1-D_i)}{1-e(X_i)}}$$

The Hajek estimator is preferred in practice for several reasons:

1. **Lower variance.** The normalization stabilizes the estimator, especially when the propensity score is estimated (as it always is in practice).
2. **Robustness to weight scaling.** Even if the propensity score model is slightly misspecified such that the weights do not sum to the correct totals, the Hajek estimator self-corrects through normalization.
3. **Interpretation.** Each term is a weighted average of outcomes — a quantity that is inherently easier to interpret and more stable than a weighted sum divided by n.

The cost of normalization is a small bias: the Hajek estimator is not exactly unbiased in finite samples, unlike the HT estimator. But this bias vanishes as $n \to \infty$, and the variance reduction typically dominates.

5.4.5 The Extreme Weight Problem

IPW has an Achilles' heel: **extreme weights**. When $e(X_i) \approx 0$ or $e(X_i) \approx 1$, the weights $1/e(X_i)$ or $1/(1 - e(X_i))$ explode. A single trainer with $e(X_i) = 0.01$ receives a weight of 100, effectively determining the entire estimate.

This is not merely a theoretical concern. In the Game Corner setting, suppose Team Rocket never subsidizes very weak trainers — those with zero badges and level-5 Pokemon. If such a trainer somehow *does* receive a subsidy (perhaps by accident), their propensity score is near zero, and their IPW weight is enormous. The entire ATE estimate hinges on this one unusual case.

Extreme weights cause:

- **High variance.** The variance of the IPW estimator is dominated by units with large weights.
- **Sensitivity to outliers.** A single misclassified or unusual observation can swing the estimate dramatically.
- **Instability.** Small changes in the propensity score model (e.g., adding a covariate) can drastically change which units receive extreme weights.

5.4.6 Weight Trimming and Truncation

Several strategies address the extreme weight problem:

Trimming. Discard observations with propensity scores outside $[\alpha, 1 - \alpha]$ for some threshold α (e.g., $\alpha = 0.05$ or $\alpha = 0.10$). This changes the estimand from the ATE to the ATE within the trimmed population — trainers for whom both treatment and control are plausible.

Truncation (Winsorization). Cap the weights at some maximum value c:

$$w_i^{\text{trunc}} = \min(w_i, c)$$

This introduces bias (we are no longer perfectly reweighting) but reduces variance. The bias-variance tradeoff depends on c.

Stabilized weights. Replace $1/e(X_i)$ with $P(D = 1)/e(X_i)$, which brings the average weight closer to 1 and reduces variability:

$$w_i^{\text{stab}} = \frac{D_i \cdot P(D = 1)}{e(X_i)} + \frac{(1 - D_i) \cdot P(D = 0)}{1 - e(X_i)}$$

Overlap weights. Weight each unit by $e(X_i)(1 - e(X_i))$ times the IPW weight, which effectively gives the most weight to units in the region of best overlap. This targets the **ATO** (average treatment effect on the overlap population) rather than the ATE.

5.4.7 Worked Example: The Game Corner Subsidies

Suppose we observe eight trainers in Celadon City. Team Rocket subsidized items for four of them $(D = 1)$, while the other four purchased items at full price $(D = 0)$:

Trainer	Badges (Y)	Subsidized (D)	Badges Pre (X_1)	Level (X_2)	$\hat{e}(X)$
A	7	1	4	35	0.85
B	5	1	2	20	0.30
C	8	1	5	40	0.90
D	6	1	3	30	0.60
E	3	0	1	15	0.15
F	5	0	3	25	0.55
G	4	0	2	18	0.25
H	6	0	4	32	0.80

The propensity scores $\hat{e}(X)$ were estimated from a logistic regression of D on X_1 and X_2. Notice that trainers A and C have high propensity scores (Team Rocket clearly targeted them), while trainer E has a very low score (an unlikely target).

Naive comparison:

$$\bar{Y}_{D=1} - \bar{Y}_{D=0} = \frac{7 + 5 + 8 + 6}{4} - \frac{3 + 5 + 4 + 6}{4} = 6.50 - 4.50 = 2.00$$

Horvitz-Thompson estimator:

$$\hat{\tau}_{HT} = \frac{1}{8}\left[\frac{7}{0.85} + \frac{5}{0.30} + \frac{8}{0.90} + \frac{6}{0.60}\right] - \frac{1}{8}\left[\frac{3}{0.15} + \frac{5}{0.45} + \frac{4}{0.75} + \frac{6}{0.20}\right]$$

Note: $(1 - e)$ for trainers E, F, G, H are $0.85, 0.45, 0.75, 0.20$ respectively.

Computing each term:

Treated sum: $8.24 + 16.67 + 8.89 + 10.00 = 43.80$

Control sum: $20.00 + 11.11 + 5.33 + 30.00 = 66.44$

$$\hat{\tau}_{HT} = \frac{43.80}{8} - \frac{66.44}{8} = 5.475 - 8.306 = -2.83$$

The HT estimate is -2.83 — a large *negative* effect, strikingly different from the naive $+2.00$! After reweighting to account for Team Rocket's targeting, the subsidized items appear to *hurt* outcomes, or at least not help. But notice the instability: trainer E (with $e(X) = 0.15$, weight $= 1/0.85 \approx 1.18$ as control) and trainer H (with $e(X) = 0.80$, weight $= 1/0.20 = 5$ as control) are dominating the control sum. Trainer H, a strong trainer who happened to *not* receive a subsidy despite being a prime target, receives enormous weight.

Hajek estimator:

$$\hat{\tau}_{H} = \frac{8.24 + 16.67 + 8.89 + 10.00}{1/0.85 + 1/0.30 + 1/0.90 + 1/0.60} - \frac{20.00 + 11.11 + 5.33 + 30.00}{1/0.85 + 1/0.45 + 1/0.75 + 1/0.20}$$

Treated weights sum: $1.18 + 3.33 + 1.11 + 1.67 = 7.29$

Control weights sum: $1.18 + 2.22 + 1.33 + 5.00 = 9.73$

$$\hat{\tau}_{H} = \frac{43.80}{7.29} - \frac{66.44}{9.73} = 6.01 - 6.83 = -0.82$$

The Hajek estimate of -0.82 is much more moderate. The normalization tempers the influence of extreme weights, yielding a more stable (though still negative) estimate. This is why the Hajek estimator is the standard choice in practice.

> ***Key Insight.*** *The naive comparison suggested a $+2.0$ advantage for subsidized trainers. After IPW adjustment, the effect is negative or near zero. The apparent benefit was entirely driven by selection: Team Rocket gave items to trainers who would have performed well regardless. Once we reweight to account for this targeting, the items themselves provide little or no benefit — and may even distract trainers from effective training.*

5.5 Augmented IPW / Doubly Robust Estimation

5.5.1 The Best of Both Worlds

We now have two approaches to estimating causal effects from observational data:

1. **Outcome modeling (regression)**: Model $E[Y \mid X, D]$, then compare predicted outcomes under treatment and control. This requires correct specification of the outcome model.
2. **Treatment modeling (IPW)**: Model $P(D = 1 \mid X)$, then reweight observations to create balance. This requires correct specification of the propensity score model.

Each approach is consistent (converges to the truth as $n \to \infty$) if its respective model is correctly specified. But in practice, we can never be certain that either model is correct. What if we could combine both approaches and obtain a estimator that is consistent if *either* model is correct?

This is exactly what the **augmented inverse probability weighting (AIPW)** estimator — also called the **doubly robust (DR)** estimator — achieves.

5.5.2 The AIPW Estimator

> ***Definition 5.6 (Augmented Inverse Probability Weighting).*** *Let* $\hat{e}(X_i)$ *be an estimate of the propensity score, and let* $\hat{\mu}_d(X_i) = \hat{E}[Y \mid X = X_i, D = d]$ *for* $d \in \{0,1\}$ *be estimates of the conditional mean outcomes. The AIPW estimator of the ATE is:*
>
> $$\hat{\tau}_{DR} = \frac{1}{n}\sum_{i=1}^{n}\left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{D_i(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} - \frac{(1-D_i)(Y_i - \hat{\mu}_0(X_i))}{1-\hat{e}(X_i)}\right]$$

This formula can be decomposed into three intuitive pieces:

Piece 1: The outcome model prediction. $\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)$ is the predicted treatment effect based on the outcome model alone. If the outcome model is correct, this by itself estimates the ATE.

Piece 2: The bias correction for treated units. $\frac{D_i(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)}$ corrects the outcome model's predictions using the actual outcomes of treated units, weighted by the inverse propensity score. If $\hat{\mu}_1$ is correct, the residual $Y_i - \hat{\mu}_1(X_i)$ has mean zero and this term vanishes in expectation. If $\hat{\mu}_1$ is wrong, the IPW weights ensure the correction is properly calibrated.

Piece 3: The bias correction for control units. $\frac{(1-D_i)(Y_i - \hat{\mu}_0(X_i))}{1-\hat{e}(X_i)}$ plays the same corrective role for the control group.

An equivalent and sometimes more transparent formulation writes the estimator as two separate potential outcome means:

$$\hat{\mu}_1^{DR} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{D_iY_i}{\hat{e}(X_i)} - \frac{D_i - \hat{e}(X_i)}{\hat{e}(X_i)}\hat{\mu}_1(X_i)\right]$$

$$\hat{\mu}_0^{DR} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{(1-D_i)Y_i}{1-\hat{e}(X_i)} + \frac{D_i - \hat{e}(X_i)}{1-\hat{e}(X_i)}\hat{\mu}_0(X_i)\right]$$

$$\hat{\tau}_{DR} = \hat{\mu}_1^{DR} - \hat{\mu}_0^{DR}$$

5.5.3 The Double Robustness Property

The defining property of the AIPW estimator is its **double robustness**:

> ***Theorem 5.3 (Double Robustness).*** *The AIPW estimator* $\hat{\tau}_{DR}$ *is consistent for the ATE if either:*
>
> *a. the propensity score model* $\hat{e}(X)$ *is correctly specified, OR*
>
> *b. the outcome model* $\hat{\mu}_d(X)$ *is correctly specified for both* $d = 0$ *and* $d = 1$,
>
> *but not necessarily both.*

Proof sketch. We show that the bias of the AIPW estimator is proportional to the *product* of the errors in both models, so if either error is zero, the bias vanishes.

Consider the estimand for $E[Y(1)]$. Define the error in the propensity score as $\Delta_e(X) = \hat{e}(X) - e(X)$ and the error in the outcome model as $\Delta_\mu(X) = \hat{\mu}_1(X) - \mu_1(X)$, where $e(X)$ and $\mu_1(X)$ are the true values.

The bias of the AIPW estimator for $E[Y(1)]$ can be shown to be:

$$\text{Bias} = E\left[\frac{\Delta_e(X) \cdot \Delta_\mu(X)}{\hat{e}(X)}\right]$$

This is a product of two error terms. If $\Delta_e(X) = 0$ for all X (propensity score is correct), the bias is zero regardless of Δ_μ. If $\Delta_\mu(X) = 0$ for all X (outcome model is correct), the bias is zero regardless of Δ_e. Only when *both* models are misspecified is the bias nonzero — and even then, it is a second-order term (the product of two errors), which tends to be small.

Case (a): Correct propensity score, wrong outcome model. Set $\hat{e}(X) = e(X)$. The AIPW estimator reduces to a standard IPW estimator plus a mean-zero correction term:

$$\hat{\mu}_1^{DR} = \frac{1}{n}\sum_i \frac{D_iY_i}{e(X_i)} - \frac{1}{n}\sum_i \frac{D_i - e(X_i)}{e(X_i)}\hat{\mu}_1(X_i)$$

The second term has expectation zero because $E[D_i - e(X_i) \mid X_i] = 0$. So the estimator is consistent by the consistency of the IPW estimator.

Case (b): Correct outcome model, wrong propensity score. Set $\hat{\mu}_1(X) = \mu_1(X)$. Then:

$$\hat{\mu}_1^{DR} = \frac{1}{n}\sum_i \mu_1(X_i) + \frac{1}{n}\sum_i \frac{D_i(Y_i - \mu_1(X_i))}{\hat{e}(X_i)}$$

The first term converges to $E[\mu_1(X)] = E[Y(1)]$ by the law of large numbers. The second term has expectation zero because $E[Y_i - \mu_1(X_i) \mid X_i, D_i = 1] = 0$ when the outcome model is correct. Even though $\hat{e}(X_i)$ may be wrong, it is multiplied by a term with conditional mean zero, so the product has unconditional mean zero. □

> ***Professor Oak Explains: Why "Doubly Robust" Matters***
>
> *"In observational studies, you can never be certain that your statistical models are correctly specified. Are the true relationships linear? Did you include the right covariates? Is your propensity score logistic model capturing the true treatment assignment mechanism? These are difficult questions.*
>
> *Doubly robust estimation gives you two chances to get it right. If your outcome model captures the true relationship between covariates and outcomes — even if your propensity score model is wrong — the AIPW estimator still converges to the truth. If your propensity score correctly captures treatment assignment — even if your outcome model is misspecified — it still works.*
>
> *This does not mean you can be careless. If both models are wrong, you still get bias. And in finite samples, having one badly wrong model can inflate your variance even if the other is correct. But the double robustness property provides a meaningful safety net that neither regression nor IPW alone can offer."*

5.5.4 Semiparametric Efficiency

Beyond double robustness, the AIPW estimator has another remarkable property: it achieves the **semiparametric efficiency bound** for the ATE (Hahn, 1998).

> ***Theorem 5.4 (Semiparametric Efficiency Bound, Hahn 1998).*** *Under unconfoundedness and overlap, the efficient influence function for the ATE is:*
>
> $$\psi(Y, D, X) = \frac{D(Y - \mu_1(X))}{e(X)} - \frac{(1 - D)(Y - \mu_0(X))}{1 - e(X)} + \mu_1(X) - \mu_0(X) - \tau$$
>
> *and the semiparametric variance bound is:*
>
> $$V_{eff} = E\left[\frac{\sigma_1^2(X)}{e(X)} + \frac{\sigma_0^2(X)}{1 - e(X)} + (\tau(X) - \tau)^2\right]$$
>
> *where* $\sigma_d^2(X) = \mathrm{Var}(Y(d) \mid X)$ *and* $\tau(X) = E[Y(1) - Y(0) \mid X]$.

The AIPW estimator, when both models are correctly specified, achieves this bound. No other regular estimator can have lower asymptotic variance. This makes AIPW not just robust but also *efficient* — the gold standard in modern applied work.

The efficiency bound also reveals an important insight: estimation is hardest (variance is highest) when propensity scores are extreme — when $e(X) \approx 0$ or $e(X) \approx 1$. This connects back to the overlap assumption and the extreme weight problem discussed in Section 5.4.5.

5.5.5 Worked Example: Deliberate Misspecification

To demonstrate double robustness in action, we return to a simplified version of the Celadon City data and deliberately misspecify one model at a time.

Setup. The true data-generating process is:

$$Y_i = 2 + 1.0 \cdot D_i + 0.5 \cdot X_i + \epsilon_i, \quad \epsilon_i \sim N(0, 1)$$

$$D_i \sim \mathrm{Bernoulli}(\Lambda(-1 + 0.8 \cdot X_i))$$

where $\Lambda(\cdot)$ is the logistic function and $X_i \sim N(3, 1)$. The true ATE is $\tau = 1.0$.

We consider four scenarios with $n = 500$:

Scenario 1: Both models correctly specified.

- Outcome model: $\hat{\mu}_d(X) = \hat{\alpha} + d \cdot \hat{\tau} + \hat{\beta}X$ (linear in X). Correct.
- Propensity model: $\hat{e}(X) = \Lambda(\hat{\gamma}_0 + \hat{\gamma}_1 X)$ (logistic in X). Correct.
- AIPW estimate: $\hat{\tau}_{DR} \approx 1.02$. Close to truth.

Scenario 2: Wrong propensity score, correct outcome model.

- Outcome model: Correctly specified as linear in X.
- Propensity model: Misspecified — we omit X and fit a constant: $\hat{e} = \bar{D}$ for all units.
- Pure IPW estimate: $\hat{\tau}_{IPW} \approx 1.42$. Biased (propensity model ignores confounding).
- AIPW estimate: $\hat{\tau}_{DR} \approx 1.01$. Still close to truth! The correct outcome model rescues the estimator.

Scenario 3: Correct propensity score, wrong outcome model.

- Outcome model: Misspecified — we predict $\hat{\mu}_d(X) = \bar{Y}_d$ (ignoring X entirely, just using group means).
- Propensity model: Correctly specified as logistic in X.
- Pure regression estimate: $\hat{\tau}_{reg} \approx 1.38$. Biased (outcome model ignores confounding).
- AIPW estimate: $\hat{\tau}_{DR} \approx 0.98$. Still close to truth! The correct propensity model rescues the estimator.

Scenario 4: Both models misspecified.

- Outcome model: $\hat{\mu}_d(X) = \bar{Y}_d$ (ignoring X).
- Propensity model: $\hat{e} = \bar{D}$ (ignoring X).
- AIPW estimate: $\hat{\tau}_{DR} \approx 1.42$. Biased. With both models wrong, double robustness cannot save us.

Scenario	Outcome Model	Propensity Model	$\hat{\tau}_{DR}$	Consistent?
1	Correct	Correct	1.02	Yes
2	Correct	Wrong	1.01	Yes
3	Wrong	Correct	0.98	Yes
4	Wrong	Wrong	1.42	No

The pattern is clear: as long as at least one model is correct, the AIPW estimator delivers a consistent estimate. Only when both fail does the estimator break down.

Blue's Mistake: Trusting a Single Model

Blue runs a regression of badges on Department Store spending, controlling for wealth but omitting trainer experience. His regression estimate is biased upward.

"No problem," he says. "I'll just use IPW instead." But he estimates the propensity score using the same incomplete set of covariates — wealth only, no experience.

His IPW estimate is also biased, for the same reason: the model omits a key confounder.

"Fine, I'll use doubly robust estimation," Blue declares. But he uses the same misspecified set of covariates for both the outcome model and the propensity model. The AIPW estimator is doubly robust to misspecification of functional form, but it cannot overcome missing confounders that are absent from both models. Double robustness gives you two chances to get the model form right — it does not create information about confounders you never measured.

The lesson*: Double robustness is not a substitute for careful thinking about which variables to include. It protects against getting the functional form wrong, not against omitting key confounders entirely.*

5.5.6 Connection to Targeted Maximum Likelihood Estimation (TMLE)

The AIPW estimator is closely related to **Targeted Maximum Likelihood Estimation (TMLE)**, developed by van der Laan and Rubin (2006). TMLE also combines outcome modeling and propensity score weighting but does so through a different algorithmic procedure:

1. Fit an initial outcome model $\hat{\mu}_d(X)$ (potentially using machine learning).
2. Compute a "clever covariate" $H(D, X) = D/\hat{e}(X) - (1 - D)/(1 - \hat{e}(X))$.
3. Update the initial outcome model by regressing Y on $H(D, X)$ with an offset of $\text{logit}(\hat{\mu}(X))$.
4. Use the updated model to compute predicted potential outcomes and the ATE.

TMLE has the same double robustness property as AIPW and also achieves the semiparametric efficiency bound. Its main advantage is that the "targeting" step ensures the final estimate respects the bounds of the outcome variable (e.g., for binary outcomes, the predicted probabilities remain in $[0, 1]$). We will explore TMLE in greater detail in Chapter 8.

5.5.7 Practical Recommendations

For applied researchers, we offer the following guidance:

1. **Start with regression** as a baseline. It is simple, well-understood, and often sufficient when the outcome model is approximately linear and treatment effects are roughly homogeneous.
2. **Estimate the propensity score** and check covariate balance in the weighted sample. If balance is poor, reconsider your model or the plausibility of the overlap assumption.
3. **Use AIPW as your primary estimator** for the final analysis. It provides double robustness and semiparametric efficiency.
4. **Report all three estimates** (regression, IPW, AIPW). Agreement across estimators is reassuring. Disagreement is informative — it suggests at least one model is misspecified and prompts further investigation.
5. **Conduct sensitivity analysis** (Oster's δ, E-values) to assess robustness to unobserved confounding.
6. **Examine the propensity score distribution** for extreme values. If many units have $\hat{e}(X) < 0.05$ or $\hat{e}(X) > 0.95$, consider trimming, truncation, or overlap weights.

Chapter Summary

Rainbow Badge earned!

In this chapter, we developed three progressively more sophisticated tools for estimating causal effects from observational data:

Regression (Section 5.1) is the workhorse of applied social science. Under conditional mean independence, linearity, and homogeneous treatment effects, OLS recovers the ATE. The Frisch-Waugh-Lovell theorem reveals that regression works by "partialling out" the confounders — correlating the residual variation in treatment with the residual variation in outcomes. Regression is powerful but vulnerable to functional form misspecification and the temptation to extrapolate beyond the data.

Omitted Variable Bias (Section 5.2) formalizes what happens when a confounder is left out of the regression. The OVB formula $\hat{\tau}_{short} - \hat{\tau}_{long} = \hat{\delta} \cdot \hat{\gamma}$ decomposes the bias into two pieces: how much the omitted variable correlates with treatment and how much it affects the outcome. Oster's (2019) coefficient stability approach extends this logic to ask: how important would unobservables need to be, relative to observables, to explain away the estimated effect?

Bad Controls (Section 5.3) — the most common mistake in applied work — arise when researchers control for variables that are affected by treatment. This blocks causal pathways, introduces collider bias, or both. The DAG-based backdoor criterion provides the correct framework for variable selection: control for confounders, never for descendants of treatment.

Inverse Probability Weighting (Section 5.4) approaches the problem from the treatment side, modeling $P(D \mid X)$ rather than $E[Y \mid X, D]$. By reweighting observations, IPW creates a pseudo-population in which treatment is independent of covariates. The Hajek (normalized) estimator is preferred in practice for its lower variance and stability, though extreme propensity scores remain a persistent challenge.

Doubly Robust Estimation (Section 5.5) combines outcome modeling and propensity weighting. The AIPW estimator is consistent if *either* model is correctly specified — providing two chances to get the analysis right. It also achieves the semiparametric efficiency bound, making it the gold standard for modern applied causal inference.

Erika nods approvingly. You have demonstrated rigor, acknowledged uncertainty, and deployed the most defensible estimators available. The Rainbow Badge is yours.

Professor Oak's Review Questions

1. **Conditional mean independence.** State the conditional mean independence assumption precisely. How does it differ from the full conditional independence assumption used in matching (Chapter 4)? Under what additional conditions does OLS recover the ATE?
2. **FWL intuition.** Explain the Frisch-Waugh-Lovell theorem in your own words. A trainer asks: "Why can't I just compare average outcomes between treated and control, instead of doing this complicated residual-on-residual regression?" How would you respond?

3. **OVB direction.** Suppose we are estimating the effect of Rare Candy usage on battle performance, but we cannot observe a trainer's "natural talent." If more talented trainers are both more likely to use Rare Candies (because they optimize their strategy) and more likely to win battles (because they are talented), what is the sign of the omitted variable bias? Is our estimated effect of Rare Candies too large or too small?
4. **Bad controls.** Consider a researcher who wants to estimate the effect of TM usage on gym battle outcomes and controls for "number of super-effective moves available." Draw the DAG and explain why this is a bad control. What would the bias look like?
5. **IPW mechanics.** A trainer has a propensity score of $\hat{e}(X) = 0.10$ and was treated ($D = 1$). What is their IPW weight? Interpret this weight in words. Why might this extreme weight be problematic?
6. **Double robustness.** Explain the double robustness property of the AIPW estimator. A colleague says: "If I use AIPW, I don't need to worry about model specification at all." Is this correct? What are the limitations of double robustness?

Trainer Challenge Exercises

Exercise 5.1: OVB Calculation

You are estimating the effect of Department Store spending (D) on Pokemon League ranking (Y). You have two regressions:

- **Short**: $Y_i = 120 - 15D_i + \epsilon_i$, with $R^2 = 0.08$
- **Long** (adding wealth W and training hours H): $Y_i = 95 - 4D_i + 0.3W_i - 0.8H_i + \epsilon_i$, with $R^2 = 0.52$

a. Compute the total OVB in the short regression.

b. The auxiliary regression of W on D gives $\hat{\delta}_W = 50$ (big spenders are wealthier), and the auxiliary regression of H on D gives $\hat{\delta}_H = 8$ (big spenders train more hours). Use the OVB formula to decompose the bias into contributions from each omitted variable. Verify that they approximately sum to the total OVB.

c. Compute Oster's δ using $R^2_{max} = 1.0$. Would you consider the long-regression estimate of $\hat{\tau} = -4$ to be robust to unobserved confounding?

Exercise 5.2: Identify Bad Controls from a DAG

Consider the following DAG for the effect of "training with a professional coach" (D) on "Elite Four wins" (Y):

```
SES ---------> D ---------> Team Strength ---------> Y
 |                               |                   ^
 |                               v                   |
 +-----------> Battle IQ       Moves Learned -------->+
                   |
                   +----------------------------------->+
```

where SES (socioeconomic status) affects both D and Battle IQ; D affects Team Strength; Team Strength affects both Moves Learned and Y; Moves Learned also affects Y; and Battle IQ directly affects Y.

a. List all paths from D to Y. Which are causal (front-door) paths and which are backdoor paths?

b. Which variables are "good controls"? Which are "bad controls"? Justify each.

c. Write the correctly specified regression model. What variables should you include?

d. Blue controls for SES, Team Strength, Moves Learned, *and* Battle IQ. What goes wrong?

Exercise 5.3: Compute IPW Estimates

You observe the following data on six trainers:

Trainer	Win Rate (Y)	Treated (D)	$\hat{e}(X)$
1	0.70	1	0.80
2	0.45	1	0.25
3	0.60	1	0.60
4	0.30	0	0.70
5	0.50	0	0.40
6	0.35	0	0.15

a. Compute the naive difference in means $\bar{Y}_{D=1} - \bar{Y}_{D=0}$.

b. Compute the Horvitz-Thompson IPW estimate $\hat{\tau}_{HT}$.

c. Compute the Hajek (normalized) IPW estimate $\hat{\tau}_{H}$.

d. Trainer 2 has $\hat{e}(X) = 0.25$ and is treated. What is their IPW weight? Now suppose $\hat{e}(X)$ were instead 0.05. How would this change the weight, and what concern does this raise?

Exercise 5.4: Compare Regression, IPW, and Doubly Robust

Using the companion notebook (`ch05_celadon_city.ipynb`) and the `kanto_trainers.csv` dataset:

a. Estimate the effect of Department Store spending on gym badges using OLS regression, controlling for wealth, experience, and starter type. Report $\hat{\tau}_{OLS}$ and its standard error.

b. Estimate the propensity score using logistic regression with the same covariates. Plot the distribution of propensity scores for treated and control groups. Assess overlap.

c. Compute the Hajek IPW estimate using the estimated propensity scores. Examine the distribution of weights: are any extreme?

d. Compute the AIPW estimate. Compare all three estimates ($\hat{\tau}_{OLS}$, $\hat{\tau}_{IPW}$, $\hat{\tau}_{DR}$). Do they agree? If not, what does the disagreement suggest about your models?

e. Deliberately misspecify the propensity score model (e.g., omit a key covariate) and re-estimate all three. Which estimator is most sensitive to the misspecification? Which is most robust?

Further Reading

- **Angrist, J. D. & Pischke, J.-S.** (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press. Chapters 3 (regression) and the discussion of OVB are foundational for the regression-based approach to causal inference.
- **Oster, E.** (2019). "Unobservable Selection and Coefficient Stability: Theory and Evidence." *Journal of Business & Economic Statistics*, 37(2), 187-204. The definitive modern treatment of coefficient stability as a tool for assessing robustness to unobserved confounding.
- **Horvitz, D. G. & Thompson, D. J.** (1952). "A Generalization of Sampling Without Replacement from a Finite Universe." *Journal of the American Statistical Association*, 47(260), 663-685. The original paper introducing inverse probability weighting for survey estimation.
- **Robins, J. M., Rotnitzky, A., & Zhao, L. P.** (1994). "Estimation of Regression Coefficients When Some Regressors Are Not Always Observed." *Journal of the American Statistical Association*, 89(427), 846-866. Introduced the augmented IPW framework and the double robustness concept.
- **Hahn, J.** (1998). "On the Role of the Propensity Score in Efficient Semiparametric Estimation of Average Treatment Effects." *Econometrica*, 66(2), 315-331. Derives the semiparametric efficiency bound for the ATE and shows that knowing the propensity score does not improve efficiency.
- **Lunceford, J. K. & Davidian, M.** (2004). "Stratification and Weighting Via the Propensity Score in Estimation of Causal Treatment Effects: A Comparative Study." *Statistics in Medicine*, 23(19), 2937-2960. An accessible review comparing propensity score methods, including regression, stratification, IPW, and doubly robust estimation.
- **Bang, H. & Robins, J. M.** (2005). "Doubly Robust Estimation in Missing Data and Causal Inference Models." *Biometrics*, 61(4), 962-973. A clear exposition of doubly robust estimators with practical guidance on implementation.

Skills to Practice in the Notebook

The notebook `notebooks/ch05_celadon_city.ipynb` is where the three estimators (regression, IPW, and doubly robust) become yours to compare and contrast. Before claiming the Rainbow Badge, you should be able to do the following fluently:

1. **Fit a causal regression.** On the Department Store dataset, regress the outcome on treatment and covariates. Extract the treatment coefficient, its standard error, and a 95% CI. Then refit with a polynomial/interaction specification and see how the estimate moves. Know which specification changes you're allowed to try.
2. **Use the Frisch-Waugh-Lovell theorem in code.** Residualize D and Y against X separately, regress the residuals, and verify the coefficient matches the full regression's treatment coefficient. This is the best single exercise for internalizing what "controlling for X " actually does.

3. **Run a signed OVB analysis.** For a simulated dataset where you know the true confounder, omit it intentionally and confirm the direction of bias matches $\text{sign}(\gamma) \cdot \text{sign}(\delta)$.
4. **Demonstrate the "bad control" failure.** Simulate a dataset with a post-treatment mediator. Fit a regression that (a) controls for the mediator and (b) does not. Show that (a) is biased toward zero and (b) recovers the truth. Be able to explain exactly why the "good" variable breaks things.
5. **Estimate propensity scores and compute IPW.** Fit `e_hat` via logistic regression, compute both the Horvitz-Thompson and Hajek estimates, and inspect the weight distribution. Flag any weights above some threshold (e.g., 10) and explain what would happen if you trimmed or Winsorized them.
6. **Build an augmented IPW / doubly robust estimator.** Combine a fitted outcome model $\hat{m}_d(X)$ and a propensity model $\hat{e}(X)$ into the AIPW formula. Then *break* one of the two models on purpose — misspecify the outcome model with a wrong functional form — and verify the AIPW estimate is still close to the truth. This is the double-robustness demo.
7. **Complete the Trainer Challenge Exercises.** The notebook walks through: (a) an OVB simulation matching the chapter, (b) a bad-control trap on the `team_level` mediator, (c) an IPW vs. regression shootout on a confounded dataset, and (d) a doubly robust stress test.

Check Your Understanding

Before earning the Rainbow Badge, make sure these are solid. Chapters 6–8 all assume fluency with the three Celadon tools.

Questions you should be able to answer out loud, without notes:

- State the causal regression model $Y_i = \alpha + \tau D_i + X_i'\beta + \epsilon_i$. What does τ represent *causally*, and what assumption must hold for OLS to recover it?
- What is conditional mean independence, and how does it differ from full conditional independence?
- State the Frisch-Waugh-Lovell theorem informally. What does it tell you about what a regression coefficient "really" is?
- Define a "bad control" and explain, via a DAG, why controlling for a post-treatment variable biases the estimate.
- Write the Horvitz-Thompson and Hajek IPW estimators. Why is Hajek preferred in practice?
- What is the "pseudo-population" interpretation of IPW? Why does reweighting by $1/e(X)$ make the treated group look like the full population?
- What is the extreme weight problem? Name three ways to address it (trimming, truncation, stabilization, overlap weights) and state the bias-variance tradeoff for each.
- State the double robustness property. What exactly does the AIPW estimator guarantee? When does the guarantee *fail*?
- In what sense are regression, IPW, and AIPW all estimating the same thing? In what sense are they different?
- Give one scenario where you would prefer regression, one where you would prefer IPW, and one where AIPW is clearly the winner.

Tasks you should be able to perform in code:

- Fit an OLS regression, extract a treatment coefficient with robust standard errors, and compute a 95% CI.
- Run the FWL residualization and verify numerically that the two-step regression reproduces the full regression's coefficient.
- Fit a logistic propensity score model, compute HT and Hajek IPW estimates, and plot the weight distribution.
- Apply trimming / truncation / stabilized weights and report how the ATE estimate moves.
- Implement the AIPW formula from $\hat{m}_d(X)$ and $\hat{e}(X)$, and verify double robustness by breaking one model at a time.
- Run a bootstrap or sandwich variance estimator for any of the above and produce a defensible 95% CI.

Nail these and the Rainbow Badge is yours.

Badge Earned: Rainbow Badge

```
╔══════════════════════════════════════════╗
║                                          ║
║          🌈  RAINBOW BADGE  🌈           ║
║                                          ║
║   Awarded for mastery of:                ║
║   • Regression for causal inference      ║
║   • Omitted variable bias                ║
║   • Bad controls & post-treatment bias   ║
║   • Inverse probability weighting        ║
║   • Doubly robust estimation             ║
║                                          ║
║   "The Pokemon user with the best        ║
║    status moves in Celadon City."        ║
║                 — Erika                  ║
║                                          ║
╚══════════════════════════════════════════╝
```

Next Chapter Preview

> *Fuchsia City's Safari Zone operates a lottery to allocate rare Pokemon encounters — and a clever trainer recognizes this as an **instrumental variable** that isolates exogenous variation in team composition. Meanwhile, on Cinnabar Island, the Pokemon Lab uses a sharp **threshold** in a Pokemon's base stat total to determine evolution eligibility, creating a natural **regression discontinuity** design. These two strategies share a profound advantage: they work even when unobserved confounders lurk in the tall grass…*
>
> ***Chapter 6: Fuchsia City & Cinnabar Island — Instrumental Variables and Regression Discontinuity***

Chapter 6: Fuchsia City & Cinnabar Island — Instrumental Variables & Regression Discontinuity

Poison-type — Koga's specialty

Koga, Fuchsia Gym Leader

> *"Sometimes the path forward is not straight. The Safari Zone lottery ticket in your pocket and the happiness meter on your Pokedex — these indirect tools are your keys to causal truth." — Professor Oak*

Your journey through Kanto has taken you from the clean experiments of Pewter City to the observational thickets of Celadon City. You have learned to draw DAGs, match on observables, weight by propensity scores, and build doubly robust estimators. But every method you have used so far shares a common assumption: that you can measure and adjust for *all* the confounders that stand between you and a causal estimate.

What happens when you cannot?

This chapter takes you to two new destinations that answer that question. In **Fuchsia City**, the Safari Zone runs a lottery that creates a natural experiment — an *instrumental variable* that lets you estimate causal effects even when critical confounders are unobserved. Then on **Cinnabar Island**, the happiness evolution threshold at 220 creates a sharp discontinuity — a *regression discontinuity design* that exploits the arbitrary cutoff to identify causal effects for Pokemon near the boundary.

These two methods — IV and RDD — are the workhorses of modern applied causal inference. They are what allow researchers to make credible causal claims from observational data when selection on observables fails. Together, they earn you two badges: the **Soul Badge** from Fuchsia and the **Volcano Badge** from Cinnabar.

Let us begin.

Part I: Instrumental Variables (Fuchsia City)

You arrive in Fuchsia City, home of the Safari Zone. The warden tells you that admission is expensive — 500 Pokedollars per visit — but each week, a lottery awards free passes to randomly selected trainers. Some winners go, some do not. Some losers pay their own way in. The question on every trainer's mind: does Safari Zone training actually improve battle performance?

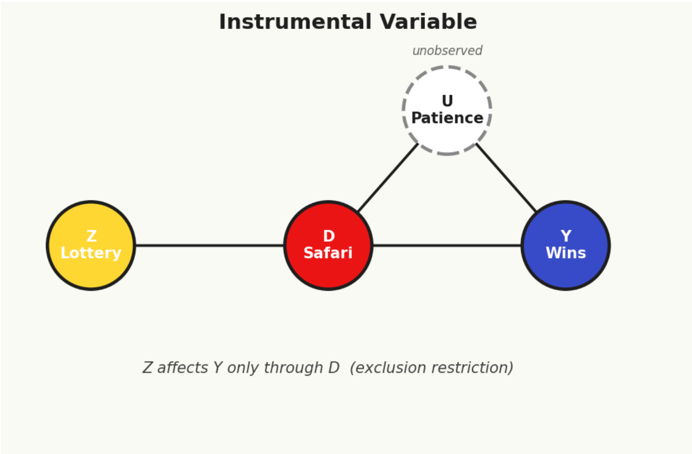
Notation at a Glance: IV and RDD Symbols

This chapter has a lot of characters. Keep this table nearby — it is the fastest way to stay oriented when formulas pile up.

Symbol	***Plain-English reading***
D_i	*Treatment (e.g., Safari Zone attendance). Can be "self-chosen" → confounded.*
Y_i	*Outcome (e.g., battle win rate).*
U_i	***Unobserved** confounder — exactly the thing regression can't fix.*
Z_i	***Instrument** — an external nudge that affects D but not Y directly.*
First stage: $Z \to D$	*"The instrument actually moves the treatment." Must be strong ($F > 10$ rule).*
Exclusion restriction: $Z \to Y$ *only via* D	*"The only channel from Z to Y runs through D." Untestable — defend it with theory.*
Independence (exogeneity)	*"The instrument is as-if randomly assigned" — no backdoor paths into Z.*
$\hat{\tau}_{IV} = \dfrac{\mathrm{Cov}(Z,Y)}{\mathrm{Cov}(Z,D)}$	*The **Wald estimator** — reduced-form effect divided by first-stage effect.*
2SLS	*Two-Stage Least Squares: regress D on Z, predict $\hat{D}$, then regress Y on $\hat{D}$.*
LATE	*Local Average Treatment Effect — the effect for compliers, not the whole population.*
compliers / always-takers / never-takers / defiers	*The four "types" of units in an IV world. LATE is only about compliers.*
monotonicity	*"No defiers" — Z never decreases anyone's treatment probability. Required for LATE interpretation.*
***RDD running variable** X_i*	*A continuous score (e.g., happiness) that determines treatment by crossing a cutoff.*
cutoff c	*The threshold: $D_i = 1$ iff $X_i \geq c$ (sharp) or $P(D_i = 1)$ jumps at c (fuzzy).*
$\lim_{x \downarrow c}$, $\lim_{x \uparrow c}$	*The limits of the outcome as X approaches the cutoff from above / below.*
$\tau_{RDD} = \lim_{x \downarrow c} E[Y \mid X = x] - \lim_{x \uparrow c} E[Y \mid X = x]$	*The RDD estimand — the jump in Y exactly at the cutoff.*
bandwidth h	*The window around the cutoff used for local regression. Small = less bias, more noise.*
continuity assumption	*"Nothing else jumps at the cutoff." The core RDD identifying assumption.*
McCrary density test	*A check for manipulation — does the density of X itself jump at c?*

***The big shift from Chapters 4–5**: IV and RDD do not require you to measure confounders. They require you to find (IV) or exploit (RDD) a source of variation in D that is "as-if random." The price you pay is narrower identification: IV recovers LATE (effect for compliers), RDD recovers the effect at the cutoff, not the ATE.*

6.1 The Instrumental Variables Idea



Instrumental variables: Z affects Y only through D.

The Problem: Unobserved Confounding

Suppose you want to estimate the causal effect of attending Safari Zone training (D_i) on battle performance (Y_i, measured as win rate in the next 20 battles). You have observational data on hundreds of trainers in Fuchsia City. Some attended the Safari Zone; others did not. The naive comparison of mean outcomes is:

$$\bar{Y}_{\text{attendees}} - \bar{Y}_{\text{non-attendees}} = 0.68 - 0.45 = 0.23$$

A 23-percentage-point advantage! But should you believe this is causal?

Consider the following DAG:

```
    Patience (U)
      /       \
     v         v
Safari Zone → Battle
Attendance    Performance
   (D)           (Y)
```

Patience — the willingness to grind, to wait in tall grass, to carefully plan strategy — is an *unobserved confounder*. Patient trainers are both more likely to attend the Safari Zone (they enjoy the slow-paced catching game) and more likely to win battles (patience helps in competitive play). Patience is not recorded in any dataset. You cannot measure it, match on it, or include it in a regression.

This is the fundamental problem. No matter how sophisticated your regression model, no matter how carefully you construct propensity scores, you cannot adjust for what you cannot see. The backdoor path $D \leftarrow U \rightarrow Y$ remains open, and your estimate is biased.

Let us formalize this. Write the outcome model as:

$$Y_i = \alpha + \tau D_i + \gamma U_i + \epsilon_i$$

where U_i is patience (unobserved). OLS of Y on D omits U. By the omitted variable bias formula:

$$\hat{\tau}_{OLS} \xrightarrow{p} \tau + \gamma \cdot \frac{\text{Cov}(D, U)}{\text{Var}(D)}$$

Since $\gamma > 0$ (patience helps win battles) and $\text{Cov}(D, U) > 0$ (patient people attend more), the bias term $\gamma \cdot \frac{\text{Cov}(D,U)}{\text{Var}(D)} > 0$. OLS *overstates* the benefit of Safari Zone training. The 23-percentage-point gap is inflated by selection bias.

Previous chapters taught you to handle confounders by conditioning on them. In Chapter 4, you learned to match patient trainers to patient non-trainers. In Chapter 5, you learned to weight by the propensity score. But all those methods require that patience is *measured*. What if your dataset contains only attendance, win rate, trainer level, and badges — but no measure of patience? Then matching on observables misses the key confounder, the propensity score omits a critical variable, and doubly robust estimation is only as good as its component models, both of which are misspecified.

The methods of Chapters 3-5 are sometimes called **selection on observables** strategies. They work beautifully when all confounders are measured, but they are powerless against unobserved confounding. This is not a minor limitation — in practice, the most important confounders are often the ones you cannot measure: motivation, ability, risk tolerance, or, in our case, patience.

You need a fundamentally different strategy. You need an **instrument**.

The Instrument: The Safari Zone Lottery

Every week, the Fuchsia City Warden draws names from the trainer registry for free Safari Zone passes. The lottery is random — it does not depend on patience, skill, or any other characteristic of the trainer. Let Z_i denote whether trainer i won the lottery:

$$Z_i = \begin{cases} 1 & \text{if trainer } i \text{ won a free pass} \\ 0 & \text{otherwise} \end{cases}$$

Now consider the updated DAG:

```
               Patience (U)
                 /       \
                v         v
Lottery → Safari Zone → Battle
 (Z)     Attendance    Performance
            (D)          (Y)
```

The lottery Z affects attendance D (winners are more likely to go), but the lottery has no *direct* effect on battle outcomes. Winning a lottery ticket does not make you a better battler. The only way the lottery could affect your win rate is *through* its effect on whether you actually attend the Safari Zone.

This is the instrumental variables idea: find a variable that shifts the treatment but has no other pathway to the outcome.

The Three IV Assumptions

For the lottery Z to serve as a valid instrument for the effect of attendance D on performance Y, three conditions must hold:

> ***Definition 6.1 (Instrumental Variable Assumptions).*** *A variable Z is a valid instrument for the causal effect of treatment D on outcome Y if:*
>
> 1. ***Relevance (First Stage):*** $\mathrm{Cov}(Z, D) \neq 0$. *The instrument must actually affect the treatment. The lottery must change attendance behavior.*
> 2. ***Independence (Exogeneity):*** $Z \perp\!\!\!\perp (Y(1), Y(0), D(1), D(0))$. *The instrument is independent of potential outcomes and potential treatment assignments. The lottery is as-if randomly assigned.*
> 3. ***Exclusion Restriction:*** *Z affects Y only through D. There is no direct arrow from Z to Y. Winning the lottery does not improve your battle performance except by getting you into the Safari Zone.*

Let us examine each assumption in detail.

Relevance is the most straightforward. If the lottery does not change anyone's behavior — if the same people would attend regardless of whether they win — then the instrument is useless. We need the lottery to actually move the needle on attendance. This is *testable*: we can regress D on Z and check whether the coefficient is statistically and economically significant.

Independence (also called exogeneity or the "as-if random" condition) requires that the instrument is not correlated with any confounders — observed or unobserved. Because the lottery is literally random, this is highly plausible. Random assignment of Z guarantees that winners and losers are comparable in expectation on all observed and unobserved characteristics, including patience. This assumption would be violated if, for example, patient trainers were more likely to enter the lottery — but we assume all registered trainers are automatically entered.

Note the key difference from the "selection on observables" assumption. In Chapters 4-5, you needed $D \perp\!\!\!\perp (Y(1), Y(0)) \mid X$ — independence of treatment and potential outcomes conditional on observables. Here, you need $Z \perp\!\!\!\perp (Y(1), Y(0), D(1), D(0))$ — independence of the *instrument* and potential outcomes/treatments, but with *no conditioning required* (because the instrument itself is randomly assigned). You have traded the problem of conditioning on the right variables for the problem of finding a valid instrument.

Exclusion restriction is the most subtle and the most consequential. It requires that the *only* channel through which the lottery affects battle performance is through Safari Zone attendance. This would be violated if, say, winning the lottery made trainers feel lucky and therefore more confident in battle — a psychological boost unrelated to Safari Zone training. Or if lottery winners celebrated by buying better healing items with the money they saved on admission. These are "direct effects" of Z on Y that bypass D.

> ***Professor Oak Explains: The Exclusion Restriction***
>
> *"The exclusion restriction is what makes IV powerful — and what makes it fragile. It says the instrument is like a key that only fits one lock. The lottery key opens the Safari Zone door, and that door is the only way it could possibly affect your battle record. If the key also opens a second door — say, a door to a free Rare Candy shop — then the instrument is invalid.*
>
> *"You cannot test the exclusion restriction with data. It is a substantive assumption that must be defended on theoretical and institutional grounds. This is why the choice of instrument is as much an art as a science."*

The Wald Estimator

Given a valid instrument, how do we actually compute the causal effect? The key insight is to decompose the problem into two observable relationships.

The reduced form is the effect of the instrument on the outcome:

$$\text{Reduced Form} = E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]$$

This tells us: how much better do lottery winners perform, on average, compared to losers? This is the *intent-to-treat* (ITT) effect — the effect of being *assigned* to the instrument, regardless of whether you actually attended the Safari Zone.

The first stage is the effect of the instrument on the treatment:

$$\text{First Stage} = E[D_i|Z_i = 1] - E[D_i|Z_i = 0]$$

This tells us: how much more likely are lottery winners to actually attend, compared to losers? This is the *compliance rate* — the fraction of people whose behavior is changed by the instrument.

The **Wald estimator** is their ratio:

> ***Definition 6.2 (Wald Estimator).*** *The instrumental variables estimate of the causal effect of* D *on* Y *using instrument* Z *is:*
>
> $$\hat{\tau}_{IV} = \frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[D_i|Z_i = 1] - E[D_i|Z_i = 0]} = \frac{\text{Reduced Form (ITT)}}{\text{First Stage (Compliance)}}$$

The intuition is beautiful. The reduced form captures the total effect of the lottery on outcomes — but this effect is "diluted" because not everyone who wins the lottery actually attends, and some who lose attend anyway. The first stage measures exactly how much dilution there is. Dividing by the first stage rescales the ITT effect to reflect the causal effect *per unit of treatment actually induced by the instrument.*

Think of it this way: if the lottery increases attendance by 30 percentage points and improves outcomes by 6 percentage points, then each unit of attendance induced by the lottery is worth $0.06/0.30 = 0.20$ — a 20-percentage-point improvement in win rate.

The Wald estimator can also be derived algebraically from the IV moment condition. Under the three assumptions, we have:

$$\text{Cov}(Z, \epsilon) = 0 \quad (\text{exogeneity} + \text{exclusion})$$

Starting from $Y_i = \beta_0 + \tau D_i + \epsilon_i$ and using the moment condition:

$$\hat{\tau}_{IV} = \frac{\text{Cov}(Z, Y)}{\text{Cov}(Z, D)} = \frac{\frac{1}{n}\sum_i (Z_i - \bar{Z})(Y_i - \bar{Y})}{\frac{1}{n}\sum_i (Z_i - \bar{Z})(D_i - \bar{D})}$$

When Z is binary, this simplifies exactly to the Wald ratio above. This moment condition perspective generalizes naturally to the case of continuous instruments and multiple regressors, which is the foundation of 2SLS.

Worked Example: Safari Zone Lottery

Suppose we observe the following data from the `safari_zone_lottery.csv` dataset, summarized by lottery status:

	$Z = 1$ **(Winners)**	$Z = 0$ **(Losers)**
n	200	300
Mean attendance ($\bar{D}$)	0.72	0.30
Mean win rate ($\bar{Y}$)	0.58	0.50

First stage:

$$E[D|Z = 1] - E[D|Z = 0] = 0.72 - 0.30 = 0.42$$

Lottery winners are 42 percentage points more likely to attend the Safari Zone. The instrument is strong.

Reduced form (ITT):

$$E[Y|Z = 1] - E[Y|Z = 0] = 0.58 - 0.50 = 0.08$$

Lottery winners have a win rate that is 8 percentage points higher. But this is the ITT — the effect of the lottery, not the effect of attendance.

Wald estimator:

$$\hat{\tau}_{IV} = \frac{0.08}{0.42} = 0.190$$

The IV estimate of the causal effect of Safari Zone attendance on battle win rate is **19.0 percentage points**. Compare this to the naive OLS estimate of 23 percentage points. The difference — about 4 percentage points — is the upward bias from unobserved patience confounding.

Let us trace through the logic once more to make sure the intuition is fully clear. The reduced form of 0.08 tells us that lottery winners have better outcomes — but not *dramatically* better, because only 42% of winners actually changed their behavior due to the lottery. The rest are always-takers (who would have attended anyway) and never-takers (who ignored the free pass). The 0.08 is a "watered-down" version of the true causal effect, watered down by the fact that most people were not affected by the instrument. Dividing by 0.42 "concentrates" the effect back onto the people whose behavior actually changed — the compliers — giving us the 19.0-percentage-point effect for this group.

> ***Blue's Mistake: "Why Not Just Compare Attendees?"***
>
> *Blue looks at the data and scoffs. "Why bother with this lottery nonsense? Just compare people who went to the Safari Zone with people who didn't. Attendees win 23% more — that's the answer."*
>
> *Blue's mistake is selection bias from unobserved confounding. Patient trainers self-select into the Safari Zone and independently perform better in battles. By comparing attendees to non-attendees, Blue is conflating the causal effect of training with the pre-existing advantage of patient people. The IV estimate of 19 percentage points strips away this selection bias by using only the variation in attendance that was induced by the lottery — variation that is, by construction, uncorrelated with patience.*

6.2 Two-Stage Least Squares (2SLS)

The Wald estimator is elegant but limited: it handles only a single binary instrument and a single binary treatment, with no covariates. In practice, we often want to include control variables for efficiency, handle continuous instruments or treatments, or use multiple instruments simultaneously. **Two-Stage Least Squares (2SLS)** is the generalization that handles all of these cases.

The Two Stages

First stage: Regress the endogenous treatment D on the instrument Z and any exogenous covariates X:

$$D_i = \pi_0 + \pi_1 Z_i + X_i' \pi_2 + v_i \tag{First Stage}$$

Save the predicted values $\hat{D}_i = \hat{\pi}_0 + \hat{\pi}_1 Z_i + X_i' \hat{\pi}_2$. These fitted values represent the part of treatment variation that is *explained by the instrument* (and covariates), stripped of the endogenous component driven by unobserved confounders.

Second stage: Regress the outcome Y on the predicted treatment $\hat{D}$ and the same covariates X:

$$Y_i = \beta_0 + \beta_1 \hat{D}_i + X_i' \beta_2 + \epsilon_i \tag{Second Stage}$$

The coefficient $\hat{\beta}_1$ is the 2SLS estimate of the causal effect.

Why Does This Work?

The key insight is that $\hat{D}_i$ contains only the variation in D that comes from the instrument Z — the exogenous, "as-if random" variation. All the endogenous variation — the part of attendance driven by patience and other unobserved confounders — is absorbed into the first-stage residual v_i and discarded. By regressing Y on $\hat{D}$ instead of D, we isolate the causal effect.

Formally, the OLS estimator using D directly is biased because $\mathrm{Cov}(D, \epsilon) \neq 0$ — treatment is correlated with the error term (which contains the unobserved confounder). But $\hat{D}$ is a function only of Z and X, both of which are exogenous. So $\mathrm{Cov}(\hat{D}, \epsilon) = 0$, and OLS on the second stage is consistent.

To see this more concretely, decompose D into its fitted value and residual: $D_i = \hat{D}_i + \hat{v}_i$. The fitted component $\hat{D}_i$ is the "good" variation — the part predicted by the exogenous instrument. The residual $\hat{v}_i$ is the "bad" variation — the part driven by unobserved confounders like patience. By substituting $\hat{D}$ for D in the second stage, we use only the good variation and throw away the bad. This is sometimes described as IV "purifying" the treatment variable by stripping out the endogenous component.

An equivalent perspective comes from the **control function** approach. Instead of replacing D with $\hat{D}$, you can include the first-stage residual $\hat{v}_i$ as an additional regressor in the outcome equation:

$$Y_i = \beta_0 + \beta_1 D_i + X_i' \beta_2 + \rho \hat{v}_i + \eta_i$$

The coefficient β_1 in this specification is numerically identical to the 2SLS estimate. Including $\hat{v}_i$ "controls for" the endogenous component of D, allowing β_1 to capture only the exogenous variation. The control function approach also provides a direct test of endogeneity: if $\hat{\rho} = 0$, then D is exogenous and OLS is consistent. This is the **Hausman test** (or Durbin-Wu-Hausman test) for endogeneity.

Including Covariates

Even when the instrument is randomly assigned, including covariates X can be valuable for two reasons:

1. **Efficiency:** If covariates explain substantial variation in Y, including them reduces the residual variance and tightens confidence intervals.
2. **Conditional IV:** Sometimes the instrument is only valid *conditional* on covariates. For example, if the lottery probability varies by trainer level (higher-level trainers get more entries), then the lottery is only as-if random within level groups. Including trainer level as a covariate restores validity.

A critical rule: **any covariate included in the second stage must also be included in the first stage.** If you control for trainer level in the outcome equation, you must also control for it in the first-stage regression. Omitting it from the first stage would contaminate $\hat{D}$ with level-related variation that is not instrumented.

Overidentification: Multiple Instruments

Sometimes you have more instruments than endogenous variables. Suppose the Safari Zone runs *two* lotteries — a weekly drawing (Z_1) and a monthly grand prize (Z_2). Both are valid instruments for attendance. With two instruments and one endogenous variable, the model is **overidentified**.

The first stage becomes:

$$D_i = \pi_0 + \pi_1 Z_{1i} + \pi_2 Z_{2i} + X_i' \pi_3 + v_i$$

Overidentification brings two advantages:

1. **Efficiency:** More instruments generally yield more precise estimates (tighter standard errors) because they capture more exogenous variation in D.
2. **Testability:** With multiple instruments, you can partially test the exclusion restriction using the **Sargan-Hansen J test**. The null hypothesis is that all instruments are valid. Rejection suggests at least one instrument violates the exclusion restriction. However, the test has limited power and cannot identify *which* instrument is invalid.

> ***Professor Oak Explains: The J Test's Limitations***
> *"The Sargan-Hansen J test is useful but imperfect. It tests whether all your instruments agree on the causal effect. If they do, it fails to reject — but this does not prove validity. All instruments could be invalid in the same way, yielding the same biased estimate. The J test catches disagreement, not invalidity per se. It is a necessary condition, not a sufficient one."*

Standard Errors: A Critical Warning

A common pitfall: if you manually run OLS in the first stage, save $\hat{D}$, and then run OLS in the second stage, the coefficient $\hat{\beta}_1$ will be correct, but the **standard errors will be wrong**. The naive second-stage OLS standard errors fail to account for the fact that $\hat{D}$ is itself an estimate (with sampling uncertainty from the first stage). Proper 2SLS standard errors are larger than the naive ones — they incorporate the estimation error from both stages.

Always use a dedicated 2SLS procedure (such as `linearmodels.IV2SLS` in Python or `ivreg` in R) that computes correct standard errors automatically.

Worked Example: Full 2SLS with Safari Zone Data

Using the `safari_zone_lottery.csv` dataset with 500 trainers, we include trainer level and number of badges as covariates:

First stage:

$$D_i = -0.15 + 0.43 \cdot Z_i + 0.02 \cdot \text{Level}_i + 0.05 \cdot \text{Badges}_i + v_i$$

Variable	Coefficient	Std. Error	t-statistic	p-value
Z (Lottery)	0.430	0.038	11.32	< 0.001
Level	0.021	0.005	4.20	< 0.001
Badges	0.048	0.019	2.53	0.012
Constant	−0.152	0.071	−2.14	0.033

The first-stage F-statistic on Z is $11.32^2 = 128.1$, well above the critical threshold of 10 (more on this shortly).

Second stage:

$$Y_i = 0.22 + 0.191 \cdot \hat{D}_i + 0.008 \cdot \text{Level}_i + 0.031 \cdot \text{Badges}_i + \epsilon_i$$

Variable	Coefficient	2SLS Std. Error	t-statistic	p-value
$\hat{D}$ (Safari attendance)	0.191	0.049	3.90	< 0.001
Level	0.008	0.004	2.00	0.046
Badges	0.031	0.016	1.94	0.053
Constant	0.220	0.062	3.55	< 0.001

The 2SLS estimate is $\hat{\tau}_{2SLS} = 0.191$ — virtually identical to the Wald estimate of 0.190, as expected when covariates are included in a well-specified model. Safari Zone attendance causes a 19.1-percentage-point increase in win rate, after removing the bias from unobserved patience.

6.3 Instrument Validity and Diagnostics

The power of IV comes with a price: the estimates are only as good as the instrument. This section covers the diagnostic toolkit every researcher must apply before trusting an IV estimate.

Weak Instruments

An instrument is **weak** if it has only a small effect on the treatment — that is, if the first-stage relationship between Z and D is statistically significant but practically small. Weak instruments cause two severe problems:

1. **Bias:** The 2SLS estimator is biased toward the OLS estimate in finite samples. With a weak instrument, 2SLS can be nearly as biased as naive OLS — defeating the entire purpose of IV.
2. **Size distortion:** Standard confidence intervals and hypothesis tests based on normal approximations become unreliable. Confidence intervals may be far too narrow, leading to false rejections.

> ***Definition 6.3 (Stock and Yogo, 2005, Rule of Thumb).*** *An instrument is considered sufficiently strong if the first-stage F-statistic exceeds 10. For a single endogenous regressor with a single instrument, this corresponds to a first-stage t-statistic exceeding approximately 3.16.*

In our Safari Zone example, the first-stage F-statistic was 128.1 — well above 10. The lottery is a strong instrument.

But what if you were using a different instrument — say, distance from a trainer's home to the Safari Zone? If trainers who live closer are only slightly more likely to attend, the first-stage F might be, say, 4.2. With such a weak instrument, the 2SLS estimate would be unreliable.

To understand the finite-sample bias intuitively, recall that the Wald estimator is a *ratio*: reduced form divided by first stage. When the first stage is near zero (a weak instrument), you are dividing by a small and noisy number. Small estimation errors in the denominator get amplified dramatically, pushing the ratio toward the OLS estimate. In the extreme case of a completely irrelevant instrument (first stage exactly zero), the IV estimator is undefined — you would be dividing by zero.

The Stock and Yogo (2005) framework provides precise critical values for the first-stage F-statistic based on two criteria: (1) the maximum relative bias of 2SLS relative to OLS that the researcher is willing to tolerate, and (2) the maximum size distortion of the Wald test that is acceptable. The $F > 10$ rule of thumb corresponds roughly to ensuring that the 2SLS bias is no more than 10% of the OLS bias. For stricter standards (e.g., 5% bias), higher F-statistics are required.

Lee, McCrary, Moreira, and Porter (2022) have recently proposed a new framework for weak IV inference that conditions on the first-stage F-statistic directly, providing valid confidence intervals for any value of the first stage — not just values above 10. Their `tF` procedure is an attractive alternative when the researcher is concerned about weak instruments but wants to use the 2SLS framework.

Anderson-Rubin confidence sets provide a solution for weak instruments. Unlike standard Wald-based confidence intervals, the Anderson-Rubin (AR) test is valid regardless of instrument strength. It inverts a test of the reduced-form hypothesis to construct a confidence set for the causal parameter. When the instrument is strong, the AR confidence set closely matches the standard 2SLS interval. When the instrument is weak, the AR set may be much wider (or even unbounded), honestly reflecting the true uncertainty.

The Exclusion Restriction: Untestable and Unforgiving

The exclusion restriction — that Z affects Y only through D — is the Achilles' heel of instrumental variables. It cannot be tested with data. No statistical procedure can determine whether the lottery has a direct effect on battle performance that bypasses the Safari Zone.

Consider a potential violation: **What if lottery winners feel lucky, and this psychological boost makes them fight more boldly and win more battles — independent of whether they actually attend the Safari Zone?**

If this "luck effect" exists, the reduced form captures both the indirect effect (through attendance) and the direct effect (through confidence). The Wald estimator becomes:

$$\frac{E[Y|Z=1]-E[Y|Z=0]}{E[D|Z=1]-E[D|Z=0]}=\tau+\frac{\text{direct effect of Z on Y}}{\text{first stage}}$$

The IV estimate is biased upward by the ratio of the direct effect to the first stage. If the direct "luck" effect is small relative to the first stage, the bias is small. But you cannot know this from the data.

> ***Blue's Mistake: "My Instrument Is Fine Because It's Significant"***
>
> *Blue claims he has found a great instrument: whether a trainer owns a Fishing Rod. "Fishing Rod owners are more likely to visit the Safari Zone — the first stage is super strong! And it's clearly relevant. So IV is valid."*
>
> *But does owning a Fishing Rod satisfy the exclusion restriction? Trainers who own Fishing Rods are likely outdoorsy, patient, and experienced — all traits that directly affect battle performance. The Fishing Rod is correlated with unobserved confounders, so the "independence" assumption fails. A strong first stage does not make an instrument valid. The exclusion restriction is the binding constraint, and no amount of statistical significance can substitute for a convincing argument that the instrument only affects the outcome through the treatment.*

Monotonicity and the Absence of Defiers

A fourth assumption, not always listed among the "big three," is critical for the LATE interpretation (Section 6.4):

> ***Definition 6.4 (Monotonicity).*** *For all individuals i:*
>
> $$D_i(1) \geq D_i(0)$$
>
> *That is, winning the lottery weakly increases attendance for every trainer. No one who would attend without winning the lottery would refuse to attend because they won.*

Monotonicity rules out **defiers** — perverse individuals who do the opposite of what the instrument encourages. In the Safari Zone context, a defier would be someone who thinks, "I won a free pass? That's too mainstream. I refuse to go now." While such behavior is logically possible, it is behaviorally implausible in most settings, and monotonicity is usually considered a mild assumption.

Partial Identification Under Weaker Assumptions

If you are unwilling to assume the exclusion restriction holds exactly, you can still obtain **bounds** on the causal effect. Nevo and Rosen (2012) show that if you know the *direction* of the exclusion restriction violation (e.g., the direct effect of Z on Y is positive), you can tighten the bounds substantially. Conley, Hansen, and Rossi (2012) propose a framework where the exclusion restriction holds "approximately" — the direct effect of Z on Y is small but not exactly zero — and derive confidence intervals that incorporate this uncertainty.

These partial identification approaches trade precision for honesty. They acknowledge that the exclusion restriction is an assumption, not a fact, and they show how sensitive the conclusions are to violations.

6.4 LATE and the Complier Taxonomy

The IV estimate does not recover the average treatment effect for the entire population. It recovers the effect for a very specific subgroup. Understanding which subgroup — and why — is essential for interpreting IV results correctly.

Four Types of Trainers

When we have a binary instrument Z (lottery win/loss) and a binary treatment D (attend/not attend), each trainer has two potential treatment values: $D_i(1)$ (what they would do if they won the lottery) and $D_i(0)$ (what they would do if they lost). This creates a 2×2 classification:

Definition 6.5 (Compliance Types). *Based on potential treatments $D_i(1)$ and $D_i(0)$:*

Type	$D_i(0)$	$D_i(1)$	***Behavior***
Complier	*0*	*1*	*Attends only if wins lottery*
Always-Taker	*1*	*1*	*Attends regardless of lottery*
Never-Taker	*0*	*0*	*Never attends regardless*
Defier	*1*	*0*	*Attends only if loses lottery*

Let us put faces to these types in Fuchsia City.

Compliers are trainers who would not pay the 500-Pokedollar admission on their own but will happily go if it is free. The lottery *changes their behavior*. These are the trainers for whom the instrument makes a difference.

Always-Takers are wealthy or devoted trainers who attend the Safari Zone every week regardless. They buy their own ticket even if they lose the lottery. The lottery does not change their behavior — they attend either way.

Never-Takers are trainers who have no interest in the Safari Zone. Even a free pass does not entice them — maybe they find it boring, or they are too busy training at the Fuchsia Gym. The lottery does not change their behavior — they stay away either way.

Defiers would be trainers who attend when they lose but refuse to attend when they win. Under the monotonicity assumption, we rule out this group. (And indeed, it is hard to imagine why anyone would behave this way with a free Safari Zone pass.)

A crucial observation: we can never directly observe any individual's type. We observe Z_i and D_i for each trainer, but this only tells us:

- If $Z_i = 1$ and $D_i = 1$: the trainer is either a Complier or an Always-Taker.
- If $Z_i = 1$ and $D_i = 0$: the trainer is a Never-Taker.
- If $Z_i = 0$ and $D_i = 1$: the trainer is an Always-Taker.
- If $Z_i = 0$ and $D_i = 0$: the trainer is either a Complier or a Never-Taker.

We can identify Never-Takers (among winners) and Always-Takers (among losers) individually, but we can never point to a single person and say with certainty "this person is a Complier."

The Local Average Treatment Effect (LATE)

The IV/Wald estimator recovers the average treatment effect *for Compliers only*:

Definition 6.6 (Local Average Treatment Effect). *Under the assumptions of relevance, independence, exclusion restriction, and monotonicity:*

$$\hat{\tau}_{IV} \xrightarrow{p} \text{LATE} = E[Y_i(1) - Y_i(0) \mid \text{Complier}]$$

The IV estimate is the causal effect of treatment for the subpopulation whose treatment status is changed by the instrument.

This is the landmark result of Imbens and Angrist (1994). It tells us precisely what IV identifies and for whom.

Why LATE ≠ ATE

The LATE is a *local* effect — local to the complier subpopulation. It need not equal the average treatment effect for the whole population, the effect for always-takers, or the effect for never-takers.

Consider why compliers might differ from the general population in the Safari Zone setting. Compliers are trainers on the margin: interested enough to attend if free but not willing to pay. These are likely mid-level trainers — not the most dedicated (who are always-takers) and not the least interested (who are never-takers). The effect of Safari Zone training on these marginal trainers may differ from its effect on devoted trainers who attend regardless.

If the treatment effect is *homogeneous* — the same for everyone — then LATE = ATE, and this distinction is moot. But with heterogeneous treatment effects, the distinction matters greatly. A LATE of 19 percentage points for compliers does not mean the ATE is 19 percentage points for the entire population.

This also means that **different instruments identify different LATEs**. Suppose a second instrument — a Pokedollar rebate coupon — also affects Safari Zone attendance. The compliers for the coupon (trainers who attend because of the rebate but would not otherwise) are a different subpopulation than the compliers for the lottery (trainers who attend because of the free pass). If the treatment effect is heterogeneous, these two instruments will produce different IV estimates — not because one is "wrong," but because they identify effects for different groups. This is not a bug; it is a feature of the LATE framework. But it does mean that comparing IV estimates across studies using different instruments requires caution.

Angrist and Fernandez-Val (2013) provide methods for extrapolating from the LATE to broader populations under additional assumptions about how the treatment effect varies with observable characteristics. Mogstad, Santos, and Torgovitsky (2018) develop a more general framework for using multiple instruments to learn about the distribution of treatment effects across compliance types.

> ***Professor Oak Explains: When LATE Is What You Want***
>
> *"LATE is often dismissed as a 'second-best' estimand — people want the ATE but settle for the LATE. But in many policy contexts, the LATE is exactly the right quantity. If the Fuchsia City Warden is considering making the Safari Zone free for everyone, the relevant question is: how much would marginal trainers benefit? These are precisely the compliers — the trainers whose behavior would be changed by the policy. Always-takers already attend, and never-takers will not attend even if it is free. The compliers are the policy-relevant population."*

Characterizing Compliers

Although we cannot identify individual compliers, we can describe their characteristics as a group. Abadie (2003) showed that you can estimate the distribution of observed covariates among compliers using a reweighting approach. For any covariate X:

$$E[X|\text{Complier}] = \frac{E[X \cdot D|Z = 1] - E[X \cdot D|Z = 0]}{E[D|Z = 1] - E[D|Z = 0]}$$

In our data, suppose we compute:

Covariate	Population Mean	Complier Mean
Trainer Level	38.2	33.7
Badges	5.1	4.4
Party Size	4.8	4.2

Compliers tend to be somewhat lower-level trainers with fewer badges and smaller parties — consistent with the "marginal trainer" interpretation.

The ITT-LATE Connection

The relationship between the intent-to-treat effect and the LATE provides useful intuition and connects back to the experimental design ideas from Chapter 2:

$$ITT = LATE \times P(\text{Complier})$$

The ITT is the "diluted" version of the LATE — diluted by the fraction of the population that are compliers. In our example:

$$0.08 = 0.19 \times 0.42$$

The ITT of 8 percentage points equals the LATE of 19 percentage points times the compliance rate of 42%. This is why we divide the ITT by the first stage to recover the LATE: we are "undiluting" the effect.

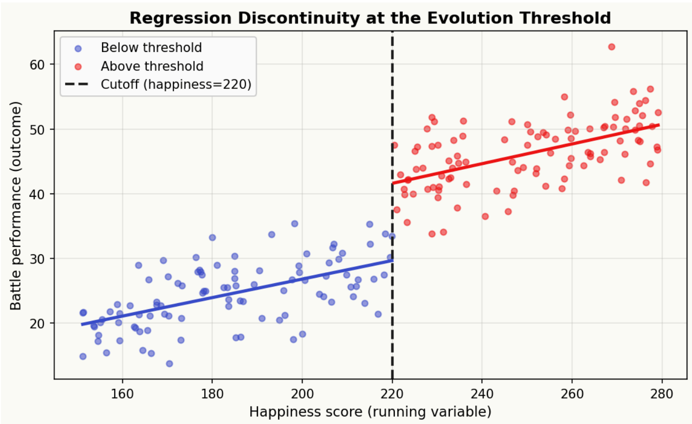
Part II: Regression Discontinuity (Cinnabar Island)

You take the ferry from Fuchsia City to Cinnabar Island. The volcanic island is famous for its research lab, where scientists study Pokemon evolution. A particularly interesting phenomenon catches your eye: Pokemon with a happiness score of 220 or above evolve into a stronger form, while those below 220 remain unevolved. This sharp threshold creates a natural experiment waiting to be exploited.

6.5 Sharp Regression Discontinuity Design

#010 Caterpie *#011 Metapod* *#012 Butterfree*

An evolution chain — the happiness threshold is the RDD running variable.



Sharp Regression Discontinuity at the evolution threshold.

The Setting

On Cinnabar Island, researchers have assembled the `happiness_evolution.csv` dataset tracking Pokemon and their trainers. Each Pokemon has a **happiness score** — a continuous measure from 0 to 255 based on time spent with the trainer, battles fought, items used, and other factors. The game mechanics dictate that when a Pokemon's happiness reaches 220, it evolves.

Let X_i denote the happiness score (the **running variable** or **forcing variable**), $c = 220$ the cutoff, and D_i the treatment (evolution):

$$D_i = \begin{cases} 1 & \text{if } X_i \geq 220 \\ 0 & \text{if } X_i < 220 \end{cases}$$

The outcome Y_i is the Pokemon's battle performance after evolution (or non-evolution), measured as a standardized combat power index.

The question: does evolution *cause* improved battle performance? Or do Pokemon that are about to evolve already have high battle stats simply because their trainers have invested heavily in them?

This is another instance of selection bias. Pokemon with high happiness have dedicated trainers who invest in every aspect of their Pokemon's development — not just happiness, but also training, move selection, and strategic battle preparation. Simply comparing evolved to unevolved Pokemon confounds the effect of evolution itself with the effect of having a dedicated trainer. We need a design that isolates the causal effect of evolution from the selection effect of trainer quality.

The Identification Idea

The RDD identification strategy is simple and elegant: **compare Pokemon just above the cutoff to Pokemon just below it.**

A Pokemon with a happiness score of 221 evolved. A Pokemon with a happiness score of 219 did not. But these two Pokemon are essentially identical in every other respect — their trainers invested almost the same amount of effort, they have nearly the same experience, and they differ by a mere 2 points on a 0-255 scale. Yet one evolved and the other did not.

The key assumption is that all factors other than treatment vary **smoothly** (continuously) at the cutoff. There is no reason to expect a sudden jump in trainer skill, Pokemon genetics, or any other determinant of battle performance at exactly 220. The only thing that jumps at 220 is evolution status.

Definition 6.7 (Sharp RDD Estimand). *The causal effect of treatment at the cutoff is:*

$$\tau_{RDD} = \lim_{x \downarrow c} E[Y_i | X_i = x] - \lim_{x \uparrow c} E[Y_i | X_i = x] = E[Y_i(1) - Y_i(0) | X_i = c]$$

The RDD estimates the average treatment effect for units at the cutoff — those with $X_i = c$.

This is a *local* estimate, just like the LATE from IV. It applies to Pokemon right at the boundary of evolution. Pokemon with happiness scores of 100 or 250 are far from the cutoff, and the RDD tells us nothing about the effect of evolution for them.

The formal identification relies on a **continuity assumption**: the conditional expectation functions $E[Y(1)|X = x]$ and $E[Y(0)|X = x]$ are both continuous at $x = c$. This means that in the absence of treatment, there would be no jump in outcomes at the cutoff. The only reason we observe a jump in $E[Y|X = x]$ at c is because treatment switches on at that point. Formally:

$$\begin{aligned}
\tau_{RDD} &= \lim_{x \downarrow c} E[Y|X = x] - \lim_{x \uparrow c} E[Y|X = x] \\
&= \lim_{x \downarrow c} E[Y(1)|X = x] - \lim_{x \uparrow c} E[Y(0)|X = x] \\
&= E[Y(1)|X = c] - E[Y(0)|X = c] \quad \text{(by continuity)} \\
&= E[Y(1) - Y(0)|X = c]
\end{aligned}$$

This derivation, due to Hahn, Todd, and van der Klaauw (2001), shows that the RDD identifies the treatment effect at the cutoff under the continuity assumption alone — no functional form restrictions, no model for selection, no conditional independence.

The Local Randomization Interpretation

Near the cutoff, treatment assignment is "as good as random." A Pokemon with a happiness score of 219 versus 221 is separated by such a tiny margin that the variation is essentially noise — the result of one extra pat on the head or one fewer potion used. This local randomization perspective makes the RDD especially credible: near the cutoff, it mimics a randomized experiment.

This interpretation requires that units cannot *precisely manipulate* the running variable to be just above or below the cutoff. If trainers could set their Pokemon's happiness to exactly 221 to guarantee evolution, the "as good as random" argument collapses. We will return to this concern in Section 6.7 on diagnostics.

Estimation: Local Linear Regression

The most common approach to estimating the RDD effect is **local linear regression**: fit separate linear regressions on each side of the cutoff using only observations within a bandwidth h of the cutoff.

For observations with $X_i \in [c - h, c)$ (below the cutoff):

$$Y_i = \alpha_l + \beta_l (X_i - c) + \epsilon_i^l$$

For observations with $X_i \in [c, c + h]$ (above the cutoff):

$$Y_i = \alpha_r + \beta_r (X_i - c) + \epsilon_i^r$$

The RDD estimate is the difference in intercepts:

$$\hat{\tau}_{RDD} = \hat{\alpha}_r - \hat{\alpha}_l$$

This estimates the jump in the conditional expectation function at the cutoff.

Why *local linear* and not global polynomial? Global polynomial regressions (fitting a single high-order polynomial across the entire range of X) are seductive but dangerous. Gelman and Imbens (2019) argue convincingly that high-order global polynomials are sensitive to observations far from the cutoff, can produce wild extrapolations, and generate confidence intervals with poor coverage. For example, fitting a fourth-degree polynomial to our happiness-performance data would allow observations with happiness scores of 150 to influence the estimated treatment effect at 220 — even though those observations are 70 points away from the cutoff and tell us almost nothing about the local behavior near the threshold.

Local linear regression avoids these problems by focusing only on data near the cutoff, where the linear approximation is most plausible. The key mathematical property is that local linear regression has a **boundary bias** that is of order h^2 (where h is the bandwidth), while local constant (Nadaraya-Watson) regression has boundary bias of order h. This means local linear regression provides a better approximation at the cutoff — precisely where we need accuracy.

> ***Blue's Mistake: "I'll Fit a Sixth-Degree Polynomial"***
>
> *Blue decides to estimate the RDD by fitting a sixth-degree polynomial on each side of the cutoff using all the data. "More flexible is better, right? My R^2 is 0.94!" But his treatment effect estimate is 22.7 — nearly double the local linear estimate. The high-order polynomial is being pulled by a handful of influential observations with happiness scores near 150 and 255, creating Runge-phenomenon-like oscillations that distort the extrapolation to the cutoff. When he drops observations below 180 or above 250, his estimate changes dramatically. This sensitivity to faraway data is precisely why Gelman and Imbens recommend against high-order polynomials. Stick with local linear regression.*

Kernel Choice and Bandwidth

In practice, local linear regression is implemented using a **kernel function** $K(\cdot)$ that assigns weights to observations based on their distance from the cutoff. Common choices include:

- **Triangular kernel:** $K(u) = (1 - |u|) \cdot 1(|u| \leq 1)$ — gives the most weight to observations closest to the cutoff. This is optimal in a minimax mean-squared-error sense (Fan, 1992).
- **Uniform (rectangular) kernel:** $K(u) = \frac{1}{2} \cdot 1(|u| \leq 1)$ — gives equal weight to all observations within the bandwidth. Equivalent to unweighted local linear regression within the bandwidth window.
- **Epanechnikov kernel:** $K(u) = \frac{3}{4}(1 - u^2) \cdot 1(|u| \leq 1)$ — a smooth compromise.

The **bandwidth** h controls the bias-variance tradeoff:

- **Small h:** Uses fewer observations near the cutoff. Low bias (the linear approximation is excellent over a narrow range) but high variance (few data points).
- **Large h:** Uses more observations further from the cutoff. Lower variance but higher bias (the linear approximation may be poor over a wide range).

Optimal bandwidth selection methods, such as those of Imbens and Kalyanaraman (2012) and Cattaneo, Idrobo, and Titiunik (2020), balance this tradeoff by minimizing mean squared error. The `rdrobust` package in R and Python implements these procedures. We will discuss optimal bandwidth selection further in Section 6.7.

Worked Example: The Effect of Evolution on Battle Performance

Using the `happiness_evolution.csv` dataset with 800 Pokemon, we estimate the effect of evolution on the standardized combat power index.

Setup: - Running variable: Happiness score (X), ranging from 150 to 255 - Cutoff: $c = 220$ - Treatment: Evolution $(D = 1(X \geq 220))$ - Outcome: Standardized combat power index (Y) - Bandwidth: $h = 15$ (selected via Cattaneo-Idrobo-Titiunik optimal bandwidth selector)

Observations within bandwidth: 312 Pokemon with happiness scores in $[205, 235]$

Left-side regression $(X \in [205, 220), n = 148)$:

$$\hat{Y}_i = 52.3 + 0.41 \cdot (X_i - 220)$$

At $X = 220$: predicted $Y = 52.3$

Right-side regression $(X \in [220, 235], n = 164)$:

$$\hat{Y}_i = 63.8 + 0.38 \cdot (X_i - 220)$$

At $X = 220$: predicted $Y = 63.8$

RDD estimate:

$$\hat{\tau}_{RDD} = 63.8 - 52.3 = 11.5$$

with a robust standard error of 3.2, yielding a 95% confidence interval of $[5.2, 17.8]$.

Evolution causes an **11.5-point increase** in the standardized combat power index for Pokemon at the happiness threshold of 220. This is a substantial and statistically significant effect.

> ***Professor Oak Explains: What the RDD Graph Shows***
>
> *"The classic RDD visualization shows a scatter plot of the outcome against the running variable, with a vertical line at the cutoff. You should see a clear jump in the outcome at the cutoff — the treatment effect. On either side of the cutoff, the relationship between happiness and combat power should be smooth and continuous. The discontinuity at the cutoff is the causal effect.*
>
> *"If you see a smooth curve with no jump, there is no treatment effect. If you see a jump but the data are noisy, the treatment effect is uncertain. If you see jumps at places other than the cutoff, something may be wrong with the design."*

6.6 Fuzzy Regression Discontinuity Design

When the Cutoff Is Not Sharp

In the sharp RDD, crossing the threshold deterministically assigns treatment: every Pokemon at or above 220 evolves, and every Pokemon below 220 does not. But in practice, the world is rarely so clean.

On Cinnabar Island, some trainers **press the B button** during the evolution animation, canceling the evolution. They might prefer the unevolved form's aesthetic, its moveset, or its type. This means that some Pokemon above the happiness threshold of 220 *do not evolve*, even though they are eligible.

Formally, let D_i now denote actual evolution status. In the **fuzzy RDD**:

$$\lim_{x \downarrow c} E[D|X = x] - \lim_{x \uparrow c} E[D|X = x] = \varkappa, \quad 0 < \varkappa < 1$$

The probability of treatment *jumps* at the cutoff but does not go from 0 to 1. In our data, suppose:

- Among Pokemon just above 220: 82% evolved ($D = 1$)
- Among Pokemon just below 220: 0% evolved ($D = 0$, no one below the threshold can evolve)

The jump in treatment probability is $\varkappa = 0.82 - 0.00 = 0.82$.

Fuzzy RDD as Instrumental Variables

The fuzzy RDD is conceptually identical to instrumental variables, where the instrument is **whether the running variable exceeds the cutoff**:

$$Z_i = 1(X_i \geq c)$$

This indicator Z_i serves as an instrument for actual treatment D_i. The three IV assumptions hold in the RDD context:

1. **Relevance:** Crossing the cutoff increases the probability of treatment (from 0% to 82% in our example).
2. **Independence:** Near the cutoff, being just above or below is "as good as random" — the running variable's position near the cutoff is essentially noise.
3. **Exclusion restriction:** Crossing the cutoff affects outcomes *only through* its effect on treatment (evolution). The cutoff itself has no direct effect on battle performance.

The **fuzzy RDD estimand** is the ratio of the jump in outcomes to the jump in treatment at the cutoff:

> ***Definition 6.8 (Fuzzy RDD Estimand).*** *The fuzzy RDD estimate is:*
>
> $$\tau_{FRDD} = \frac{\lim_{x \downarrow c} E[Y|X = x] - \lim_{x \uparrow c} E[Y|X = x]}{\lim_{x \downarrow c} E[D|X = x] - \lim_{x \uparrow c} E[D|X = x]}$$
>
> *This is exactly the Wald estimator applied locally at the discontinuity.*

Connection to 2SLS

The fuzzy RDD can be estimated via 2SLS using only observations within the bandwidth:

First stage:

$$D_i = \pi_0 + \pi_1 \cdot 1(X_i \geq c) + \pi_2 \cdot (X_i - c) + \pi_3 \cdot 1(X_i \geq c) \cdot (X_i - c) + v_i$$

Second stage:

$$Y_i = \beta_0 + \beta_1 \hat{D}_i + \beta_2 \cdot (X_i - c) + \beta_3 \cdot 1(X_i \geq c) \cdot (X_i - c) + \epsilon_i$$

The coefficient $\hat{\beta}_1$ is the fuzzy RDD estimate. Note the inclusion of the running variable $(X_i - c)$ and its interaction with the cutoff indicator — these allow separate slopes on each side of the cutoff, as in the sharp RDD.

The Complier Interpretation in Fuzzy RDD

Just as IV recovers a LATE, the fuzzy RDD recovers the treatment effect for **compliers at the cutoff** — Pokemon whose evolution status is changed by crossing the threshold. These are the Pokemon that *would* evolve if above 220 but *would not* (could not) if below. The B-button pressers are the non-compliers — they are above the threshold but refuse treatment (evolution).

In the language of the complier taxonomy from Section 6.4: - **Compliers:** Pokemon that evolve when above 220 and would not evolve if below. These are the "normal" cases — the game triggers evolution, and the trainer allows it. - **Never-Takers (B-button pressers):** Pokemon whose trainers cancel evolution even when above 220. They are treated (crossing the threshold) but refuse the actual treatment (evolution). - **Always-Takers:** In principle, these would be Pokemon that evolve even below 220, which is impossible by game mechanics. So in this one-sided fuzzy design, there are no always-takers below the cutoff. - **Defiers:** Ruled out by monotonicity — no Pokemon evolves *only when below* the threshold.

The fuzzy RDD LATE is the effect of evolution for the compliers at the cutoff: trainers who allow evolution to proceed when the happiness threshold is crossed. This is a natural and policy-relevant quantity — it tells us what evolution does for the "typical" Pokemon that evolves at the margin.

Worked Example: Fuzzy RDD with Happiness Evolution Data

Using the same `happiness_evolution.csv` dataset, but now accounting for B-button cancellations:

Jump in outcomes at cutoff:

$$\lim_{x \downarrow 220} E[Y|X = x] - \lim_{x \uparrow 220} E[Y|X = x] = 62.1 - 52.3 = 9.8$$

Jump in treatment at cutoff:

$$\lim_{x \downarrow 220} E[D|X = x] - \lim_{x \uparrow 220} E[D|X = x] = 0.82 - 0.00 = 0.82$$

Fuzzy RDD estimate:

$$\hat{\tau}_{FRDD} = \frac{9.8}{0.82} = 11.95$$

The fuzzy RDD estimate is 11.95 — slightly larger than the sharp RDD estimate of 11.5. This makes sense: the sharp RDD "dilutes" the effect by averaging over both evolved Pokemon and B-button non-compliers above the threshold. The fuzzy RDD corrects for this dilution by dividing by the compliance rate.

> ***Blue's Mistake: "Just Compare Above vs. Below"***
>
> *Blue looks at the happiness evolution data and says, "Easy! Pokemon above 220 have an average combat power of 61.5, and those below average 48.2. The effect of evolution is $61.5 - 48.2 = 13.3$ points."*
>
> *Blue is committing two errors. First, he is comparing all Pokemon above and below the cutoff, not just those near it. Pokemon with happiness of 250 are very different from those with happiness of 180 — they have more experienced trainers, more battles, more investment. The RDD uses only observations near the cutoff to ensure comparability. Second, he is ignoring the fuzzy nature of the design — not all Pokemon above 220 actually evolved. The naive comparison conflates the effect of evolution with the effect of happiness itself.*

6.7 RDD Diagnostics

A well-executed RDD is among the most credible quasi-experimental designs. But this credibility rests on assumptions that must be carefully validated. This section presents the essential diagnostic checks.

McCrary Density Test: Checking for Manipulation

The most important threat to RDD validity is **manipulation of the running variable**. If trainers can precisely control their Pokemon's happiness to land just above 220, then the Pokemon right above the cutoff are not comparable to those right below — they are systematically different (their trainers are more skilled, more strategic, more invested).

The **McCrary (2008) density test** checks for manipulation by examining whether the density of the running variable is continuous at the cutoff. Under no manipulation, we expect approximately equal numbers of Pokemon on both sides of 220. If trainers are gaming the threshold, we would see a **bunching** — an excess mass of observations just above 220 and a corresponding hollow just below.

Formally, the test estimates the density of X separately on each side of the cutoff and tests for a discontinuity:

$$H_0 : \lim_{x \downarrow c} f(x) = \lim_{x \uparrow c} f(x) \qquad H_1 : \lim_{x \downarrow c} f(x) \neq \lim_{x \uparrow c} f(x)$$

To implement the test visually, plot a histogram of the happiness score with fine bins near the cutoff. If you see a smooth, continuous distribution, manipulation is unlikely. If you see a spike just above 220 or a dip just below, there is cause for concern.

In our data, the histogram shows a roughly uniform distribution of happiness scores between 200 and 240, with no visible bunching at 220. The McCrary test statistic is -0.08 with a p-value of 0.64 — we fail to reject the null of no manipulation. This is reassuring.

The McCrary test works by estimating the density of X on each side of the cutoff using local polynomial density estimation, then computing the log difference in the estimated densities at the cutoff. Under the null of no manipulation, this log difference is asymptotically normal. Cattaneo, Jansson, and Ma (2020) provide an updated version of the density test (`rddensity`) with improved finite-sample properties and automatic bandwidth selection.

It is worth noting what the McCrary test can and cannot tell you. It can detect *precise* manipulation — trainers pushing their Pokemon's happiness to exactly 220 or just above. But it cannot detect *imprecise* manipulation — trainers generally trying to increase happiness without precise control over the exact score. If all trainers are trying to make their Pokemon happier (a reasonable assumption), this does not threaten the RDD as long as they cannot precisely target the 220 threshold.

> ***Professor Oak Explains: When Manipulation Is Plausible***
>
> *"Not all running variables are equally susceptible to manipulation. Happiness scores in Pokemon games are a complex function of many inputs — walking steps, using items, winning battles, leveling up — and they are not displayed to trainers as a precise number. This makes precise manipulation difficult.*
>
> *"Contrast this with settings where the running variable is self-reported (income for benefit eligibility) or precisely known to the individual (test scores for program entry). In those cases, manipulation is a serious concern, and the McCrary test becomes essential. If the density test rejects, the RDD may not be credible."*

Placebo Tests: Covariate Balance at the Cutoff

The key assumption of RDD is that all factors other than treatment vary smoothly at the cutoff. We can partially test this by checking whether **pre-treatment covariates** show discontinuities at the cutoff. If the design is valid, covariates that are determined before the running variable reaches the cutoff should be continuous through it.

For each covariate W (e.g., Pokemon level, base attack, base defense, base speed), run the RDD with W as the outcome:

$$\hat{\delta}_W = \lim_{x \downarrow c} E[W|X = x] - \lim_{x \uparrow c} E[W|X = x]$$

If the design is valid, $\hat{\delta}_W \approx 0$ for all pre-treatment covariates. A statistically significant discontinuity in a covariate suggests that something other than the treatment changes at the cutoff — a red flag.

In our data:

Covariate	Estimated Discontinuity	Std. Error	p-value
Level	-0.3	1.1	0.78
Base Attack	0.8	1.5	0.59
Base Defense	-0.5	1.4	0.72
Base Speed	0.2	1.3	0.88

None of the covariates show a significant discontinuity at 220. This is consistent with the identifying assumption that all factors other than evolution are smooth through the cutoff.

Bandwidth Sensitivity

Because the RDD estimate depends on the choice of bandwidth h, it is essential to show that the results are **robust to different bandwidths**. If the estimate changes dramatically as you widen or narrow the bandwidth, this raises concerns about the specification.

Good practice is to present the estimate for a range of bandwidths: half the optimal bandwidth, the optimal bandwidth, one-and-a-half times the optimal, and twice the optimal:

Bandwidth	n within window	$\hat{\tau}_{RDD}$	Std. Error	95% CI
$h = 8$	158	12.8	4.7	$[3.6, 22.0]$
$h = 12$	247	11.9	3.6	$[4.8, 19.0]$
$h = 15$ (optimal)	312	11.5	3.2	$[5.2, 17.8]$
$h = 20$	402	10.8	2.8	$[5.3, 16.3]$
$h = 30$	548	9.6	2.4	$[4.9, 14.3]$

The estimate ranges from 9.6 to 12.8, with all confidence intervals overlapping. The point estimate decreases slightly as the bandwidth increases (because the linear approximation becomes less accurate at wider bandwidths), but the estimates are reasonably stable. This is reassuring.

Donut Hole RDD

If there is concern about manipulation *very close* to the cutoff, a **donut hole RDD** excludes observations within a small window around the cutoff. For example, exclude Pokemon with happiness scores in $[218, 222]$ and estimate the RDD using the remaining observations within the bandwidth.

If manipulation affects only the observations immediately adjacent to the cutoff, the donut hole RDD removes these contaminated observations while still estimating the discontinuity. In our data, the donut hole estimate (excluding $[218, 222]$) is 11.2 ($\text{SE} = 3.5$), very close to the baseline estimate of 11.5.

Optimal Bandwidth Selection

Cattaneo, Idrobo, and Titiunik (2020) provide the modern standard for bandwidth selection in RDD. Their approach, implemented in the `rdrobust` package, minimizes the mean squared error of the RDD estimator by balancing bias and variance. The procedure:

1. Estimates the second derivative of the conditional expectation function to quantify the bias from the linear approximation.
2. Estimates the variance of the local linear estimator.
3. Selects the bandwidth h^* that minimizes the asymptotic MSE.
4. Provides **bias-corrected confidence intervals** that account for the remaining bias at the optimal bandwidth.

The bias-corrected confidence intervals from `rdrobust` are generally wider than conventional ones because they honestly account for the approximation error inherent in the local linear specification. These are the intervals that should be reported in applied work.

> ***Blue's Mistake: "I'll Just Pick the Bandwidth That Gives the Best Result"***
>
> *Blue runs the RDD with 15 different bandwidths and reports only the one that gives the largest and most significant estimate. "Look, with a bandwidth of 7, the effect is 14.8 with a* p*-value of 0.003! Very significant!"*
>
> *This is cherry-picking — a form of* p*-hacking. The bandwidth should be chosen by a principled, data-driven procedure before looking at the results, or the full range of estimates should be reported transparently. Cattaneo, Idrobo, and Titiunik's optimal bandwidth selector provides a pre-committed choice that balances bias and variance without reference to the treatment effect estimate. Always report results for the optimal bandwidth and show sensitivity to alternative choices.*

Brief Mentions: Extensions and Related Designs

Shift-Share (Bartik) Instruments

The shift-share instrument, popularized by Bartik (1991), is a widely used IV strategy in economics. The idea is to construct an instrument that combines aggregate-level "shifts" (changes) with unit-level "shares" (exposure). The instrument for unit i takes the form:

$$B_i = \sum_k s_{ik} \cdot g_k$$

where s_{ik} is unit i's share in category k and g_k is the aggregate change in category k.

In the Kanto context, imagine constructing an instrument for changes in local Gym difficulty. Each city has a different composition of trainer types (share of Water-type specialists, Fire-type specialists, etc.). Region-wide changes in the availability of type-specific TMs differentially affect cities depending on their trainer composition. The Bartik instrument would be:

$$B_{city} = \sum_{type} (\text{share of type in city}) \times (\text{region-wide change in type-specific TM availability})$$

Recent work by Goldsmith-Pinkham, Sorkin, and Swift (2020) and Borusyak, Hull, and Jaravel (2022) has clarified the identification assumptions underlying Bartik instruments, showing that identification can come either from the exogeneity of shares (the GPSS approach) or the exogeneity of shifts (the BHJ approach), with different implications for which confounders must be addressed.

Regression Discontinuity in Time (RDiT)

When the running variable is *time* rather than a continuous score, the design is called **Regression Discontinuity in Time** (Hausman and Rapson, 2018). For example, if the Pokemon League changed its battle rules on a specific date, one could compare battle outcomes just before and just after the policy change, using date as the running variable.

RDiT requires the same continuity assumptions as standard RDD — all factors other than the policy change must vary smoothly across the implementation date. But time series data present additional challenges: seasonality (battle performance may vary by day of the week or month of the year), trends (trainers may be improving over time regardless of the policy), and autocorrelation (outcomes in adjacent time periods are correlated). These features make the continuity assumption harder to defend and require more careful specification than in a standard cross-sectional RDD.

RDiT is related to but distinct from difference-in-differences (Chapter 7). DiD exploits variation across both time and units, requiring parallel trends. RDiT exploits only the temporal discontinuity, requiring continuity at the cutoff. The two designs answer different questions and require different assumptions.

Chapter Summary

Soul Badge!

Volcano Badge!

Part I: Instrumental Variables

- **Unobserved confounding** cannot be addressed by regression, matching, or propensity scores. When important confounders are unmeasured, these methods fail.
- An **instrumental variable** Z provides an alternative route to causal identification. It must satisfy three assumptions: relevance (Z affects D), independence (Z is as-if random), and the exclusion restriction (Z affects Y only through D).
- The **Wald estimator** $\hat{\tau}_{IV} = \frac{E[Y|Z=1]-E[Y|Z=0]}{E[D|Z=1]-E[D|Z=0]}$ is the ratio of the reduced-form effect to the first-stage effect.
- **Two-Stage Least Squares (2SLS)** generalizes IV to include covariates, continuous variables, and multiple instruments.
- The **exclusion restriction** is untestable and is the primary vulnerability of any IV design.
- **Weak instruments** (first-stage $F < 10$) cause bias and unreliable inference. Anderson-Rubin confidence sets provide a weak-instrument-robust alternative.
- IV recovers the **Local Average Treatment Effect (LATE)** — the causal effect for *compliers*, the subpopulation whose treatment is changed by the instrument.
- The four compliance types are Compliers, Always-Takers, Never-Takers, and Defiers (ruled out by monotonicity).

Part II: Regression Discontinuity

- **Sharp RDD** exploits a deterministic treatment assignment rule based on a continuous running variable crossing a cutoff.
- The key assumption is that all factors other than treatment are **continuous at the cutoff** — ensuring that units just above and below the cutoff are comparable.
- **Local linear regression** with optimal bandwidth is the recommended estimation strategy. Avoid global polynomials.
- **Fuzzy RDD** arises when crossing the cutoff increases the *probability* of treatment without determining it completely. Fuzzy RDD is equivalent to IV at the discontinuity.
- **McCrary density tests** check for manipulation of the running variable. Bunching at the cutoff invalidates the design.
- **Placebo tests** on pre-treatment covariates verify that no other discontinuity exists at the cutoff.

- **Bandwidth sensitivity** analysis and **bias-corrected confidence intervals** (via `rdrobust`) ensure robust inference.

Professor Oak's Review Questions

1. **IV Conceptual:** Explain why the exclusion restriction cannot be tested with data. Give an example of a plausible violation of the exclusion restriction in the Safari Zone lottery setting that would bias the IV estimate upward.
2. **IV Technical:** Suppose you have two instruments for Safari Zone attendance: the lottery (Z_1) and a Pokedollar rebate coupon (Z_2). The first-stage F-statistics are 85.3 and 6.2, respectively. Which instrument raises concerns about weak instrument bias? What diagnostic would you use to obtain reliable inference with the weaker instrument?
3. **IV Interpretation:** A researcher estimates that the LATE of Safari Zone training on win rate is 19 percentage points. A policymaker wants to know the effect of making the Safari Zone free for *all* trainers. Under what conditions would the LATE be a good estimate of the policy-relevant treatment effect? Under what conditions might it be misleading?
4. **RDD Conceptual:** Why is the McCrary density test important for the credibility of an RDD? Describe a scenario in the happiness evolution setting where the test might reject, and explain what this would imply for the causal estimate.
5. **RDD Technical:** A researcher estimates a sharp RDD with optimal bandwidth $h = 15$ and finds a significant effect of evolution on combat power. She then tries bandwidths of 8, 12, 20, and 30, and finds that the effect is significant for $h = 15$ and $h = 20$ but not for $h = 8$ or $h = 30$. Should she be concerned? Explain.
6. **RDD vs. IV:** Both fuzzy RDD and standard IV use the Wald estimator structure. What is the key difference in the source of identifying variation? Why is the fuzzy RDD often considered more credible than a typical IV design?

Trainer Challenge Exercises

Exercise 1: Compute the Wald Estimate

The following table summarizes data from a Safari Zone lottery:

	Winners ($Z = 1$)	Losers ($Z = 0$)
n	150	350
Mean attendance ($\bar{D}$)	0.80	0.25
Mean win rate ($\bar{Y}$)	0.62	0.51

a. Compute the first-stage effect.
b. Compute the reduced-form (ITT) effect.
c. Compute the Wald IV estimate.
d. What is the estimated compliance rate? Interpret this number in context.
e. Using the ITT-LATE relationship, verify that $\mathrm{ITT} = \mathrm{LATE} \times P(\mathrm{Complier})$.

Exercise 2: Assess Instrument Strength

A researcher proposes using "distance from a trainer's home to the nearest Safari Zone entrance" as an instrument for Safari Zone attendance. The first-stage regression yields:

$$D_i = 0.65 - 0.008 \cdot \mathrm{Distance}_i + \vartheta_i$$

with a first-stage F-statistic of 7.3.

a. According to the Stock-Yogo rule of thumb, is this instrument sufficiently strong?
b. Describe two problems that arise from using a weak instrument.
c. Propose a diagnostic that provides valid inference even with a weak instrument.
d. Separately from strength, evaluate whether "distance to Safari Zone" is likely to satisfy the exclusion restriction. What potential violations might arise?

Exercise 3: Estimate a Sharp RDD

Using the following data from the `happiness_evolution.csv` dataset (showing mean combat power for bins of happiness score), estimate the sharp RDD effect of evolution on combat power:

Happiness bin	Midpoint	Mean Y	n
[205, 210)	207.5	49.1	38
[210, 215)	212.5	50.8	42
[215, 220)	217.5	52.6	45
[220, 225)	222.5	64.1	48
[225, 230)	227.5	65.3	44
[230, 235)	232.5	66.8	40

a. Fit separate linear regressions for the three bins below and above the cutoff of 220.
b. Extrapolate each regression to the cutoff ($X = 220$) and compute the RDD estimate.
c. Plot the data and your fitted lines (sketch or software).
d. What assumption is required for this estimate to be causal? How would you test it?

Exercise 4: Conceptual McCrary Test

Suppose you are studying the effect of a new TM move on Pokemon battle performance. The TM is available only to Pokemon whose trainer has at least 1000 hours of gameplay, and trainers can see their hour count on their Trainer Card.

a. Explain why you might be concerned about manipulation of the running variable in this setting.
b. Describe what a McCrary density test would look like if manipulation were occurring. Sketch the expected histogram shape.
c. Propose two additional diagnostic checks (beyond the McCrary test) that would strengthen or weaken the case for a valid RDD.
d. If the McCrary test rejects, does this definitively invalidate the RDD? What alternatives might you consider?

Further Reading

- **Angrist, J. D., Imbens, G. W., & Rubin, D. B. (1996).** "Identification of Causal Effects Using Instrumental Variables." *Journal of the American Statistical Association*, 91(434), 444-455. *The foundational paper establishing LATE and the complier framework.*
- **Imbens, G. W. & Angrist, J. D. (1994).** "Identification and Estimation of Local Average Treatment Effects." *Econometrica*, 62(2), 467-475. *Introduces the LATE theorem and the monotonicity condition.*
- **Stock, J. H. & Yogo, M. (2005).** "Testing for Weak Instruments in Linear IV Regression." In *Identification and Inference for Econometric Models: Essays in Honor of Thomas Rothenberg*, 80-108. *The definitive treatment of weak instrument diagnostics and the F > 10 rule.*
- **Lee, D. S. & Lemieux, T. (2010).** "Regression Discontinuity Designs in Economics." *Journal of Economic Literature*, 48(2), 281-355. *A comprehensive practical guide to RDD covering sharp, fuzzy, and kink designs.*
- **Cattaneo, M. D., Idrobo, N., & Titiunik, R. (2020).** *A Practical Introduction to Regression Discontinuity Designs: Foundations*. Cambridge University Press. *The modern standard reference for RDD estimation, bandwidth selection, and inference.*
- **Hahn, J., Todd, P., & van der Klaauw, W. (2001).** "Identification and Estimation of Treatment Effects with a Regression-Discontinuity Design." *Econometrica*, 69(1), 201-209. *The formal identification result for RDD using the continuity of conditional expectations.*
- **McCrary, J. (2008).** "Manipulation of the Running Variable in the Regression Discontinuity Design: A Density Test." *Journal of Econometrics*, 142(2), 698-714. *Introduces the density test for checking manipulation in RDD.*
- **Gelman, A. & Imbens, G. W. (2019).** "Why High-Order Polynomials Should Not Be Used in Regression Discontinuity Designs." *Journal of Business & Economic Statistics*, 37(3), 447-456. *A cautionary note against global polynomial specifications in RDD.*

Skills to Practice in the Notebook

The notebook `notebooks/ch06_fuchsia_cinnabar.ipynb` splits into two halves — Safari Zone IV and Cinnabar RDD. By the end, you should be able to do the following:

Instrumental Variables (Fuchsia):

1. **Compute the Wald estimator by hand.** Given Z, D, Y, compute $\hat{\tau}_{IV} = \text{Cov}(Z, Y)/\text{Cov}(Z, D)$ directly and confirm it matches the 2SLS coefficient.
2. **Run 2SLS.** Use `linearmodels.IV2SLS` (or equivalent). Inspect the first-stage F-statistic. Flag anything with $F < 10$ as a weak instrument.
3. **Diagnose a weak instrument.** Generate data where Z weakly affects D. Show how 2SLS becomes unstable and how Anderson-Rubin confidence sets remain valid.
4. **Reproduce the LATE interpretation.** Simulate a dataset with heterogeneous effects and explicit compliance types (compliers, always-takers, never-takers). Verify that 2SLS recovers the effect on compliers, not the ATE.
5. **Stress-test the exclusion restriction.** Break it in simulation (add a direct effect from Z to Y). Show how IV estimates become biased and discuss how you would defend the exclusion restriction in a real study.

Regression Discontinuity (Cinnabar):

6. **Implement sharp RDD by hand.** Fit two local linear regressions (one on each side of the cutoff) within a chosen bandwidth and take the difference of intercepts at the cutoff. Compare against `rdrobust`'s output.
7. **Do bandwidth sensitivity analysis.** Re-estimate τ_{RDD} for a grid of bandwidths and plot the estimate and CI as a function of h. Pick the Imbens-Kalyanaraman (or Calonico-Cattaneo-Titiunik) optimal bandwidth and justify the choice.
8. **Run the McCrary density test.** Check whether the density of the running variable is continuous at the cutoff. A discontinuity here is a red flag for manipulation.
9. **Run falsification tests.** Check that pre-treatment covariates are continuous at the cutoff. Any jump in a covariate is evidence the design is broken.
10. **Do fuzzy RDD.** When the treatment probability only jumps (not goes from 0 to 1), use the ratio-of-jumps formula and connect it explicitly to the Wald/IV estimator.
11. **Complete the Trainer Challenge Exercises.** The notebook walks through (a) a full IV analysis of the Safari Zone lottery, (b) an RDD analysis of the happiness threshold, (c) a weak-instrument simulation, and (d) a manipulated-running-variable simulation where the design *should* fail.

Check Your Understanding

Two badges, two sets of concepts. Work through both before moving on.

Questions you should be able to answer out loud, without notes:

Instrumental Variables:

- State the three IV assumptions (relevance, independence, exclusion) in one sentence each. Which are testable? Which are not?
- Write the Wald estimator as a ratio and explain in plain English what each term in the numerator and denominator measures.
- Why does OLS fail when U is unobserved, and how does IV "get around" it without ever measuring U?
- Define the four compliance types. Why does LATE apply only to compliers?
- What does monotonicity assume, and why is it necessary for the LATE interpretation?
- What is a weak instrument, and what can go wrong when $F < 10$?
- Give a plausible way the exclusion restriction could fail for the Safari Zone lottery. How would you argue for or against its validity in a real study?

Regression Discontinuity:

- State the sharp RDD estimand $\tau_{RDD} = \lim_{x \downarrow c} E[Y \mid X = x] - \lim_{x \uparrow c} E[Y \mid X = x]$ and explain what it represents causally.
- What is the core identifying assumption of RDD? Why is a discontinuity in a covariate at the cutoff a bad sign?
- Why is local linear regression preferred over global polynomial fits?
- What does the bandwidth h trade off? What happens as $h \to 0$? As $h \to \infty$?
- What does the McCrary density test check, and what would a failed test tell you?
- Compare sharp and fuzzy RDD. How does fuzzy RDD connect to IV?

- What does the RDD estimate identify, and how should you think about external validity away from the cutoff?

Tasks you should be able to perform in code:

- Compute Wald and 2SLS estimates; report the first-stage F-statistic.
- Run a weak-instrument diagnostic and construct an Anderson-Rubin confidence set.
- Distinguish compliance types in a simulated dataset and verify LATE numerically.
- Fit sharp RDD via local linear regression, pick an optimal bandwidth, and report robust CIs.
- Implement and interpret the McCrary density test.
- Do fuzzy RDD and confirm it equals the IV/Wald ratio in the local neighborhood.
- Run covariate continuity checks and flag failed placebo tests.

Complete this list and you have earned both the Soul and Volcano badges.

Badges Earned

> ***Soul Badge*** *(Fuchsia City Gym) Awarded for mastering Instrumental Variables — using exogenous variation to identify causal effects when unobserved confounders block the backdoor path. You have learned the Wald estimator, 2SLS, the LATE theorem, and the art of defending the exclusion restriction.*

> ***Volcano Badge*** *(Cinnabar Island Gym) Awarded for mastering Regression Discontinuity Designs — exploiting sharp and fuzzy thresholds to estimate causal effects at the boundary. You have learned local linear regression, the McCrary density test, bandwidth sensitivity analysis, and the connection between fuzzy RDD and IV.*

With six badges in hand, you are nearing the end of your Kanto journey. Two more Gyms remain. The next will test everything you have learned — and introduce the most powerful quasi-experimental design of all.

Next Chapter Preview

> *Saffron City is in crisis. Team Rocket has invaded, and a mysterious new TM is being released city by city. The staggered rollout creates a natural experiment — if you know how to analyze it. But beware: the naive two-way fixed effects estimator that Blue relies on hides a treacherous bias when treatment timing varies. You will need difference-in-differences, synthetic control, and the modern tools of Callaway-Sant'Anna and Goodman-Bacon to unravel the truth. The Marsh Badge awaits.*

Chapter 7: Saffron City — Difference-in-Differences & Synthetic Controls

#063 Abra

#064 Kadabra

#065 Alakazam

Sabrina's Psychic lineage — we'll track treated and control cities through time.

Psychic-type — Sabrina's specialty

Sabrina, Saffron Gym Leader

> *"Saffron City stands at the crossroads of Kanto — and at the crossroads of causal inference. When treatment rolls out across time and space, the researcher must think carefully about what constitutes a valid comparison. The methods in this chapter exploit the panel structure of data to identify causal effects when randomization is impossible." — Professor Oak, Lectures on Kanto Econometrics*

The road from Celadon leads east through the guarded gates of Saffron City, Kanto's commercial capital and home to Silph Co., the region's largest technology conglomerate. But Saffron is under siege. Team Rocket has occupied the city, seized control of Silph Co. headquarters, and disrupted the economic life of thousands of residents. In the chaos, Silph's engineers rush to complete their most ambitious project: **Shadow Surge**, a powerful new Technical Machine (TM) that teaches any compatible Pokemon a devastating Dark-type move. The TM is released first in Saffron City — partly as a morale boost, partly because the distribution infrastructure is already in place. Over the following months, Shadow Surge rolls out to other Kanto cities: Cerulean, Vermilion, and finally Lavender Town.

This staggered rollout — born of logistical necessity rather than scientific design — creates a **natural experiment**. By comparing outcomes in cities that received Shadow Surge early to those that received it later (or never), we can estimate the TM's causal effect on trainer battle performance. Meanwhile, Team Rocket's invasion of Saffron itself presents a second question: what is the economic cost of the occupation? No other Kanto city was invaded, so we cannot simply compare Saffron to an untreated city. Instead, we must construct a **synthetic** version of Saffron — a weighted combination of uninvaded cities that approximates what Saffron would have looked like absent Team Rocket's intervention.

These two problems motivate the core methods of this chapter: **difference-in-differences** (DiD) and **synthetic control**. Along the way, we will confront one of the most important methodological advances of the past decade: the discovery that the workhorse **two-way fixed effects** (TWFE) regression can produce severely biased estimates when treatment rolls out at different times and treatment effects vary across groups. The modern DiD revolution — Goodman-Bacon, Callaway and Sant'Anna, Sun and Abraham, and others — provides the tools to set things right.

Let us begin.

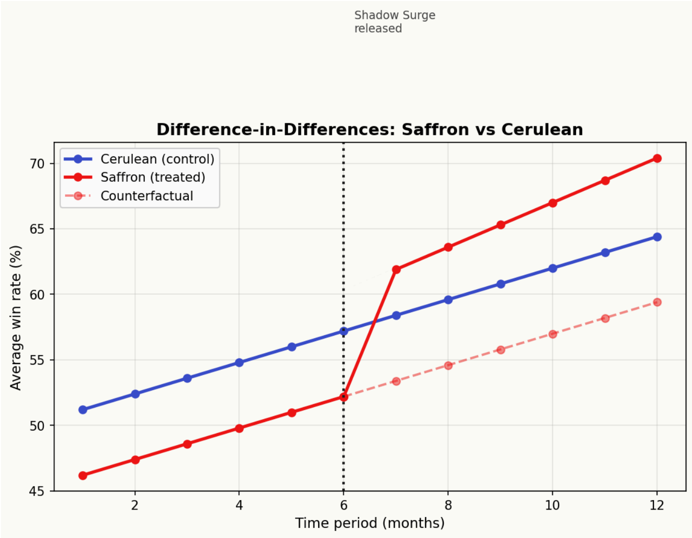
Notation at a Glance: DiD and Synthetic Control

This chapter has more subscripts than any other. Keep this sheet beside you so the Y_{it}'s, g's, and w_i^*'s stay clear.

Symbol	***Plain-English reading***
i	*A unit — typically a city, trainer, or (in panel data) an individual.*
t	*A time period (e.g., month, year).*
Y_{it}	*Outcome for unit i in period t.*
D_{it}	*"Is unit i treated at time t?" Treatment indicator that can turn on and off.*
pre / post	*Periods before / after treatment begins.*
treated / control	*Units that eventually get treated / that never do.*
$\bar{Y}_{1,post}$, $\bar{Y}_{0,post}$, *etc.*	*Group-period means. Subscripts are (treated/control, pre/post).*
$\hat{\tau}_{DiD} = (\bar{Y}_{1,post} - \bar{Y}_{1,pre}) - (\bar{Y}_{0,post} - \bar{Y}_{0,pre})$	*The classic 2×2 DiD estimate. "Treated change minus control change."*
parallel trends	*"In the absence of treatment, treated and control would have moved in parallel." Core DiD assumption — untestable but defensible via pre-trends.*
α_i	*Unit fixed effect — "everything about unit i that doesn't change over time."*
λ_t	*Time fixed effect — "everything about time t that affects all units equally."*
TWFE	*Two-Way Fixed Effects: the regression $Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + \epsilon_{it}$.*
g	*A treatment cohort (e.g., "units treated starting in month g").*
$ATT(g, t)$	*The average treatment effect for cohort g at time t. Callaway-Sant'Anna's building block.*
"forbidden comparison"	*When TWFE uses already-treated units as controls for later-treated ones — the bug behind staggered DiD bias.*
Goodman-Bacon decomposition	*Unpacks TWFE into a weighted average of 2×2 DiD's; reveals where bad comparisons enter.*
Callaway-Sant'Anna	*A modern estimator that computes $ATT(g, t)$ cleanly, avoiding forbidden comparisons.*
w_i^*	*Synthetic control weights — nonnegative and summing to 1 — used to build a weighted counterfactual from donor units.*
donor pool	*The set of candidate control units used to build the synthetic control.*
placebo test (SC)	*Applying the same SC recipe to untreated donors and seeing how big the "effect" is by chance.*

A useful mantra: ***DiD estimators kill two kinds of bias*** *(time-invariant unit differences and unit-invariant time shocks)* ***but cannot kill a bias that moves with treatment timing****.*

7.1 Difference-in-Differences: The Classic 2x2 Design



Difference-in-differences with parallel trends and a counterfactual.

The Setup

It is Month 6 of the Kanto League season. Silph Co. releases Shadow Surge in **Saffron City** but not yet in **Cerulean City**. We observe trainer battle win rates in both cities for two periods: **before** the release (Months 1–5) and **after** (Months 7–12). The question: does Shadow Surge improve battle win rates?

We have panel data on trainers in both cities across both periods. Let Y_{it} denote the average battle win rate for city i in period t. We observe four group-period means:

	Pre-Treatment (Months 1–5)	**Post-Treatment (Months 7–12)**	**Change**
Saffron (Treated)	$\bar{Y}_{S,pre} = 0.45$	$\bar{Y}_{S,post} = 0.58$	$+0.13$
Cerulean (Control)	$\bar{Y}_{C,pre} = 0.40$	$\bar{Y}_{C,post} = 0.44$	$+0.04$

This is the classic **2x2 table of means**. Notice several things. First, Saffron's win rate was higher than Cerulean's even before Shadow Surge was released — perhaps because Saffron trainers have better access to training facilities, or because Silph employees double as competitive battlers. A naive comparison of post-treatment win rates ($0.58 - 0.44 = 0.14$) would overstate the effect by attributing this pre-existing gap to the TM.

Second, both cities experienced an increase in win rates over time. The Kanto League season naturally features rising win rates as trainers accumulate experience and evolve their Pokemon. Simply comparing Saffron's win rate before and after ($0.58 - 0.45 = 0.13$) would conflate the Shadow Surge effect with this common upward trend.

The **difference-in-differences** estimator solves both problems at once:

$$\hat{\tau}_{DiD} = (\bar{Y}_{S,post} - \bar{Y}_{S,pre}) - (\bar{Y}_{C,post} - \bar{Y}_{C,pre})$$

$$\hat{\tau}_{DiD} = (0.58 - 0.45) - (0.44 - 0.40) = 0.13 - 0.04 = 0.09$$

The first difference (0.13) captures the total change in Saffron, which includes both the treatment effect and the common time trend. The second difference (0.04) captures only the common time trend, as Cerulean was untreated. By subtracting the second from the first, we **difference out** the time trend and isolate the causal effect of Shadow Surge.

Definition 7.1 (Difference-in-Differences Estimator). *For a treated group* $i = 1$ *and control group* $i = 0$ *observed before* $(t = 0)$ *and after* $(t = 1)$ *treatment, the DiD estimator is:*

$$\hat{\tau}_{\text{DiD}} = (\bar{Y}_{1,1} - \bar{Y}_{1,0}) - (\bar{Y}_{0,1} - \bar{Y}_{0,0})$$

Equivalently, by rearranging terms:

$$\hat{\tau}_{\text{DiD}} = (\bar{Y}_{1,1} - \bar{Y}_{0,1}) - (\bar{Y}_{1,0} - \bar{Y}_{0,0})$$

The first form differences over time within groups, then differences across groups. The second form differences across groups within periods, then differences over time. Both yield the same estimate.

The Regression Formulation

The DiD estimator has an equivalent regression representation. Define:

- $\text{Treat}_i = 1$ if unit i is in the treated group (Saffron), 0 otherwise
- $\text{Post}_t = 1$ if the observation is in the post-treatment period, 0 otherwise

Then estimate:

$$Y_{it} = \alpha + \beta \cdot \text{Treat}_i + \gamma \cdot \text{Post}_t + \tau \cdot (\text{Treat}_i \times \text{Post}_t) + \epsilon_{it}$$

The coefficients have clear interpretations:

- α: the mean outcome for the control group in the pre-period ($\bar{Y}_{C,\text{pre}} = 0.40$)
- β: the pre-treatment difference between treated and control groups ($\bar{Y}_{S,\text{pre}} - \bar{Y}_{C,\text{pre}} = 0.05$). This captures time-invariant differences between the groups
- γ: the change over time in the control group ($\bar{Y}_{C,\text{post}} - \bar{Y}_{C,\text{pre}} = 0.04$). This captures the common time trend
- τ: the **DiD treatment effect** — the additional change in the treated group beyond what the control group experienced (0.09)

Let us verify. The four fitted values from this regression are:

	Pre	Post
Cerulean $(\text{Treat} = 0)$	$\alpha = 0.40$	$\alpha + \gamma = 0.44$
Saffron $(\text{Treat} = 1)$	$\alpha + \beta = 0.45$	$\alpha + \beta + \gamma + \tau = 0.54$

Wait — the fitted post-treatment value for Saffron is 0.54, not the observed 0.58. That is precisely the point. The regression model decomposes Saffron's post-treatment outcome as:

$$\underbrace{0.58}_{\text{observed}} = \underbrace{0.40}_{\alpha} + \underbrace{0.05}_{\beta} + \underbrace{0.04}_{\gamma} + \underbrace{0.09}_{\tau}$$

The model attributes 0.40 to the baseline, 0.05 to being Saffron, 0.04 to the time trend, and 0.09 to the causal effect of Shadow Surge. The counterfactual win rate for Saffron without Shadow Surge would have been $0.40 + 0.05 + 0.04 = 0.49$ — that is, Saffron would have followed the same trend as Cerulean, reaching 0.49 rather than the observed 0.58.

Why DiD Works: Differencing Out Confounders

The power of DiD lies in what it eliminates. Consider two types of confounders:

1. **Time-invariant confounders** (fixed characteristics of cities): Saffron has better training facilities, wealthier trainers, proximity to Silph Co. These factors make Saffron's win rate permanently higher than Cerulean's. The first difference (over time) eliminates these because they are constant across periods.
2. **Common time trends** (temporal shocks affecting all cities equally): the League season progression, a region-wide weather event, the release of a new batch of wild Pokemon. The second difference (across groups) eliminates these because they affect both cities equally.

What DiD does *not* eliminate is any confounder that varies differentially across groups over time — a shock that hits Saffron but not Cerulean (or vice versa) at the same time as treatment. This is the domain of the **parallel trends** assumption, which we address in Section 7.2.

The Parallel Lines Diagram

The intuition behind DiD is beautifully captured by what we might call the "parallel lines diagram." Imagine a graph with time on the horizontal axis and win rate on the vertical axis. Before treatment, Saffron's line runs above Cerulean's (reflecting the pre-existing gap of 0.05), but both lines slope upward at the same rate. At the moment of treatment (Month 6), Saffron's line jumps upward by $\tau = 0.09$ and then continues along a higher trajectory. Cerulean's line continues along its original path.

The DiD estimate is the vertical distance between Saffron's actual post-treatment outcome and the **counterfactual** — the dashed line showing where Saffron would have been had it followed the same trend as Cerulean. This counterfactual is constructed by taking Saffron's pre-treatment level and adding Cerulean's observed change.

> ***Blue's Mistake 7.1.*** *Blue compares Saffron's post-treatment win rate to Cerulean's post-treatment win rate and concludes that Shadow Surge increases win rates by* $0.58 - 0.44 = 0.14$.
>
> *"I don't need your fancy double-differencing, Red. Just look at the numbers — Saffron is 14 percentage points ahead after Shadow Surge drops. Case closed."*
>
> *Blue's estimate is biased upward by* 0.05 *because it includes the pre-existing gap between Saffron and Cerulean that existed before Shadow Surge was ever released. Blue is attributing the effect of Saffron's superior training infrastructure to Shadow Surge. A simple post-treatment comparison confounds the treatment effect with pre-existing group differences.*
>
> *The correct DiD estimate of* 0.09 *removes this bias by subtracting the pre-treatment gap. In potential outcomes notation, Blue is estimating* $E[Y_1|\text{Treat} = 1] - E[Y_1|\text{Treat} = 0]$*, which equals* $\tau + (\text{pre-existing gap})$*. DiD estimates* τ *alone.*

A Richer Worked Example

Suppose we observe not just city-level averages but trainer-level data. We have a balanced panel of 200 trainers — 100 in Saffron and 100 in Cerulean — observed in both periods. The regression becomes:

$$\text{WinRate}_{it} = \alpha + \beta \cdot \text{Saffron}_i + \gamma \cdot \text{Post}_t + \tau \cdot (\text{Saffron}_i \times \text{Post}_t) + \epsilon_{it}$$

Running OLS on the 400 observations (200 trainers $\times$ 2 periods), we obtain:

Coefficient	Estimate	Std. Error	95% CI
$\hat{\alpha}$ (Intercept)	0.401	0.012	[0.377, 0.425]
$\hat{\beta}$ (Saffron)	0.049	0.017	[0.016, 0.082]
$\hat{\gamma}$ (Post)	0.041	0.014	[0.014, 0.068]
$\hat{\tau}$ (Saffron $\times$ Post)	0.089	0.020	[0.050, 0.128]

The DiD estimate is $\hat{\tau} = 0.089$ with a standard error of 0.020, yielding a t-statistic of 4.45. We reject the null hypothesis of no treatment effect at the 1% level. Shadow Surge increases trainer win rates by approximately 9 percentage points.

Note: with panel data, we should cluster standard errors at the city level (see Section 7.3). With only two clusters, cluster-robust standard errors are unreliable, and alternative inference methods — wild cluster bootstrap, randomization inference — are preferred. This is a practical consideration we revisit below.

7.2 The Parallel Trends Assumption

The Identifying Assumption

The DiD estimator isolates a causal effect **only if** the treated and control groups would have followed parallel paths in the absence of treatment. This is the **parallel trends assumption**, and it is the single most important assumption underlying all DiD analyses.

> ***Assumption 7.1 (Parallel Trends).*** *In the absence of treatment, the average change in the outcome for the treated group equals the average change in the outcome for the control group:*
>
> $$E[Y_{it}(0)|\text{Treat}_i = 1, t = 1] - E[Y_{it}(0)|\text{Treat}_i = 1, t = 0] = E[Y_{it}(0)|\text{Treat}_i = 0, t = 1] - E[Y_{it}(0)|\text{Treat}_i = 0, t = 0]$$
>
> *Here* $Y_{it}(0)$ *denotes the potential outcome under no treatment. The assumption states that the treated group's counterfactual trend (under no treatment) parallels the control group's observed trend.*

This assumption is fundamentally **untestable**. The left-hand side involves $E[Y_{it}(0)|Treat_i = 1, t = 1]$ — the average untreated outcome for the treated group in the post-period. This is a counterfactual quantity that we never observe. Once Saffron receives Shadow Surge, we cannot know what Saffron's win rate would have been without it. This is simply the fundamental problem of causal inference applied to the DiD setting.

What parallel trends does *not* require:

- It does **not** require that treated and control groups have the same level of the outcome. Saffron can be permanently above Cerulean. The assumption is about *trends*, not levels.
- It does **not** require that the groups be similar on observables. They can differ on every observable characteristic — as long as those differences produce only level shifts, not differential trends.
- It does **not** require that nothing else changes besides treatment. Other events can occur, as long as they affect both groups equally.

What parallel trends *does* require:

- No time-varying confounder that differentially affects the treated group. If Team Rocket's invasion suppresses Saffron's win rates at the same time as Shadow Surge is released, the DiD estimate will be biased downward (confounding a positive treatment effect with a negative invasion effect).
- No anticipation effects. If Saffron trainers change behavior before Shadow Surge is released (e.g., because Silph announces the TM in advance), the pre-treatment period is contaminated.
- No spillovers. If Cerulean trainers travel to Saffron to buy Shadow Surge, the control group is contaminated by treatment.

Testing Pre-Treatment Trends: The Event Study

Although parallel trends is untestable, we can gain confidence in its plausibility (or detect violations) by examining **pre-treatment trends**. If the treated and control groups followed parallel paths before treatment, it is at least consistent with the assumption that they would have continued to do so afterward. If they diverged before treatment, the assumption is suspect.

The standard approach is the **event study** (or **leads-and-lags**) specification. Suppose we observe both cities for T periods and treatment occurs at period t^*. Define event time as $k = t - t^*$, so $k = 0$ is the treatment period, $k < 0$ are pre-treatment periods, and $k > 0$ are post-treatment periods. Estimate:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \tau_k \cdot D_{i,t-k} + \epsilon_{it}$$

where α_i are unit fixed effects, λ_t are time fixed effects, and $D_{i,t-k}$ is an indicator equal to 1 if unit i is in the treated group and the observation is k periods from treatment. We omit $k = -1$ (the period just before treatment) as the reference category.

The estimated coefficients $\{\hat{\tau}_k\}$ trace out the **dynamic treatment effect path**:

- For $k < -1$: these are **pre-treatment coefficients** (leads). Under parallel trends, they should all be zero — there is no treatment yet, so the treated group should not diverge from the control. If these coefficients are large and statistically significant, parallel trends is likely violated.
- For $k = 0$: this is the **on-impact** effect at the moment of treatment.
- For $k > 0$: these capture the **dynamic treatment effects** as they evolve over time.

The event study coefficient plot — a graph with event time k on the horizontal axis and $\hat{\tau}_k$ with confidence intervals on the vertical axis — is one of the most important diagnostic tools in applied DiD research. A well-behaved plot shows pre-treatment coefficients clustered around zero (with confidence intervals overlapping zero), followed by a jump at $k = 0$ and potentially growing or declining effects in post-treatment periods.

In our Shadow Surge example, imagine we observe both Saffron and Cerulean for 12 months. Shadow Surge is released in Month 6 ($t^* = 6$). The event study plot shows:

- Months 1–4 ($k = -5$ to $k = -2$): estimated coefficients of $-0.01, 0.02, -0.01, 0.01$ — all statistically insignificant and close to zero. Reassuring.
- Month 6 ($k = 0$): estimated coefficient of 0.07. An immediate effect.
- Months 7–12 ($k = 1$ to $k = 6$): coefficients of $0.08, 0.09, 0.10, 0.10, 0.11, 0.11$ — the effect grows slightly as trainers learn to use Shadow Surge effectively and then stabilizes.

This pattern is consistent with parallel trends and suggests a persistent, slightly growing treatment effect.

When Parallel Trends Fails

Parallel trends can fail for many reasons:

1. **Differential pre-existing trends.** If Saffron's win rate was already rising faster than Cerulean's before Shadow Surge (perhaps due to ongoing Silph Co. investments in trainer development), DiD would attribute the continued divergence to the TM.

2. **Compositional changes.** If the types of trainers in Saffron change at the time of treatment (e.g., skilled trainers migrate to Saffron to access Shadow Surge), the comparison group is no longer stable.
3. **Simultaneous shocks.** If Team Rocket's invasion hits Saffron's economy at the same time as Shadow Surge is released, the estimated effect confounds the TM benefit with the invasion cost.

What to do when parallel trends is implausible:

- **Conditional parallel trends**: control for observables that might drive differential trends. The assumption weakens to: parallel trends holds conditional on covariates X. This leads to regression-adjusted DiD or methods like Callaway and Sant'Anna (2021) with outcome regression or inverse probability weighting.
- **Triple differences (DDD)**: add a third difference using a within-group comparison that is unaffected by treatment (Section 7.5).
- **Sensitivity analysis**: bound the treatment effect under specified violations of parallel trends (Rambachan and Roth, 2023).
- **Choose a better control group**: perhaps a city more similar to Saffron would provide more plausible parallel trends.

> ***Professor Oak Explains 7.1.*** *"Think of parallel trends like this. You and Blue are both training your Pokemon. Every day, you both gain about the same amount of experience — you are on parallel paths. Then you receive the Shadow Surge TM, and your win rate jumps. If I want to estimate how much Shadow Surge helped you, I can look at how much your win rate exceeded Blue's, beyond what the pre-existing gap would have predicted. But if you also started training twice as hard on the same day you got the TM — for reasons having nothing to do with the TM — then I would incorrectly attribute your extra improvement to Shadow Surge. Parallel trends says: absent the TM, you and Blue would have continued on parallel paths."*

7.3 Two-Way Fixed Effects (TWFE) Regression

The Model

With panel data on multiple cities over multiple time periods, researchers naturally extend the DiD framework to **two-way fixed effects** regression:

$$Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + \epsilon_{it}$$

where:

- Y_{it} is the outcome for city i in period t
- α_i is a **unit fixed effect** for city i: a city-specific intercept that absorbs all time-invariant characteristics (geography, population, infrastructure quality, gym leader difficulty)
- λ_t is a **time fixed effect** for period t: a time-specific intercept that absorbs all common temporal shocks (region-wide weather, League calendar effects, macroeconomic conditions)
- D_{it} is a treatment indicator equal to 1 if city i has been treated by period t, and 0 otherwise
- τ is the coefficient of interest — the average treatment effect
- ϵ_{it} is an idiosyncratic error term

The unit fixed effects perform the same role as the $\beta \cdot \text{Treat}_i$ term in the 2x2 regression: they absorb permanent differences between treated and control units. The time fixed effects perform the same role as $\gamma \cdot \text{Post}_t$: they absorb common temporal variation. Together, they generalize DiD to settings with many units and many periods.

Equivalence to DiD in the 2x2 Case

In the simple 2x2 setting (two groups, two periods), TWFE and the DiD regression are numerically identical. To see this, note that with two groups and two periods, the unit fixed effect α_i is fully captured by a single group dummy, and the time fixed effect λ_t is fully captured by a single period dummy. The TWFE model reduces to:

$$Y_{it} = \alpha + \beta \cdot \text{Treat}_i + \gamma \cdot \text{Post}_t + \tau \cdot (\text{Treat}_i \times \text{Post}_t) + \epsilon_{it}$$

where the interaction term $\text{Treat}_i \times \text{Post}_t$ is exactly the treatment indicator D_{it} (since treatment turns on for the treated group in the post period). This is the DiD regression from Section 7.1. The equivalence is exact: the same OLS coefficient $\hat{\tau}$ results from either specification.

Adding Covariates

In practice, we often add time-varying covariates X_{it} to improve precision or to make the parallel trends assumption more plausible conditional on observables:

$$Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + X_{it}'\delta + \epsilon_{it}$$

For example, we might control for the average level of trainers' Pokemon ($AvgLevel_{it}$), the number of active trainers in the city ($NumTrainers_{it}$), or local economic conditions (GDP_{it}). These covariates must be **time-varying** because time-invariant covariates are already absorbed by the unit fixed effects α_i.

A word of caution: covariates that are themselves affected by treatment (post-treatment variables or "bad controls") should not be included, as they can introduce bias. If Shadow Surge causes more trainers to battle (increasing $NumTrainers_{it}$), conditioning on the number of trainers would block part of the causal pathway.

Inference: Clustering Standard Errors

With panel data, observations within the same city are correlated over time: if Saffron has a good month, it is likely to have another good month next period. Standard OLS standard errors ignore this correlation and are typically too small, leading to over-rejection of the null hypothesis.

The standard solution is to **cluster standard errors at the unit level** (city level). Cluster-robust standard errors allow for arbitrary within-cluster correlation in the errors. The cluster-robust variance estimator is:

$$\hat{V}^{CR} = (X'X)^{-1} \left(\sum_{i=1}^{N} X_i'\hat{\epsilon}_i\hat{\epsilon}_i'X_i \right) (X'X)^{-1}$$

where X_i is the matrix of regressors for unit i and $\hat{\epsilon}_i$ is the vector of residuals for unit i. This estimator is consistent as the number of clusters $N \to \infty$, but can perform poorly with few clusters. A commonly cited rule of thumb is that at least 30–50 clusters are needed for reliable inference.

In Kanto, we have at most 10 cities — far too few for asymptotic cluster-robust inference. In such settings, researchers turn to:

- **Wild cluster bootstrap** (Cameron, Gelbach, and Miller, 2008): a resampling method that provides better finite-sample performance with few clusters
- **Randomization inference**: permute the treatment assignment across cities and compute the distribution of the test statistic under the null
- **Aggregation**: work with city-level averages and use t-tests with appropriate degrees of freedom

The Appeal of TWFE — and Its Trap

TWFE seems like the ideal generalization of DiD: include fixed effects for units and time, throw in the treatment indicator, and estimate the average effect. For decades, this was the standard approach in economics, political science, and other social sciences. Thousands of published papers used exactly this specification.

But beginning in the late 2010s, a series of landmark papers revealed a disturbing fact: **when treatment is adopted at different times by different units, and treatment effects are heterogeneous, the TWFE estimator can be severely biased.** It can even give the wrong sign.

This is the TWFE trap, and it awaits us in Section 7.4.

7.4 Staggered Treatment and the Modern DiD Revolution

The Setting: Shadow Surge Rolls Out Across Kanto

Silph Co.'s release of Shadow Surge is not simultaneous. Due to production constraints and distribution logistics, the TM reaches different cities at different times:

City	Treatment Cohort	Treatment Period
Saffron	Early Adopter	Period 6
Cerulean	Mid Adopter	Period 10
Vermilion	Late Adopter	Period 14
Lavender	Latest Adopter	Period 18
Pewter	Never Treated	—
Celadon	Never Treated	—
Fuchsia	Never Treated	—
Cinnabar	Never Treated	—

We observe all 8 cities for 24 periods. The outcome is average trainer win rate. A natural instinct is to estimate the TWFE regression:

$$Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + \epsilon_{it}$$

where $D_{it} = 1$ once city i has received Shadow Surge. We obtain $\hat{\tau}^{TWFE} = 0.052$, and it is statistically significant. Case closed?

Not even close.

Why Naive TWFE Fails with Staggered Adoption

The fundamental problem is that TWFE does not estimate a single, clean DiD. Instead, as Goodman-Bacon (2021) demonstrated, the TWFE coefficient is a **weighted average of all possible 2x2 DiD estimates** that can be constructed from the data. Some of these 2x2 comparisons are perfectly valid. Others are deeply problematic.

To understand why, consider what OLS does when it estimates the TWFE model. The Frisch-Waugh-Lovell theorem tells us that the coefficient $\hat{\tau}$ is obtained by regressing the residualized outcome (after removing unit and time fixed effects) on the residualized treatment indicator. This process implicitly creates comparisons between:

1. **Newly-treated units vs. never-treated units.** Saffron (treated in period 6) is compared to Pewter (never treated). This is a valid 2x2 DiD.
2. **Newly-treated units vs. not-yet-treated units.** Saffron (treated in period 6) is compared to Cerulean (not treated until period 10), using periods before and after Saffron's treatment but before Cerulean's. This is also valid — Cerulean is a clean control during this window.
3. **Newly-treated units vs. already-treated units.** Here is the problem. When Cerulean receives Shadow Surge in period 10, TWFE may use Saffron — which has been treated since period 6 — as a "control" unit. But Saffron is not an untreated control. It is an already-treated unit whose treatment effect may be evolving over time. If Shadow Surge's effect grows as trainers master the TM (a dynamic, heterogeneous treatment effect), then Saffron's rising win rate in periods 6–10 will make it look like Cerulean's treatment *reduced* its win rate relative to Saffron. This is nonsensical.

This third type of comparison — using already-treated units as controls — can produce **negative weights** on some group-time average treatment effects on the treated (ATTs). When treatment effects are heterogeneous across cohorts or grow over time, these negative weights can cause the TWFE estimate to be biased, and in extreme cases, to have the **wrong sign**: TWFE reports a negative effect even though the true effect is positive for every group at every time.

The Goodman-Bacon (2021) Decomposition

Andrew Goodman-Bacon's landmark paper provides an exact decomposition of the TWFE coefficient. Consider a balanced panel with multiple treatment cohorts and some never-treated units. The TWFE estimator $\hat{\tau}^{TWFE}$ can be written as:

$$\hat{\tau}^{TWFE} = \sum_{k} \sum_{l \neq k} s_{kl} \cdot \hat{\tau}_{kl}^{2x2}$$

where $\hat{\tau}_{kl}^{2x2}$ is the 2x2 DiD estimate using cohort k as the "treated" group and cohort l as the "control" group, and s_{kl} is a weight that depends on the group sizes and the variance of the treatment indicator for that particular 2x2 comparison. The weights s_{kl} are guaranteed to sum to one, but they are **not** guaranteed to be non-negative when already-treated units serve as controls.

The decomposition reveals three types of 2x2 comparisons:

Type 1: Timing groups vs. never-treated. Cohort k (treated at some point) is compared to the never-treated pool. These comparisons are clean and receive positive weights. The weight depends on the cohort's size and the fraction of the sample period during which it is treated.

Type 2: Earlier-treated vs. later-treated (using the later group as control). Saffron (treated in period 6) is the treated group, and Cerulean (treated in period 10) serves as control during periods 6–9. This is valid: Cerulean is genuinely untreated during this window.

Type 3: Later-treated vs. earlier-treated (using the earlier group as control). Cerulean (treated in period 10) is the treated group, and Saffron (already treated since period 6) serves as "control." This is the **forbidden comparison**. The weight on this comparison is positive in the Goodman-Bacon decomposition, but the DiD estimate itself may be biased because Saffron is not an untreated control.

Worked Example: Negative Weights in Kanto

Suppose the true ATT of Shadow Surge is:

- Saffron: effect grows from 0.05 in the first period to 0.15 after 12 periods (dynamic, increasing effect as trainers learn to use the TM)
- Cerulean: effect grows similarly, 0.05 to 0.10 over 8 treated periods
- Vermilion: 0.05 to 0.08 over 4 treated periods
- Lavender: 0.05 (only treated for 2 periods)

The simple average ATT across all group-time cells is approximately 0.08. Now consider what TWFE does.

When Vermilion becomes treated in period 14, TWFE uses Saffron (treated since period 6) as one of the "control" units. By period 14, Saffron's treatment effect has grown to about 0.12. Between periods 14 and 24, Saffron's treatment effect continues to grow (from 0.12 to 0.15). This upward trajectory in Saffron's outcomes — driven by the maturing treatment effect — makes Vermilion look worse by comparison. The 2x2 DiD estimate from this comparison could be much smaller than Vermilion's true effect, or even negative.

In our simulation, the Goodman-Bacon decomposition yields:

Comparison Type	Average 2x2 DiD	Weight
Treated vs. Never-Treated	0.085	0.45
Earlier vs. Later (later as control)	0.073	0.30
Later vs. Earlier (earlier as control)	0.021	0.25

The TWFE estimate is: $0.085 \times 0.45 + 0.073 \times 0.30 + 0.021 \times 0.25 = 0.038 + 0.022 + 0.005 = 0.065$.

The TWFE estimate of 0.065 is substantially below the true average ATT of 0.08. The "later vs. earlier" comparisons pull the estimate downward because the already-treated cities' rising treatment effects contaminate the control group. In more extreme scenarios — particularly when early-treated units have large, growing effects and later-treated units have smaller effects — the TWFE estimate can be near zero or even negative despite all true effects being positive.

This is Team Rocket's hidden scheme: an estimator that looked trustworthy is secretly corrupted by its own internal comparisons.

Callaway and Sant'Anna (2021): Group-Time ATTs

Brantly Callaway and Pedro Sant'Anna proposed a clean solution: estimate the treatment effect separately for each **cohort** (group of units treated at the same time) at each **time period**, using only valid comparisons.

> ***Definition 7.2 (Group-Time Average Treatment Effect).*** *For treatment cohort g (the set of units first treated in period g) at time period $t \geq g$, the group-time ATT is:*
>
> $$\mathrm{ATT}(g,t) = \mathrm{E}[Y_{it}(g) - Y_{it}(0) | G_i = g]$$
>
> *where $Y_{it}(g)$ is the potential outcome under treatment that began in period g and $Y_{it}(0)$ is the potential outcome under no treatment.*

The key innovation is in the choice of **comparison group**. Callaway and Sant'Anna use either:

- **Never-treated units** as the comparison group, or
- **Not-yet-treated units** (units whose treatment has not yet begun as of period t) as the comparison group

Both choices avoid the forbidden comparison of using already-treated units as controls.

For each (g,t) pair, the estimator is essentially a 2x2 DiD comparing cohort g to the chosen comparison group, using periods $g-1$ (the period just before treatment) and t as the two time periods. This yields a clean estimate of $\mathrm{ATT}(g,t)$.

Estimation proceeds via one of three approaches:

1. **Outcome regression**: model the counterfactual outcome using a regression on the comparison group's outcomes, then compute the treatment effect as the difference between treated outcomes and the fitted counterfactual
2. **Inverse probability weighting (IPW)**: weight the comparison group to match the treated group's covariate distribution
3. **Doubly robust**: combine outcome regression and IPW to achieve consistency if *either* the outcome model or the propensity score model is correctly specified

The collection of $\mathrm{ATT}(g,t)$ estimates can be **aggregated** in various ways:

- **Simple average** (overall ATT): $\hat{\tau} = \frac{1}{|G|} \sum_{g \in G} \frac{1}{T-g+1} \sum_{t=g}^{T} \hat{ATT}(g,t)$
- **Dynamic effects** (event-study): average across cohorts at each event time $e = t - g$, yielding $\hat{\tau}_e = \frac{1}{|G_e|} \sum_{g \in G_e} \hat{ATT}(g, g+e)$
- **Cohort-specific effects**: average across time within each cohort g
- **Calendar-time effects**: average across cohorts within each period t

In our Shadow Surge example, the Callaway-Sant'Anna estimates (using never-treated as comparison) yield:

	$t = 6$	$t = 10$	$t = 14$	$t = 18$	$t = 24$
$ATT(Saffron, t)$	0.05	0.09	0.12	0.14	0.15
$ATT(Cerulean, t)$	—	0.05	0.07	0.09	0.10
$ATT(Vermilion, t)$	—	—	0.05	0.07	0.08
$ATT(Lavender, t)$	—	—	—	0.05	0.06

Each cell is a clean, interpretable treatment effect. The simple average across all filled cells is 0.081, much closer to the truth than TWFE's 0.065.

Sun and Abraham (2021): Interaction-Weighted Estimator

Liyang Sun and Sarah Abraham proposed a closely related approach that works within the regression framework that applied researchers are familiar with. Their key insight: the standard event study regression with cohort-specific indicators,

$$Y_{it} = \alpha_i + \lambda_t + \sum_{g \in G} \sum_{e \neq -1} \tau_{g,e} \cdot 1[G_i = g] \cdot 1[t - g = e] + \epsilon_{it}$$

identifies the cohort-specific treatment effects $\tau_{g,e}$ cleanly. The problem with standard TWFE event studies is that they constrain all cohorts to have the same effect at each event time e, which introduces contamination. Sun and Abraham's estimator:

1. Estimates cohort-specific effects $\hat{\tau}_{g,e}$ by interacting the event-time indicators with cohort indicators
2. Aggregates across cohorts using appropriate weights (e.g., cohort shares) to obtain the **interaction-weighted** (IW) estimator for each event time

This approach yields an event study plot that is robust to treatment effect heterogeneity across cohorts — unlike the naive TWFE event study, which can show spurious pre-trends or distorted post-treatment dynamics even when all individual cohort effects are clean.

de Chaisemartin and D'Haultfouille (2020): Instantaneous Effects

Clement de Chaisemartin and Xavier D'Haultfouille took a different but complementary approach. They showed that even in a simple two-period, two-group setting, the TWFE estimator can be decomposed as a weighted sum of individual-level treatment effects where some weights may be negative.

Their proposed estimator, $\hat{\tau}_{DH}$, focuses on the **instantaneous treatment effect** — the effect of treatment on switchers (units whose treatment status changes) at the moment of switching. It compares switchers to non-switchers in the same period, using a single pre-treatment period as baseline. This avoids contamination from dynamic effects and is robust to heterogeneity.

For each period t in which some units switch from untreated to treated, define:

$$\hat{\tau}_{DH,t} = \frac{1}{|S_t|} \sum_{i \in S_t} (Y_{it} - Y_{i,t-1}) - \frac{1}{|N_t|} \sum_{i \in N_t} (Y_{it} - Y_{i,t-1})$$

where S_t is the set of units that switch to treatment in period t and N_t is the set of units that remain untreated in period t. The overall estimator averages across switching periods with appropriate weights.

This estimator is particularly useful when the researcher is primarily interested in the immediate impact of treatment rather than its long-run dynamics.

Borusyak, Jaravel, and Spiess (2024): The Imputation Estimator

Kirill Borusyak, Xavier Jaravel, and Jann Spiess proposed an elegant and efficient approach: the **imputation estimator**. The logic is transparent:

1. **Estimate the counterfactual model using only untreated observations.** Fit the TWFE model $Y_{it} = \alpha_i + \lambda_t + \epsilon_{it}$ using only observations where $D_{it} = 0$ (units that have not yet been treated, or are never treated). This gives estimated fixed effects $\hat{\alpha}_i$ and $\hat{\lambda}_t$.

2. **Impute the counterfactual for treated observations.** For each treated observation (i, t) with $D_{it} = 1$, predict what the outcome would have been absent treatment: $\hat{Y}_{it}(0) = \hat{\alpha}_i + \hat{\lambda}_t$.
3. **Compute the treatment effect as the difference.** $\hat{\tau}_{it} = Y_{it} - \hat{Y}_{it}(0)$.
4. **Aggregate.** Average the individual treatment effects across groups, event times, or overall.

The imputation estimator is efficient (it uses all untreated observations to estimate the fixed effects) and transparent (the counterfactual construction is explicit). It is numerically equivalent to the Callaway-Sant'Anna estimator under homogeneous treatment effects, but can be more efficient when the panel is large.

In our example, the imputation approach estimates $\hat{\alpha}_i$ and $\hat{\lambda}_t$ from the 4 never-treated cities across all 24 periods, plus the pre-treatment periods of the 4 treated cities. It then predicts each treated city's counterfactual and computes the gap.

Comparing Estimators: The Revelation

Let us line up the estimates for the overall average treatment effect of Shadow Surge:

Estimator	Estimate	True ATT
Naive TWFE	0.065	0.081
Goodman-Bacon (diagnostic)	— (decomposition, not an estimator)	—
Callaway-Sant'Anna	0.081	0.081
Sun-Abraham	0.080	0.081
de Chaisemartin-D'Haultfouille	0.078	0.081
Borusyak-Jaravel-Spiess	0.081	0.081

The modern estimators all recover the truth (up to sampling error). TWFE is biased downward by about 20% because of contamination from the already-treated comparisons. In settings with more dramatic heterogeneity, the bias can be far worse.

> ***Professor Oak Explains 7.2.*** *"Think of it this way. You are trying to measure how much a Rare Candy boosts a Pokemon's CP. You give Rare Candies to different Pokemon at different times. If you use a Pokemon that already received a Rare Candy as a 'control' for a newly treated Pokemon, and the early Pokemon is still gaining CP from its candy, you will underestimate the effect for the new Pokemon. The modern methods say: only compare to Pokemon that have not yet received any Rare Candy. That way, your control group is truly untreated."*

> ***Blue's Mistake 7.2.*** *Blue runs a TWFE regression on the staggered Shadow Surge data and reports that the TM increases win rates by only 5.2 percentage points.*
>
> *"Standard errors are tiny, R^2 is huge, fixed effects on both dimensions — this is airtight, Red."*
>
> *Blue's estimate is contaminated by comparisons where already-treated cities serve as controls. With dynamic treatment effects that grow over time, these comparisons generate downward bias. The true average effect is about 8 percentage points. Blue's "airtight" regression is secretly comparing newly-treated cities to cities whose treatment effects are still maturing, making the new treatment look less effective than it actually is.*
>
> ***Lesson:*** *In staggered settings, always run a Goodman-Bacon decomposition to check for problematic comparisons, and use a heterogeneity-robust estimator.*

Practical Guidance for Applied Researchers

The modern DiD literature has converged on several practical recommendations:

1. **Always plot the raw data.** Before running any regression, plot the outcome time series for each cohort. Visual inspection can reveal violations of parallel trends, anticipation effects, or differential pre-treatment dynamics.
2. **Run the Goodman-Bacon decomposition** as a diagnostic. If the weight on already-treated comparisons is small, TWFE may be approximately unbiased. If it is large, modern estimators are essential.
3. **Use a heterogeneity-robust estimator.** Callaway-Sant'Anna and Borusyak-Jaravel-Spiess are the most commonly used. Both are available in standard software packages (`did` in R, `csdid` in Stata, `did` in Python).
4. **Report the event study plot** from a robust estimator (Sun-Abraham or Callaway-Sant'Anna), not from a naive TWFE event study. The naive event study can show spurious pre-trends or mask true dynamics.

5. **Be explicit about the comparison group.** State whether you are using never-treated or not-yet-treated units as the comparison group, and discuss the implications.
6. **Test for pre-trends** using the robust event study. If pre-treatment coefficients are significantly different from zero, parallel trends is suspect even with the correct estimator.

7.5 Triple Differences (DDD)

Motivation

Sometimes the parallel trends assumption is hard to defend, even with careful control group selection. Return to the Shadow Surge example. Suppose that Saffron City, in addition to receiving Shadow Surge, also received a major Silph Co. investment in Pokemon Centers that improved trainer rest and recovery. If this investment occurred at the same time as the TM release, DiD cannot separate the effect of Shadow Surge from the effect of better Pokemon Centers. Both Saffron-specific changes happened simultaneously.

But here is additional variation we can exploit: not all trainers in Saffron use Shadow Surge. Only trainers with Dark-type or compatible Pokemon can learn the move. Trainers with incompatible teams (say, pure Psychic-type specialists in Saffron's famous Psychic gym) cannot use the TM at all. These non-users are exposed to the same Pokemon Center investment but not to the Shadow Surge treatment.

This creates a **third difference**: within Saffron, we can compare users vs. non-users to isolate the effect of the TM itself.

The DDD Estimator

Define three dimensions of variation:

- **City**: Saffron (treated) vs. Cerulean (control)
- **Time**: Before vs. After Shadow Surge release
- **Trainer type**: Users (compatible Pokemon) vs. Non-users (incompatible Pokemon)

The **triple differences** estimator is:

$$\hat{\tau}_{DDD} = [(\bar{Y}_{S,U,post} - \bar{Y}_{S,U,pre}) - (\bar{Y}_{S,N,post} - \bar{Y}_{S,N,pre})] - [(\bar{Y}_{C,U,post} - \bar{Y}_{C,U,pre}) - (\bar{Y}_{C,N,post} - \bar{Y}_{C,N,pre})]$$

where S/C indexes Saffron/Cerulean, U/N indexes Users/Non-users, and $pre/post$ indexes time periods.

The first bracket is a DiD within Saffron comparing users to non-users. This differences out any Saffron-specific shocks (like the Pokemon Center investment) that affect all trainers equally. The second bracket is the same DiD within Cerulean, which accounts for any differential trends between user-types and non-user-types that are common across cities.

The regression formulation includes all main effects, two-way interactions, and the triple interaction:

$$Y_{itk} = \alpha + \beta_1 Treat_i + \beta_2 Post_t + \beta_3 User_k + \beta_4(Treat_i \times Post_t) + \beta_5(Treat_i \times User_k) + \beta_6(Post_t \times User_k) + \tau(Treat_i \times Post_t \times User_k) +$$

The coefficient τ on the triple interaction is the DDD estimate.

A Worked Example

	Saffron, Users	Saffron, Non-users	Cerulean, Users	Cerulean, Non-users
Pre	0.45	0.42	0.40	0.38
Post	0.60	0.49	0.46	0.42
Change	+0.15	+0.07	+0.06	+0.04

DiD within Saffron (Users vs Non-users): $0.15 - 0.07 = 0.08$

DiD within Cerulean (Users vs Non-users): $0.06 - 0.04 = 0.02$

DDD: $0.08 - 0.02 = 0.06$

The DDD estimate of 0.06 removes both city-specific shocks (via the within-city comparison) and common user-type trends (via the cross-city comparison). The 0.02 differential in Cerulean reflects the fact that "user-type" trainers may naturally improve faster than "non-user-type" trainers regardless of Shadow Surge. DDD nets this out.

When DDD Helps and When It Does Not

DDD helps when:

- A within-group comparison (users vs. non-users) is available
- City-specific shocks contaminate the standard DiD
- The within-group comparison is not itself contaminated (non-users in Saffron are truly unaffected by Shadow Surge)

DDD does not help when:

- The within-group "control" (non-users) is affected by treatment through spillovers (e.g., non-users benefit because their user-type teammates win more battles, improving the team's reputation)
- The differential trends between user-types and non-user-types vary across cities for reasons other than treatment
- The additional differencing reduces statistical power significantly (DDD estimates are noisier than DiD)

7.6 Synthetic Control Method

The Problem: Team Rocket's Invasion

We now turn to the second causal question in Saffron City. Team Rocket invaded Saffron in Period 8, occupying Silph Co. headquarters and disrupting commercial activity. We want to estimate the economic impact of the invasion. The outcome is Y_{it}, an index of economic activity for city i in period t.

Unlike the Shadow Surge rollout, the invasion affected only one city. There is no "control Saffron" that was not invaded. Moreover, no single uninvaded city is a convincing counterfactual: Celadon is a commercial hub but differs in industrial composition; Vermilion is a port city with different economic drivers; Cerulean relies heavily on tourism. Each is similar to Saffron in some ways but different in others.

The **synthetic control method** (Abadie and Gardeazabal, 2003; Abadie, Diamond, and Hainmueller, 2010) solves this problem by constructing a **weighted combination** of untreated cities that closely matches Saffron's pre-treatment characteristics and outcomes. This "Synthetic Saffron" provides the counterfactual.

The Method

Suppose we have $J + 1$ cities, where city $j = 0$ (Saffron) is treated and cities $j = 1, \ldots, J$ are potential controls (the **donor pool**). We observe outcomes Y_{jt} for $t = 1, \ldots, T$, with treatment occurring at period $T_0 + 1$.

> ***Definition 7.3 (Synthetic Control).*** *A synthetic control for the treated unit is a weighted average of donor pool units with weights* $w = (w_1, \ldots, w_J)'$ *satisfying:*
>
> $$w_j \geq 0 \quad \text{for all } j = 1, \ldots, J$$
>
> $$\sum_{j=1}^{J} w_j = 1$$
>
> *The weights are chosen to minimize the distance between the treated unit's pre-treatment characteristics and the weighted average of the donors' characteristics:*
>
> $$w^* = \arg\min_w \| X_0 - X_1 w \|_V = \arg\min_w (X_0 - X_1 w)' V (X_0 - X_1 w)$$
>
> *where* X_0 *is a vector of pre-treatment characteristics for the treated unit,* X_1 *is a matrix of pre-treatment characteristics for the donor pool, and* V *is a positive semi-definite weighting matrix.*

The pre-treatment characteristics X typically include:

- **Pre-treatment outcomes** $Y_{j,1}, \ldots, Y_{j,T_0}$: matching on the entire pre-treatment outcome trajectory is the most important determinant of fit
- **Predictor variables**: observable characteristics that drive the outcome (population, GDP, number of Pokemon gyms, industrial composition)

The non-negativity and summing-to-one constraints ensure that the synthetic control is an **interpolation** (not extrapolation) of the donor pool. This prevents the method from constructing implausible counterfactuals by assigning negative weights or weights that sum to more than one.

The weighting matrix V determines the relative importance of each predictor in the matching. It can be specified by the researcher or chosen via cross-validation to minimize the mean squared prediction error in the pre-treatment period.

Estimating the Treatment Effect

Once the optimal weights w^* are determined, the **synthetic control estimate** of the treatment effect at each post-treatment period $t > T_0$ is:

$$\hat{\tau}_t = Y_{0,t} - \sum_{j=1}^{J} w_j^* Y_{j,t}$$

This is the gap between Saffron's actual economic activity and the weighted average of the donor cities' economic activity. If the synthetic control closely matches Saffron in the pre-treatment period, the post-treatment gap is a credible estimate of the causal effect of Team Rocket's invasion.

Constructing Synthetic Saffron

We have seven potential donor cities. The pre-treatment characteristics include average economic activity (Periods 1–7), population, number of registered trainers, and distance to the nearest port. The optimization finds:

Donor City	Weight
Celadon	0.42
Vermilion	0.31
Cerulean	0.18
Pewter	0.09
Fuchsia	0.00
Cinnabar	0.00
Lavender	0.00

Synthetic Saffron is primarily a blend of Celadon (the other major commercial center), Vermilion (a port city with substantial economic activity), and Cerulean (a mid-sized city with tourism). Fuchsia, Cinnabar, and Lavender receive zero weight — they are too dissimilar to be useful.

Let us examine the pre-treatment fit:

Characteristic	Saffron (Actual)	Synthetic Saffron
Avg. Econ. Activity (Per. 1–7)	87.3	86.8
Population (thousands)	145	141
Registered Trainers	2,340	2,290
Distance to Port (km)	35	38

The pre-treatment match is excellent. More importantly, the pre-treatment outcome trajectory of Synthetic Saffron closely tracks actual Saffron:

Period	Saffron	Synthetic Saffron	Gap
1	85.2	84.9	0.3
2	86.1	85.8	0.3
3	86.8	86.5	0.3
4	87.5	87.1	0.4
5	88.0	87.7	0.3
6	88.4	88.2	0.2
7	88.9	88.6	0.3
8 (Invasion)	**84.1**	**89.1**	**-5.0**
9	80.5	89.5	-9.0
10	78.2	89.9	-11.7
11	76.8	90.2	-13.4
12	77.5	90.6	-13.1

The pre-treatment gaps are tiny (around 0.3), confirming a good fit. After the invasion, Saffron's economic activity plummets while Synthetic Saffron continues its upward trajectory. The estimated effect of the invasion is devastating: by Period 12, Saffron's economy is 13.1 index points below its synthetic counterfactual, a decline of roughly 14.5%.

Inference via Placebo Tests

Standard inference (confidence intervals, p-values) is not straightforward for synthetic control because there is typically only one treated unit. Instead, Abadie, Diamond, and Hainmueller propose **placebo tests** (or "in-space" placebos):

1. **Apply the synthetic control method to every donor city in turn**, pretending each is the "treated" unit and using the remaining donors (plus, optionally, the actual treated unit in the pre-treatment period) as the donor pool.
2. **Compute the post-treatment gap** for each placebo unit. If the method is well-calibrated, placebo units should show small post-treatment gaps (since they were not actually treated).
3. **Compare Saffron's gap to the distribution of placebo gaps.** If Saffron's gap is unusually large relative to the placebo distribution, the effect is statistically significant.

A common refinement uses the **ratio of post-treatment RMSPE to pre-treatment RMSPE** for each unit:

$$\text{Ratio}_j = \frac{\text{RMSPE}_{j,\text{post}}}{\text{RMSPE}_{j,\text{pre}}}$$

where $\text{RMSPE}_{j,\text{post}} = \sqrt{\frac{1}{T-T_0}\sum_{t=T_0+1}^{T}(Y_{jt} - \hat{Y}_{jt}^{\text{synth}})^2}$ and similarly for the pre-treatment period. A large ratio indicates a genuine treatment effect (large post-treatment gap) rather than poor pre-treatment fit (which would inflate both numerator and denominator).

In our example, Saffron's RMSPE ratio is 38.7. The largest placebo ratio is 4.2 (for Celadon). Saffron's ratio dwarfs all placebos. The implied p-value is $1/8 = 0.125$ if we simply rank the ratios (with 8 total units), which is not significant at the 5% level. This highlights a limitation of synthetic control: with few donor units, the permutation distribution has limited resolution. With 20 or more donors, a p-value of $1/20 = 0.05$ is achievable.

> ***Professor Oak Explains 7.3.*** *"The synthetic control method is like constructing a custom Pokemon team. No single Pokemon can perfectly replicate the strengths of your Charizard. But perhaps a team of Arcanine (42%), Gyarados (31%), Vaporeon (18%), and Golem (9%) comes very close in terms of stats and type coverage. If your Charizard suddenly loses a battle it should have won, and this team of substitutes performs normally, you know something happened specifically to Charizard — it was not just bad luck affecting all Fire-types."*

7.7 Extensions and Frontiers

Augmented Synthetic Control (Ben-Michael, Feller, and Rothstein, 2021)

Standard synthetic control can struggle when no convex combination of donor units closely matches the treated unit's pre-treatment trajectory. The **Augmented Synthetic Control Method (ASCM)** addresses this by adding a **bias correction** term based on an outcome model.

The ASCM estimate is:

$$\hat{\tau}_t^{\text{ASCM}} = \hat{\tau}_t^{\text{SC}} + \sum_{j=1}^{J}\left(w_j^* - \frac{1}{J}\right)\cdot\left(\hat{m}(X_j, t) - \hat{m}(X_0, t)\right)$$

where $\hat{m}(\cdot)$ is a fitted outcome model (e.g., a ridge regression of Y_{jt} on pre-treatment covariates) and w_j^* are the synthetic control weights. The correction term adjusts for residual imbalance between the synthetic control and the treated unit. If the standard synthetic control achieves perfect balance, the correction is zero and ASCM reduces to standard SC. If balance is imperfect, the outcome model fills the gap.

ASCM is doubly robust in spirit: it is consistent if either the synthetic control provides a good match or the outcome model is correctly specified. It also provides valid confidence intervals based on asymptotic theory, avoiding the need for permutation-based inference.

Synthetic Difference-in-Differences (Arkhangelsky et al., 2021)

Dmitry Arkhangelsky, Susan Athey, David Hirshberg, Guido Imbens, and Stefan Wager proposed the **Synthetic Difference-in-Differences (SDiD)** estimator, which elegantly combines the strengths of DiD and synthetic control.

Standard DiD uses equal weights on all control units but requires parallel trends. Standard SC uses optimized unit weights but does not adjust for time trends. SDiD uses **both** unit weights (like SC) and time weights (novel) to achieve a "best of both worlds" estimator.

The SDiD estimator solves:

$$(\hat{\tau}^{SDiD}, \hat{\mu}, \hat{\alpha}, \hat{\lambda}) = \arg\min_{\tau,\mu,\alpha,\lambda} \sum_i \sum_t (\hat{\omega}_i^{unit} \cdot \hat{\omega}_t^{time}) \cdot (Y_{it} - \mu - \alpha_i - \lambda_t - \tau D_{it})^2$$

where $\hat{\omega}_i^{unit}$ are unit weights (constructed similarly to SC weights) and $\hat{\omega}_t^{time}$ are time weights that upweight pre-treatment periods most similar to the post-treatment period. This combination makes SDiD robust to failures of parallel trends (via unit weights) and to poor pre-treatment fit (via time weights that focus on the most relevant pre-treatment periods).

In simulations, SDiD tends to have lower variance than SC (because it leverages the panel structure more efficiently) and lower bias than DiD (because it does not require strict parallel trends).

Penalized Synthetic Control (Abadie and L'Hour, 2021)

Standard synthetic control imposes strict non-negativity and summing-to-one constraints, which may exclude useful donor units or produce sparse weight vectors. **Penalized synthetic control** relaxes these constraints by adding a penalty term that encourages sparsity while allowing more flexible weight structures:

$$w^* = \arg\min_w \| X_0 - X_1 w \|^2 + \lambda \| w \|_p$$

where λ is a regularization parameter and $\| \cdot \|_p$ is a norm penalty ($p = 1$ for lasso-type sparsity, $p = 2$ for ridge-type shrinkage). The penalty helps in settings with many donor units where overfitting is a concern, and it can improve the method's performance when the donor pool is large relative to the pre-treatment periods.

Matrix Completion Methods (Athey et al., 2021)

An alternative framework treats the causal inference problem as a **matrix completion** problem. Arrange the panel data as a matrix with units as rows and time periods as columns. The treated observations are "missing" in the sense that we observe $Y_{it}(1)$ but want $Y_{it}(0)$. Under the assumption that the matrix of untreated potential outcomes has low rank (i.e., outcomes are driven by a small number of latent factors), techniques from the matrix completion literature — such as nuclear norm minimization — can "fill in" the missing entries.

The matrix completion approach is particularly attractive when:

- The number of treated units is large relative to the donor pool
- Treatment effects vary across units and time in complex ways
- The researcher wants to avoid specifying a particular model for the counterfactual

This method nests both DiD (which assumes a rank-2 structure: unit effects plus time effects) and synthetic control (which constructs the counterfactual as a weighted average of rows) as special cases.

Difference-in-Discontinuities

When treatment is assigned by a threshold rule that changes over time, the **difference-in-discontinuities** design combines RDD and DiD. For example, suppose Silph Co. distributes Shadow Surge to trainers with 5 or more gym badges, and this threshold is lowered to 3 badges in the post-period. By comparing the RDD estimate at the threshold before and after the policy change, researchers can difference out any time-invariant confounders at the cutoff.

This method is useful in settings where neither RDD alone (because the cutoff changes) nor DiD alone (because treatment assignment is not cleanly separated across groups) provides identification, but their combination does.

Chapter Summary

Marsh Badge earned!

This chapter covered the family of methods that exploit **panel data** — repeated observations on the same units over time — to identify causal effects. We began with the classic **2x2 difference-in-differences** design, which eliminates time-invariant confounders (via within-unit differencing) and common time trends (via cross-group differencing) to isolate treatment effects. The parallel trends assumption — that treated and control units would have followed the same trajectory absent treatment — is the key identifying assumption, and while it is fundamentally untestable, event study plots provide crucial diagnostic evidence.

We then examined **two-way fixed effects** regression as the natural panel data generalization of DiD. While TWFE is equivalent to DiD in the simple 2x2 case, it produces biased estimates when treatment is **staggered** across units and treatment effects are heterogeneous. The Goodman-Bacon decomposition reveals why: TWFE implicitly uses already-treated units as controls, generating contaminated comparisons. The modern DiD revolution — **Callaway and Sant'Anna**, **Sun and Abraham**, **de Chaisemartin and D'Haultfouille**, **Borusyak, Jaravel, and Spiess** — provides heterogeneity-robust estimators that use only clean comparisons.

For settings with a single treated unit and no obvious control, the **synthetic control method** constructs a data-driven counterfactual from a weighted combination of donor units. Inference proceeds via placebo tests. Extensions including **augmented synthetic control**, **synthetic difference-in-differences**, and **matrix completion** further enrich the toolkit.

These methods are the workhorses of modern policy evaluation, program evaluation, and empirical social science. Mastering them equips the researcher to extract causal knowledge from the rich panel data structures that arise naturally in observational settings.

Professor Oak's Review Questions

1. **Explain in your own words** why the DiD estimator requires two differences rather than one. What specific type of bias does each difference eliminate? Illustrate with the Shadow Surge example.
2. **State the parallel trends assumption** formally. Why is it untestable? What is the best available diagnostic, and what are its limitations?
3. **In the 2x2 setting**, show algebraically that the DiD estimator from the table of means is identical to the OLS coefficient $\hat{\tau}$ on the interaction term $\text{Treat}_i \times \text{Post}_t$.
4. **Explain the Goodman-Bacon decomposition** intuitively. What are the three types of 2x2 comparisons, and why is the "already-treated as control" comparison problematic?
5. **Compare and contrast** the Callaway-Sant'Anna and Borusyak-Jaravel-Spiess estimators. How do they handle the staggered treatment problem differently, and under what conditions might one be preferred over the other?
6. **Describe the synthetic control method's** approach to inference. Why can't we simply compute standard errors as in a regression? What determines the smallest achievable p-value in a placebo test with J donor units?

Trainer Challenge Exercises

Exercise 7.1: Compute a 2x2 DiD

The Kanto Pokemon League introduced a new battle format ("Double Battles") in Vermilion City in Season 3 but not in Pewter City. Average trainer participation rates (in battles per month) are shown below:

	Season 2 (Pre)	Season 4 (Post)
Vermilion (Treated)	12.4	18.7
Pewter (Control)	9.8	13.2

(a) Compute the DiD estimate of the effect of Double Battles on participation.

(b) Write out the corresponding regression equation and state the value of each coefficient $(\alpha, \beta, \gamma, \tau)$.

(c) What is the counterfactual participation rate for Vermilion in Season 4 under the parallel trends assumption?

(d) Blue claims the effect is $18.7 - 13.2 = 5.5$. Explain his mistake.

Exercise 7.2: Event Study Interpretation

A researcher estimates an event study for the effect of Shadow Surge on trainer win rates across Kanto cities. The estimated coefficients (with 95% confidence intervals) at each event time relative to treatment ($k = -1$ omitted) are:

Event Time k	$\hat{\tau}_k$	95% CI
-5	-0.02	$[-0.06, 0.02]$
-4	0.01	$[-0.03, 0.05]$
-3	-0.01	$[-0.05, 0.03]$
-2	0.03	$[-0.01, 0.07]$
-1	(omitted)	—
0	0.08	$[0.04, 0.12]$
1	0.10	$[0.06, 0.14]$
2	0.11	$[0.07, 0.15]$
3	0.09	$[0.05, 0.13]$
4	0.10	$[0.06, 0.14]$

(a) Sketch the event study coefficient plot with confidence intervals.

(b) Do the pre-treatment coefficients support the parallel trends assumption? Explain.

(c) Describe the dynamic treatment effect pattern. Is the effect immediate or gradual? Does it persist, grow, or fade?

(d) A skeptic notes that $\hat{\tau}_{-2} = 0.03$ is the largest pre-treatment coefficient and worries about anticipation effects. How would you respond?

Exercise 7.3: Identifying TWFE Bias

Consider a staggered DiD setting with three treatment cohorts (treated in periods 4, 8, and 12) and one never-treated group. The true treatment effect is constant at $\tau = 5$ for all groups in all post-treatment periods (homogeneous effects). A researcher runs TWFE and obtains $\hat{\tau}^{TWFE} = 5.0$.

(a) Is the TWFE estimate biased in this case? Explain why or why not.

(b) Now suppose the true effect grows over time: $\tau_k = 2 + 0.5k$ where k is the number of periods since treatment. The researcher again runs TWFE and obtains $\hat{\tau}^{TWFE} = 3.8$, while the true average ATT across all treated unit-periods is 5.2. Explain the source and direction of the bias.

(c) The researcher runs a Goodman-Bacon decomposition and finds that 35% of the TWFE weight comes from comparisons where already-treated units serve as controls. What does this tell you?

(d) Recommend a specific modern estimator and explain how it would fix the problem.

Exercise 7.4: Construct a Synthetic Control

Team Rocket briefly occupied Celadon City's Game Corner in Period 10, causing economic disruption. You observe economic activity indices for five cities over 16 periods (treatment at period 10 for Celadon only):

Pre-treatment averages (Periods 1–9):

City	Econ. Index	Population	Trainer Count
Celadon (Treated)	92.0	130	2,100
Cerulean	78.0	95	1,500
Vermilion	85.0	110	1,800
Fuchsia	72.0	80	1,200
Lavender	65.0	60	900

(a) Explain intuitively why no single control city is a good match for Celadon. Which city or cities seem closest?

(b) Suppose the optimal synthetic control weights are: Vermilion (0.55), Cerulean (0.30), Fuchsia (0.15), Lavender (0.00). Compute the synthetic Celadon's pre-treatment average economic index. How does it compare to Celadon's actual value?

(c) In Period 12, Celadon's economic index is 83.0. The synthetic control's predicted value is 93.5. What is the estimated treatment effect of the Team Rocket occupation in Period 12?

(d) You run placebo tests by applying synthetic control to each donor city. The post/pre RMSPE ratios are: Celadon (15.2), Vermilion (2.1), Cerulean (3.5), Fuchsia (1.8), Lavender (1.2). What can you conclude about the statistical significance of the effect?

Further Reading

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- **Rambachan, A. and Roth, J.** (2023). "A More Credible Approach to Parallel Trends." *Review of Economic Studies*, 90(5), 2555–2591. Sensitivity analysis for violations of parallel trends.

Skills to Practice in the Notebook

The notebook `notebooks/ch07_saffron_city.ipynb` turns the Shadow Surge story into a full DiD and synthetic control pipeline. Before you earn the Marsh Badge, you should be comfortable doing the following end-to-end:

1. **Compute the 2×2 DiD by hand.** From a 2-city, 2-period dataset, compute the four group-period means, take the double difference, and confirm you get the same number from an interaction regression $Y \sim \mathrm{Treat} + \mathrm{Post} + \mathrm{Treat} \times \mathrm{Post}$.
2. **Fit a two-way fixed effects regression.** Use `linearmodels.PanelOLS` (or equivalent) and extract the treatment coefficient and clustered standard errors.
3. **Plot the parallel trends diagram.** Overlay observed treated/control means over time with the implied counterfactual dashed line. This plot is the single best visual check of parallel trends.
4. **Run an event study.** For staggered treatment, estimate leads and lags around the event and plot them. Verify that pre-treatment leads are close to zero (pre-trends test) and use post-treatment coefficients as dynamic treatment effects.
5. **Reproduce the "forbidden comparison" bug.** Simulate a staggered adoption dataset where later-treated units use already-treated units as controls. Compare naive TWFE against Callaway-Sant'Anna and show that TWFE is biased when effects are heterogeneous across cohorts.
6. **Do a Goodman-Bacon decomposition.** Use `bacondecomp` (or similar) to break TWFE into its component 2×2 comparisons. Identify any forbidden comparisons by weight.
7. **Implement Callaway-Sant'Anna.** Use `differences` (or `csdid`) to estimate $\mathrm{ATT}(g, t)$ for each cohort-period and aggregate into overall, group-specific, and event-study summaries.
8. **Build a synthetic control.** For Saffron, construct weights w_i^* over donor cities that match pre-treatment outcomes. Plot the actual vs. synthetic path, mark treatment time, and compute the gap.

9. **Run placebo-in-space tests.** Re-run the SC pipeline on each donor city as if it were treated. Compare the "real" treatment effect against the distribution of placebo effects to get a non-parametric p-value.
10. **Complete the Trainer Challenge Exercises** — (a) a full 2×2 DiD, (b) an event study with pre-trends test, (c) a Callaway-Sant'Anna vs. TWFE comparison on a biased DGP, and (d) a synthetic control with placebo inference.

Check Your Understanding

Before facing Sabrina, run through this checklist. DiD is one of the most commonly misused methods in applied work — fluency here is not optional.

Questions you should be able to answer out loud, without notes:

- Write the 2×2 DiD estimator in plain-English terms: "treated change minus control change."
- State the parallel trends assumption in one sentence. Is it testable? What is a *pre-trends* check, and what is it actually testing?
- Why does DiD eliminate time-invariant unit differences and common time shocks, but not shocks that hit only the treated group at the treatment time?
- Write the TWFE regression $Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + \epsilon_{it}$ and explain what each fixed effect absorbs.
- Under what conditions does TWFE recover a causal effect? Under what conditions does it fail?
- Explain the "forbidden comparison" problem in staggered DiD. Why does using already-treated units as controls cause bias?
- What does the Goodman-Bacon decomposition tell you, and how do you use it diagnostically?
- What does Callaway-Sant'Anna estimate? How is its building block $ATT(g, t)$ different from the TWFE $\hat{\tau}$?
- What is an event study, and how does it differ from a single-coefficient DiD?
- What is the synthetic control method? What exactly are the weights w_i^* trying to match?
- How do you do inference for synthetic control when there's only one treated unit? (Permutation / placebo tests.)
- When should you reach for DiD versus synthetic control? What do you need for each?

Tasks you should be able to perform in code:

- Compute 2×2 DiD both from group means and via an interaction regression; verify they agree.
- Fit TWFE with clustered standard errors on a panel dataset.
- Produce an event study plot with pre-period leads and post-period lags.
- Simulate staggered adoption with heterogeneous effects and confirm TWFE is biased.
- Run Callaway-Sant'Anna (`differences` / `csdid`) and aggregate $ATT(g, t)$ into overall and dynamic effects.
- Do a Goodman-Bacon decomposition and interpret the weights.
- Construct synthetic control weights, plot the gap, and run placebo-in-space inference.
- Compute a pre-trends test and know when to fail it.

Do all this and the Marsh Badge is yours.

Badge Earned: Marsh Badge

You have defeated Sabrina, the Psychic-type Gym Leader of Saffron City, and earned the Marsh Badge. You have mastered the art of differencing — across time, across groups, and across synthetic counterfactuals. You understand why naive two-way fixed effects can deceive, and you know how to deploy the modern toolkit of heterogeneity-robust estimators. You can construct synthetic controls from weighted combinations of donor units and conduct inference through placebo tests. The Marsh Badge certifies your command of panel data methods for causal inference.

Badges earned: Boulder, Cascade, Thunder, Rainbow, Soul, Volcano, Marsh (7/8)

> ***Next Chapter Preview:*** *Victory Road leads to the Indigo Plateau, where the Elite Four await. Each champion guards a frontier of causal inference: mediation analysis reveals the mechanisms through which causes operate; heterogeneous treatment effects uncover for whom treatments work best; sensitivity analysis probes the robustness of our conclusions to hidden bias; and interference — the bane of SUTVA — forces us to confront a world where one unit's treatment affects another's outcome. The path is steep, but the summit offers a commanding view of the entire causal landscape...*

Chapter 8: Indigo Plateau & Beyond — Advanced Topics & Frontiers

Lorelei *Bruno* *Agatha* *Lance*

The road from Viridian City narrows into a winding mountain path. Rock walls rise on either side, and the air grows colder with every step. Behind you lies the entire Kanto region — eight Gym Badges earned, each one representing a different tool in your causal inference arsenal. Potential outcomes. Randomized experiments. DAGs and d-separation. Matching and propensity scores. Regression and doubly robust estimation. Instrumental variables and regression discontinuity. Difference-in-differences and synthetic control.

Ahead lies the Indigo Plateau.

You push through the gates and step into a vast, echoing hall. A League official checks your Badge case and nods. "Eight badges. You're cleared to challenge the Elite Four." She pauses. "But I should warn you — each member guards a frontier of causal inference that most trainers never reach. Lorelei will test your understanding of mechanisms. Bruno will ask whether everyone responds the same way. Agatha will haunt you with the specter of unobserved confounding. And Lance… Lance will shatter your most basic assumption about how units interact."

She steps aside. "Beyond them waits the Champion. He thinks he knows everything about causation. Prove him wrong."

You take a deep breath. You adjust your bag. You walk through the first door.

Notation at a Glance: The Frontier Toolkit

Chapter 8 introduces more notation than any other chapter — each Elite Four section has its own vocabulary. Use this sheet as a master index; every symbol you see in the body points back here.

Mediation (§8.1)

Symbol	*Plain-English reading*
M_i	*A* ***mediator*** *— a variable on the causal path* $D \to M \to Y$.
$M_i(d)$	*The mediator's value if* i *received treatment* d.
$Y_i(d, m)$	*The outcome if* i *had treatment* d *and mediator value* m.
NDE $= E[Y_i(1, M_i(0)) - Y_i(0, M_i(0))]$	***Natural Direct Effect*** *— the effect of treatment holding the mediator at its control value.*
NIE $= E[Y_i(1, M_i(1)) - Y_i(1, M_i(0))]$	***Natural Indirect Effect*** *— the effect operating through the mediator.*
TE = NDE + NIE	*Total effect decomposition.*
sequential ignorability	*The strong untestable assumption needed to identify NDE/NIE.*

Sensitivity Analysis (§8.2)

Symbol	*Plain-English reading*
Γ	*Rosenbaum's sensitivity parameter — "how much could hidden bias multiply the odds of treatment?"*
E-value	*"The minimum strength an unmeasured confounder would need to explain away the effect."*
Oster δ	*Coefficient stability: how much more influential would unobservables need to be than observables to kill the effect?*
Manski bounds	*Worst-case bounds on the causal effect under no assumptions about the unobserved counterfactual.*

Heterogeneous Effects (§8.3)

Symbol	*Plain-English reading*
$\tau(x) = E[Y(1) - Y(0) \mid X = x]$	***CATE*** *— the treatment effect for units with covariates* x.
$\hat{\tau}(x)$	*CATE estimate from a model (e.g., causal forest).*
BLP	*Best Linear Projection of* $\tau(x)$ *onto a small set of covariates.*
GATES	*Sorted Group ATEs — split units by predicted* $\hat{\tau}(x)$*, compare actual effects across quantile groups.*
CLAN	*Classification Analysis — characterize the "most affected" group's covariates.*
policy $\pi(x)$	*A rule mapping covariates to a treatment decision.*

Interference (§8.4)

Symbol	*Plain-English reading*
SUTVA	*"No interference, no hidden variations." The standard assumption.*
$Y_i(D)$	*Outcome for unit i as a function of the entire vector of treatment assignments.*
exposure mapping $e(D)$	*Compresses the vector of treatments into a summary relevant for unit i (e.g., "how many neighbors were treated").*
partial interference	*"Interference only happens inside clusters, not across them."*
direct / spillover effect	*The effect of your own treatment / the effect of a peer's treatment on you.*
two-stage randomization	*Randomize cluster-level saturation, then randomize individuals within clusters.*

Causal Discovery (§8.7)

Symbol	*Plain-English reading*
PC / FCI / GES	*Algorithms that learn DAG structure from data using independence tests or scores.*
Markov equivalence class	*The set of DAGs that cannot be distinguished using observational data alone.*

Causal Inference + ML (§8.8)

Symbol	*Plain-English reading*
DML	*Double / Debiased Machine Learning — plug ML nuisance estimates into a Neyman-orthogonal score.*
TMLE	*Targeted Maximum Likelihood Estimation — doubly robust, semiparametrically efficient.*
nuisance parameter	*Things you need to estimate but don't directly care about (e.g., $e(X)$, $m(X)$).*
$\sqrt{n}$*-consistency*	*Estimates whose error shrinks at rate $1/\sqrt{n}$ — the gold standard.*

Transportability (§8.9)

Symbol	*Plain-English reading*
S	*Indicator for "study" population (source = 0, target = 1).*
selection diagram	*A DAG with selection nodes showing where populations differ.*
reweighting to transport	*Multiply source-population estimates by density ratios to match the target population.*

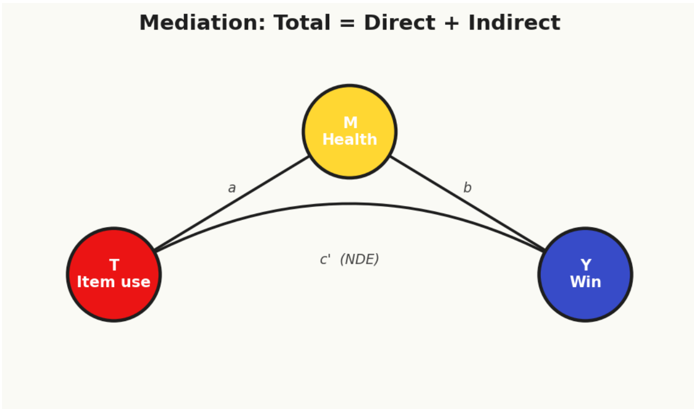
Dynamic Treatment Regimes (§8.10)

Symbol	*Plain-English reading*
A_t	*Treatment at time t.*
$\bar{A}_t = (A_1, \dots, A_t)$	*Treatment history up to time t.*
MSM	*Marginal Structural Model — parameterizes causal effects of full treatment histories.*
IPTW	*Inverse Probability of Treatment Weights — time-varying version of IPW.*
g-computation	*Iteratively impute future outcomes under hypothetical treatment sequences.*
Q-learning	*Backward induction to find optimal dynamic treatment rules.*

If at any point the body text feels opaque, jump here first. Most confusion in Chapter 8 is notation shock, not concept shock.

8.1 Mediation Analysis: Lorelei's Ice Chamber

#087 Dewgong



Mediation: total effect = direct (NDE) + indirect through M (NIE).

The chamber is cold. Ice crystals hang from the ceiling, refracting light into prismatic arcs. Lorelei sits on a throne of frozen stalagmites, a Dewgong resting at her side. She adjusts her glasses and studies you.

"Most trainers," she says, "are satisfied knowing that *a treatment works. I want to know* how. *Through what pathways does a cause produce its effect? Can you freeze out the direct path and isolate the indirect one?"*

She gestures to the battlefield. "Show me you understand mechanisms."

Total, Direct, and Indirect Effects

In Chapters 1 through 7, we focused on estimating the **total effect** of a treatment D on an outcome Y. But in many settings, we want to understand the **mechanism** — the causal pathway through which the effect operates. Mediation analysis decomposes the total effect into a **direct effect** (the pathway from D to Y that does not pass through a specified mediator M) and an **indirect effect** (the pathway that operates through M).

Consider a concrete example from the Elite Four challenge. Suppose we want to understand whether using healing items (D) during Elite Four battles affects the probability of defeating all four members (Y). One plausible mechanism is that items maintain Pokemon health (M) between battles, which in turn affects the probability of winning later rounds. The question is: does item use affect outcomes *only* through health maintenance, or does it also have a direct effect — perhaps through strategic confidence or reduced anxiety?

The causal graph is:

$$D \to M \to Y$$
$$D \to Y$$

The total effect decomposes as:

$$\text{Total Effect} = \text{Direct Effect} + \text{Indirect Effect}$$

The Baron-Kenny Approach and Its Limitations

The classic approach to mediation, introduced by Baron and Kenny (1986), proceeds in three regressions:

1. Regress Y on D: $Y = \alpha_1 + c \cdot D + \epsilon_1$
2. Regress M on D: $M = \alpha_2 + a \cdot D + \epsilon_2$
3. Regress Y on D and M: $Y = \alpha_3 + c' \cdot D + b \cdot M + \epsilon_3$

The indirect effect is estimated as $\hat{a} \times \hat{b}$ (or equivalently, $\hat{c} - \hat{c}'$), and the direct effect is $\hat{c}'$. This approach is intuitive and remains widely used, but it suffers from serious limitations:

- It assumes **linear models** and **no interaction** between D and M.
- It conflates causal mediation with statistical mediation; $c - c'$ need not equal the causal indirect effect when models are nonlinear.
- It provides no formal framework for identifying assumptions.
- It does not define what the "direct" and "indirect" effects *mean* in terms of potential outcomes.

> ***Blue's Mistake: The Baron-Kenny Trap***
>
> *Blue runs the three regressions and announces: "The indirect effect through Pokemon health is 0.15, and the direct effect is 0.08. Therefore 65% of the total effect operates through the health mechanism." But Blue's mediator — average team HP entering each battle — is also affected by the opponent's strategy, which independently affects Y. There is a confounder of the $M \to Y$ relationship that Blue has not accounted for. In the presence of such confounders, the Baron-Kenny decomposition does not recover causal mediation effects. Blue needs the formal potential outcomes framework for mediation.*

Causal Mediation: Natural Direct and Indirect Effects

The modern framework for causal mediation defines effects using **nested counterfactuals** (Robins & Greenland, 1992; Pearl, 2001; Imai, Keele & Tingley, 2010). Let $M(d)$ denote the potential value of the mediator when treatment is set to d, and let $Y(d, m)$ denote the potential outcome when treatment is d and the mediator is set to m.

The **Natural Direct Effect (NDE)** is:

$$\mathrm{NDE} = E[Y(1, M(0)) - Y(0, M(0))]$$

This is the effect of changing treatment from 0 to 1 while holding the mediator at the value it *would naturally take* under control. It captures the portion of the treatment effect that does not operate through M.

The **Natural Indirect Effect (NIE)** is:

$$\mathrm{NIE} = E[Y(1, M(1)) - Y(1, M(0))]$$

This is the effect of changing the mediator from its natural value under control to its natural value under treatment, while holding treatment fixed at 1. It captures the portion of the effect that operates through the change in M induced by treatment.

The total effect decomposes exactly:

$$TE = \mathrm{NDE} + \mathrm{NIE}$$

since $E[Y(1, M(1)) - Y(0, M(0))] = E[Y(1, M(0)) - Y(0, M(0))] + E[Y(1, M(1)) - Y(1, M(0))]$.

> ***Professor Oak Explains: Nested Counterfactuals***
>
> *The expression $Y(1, M(0))$ is a nested counterfactual: it asks what the outcome would be if treatment were set to 1 but the mediator took the value it would have taken under treatment 0. This quantity is fundamentally unobservable — even in a randomized experiment, we cannot simultaneously set $D = 1$ and force M to behave as if $D = 0$. Identification therefore requires strong assumptions beyond what randomization alone provides.*

Sequential Ignorability

Imai, Keele, and Tingley (2010) showed that identification of NDE and NIE requires **sequential ignorability**:

1. **Treatment ignorability**: $\{Y(d', m), M(d)\} \perp\!\!\!\perp D \mid X$ for all d, d', m.
2. **Mediator ignorability**: $Y(d', m) \perp\!\!\!\perp M \mid D = d, X$ for all d, d', m.

The first assumption is satisfied by randomization of treatment (or conditional on covariates X). The second is far more demanding: conditional on treatment and covariates, the mediator must be as-if randomly assigned. This rules out unobserved confounders of the $M \to Y$ relationship — a strong requirement, since the mediator is typically *not* randomly assigned.

Under sequential ignorability with a linear model, the mediation effects reduce to the familiar product-of-coefficients $a \times b$. But the framework extends naturally to nonlinear and nonparametric settings.

Nonparametric and Sensitivity Approaches

When linearity fails (e.g., binary outcomes, count data), the product-of-coefficients method is inconsistent. Imai et al. (2010) propose a general nonparametric identification strategy and provide the `mediation` R package for estimation.

Because the sequential ignorability assumption for the mediator is untestable, **sensitivity analysis for mediation** is critical. The key question is: how large would an unobserved confounder of $M \rightarrow Y$ need to be to explain away the estimated indirect effect? Imai et al. (2010) propose a sensitivity parameter ϱ — the correlation between the error terms in the mediator and outcome equations — and show how the estimated indirect effect varies as ϱ departs from zero.

Pokemon Application

In the Elite Four data, we observe:

- D: whether trainer used healing items (Full Restores, Max Potions) between battles
- M: average team HP percentage entering each subsequent battle
- Y: whether trainer defeated all four Elite Four members
- X: trainer level, number of badges, team composition

Estimating the mediation model, we find that roughly 60% of the total effect of item use operates through the health maintenance pathway (NIE), while 40% operates through other channels (NDE) — perhaps strategic confidence or the ability to use offensive items instead of healing during battle. The sensitivity analysis reveals that a confounder with $\varrho > 0.3$ would be sufficient to nullify the indirect effect, suggesting the mediation finding is moderately robust but not ironclad.

Lorelei's Dewgong falls. She nods approvingly. "You understand that knowing a treatment works is not enough. Understanding how *it works — and being honest about the assumptions required to decompose effects — is the mark of a serious researcher."*

She steps aside. The door to the next chamber opens.

8.2 Sensitivity Analysis: Agatha's Ghost Chamber

#094 Gengar

The room is dark. Purple mist coils along the floor. Somewhere in the shadows, something cackles. Then Agatha materializes from the fog, leaning on her cane, a Gengar grinning at her shoulder.

"Every study you have ever read," she whispers, "rests on the assumption that you have measured everything that matters. But what about the things you cannot *see? The ghosts in your model? The unobserved confounders that lurk behind every observational estimate?"*

Her Gengar's eyes glow red. "Can your estimates survive my ghosts?"

The Ghost of Unobserved Confounding

Throughout this book, we have relied on assumptions like conditional ignorability (selection on observables), the exclusion restriction (IV), and the parallel trends assumption (DiD). Each of these assumptions is, at its core, a claim about what we *do not need to worry about* — that there are no unobserved confounders, no direct effects of instruments, no differential trends. But these assumptions are untestable. Sensitivity analysis asks: **how robust are our conclusions to violations of these assumptions?**

Rather than treating identification assumptions as binary (satisfied or not), sensitivity analysis parameterizes the degree of violation and examines how the estimated effect changes. If the estimate survives large violations, we gain confidence; if it collapses under small perturbations, we should be cautious.

Rosenbaum Bounds

Rosenbaum (2002) developed a sensitivity analysis framework for matched observational studies. The key parameter is Γ (capital gamma), which measures how much an unobserved confounder could alter the odds of treatment assignment.

In a randomized experiment, two matched units with identical observed covariates have the same probability of treatment. An unobserved confounder could distort this. Rosenbaum's framework asks: if two matched units could differ in treatment odds by a factor of at most Γ, would our conclusion still hold?

Formally, for matched pair (i, j) with identical observed covariates:

$$\frac{1}{\Gamma} \leq \frac{P(D_i = 1 \mid X_i)/P(D_i = 0 \mid X_i)}{P(D_j = 1 \mid X_j)/P(D_j = 0 \mid X_j)} \leq \Gamma$$

When $\Gamma = 1$, there is no hidden bias (equivalent to a randomized experiment). As Γ increases, we allow more hidden bias. For each value of Γ, we compute the worst-case p-value. The **sensitivity value** is the smallest Γ at which the result becomes statistically insignificant.

> ***Professor Oak Explains: Interpreting*** Γ
>
> *A sensitivity value of $\Gamma = 2$ means that an unobserved confounder would need to double the odds of treatment for one unit relative to its match in order to explain away the observed effect. Whether $\Gamma = 2$ is "large" depends on context — in a well-designed study with rich covariates, a confounder doubling the odds of treatment (beyond what observed variables already explain) may be implausible. In a study with sparse controls, $\Gamma = 2$ may be easy to imagine.*

The E-value

VanderWeele and Ding (2017) introduced the **E-value** as a model-free measure of the minimum strength of association that an unobserved confounder would need to have with both the treatment and the outcome to fully explain away an observed effect.

For an observed risk ratio RR, the E-value is:

$$\text{E-value} = RR + \sqrt{RR \times (RR - 1)}$$

The E-value has an elegant interpretation: it is the minimum value of the risk ratio that an unmeasured confounder would need to have with *both* the treatment and the outcome (conditional on measured covariates) to reduce the observed association to the null.

For example, if we observe $RR = 2.5$ for the effect of Exp. Share on defeating the Elite Four, the E-value is:

$$\text{E-value} = 2.5 + \sqrt{2.5 \times 1.5} = 2.5 + \sqrt{3.75} \approx 2.5 + 1.94 = 4.44$$

This means an unobserved confounder would need to have a risk ratio of at least 4.44 with both Exp. Share use and Elite Four victory to explain away the observed $RR = 2.5$. That is a very strong confounder. For reference, the E-value can also be computed for the lower bound of the confidence interval, providing a more conservative assessment.

Oster Bounds (Coefficient Stability)

Oster (2019) developed a sensitivity analysis based on **coefficient stability** — how much the treatment effect estimate changes as additional controls are added. The intuition is simple: if adding observed controls barely moves the coefficient, then unobserved confounders (which are presumably "similar" to observed ones) would also barely move it.

The key parameter is δ, which represents the ratio of selection on unobservables to selection on observables. Oster's approach assumes a maximum R^2 for the hypothetical regression that includes all confounders (observed and unobserved), denoted R^2_{max}, and computes the bias-adjusted treatment effect as a function of δ and R^2_{max}.

The identified set is constructed by varying δ. If $\delta = 1$ means unobservables are as important as observables, and the bias-adjusted effect remains meaningfully different from zero at $\delta = 1$ (with $R^2_{max} = 1.3 \times \tilde{R}^2$, where $\tilde{R}^2$ is the R^2 from the controlled regression), the result is considered robust.

Manski Partial Identification Bounds

Manski (1990, 2003) took a fundamentally different approach: rather than asking "how large must the bias be to explain the result," he asked "what can we learn with minimal assumptions?" His **worst-case bounds** place no restrictions on the relationship between treatment and potential outcomes among the untreated (or vice versa).

For a binary treatment D and bounded outcome $Y \in [y_{min}, y_{max}]$:

$$E[Y(1)|D = 1]P(D = 1) + y_{min}P(D = 0) \leq E[Y(1)] \leq E[Y(1)|D = 1]P(D = 1) + y_{max}P(D = 0)$$

$$E[Y(0)|D = 0]P(D = 0) + y_{min}P(D = 1) \leq E[Y(0)] \leq E[Y(0)|D = 0]P(D = 0) + y_{max}P(D = 1)$$

These bounds are wide — often too wide to be informative — but they are **assumption-free** (given random sampling). They represent the price of honesty.

Lee Bounds for Sample Selection

Lee (2009) developed bounds for settings with **sample selection** or attrition — where the outcome is only observed for a subset of the population, and the selection is potentially affected by treatment. For example, if item use affects whether trainers make it to the later Elite Four battles (where we measure performance), then naive estimation suffers from selection bias.

Lee bounds assume only **monotonicity of selection**: treatment can only move selection in one direction (e.g., item use can only increase the probability of reaching later battles, never decrease it). Under this assumption, bounds on the treatment effect are obtained by trimming the “excess” observations in the group with higher selection rates.

Worked Example

Suppose we estimate that Exp. Share increases the probability of defeating the Elite Four by a factor of $RR = 1.8$ (a propensity-score-matched study from Chapter 4 data). We compute:

E-value:

$$E = 1.8 + \sqrt{1.8 \times 0.8} = 1.8 + \sqrt{1.44} = 1.8 + 1.2 = 3.0$$

An unobserved confounder would need $RR \geq 3.0$ with both treatment and outcome to explain away the result.

Rosenbaum bounds: In our matched sample, the result remains significant at $\alpha = 0.05$ up to $\Gamma = 1.6$. At $\Gamma = 1.7$, the worst-case p-value crosses 0.05. This means a moderate unobserved confounder (increasing treatment odds by 70%) could potentially explain the finding.

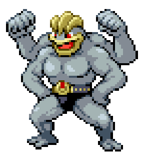
Oster bounds: The coefficient on Exp. Share moves from 0.42 (no controls) to 0.35 (full controls), while R^2 increases from 0.05 to 0.22. At $\delta = 1$ and $R^2_{max} = 0.29$, the bias-adjusted estimate is 0.30, still meaningfully different from zero.

Taken together, these analyses suggest the Exp. Share effect is moderately robust but not impervious to confounding — a common and honest conclusion.

Agatha’s Gengar fades into the shadows. The old woman studies you. “You didn’t flinch when I showed you the ghosts. Good. A researcher who pretends there are no unobserved confounders is more dangerous than one who knows they exist.”

She taps her cane on the floor. The mist parts, revealing the third door.

8.3 Heterogeneous Treatment Effects: Bruno’s Fighting Ring



#068 Machamp

The chamber shakes. Two Machamp are sparring in the center of a raised platform. Bruno sits cross-legged at the edge, shirtless, his arms folded across his chest. He opens one eye as you enter.

“You’ve spent this entire journey estimating average effects,” he says. “But not everyone is average. Different fighters respond differently to the same training. The question is not just does the treatment work — *it is* for whom does it work, and how much?”

He stands. “Show me you can see the differences.”

Beyond the Average

The Average Treatment Effect is a useful summary, but it masks potentially enormous variation. A drug that helps half the population and harms the other half has an ATE of zero — the same as a drug that does nothing for anyone. Policy-makers, clinicians, and trainers all want to know: **who benefits, who is harmed, and who is unaffected?**

The **Conditional Average Treatment Effect (CATE)** captures this heterogeneity:

$$CATE(x) = E[Y(1) - Y(0) \mid X = x]$$

where X is a vector of pre-treatment characteristics. The CATE is the average treatment effect for the subpopulation with characteristics $X = x$. If we could estimate CATE as a function of x, we would have a complete picture of treatment effect heterogeneity.

Traditional approaches to heterogeneity involve specifying interaction terms (e.g., $D \times X$) in a regression. But this requires the researcher to pre-specify which variables matter and what functional form the interactions take. With high-dimensional X, the number of possible interactions is enormous, and pre-specification becomes guesswork.

Causal Forests

Wager and Athey (2018) introduced **causal forests**, a machine learning method specifically designed to estimate CATE. The key innovation is adapting random forests — a powerful nonparametric prediction method — to the causal inference setting.

Core idea. A standard random forest predicts $E[Y \mid X = x]$ by partitioning the covariate space into leaves and averaging outcomes within each leaf. A causal forest instead estimates $E[Y(1) - Y(0) \mid X = x]$ by partitioning the covariate space into regions where treatment effects are most heterogeneous, and estimating the local treatment effect within each leaf.

Honest estimation. A critical innovation is **honesty** — the sample is split so that one subsample determines the tree structure (which covariates to split on and where) and a separate subsample estimates the treatment effects within each leaf. This separation prevents overfitting: the tree structure cannot "peek" at the outcomes used for estimation.

Formally, for a new observation with covariates x, the causal forest estimate is:

$$\hat{\tau}(x) = \sum_{i=1}^{n} \alpha_i(x) \cdot Y_i$$

where the weights $\alpha_i(x)$ are determined by how often observation i falls in the same leaf as x across the ensemble of trees. Treated and control observations within each leaf contribute with opposite signs.

Asymptotic properties. Under regularity conditions (including honesty and a minimum leaf size that grows with n), Wager and Athey (2018) prove that:

$$\frac{\hat{\tau}(x) - \tau(x)}{\hat{\sigma}(x)} \xrightarrow{d} N(0, 1)$$

This means causal forests produce valid confidence intervals for $CATE(x)$ — a remarkable result for a machine learning method. The variance estimate $\hat{\sigma}^2(x)$ is obtained via an infinitesimal jackknife.

Best Linear Projection (BLP)

Estimating $\hat{\tau}(x)$ for every x is informative but can be hard to summarize. Chernozhukov, Demirer, Duflo, and Fernandez-Val (2020) proposed the **Best Linear Projection (BLP)** as a simple summary of heterogeneity. The BLP regresses the estimated CATE on covariates:

$$\hat{\tau}(X_i) = \beta_0 + \beta_1' X_i + \epsilon_i$$

If $\beta_1 = 0$, there is no detectable heterogeneity that can be explained linearly by X. The coefficients β_1 tell us which covariates are most associated with treatment effect variation.

Sorted Group Average Treatment Effects (GATES)

The GATES approach (also from Chernozhukov et al., 2020) provides a nonparametric summary. The procedure is:

1. Estimate $\hat{\tau}(x)$ for all observations (e.g., using a causal forest).
2. Sort observations by their predicted CATE.
3. Divide into K groups (e.g., quintiles).
4. Estimate the average treatment effect within each group.

If the predicted CATE is informative, we should see a monotonically increasing pattern across groups: the group predicted to benefit most should show the largest estimated effect.

> ***Professor Oak Explains: The CLAN***
> *The **Classification Analysis (CLAN)** complements GATES by asking: what are the characteristics of the most-affected and least-affected groups? After sorting by predicted CATE, we compare covariate means across the top and bottom groups. This reveals who benefits most, not just that heterogeneity exists. For policy purposes, CLAN is often the most actionable output.*

Policy Learning and Optimal Treatment Rules

Given an estimate of CATE, the natural next question is: **who should receive treatment?** This is the problem of **optimal treatment rules** or **policy learning** (Athey & Wager, 2021; Kitagawa & Tetenov, 2018).

An optimal treatment rule $\pi^*(x)$ assigns treatment to maximize welfare:

$$\pi^* = \arg\max_{\pi} E[Y(\pi(X))]$$

where $\pi(X) \in \{0, 1\}$ is a decision rule mapping covariates to treatment assignment. Under the simplest formulation, the optimal rule is to treat everyone with $\text{CATE}(x) > 0$ and withhold treatment from everyone with $\text{CATE}(x) < 0$.

In practice, constraints matter: budget constraints (treat at most 30% of the population), fairness constraints (the rule should not discriminate on protected characteristics), or simplicity constraints (the rule should be a short decision tree that practitioners can follow).

Athey and Wager (2021) develop a framework for **policy learning from observational data** using doubly robust scores, ensuring that the estimated optimal policy converges to the true optimal policy even when nuisance parameters are estimated with ML.

Conformal Causal Inference

A recent development is the use of **conformal inference** for individual treatment effects (Lei & Candes, 2021). Traditional CATE estimation provides *average* effects conditional on X, but conformal methods can provide valid **prediction intervals** for individual treatment effects — bounds that contain the true τ_i with specified probability. This is a promising frontier for individualized treatment decisions.

Pokemon Application

Which types of trainers benefit most from Exp. Share? Using a causal forest on the Kanto trainer data, we estimate CATE as a function of trainer experience, team diversity (number of unique types), play style (offensive vs. defensive), and starting region.

The GATES analysis reveals striking heterogeneity:

Quintile	Predicted CATE	Estimated ATE	95% CI
Q1 (lowest)	-0.2 to 0.3	0.1	$[-0.3, 0.5]$
Q2	0.3 to 0.8	0.5	$[0.1, 0.9]$
Q3	0.8 to 1.2	1.0	$[0.5, 1.5]$
Q4	1.2 to 1.8	1.5	$[0.9, 2.1]$
Q5 (highest)	1.8 to 3.1	2.7	$[1.9, 3.5]$

The CLAN analysis reveals that Q5 (highest benefit) consists predominantly of trainers with diverse teams (4+ types), moderate experience, and an offensive play style. The Exp. Share allows their underleveled team members to catch up, dramatically improving coverage across type matchups. Q1 (lowest benefit) consists of trainers who already have a concentrated, high-level team — the Exp. Share dilutes experience without adding much.

The optimal treatment rule recommends Exp. Share for trainers with team diversity above the median and experience below the 75th percentile. Under this rule, expected badge count increases by 1.8 (vs. 0.9 under treat-all).

Bruno nods. "Not every fighter needs the same training regimen. The master understands that individual differences are not noise to be averaged away — they are signal to be understood."

He bows and steps aside. The third door creaks open. A chill runs down your spine.

8.4 Interference & Spillovers: Lance's Dragon Chamber

#149 Dragonite

The final Elite Four chamber is vast — an arena open to the sky, where dark clouds swirl overhead. Lance stands at the far end, his cape billowing, flanked by three Dragonite. As you enter, two of them turn to face you — and each other.

"Your entire journey," Lance says, his voice carrying across the arena, "has relied on a single, foundational assumption: that one trainer's treatment does not affect another trainer's outcome. But dragons do not fight in isolation. My Dragonite share the battlefield. When one uses Earthquake, the other feels it too."

His eyes narrow. "What happens when SUTVA fails?"

The Stable Unit Treatment Value Assumption — Revisited

Recall from Chapter 1 the **Stable Unit Treatment Value Assumption (SUTVA)**: each unit's potential outcome depends only on its own treatment assignment, not on the treatment assignments of other units. Formally, $Y_i(d_1, d_2, \ldots, d_n) = Y_i(d_i)$ for all i. SUTVA also requires no hidden variations of treatment.

SUTVA is often reasonable — whether Ash uses Exp. Share should not affect Misty's badge count. But in many settings, it fails spectacularly:

- **Double battles**: using Earthquake damages your partner Pokemon.
- **Vaccination**: my vaccination protects not only me but also those around me (herd immunity).
- **Education interventions**: a trained teacher improves outcomes for students in her class, but also for teachers who collaborate with her.
- **Social programs**: job training for some workers may worsen outcomes for untrained workers competing for the same jobs.

When SUTVA fails, the standard potential outcomes framework breaks down. The potential outcome for unit i depends on the entire treatment vector $d = (d_1, \ldots, d_n)$, not just d_i. With n units and binary treatment, there are 2^n possible treatment vectors — far too many potential outcomes to estimate.

Estimands Under Interference

To make progress, we need to define new causal estimands. Following Hudgens and Halloran (2008), the key effects under interference are:

Direct effect: the effect of changing unit i's treatment, holding neighbors' treatments fixed.

$$DE(d_{-i}) = E[Y_i(1, d_{-i}) - Y_i(0, d_{-i})]$$

Indirect (spillover) effect: the effect of changing neighbors' treatments, holding unit i's treatment fixed.

$$IE(d_i) = E[Y_i(d_i, 1_{-i}) - Y_i(d_i, 0_{-i})]$$

Total effect: the combined effect of changing both own and neighbors' treatments.

$$TE = E[Y_i(1, 1_{-i}) - Y_i(0, 0_{-i})]$$

Overall effect: the average effect of a treatment policy across the population.

Exposure Mapping

To tame the 2^n complexity, we use **exposure mappings** — functions that summarize the relevant aspects of others' treatments into a low-dimensional statistic. For unit i, define the effective exposure as:

$$g_i(d) = f(d_i, d_{N(i)})$$

where $N(i)$ is the set of unit i's neighbors and f is a known function. Common choices include:

- **Proportion treated among neighbors**: $g_i = (d_i, \bar{d}_{N(i)})$
- **Any neighbor treated**: $g_i = (d_i, 1[\exists j \in N(i) : d_j = 1])$
- **Number of treated neighbors**: $g_i = (d_i, \sum_{j \in N(i)} d_j)$

Under an exposure mapping assumption, the potential outcomes simplify: $Y_i(d) = Y_i(g_i(d))$.

Partial Interference

A common and useful simplification is **partial interference** (Sobel, 2006): units can be grouped into clusters such that interference occurs *within* clusters but not *across* clusters. In our Pokemon setting, this is natural: interference occurs within a double battle (your two Pokemon affect each other) but not across separate battles.

Under partial interference, the treatment vector that matters for unit i is only $d_{C(i)}$ — the treatments of units in i's cluster $C(i)$. If clusters are small, the number of relevant treatment vectors is manageable.

Two-Stage Randomization

Hudgens and Halloran (2008) proposed **two-stage randomization** as an experimental design for estimating effects under partial interference:

1. **Stage 1**: Randomly assign clusters to different treatment *intensities* (e.g., 30% treated vs. 70% treated).
2. **Stage 2**: Within each cluster, randomly assign individual units to treatment or control at the designated intensity.

This design identifies both direct and spillover effects. The direct effect is estimated by comparing treated and control units *within* the same cluster (same neighbors' treatment environment). The spillover effect is estimated by comparing units with the same individual treatment status *across* clusters with different treatment intensities.

Network Interference

When interference propagates through a network (social connections, trade relationships, geographic proximity), the structure matters. **Network interference** models specify how effects propagate through edges of a graph.

Key challenges include:

- **Network endogeneity**: network connections may be formed based on the same variables that affect outcomes.
- **Propagation depth**: does interference extend to friends of friends? How far?
- **Strategic interaction**: units may anticipate spillovers and adjust behavior.

Aronow and Samii (2017) develop inverse probability weighted estimators for network experiments that are unbiased under specified exposure mapping assumptions.

Double Battle Worked Example

Consider double battles in which two allied Pokemon (A and B) each independently choose whether to use a multi-target move ($D_A, D_B \in \{0, 1\}$). The outcome is total damage dealt by the pair. Multi-target moves hit the opponent but may also damage the ally (interference).

We observe data from 200 double battles with two-stage randomization. In Stage 1, battles are randomly assigned to "low multi-target" (20% probability) or "high multi-target" (80% probability) conditions. In Stage 2, within each battle, each Pokemon independently receives its multi-target assignment.

Results:

Own Treatment	Partner's Treatment	Mean Total Damage
0	0	145.2
1	0	178.6
0	1	132.8
1	1	161.4

Direct effect (partner untreated): $178.6 - 145.2 = 33.4$ (using multi-target moves increases own damage).

Spillover effect (own treatment = 0): $132.8 - 145.2 = -12.4$ (partner's multi-target move *reduces* your damage — you take splash damage).

Total effect: $161.4 - 145.2 = 16.2$ (smaller than the direct effect because the spillover partially offsets).

This decomposition has clear strategic implications: multi-target moves are individually beneficial but impose negative externalities on partners. A coordinator who accounts for interference would use multi-target moves more selectively than a naive analysis (ignoring spillovers) would suggest.

Lance recalls his Dragonite. "SUTVA is a convenient fiction. In a world where units interact — and they always do — you must account for the connections between them. Ignore interference at your peril."

He steps aside, and the final door swings wide. Beyond it, a familiar figure waits, arms crossed, smirking.

8.5 Regression Kink Design

Between the Elite Four and the Champion's chamber, there is a quiet corridor lined with bookshelves. A League librarian intercepts you. "Before you face the Champion, allow me to share a few more tools. You may need them."

Identification from Kinks

The Regression Discontinuity Design (Chapter 6) exploits a **jump** in the probability of treatment at a threshold. But what if treatment does not jump — instead, the *intensity* of treatment changes slope?

A **Regression Kink Design (RKD)** exploits a kink (change in slope) in the relationship between a running variable R and treatment intensity D at a threshold r_0 (Card, Lee, Pei & Weber, 2015). Unlike RDD, where treatment is binary and jumps at the threshold, RKD applies when treatment is continuous and its relationship to the running variable has a slope change.

The key identification result is:

$$\tau_{RKD} = \frac{\lim_{r \downarrow r_0} \frac{dE[Y|R=r]}{dr} - \lim_{r \uparrow r_0} \frac{dE[Y|R=r]}{dr}}{\lim_{r \downarrow r_0} \frac{dE[D|R=r]}{dr} - \lim_{r \uparrow r_0} \frac{dE[D|R=r]}{dr}}$$

This is the ratio of the change in slope of the outcome to the change in slope of the treatment at the kink point.

Pokemon Application

Consider the Kanto League training program, which provides subsidized training hours to trainers based on their performance score R. Below the threshold $r_0 = 50$, trainers receive a flat allocation of 10 hours per week. Above r_0, hours increase linearly with the score: $D(r) = 10 + 0.5(r - 50)$ for $r > 50$. There is no jump in training hours at the threshold — only a change in slope from 0 to 0.5.

To estimate the effect of marginal training hours on battle performance, we estimate the change in slope of outcomes at r_0 and divide by the change in slope of treatment (0.5). If the outcome slope changes by 1.2 win-rate percentage points at the kink, the estimated effect of one additional training hour is $1.2/0.5 = 2.4$ percentage points.

Bunching Estimators

A related method is the **bunching estimator** (Saez, 2010; Kleven & Waseem, 2013), which exploits excess mass at kink points in the density of the running variable. If agents optimize against a kinked budget constraint (e.g., a tax schedule), bunching at the kink reveals the behavioral response. While primarily used in public finance, the logic extends to any setting where agents respond to kinks in incentive schedules.

Validity and Diagnostics

RKD requires smoothness of potential outcomes and the density of R at the kink point — analogous to RDD validity conditions but applied to derivatives rather than levels. Key diagnostics include testing for a kink in the density of R (there should be none if the running variable is not manipulated) and checking for kinks in predetermined covariates at r_0.

8.6 Bounds and Partial Identification

What Can We Learn Without Assumptions?

Manski (1990, 2003) posed a profound question: if we are unwilling to make any assumptions about the relationship between treatment assignment and potential outcomes, what can we still learn? The answer is **partial identification** — we cannot point-identify the treatment effect, but we can bound it.

Worst-Case Bounds

For a binary treatment D and outcome $Y \in [y_{min}, y_{max}]$:

The mean potential outcome under treatment is bounded by:

$$E[Y(1)|D = 1] \cdot P(D = 1) + y_{min} \cdot P(D = 0) \leq E[Y(1)] \leq E[Y(1)|D = 1] \cdot P(D = 1) + y_{max} \cdot P(D = 0)$$

The first term, $E[Y(1)|D = 1] \cdot P(D = 1)$, is identified from data (the observed mean outcome among the treated, weighted by the probability of treatment). The second term replaces the *unidentified* component — $E[Y(1)|D = 0]$, which we cannot observe — with its worst-case values (y_{min} and y_{max}).

Similarly for the untreated potential outcome:

$$E[Y(0)|D = 0] \cdot P(D = 0) + y_{min} \cdot P(D = 1) \leq E[Y(0)] \leq E[Y(0)|D = 0] \cdot P(D = 0) + y_{max} \cdot P(D = 1)$$

The bounds on the ATE are obtained by subtracting:

$$ATE \in [E[Y|D = 1]P(D = 1) + y_{min}P(D = 0) - E[Y|D = 0]P(D = 0) - y_{max}P(D = 1), \text{ upper analog}]$$

Monotone Treatment Response

The worst-case bounds are typically very wide. Manski and Pepper (2000) showed that adding the assumption of **monotone treatment response (MTR)** — $Y_i(1) \geq Y_i(0)$ for all i (treatment cannot hurt) — can substantially tighten bounds.

Under MTR, $E[Y(1)|D = 0] \geq E[Y(0)|D = 0]$ (the treated potential outcome for the untreated is at least as large as their observed control outcome). This replaces the y_{min} lower bound with the observed $E[Y|D = 0]$, dramatically narrowing the identified set.

Lee Bounds for Sample Selection

Lee (2009) addressed the problem of **sample selection**: when the outcome is only observed for a selected subset, and selection is potentially affected by treatment. For example, wages are observed only for employed workers, and a job training program may affect employment.

Lee bounds require only **monotonicity of selection** — the treatment can only increase (or only decrease) the probability that the outcome is observed. Under this assumption, bounds on the treatment effect are obtained by trimming the group with higher selection rates to equalize selection across treatment and control.

Formally, if treatment increases selection (more treated units have observed outcomes), the trimming proportion is:

$$p_0 = 1 - \frac{P(\text{selected}|D = 0)}{P(\text{selected}|D = 1)}$$

Trim the top or bottom p_0 fraction of the treatment group's outcome distribution to get upper and lower bounds.

When Bounds Are Useful

Wide bounds may seem uninformative, but they can still rule out interesting hypotheses. If the entire identified set is positive, we can conclude that the treatment effect is beneficial even without point identification. If the bounds for a cheap intervention exclude zero, the policy implication may be clear even though we do not know the exact magnitude.

> ***Professor Oak Explains: The Virtue of Honesty***
>
> *Partial identification represents a philosophical stance: it is better to report what the data can actually support than to manufacture precision through strong (and potentially false) assumptions. As Manski (2003) wrote: "Better to have wide bounds that are honest than narrow confidence intervals that are based on incredible assumptions." When you report bounds, you are saying: "Given what I know for certain, the answer lies in this range." That is a powerful statement.*

8.7 Causal Discovery

Can We Learn the DAG from Data?

Throughout this textbook, we have *assumed* the causal graph (DAG) based on domain knowledge and theoretical reasoning. But what if we could **learn** the causal structure directly from data? This is the goal of **causal discovery** (or causal structure learning).

The motivation is clear: in complex systems with many variables, the correct DAG may not be obvious. If observational data contain enough information to infer at least some causal relationships, this could guide both research design and analysis.

The PC Algorithm

The **PC algorithm** (named after its inventors, Peter Spirtes and Clark Glymour; Spirtes, Glymour & Scheines, 2000) is a constraint-based method that uses conditional independence tests to infer the causal graph.

Step 1: Start with a complete undirected graph. Connect every pair of variables with an edge.

Step 2: Test pairwise independences. For each pair (X, Y), test whether $X \perp\!\!\!\perp Y$. If independent, remove the edge.

Step 3: Test conditional independences. For each remaining edge (X, Y), test whether $X \perp\!\!\!\perp Y \mid Z$ for each possible conditioning set Z. If conditionally independent given some Z, remove the edge and record Z as the *separating set*.

Step 4: Orient edges. Use the separating sets to identify v-structures (colliders). If $X - Z - Y$ with no edge between X and Y, and Z is *not* in the separating set of (X, Y), then orient as $X \to Z \leftarrow Y$. Apply additional orientation rules to propagate directions.

The FCI Algorithm

The PC algorithm assumes **causal sufficiency** — no unobserved common causes. When latent variables may exist, the **FCI (Fast Causal Inference) algorithm** (Spirtes et al., 2000) extends PC to handle hidden confounders. FCI produces a **Partial Ancestral Graph (PAG)**, which represents an equivalence class of graphs that are compatible with the observed conditional independences and the possible presence of latent variables.

GES: Score-Based Approach

An alternative to constraint-based methods is the **Greedy Equivalence Search (GES)** (Chickering, 2002). GES optimizes a score (e.g., BIC) over the space of DAG equivalence classes:

1. **Forward phase**: start with an empty graph and greedily add edges that most improve the score.
2. **Backward phase**: greedily remove edges that improve the score.

GES is consistent (recovers the true equivalence class in the large-sample limit) under the faithfulness assumption and is often more statistically powerful than constraint-based methods.

The Faithfulness Assumption

All causal discovery methods rely on the **faithfulness assumption**: every conditional independence in the data reflects a structural feature of the DAG (specifically, d-separation). Equivalently, there are no "accidental" cancellations of causal paths that produce conditional independences not implied by the graph.

Faithfulness can fail when two causal paths have effects that exactly cancel (e.g., $X \to Y$ with coefficient $+1$ and $X \to M \to Y$ with combined coefficient -1, yielding $X \perp\!\!\!\perp Y$ despite two active causal paths). Such cancellations are measure-zero in parameter space but can occur in practice.

Markov Equivalence Classes

A fundamental limitation of causal discovery from observational data: multiple DAGs can produce the same set of conditional independences. These observationally indistinguishable DAGs form a **Markov equivalence class**.

For example, $X \to Y$ and $X \leftarrow Y$ produce identical observational distributions (both imply $X \not\perp\!\!\!\perp Y$ and nothing else). Without additional assumptions (e.g., temporal ordering, experimental data, functional form restrictions), we cannot distinguish the causal direction.

Markov equivalence classes can be represented compactly as **Completed Partially Directed Acyclic Graphs (CPDAGs)**, where directed edges indicate orientations shared by all members of the class, and undirected edges indicate orientations that vary.

Pokemon Application

Suppose we observe five trainer statistics: Hours Trained, Strategy Score, Team Level, Badges, and Confidence. We apply the PC algorithm with $\alpha = 0.05$ conditional independence tests.

The algorithm returns the following CPDAG:

- $\text{Hours Trained} \to \text{Team Level}$ (directed: more training raises levels)
- $\text{Hours Trained} - \text{Strategy Score}$ (undirected: cannot determine direction)
- $\text{Strategy Score} \to \text{Badges}$ (directed: part of a v-structure)
- $\text{Team Level} \to \text{Badges}$ (directed: part of a v-structure)
- $\text{Badges} \to \text{Confidence}$ (directed: oriented by propagation rule)

The v-structure $\text{Strategy Score} \to \text{Badges} \leftarrow \text{Team Level}$ is identified because $\text{Strategy Score} \perp\!\!\!\perp \text{Team Level}$ marginally, but $\text{Strategy Score} \not\perp\!\!\!\perp \text{Team Level} \mid \text{Badges}$ — the hallmark of a collider.

> ***Professor Oak Explains: The Limits of Discovery***
> *Causal discovery is a powerful complement to domain knowledge, but it is not a replacement. The methods tell us which graphs are consistent with the data — they cannot uniquely determine the true graph from observational data alone. Always combine algorithmic output with substantive knowledge. If the algorithm tells you "Hours Trained causes Strategy Score," but you know that strategic trainers also train more, the undirected edge is honest: the data alone cannot resolve this.*

8.8 Causal Inference with Machine Learning

The corridor opens into a vast library. Shelves stretch to the ceiling, filled with volumes on semiparametric theory, empirical process theory, and high-dimensional statistics. A sign reads: "The Machine Learning Wing."

The Promise and the Peril

Machine learning methods — random forests, neural networks, gradient boosting, LASSO — are extraordinary at prediction. But naive application of ML to causal inference fails for a subtle reason: **regularization bias**.

When we use LASSO to estimate a treatment effect, the penalty that improves prediction accuracy also shrinks the treatment coefficient toward zero. This bias does not vanish as $n \to \infty$ because the regularization is applied to the parameter of interest. The result is biased estimates with invalid confidence intervals.

The solution is not to abandon ML but to use it *carefully*, in ways that separate the ML estimation of nuisance parameters (confounders, propensity scores) from the estimation of the causal parameter of interest.

Double/Debiased Machine Learning (DML)

Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey, and Robins (2018) developed **Double/Debiased Machine Learning (DML)**, a general framework for using ML in causal inference while maintaining valid statistical inference.

The key ideas:

1. **Neyman orthogonalization.** Construct a **moment condition** for the causal parameter that is **orthogonal** to (insensitive to) small errors in nuisance parameter estimation. This is the "debiasing" step.
2. **Cross-fitting.** Split the sample into K folds. For each fold k, estimate nuisance parameters (propensity score, outcome model) on the other $K - 1$ folds, then compute the causal estimate on fold k. Average across folds. This avoids overfitting bias.

The partially linear model. Consider:

$$Y = D\theta_0 + g_0(X) + U, \quad E[U|D, X] = 0$$

$$D = m_0(X) + V, \quad E[V|X] = 0$$

where θ_0 is the causal parameter, $g_0(X)$ is the (potentially complex) confounding function, and $m_0(X)$ is the propensity score. The DML procedure:

1. Estimate $\hat{g}$ and $\hat{m}$ using any ML method (random forest, neural net, etc.) via cross-fitting.
2. Compute residuals: $\tilde{Y}_i = Y_i - \hat{g}(X_i)$ and $\tilde{D}_i = D_i - \hat{m}(X_i)$.
3. Estimate θ_0 by regressing $\tilde{Y}$ on $\tilde{D}$:

$$\hat{\theta}_{DML} = \left(\frac{1}{n}\sum_i \tilde{D}_i^2\right)^{-1} \frac{1}{n}\sum_i \tilde{D}_i \tilde{Y}_i$$

Under regularity conditions, $\hat{\theta}_{DML}$ is $\sqrt{n}$-consistent and asymptotically normal, even when $\hat{g}$ and $\hat{m}$ converge at slower-than-$\sqrt{n}$ rates. This is the power of Neyman orthogonality: first-order errors in nuisance estimation do not contaminate the causal estimate.

> ***Blue's Mistake: Naive ML***
>
> *Blue trains a gradient-boosted model to predict badge count from all available features, including an Exp. Share indicator. He reports the coefficient on Exp. Share as the "causal effect." But the regularization in gradient boosting shrinks all coefficients, including the treatment effect, and the model optimizes for prediction accuracy, not causal estimation. The coefficient is biased, and the standard error from the boosting model is meaningless for inference. Blue needs DML.*

Targeted Learning (TMLE)

Targeted Maximum Likelihood Estimation (TMLE), developed by van der Laan and Rose (2011), takes a different approach to combining ML with causal inference:

1. **Initial estimate.** Use ML (Super Learner — an ensemble method) to estimate the outcome model $E[Y|D, X]$.
2. **Clever covariate.** Compute the "clever covariate" $H(D, X) = \frac{D}{\hat{e}(X)} - \frac{1-D}{1-\hat{e}(X)}$, where $\hat{e}(X)$ is the estimated propensity score.
3. **Targeting step.** Fit a logistic regression of Y on $H(D, X)$ with offset $\text{logit}(\hat{Q}(D, X))$, where $\hat{Q}$ is the initial outcome model. This "tilts" the initial estimate to optimize the bias-variance tradeoff for the specific causal parameter.
4. **Substitution estimator.** Compute the ATE by averaging the updated predictions under treatment and control.

TMLE achieves the **semiparametric efficiency bound** — no regular estimator can have smaller asymptotic variance. It is also **doubly robust**: consistent if either the outcome model or the propensity score model is correctly specified.

Entropy Balancing and CBPS

Two methods that use optimization to improve covariate balance deserve mention:

Entropy Balancing (Hainmueller, 2012) finds weights for the control group that exactly match specified moments (means, variances, skewness) of the treatment group's covariate distribution. Unlike propensity score methods, which achieve balance *approximately* through modeling the treatment assignment, entropy balancing achieves exact balance by construction, subject to the specified moments.

Covariate Balancing Propensity Score (CBPS) (Imai & Ratkovic, 2014) estimates the propensity score model by jointly optimizing for correct specification of the treatment model *and* covariate balance. This addresses the disconnect in standard propensity score methods, where the propensity score is estimated to predict treatment (not to balance covariates), and good prediction does not guarantee good balance.

Automatic Debiased Machine Learning

Chernozhukov, Newey, and Singh (2022) extend DML to a broader class of causal parameters — not just average treatment effects, but any functional of the data distribution that can be expressed as a linear functional of a conditional expectation. Their **Automatic Debiased Machine Learning (AutoDML)** framework automatically constructs the Riesz representer needed for debiasing, using ML methods to learn the debiasing correction itself. This makes the framework applicable to a wide range of causal estimands (CATE, dose-response curves, policy effects) without requiring the researcher to manually derive the orthogonal moment condition for each new parameter.

8.9 Transportability & External Validity

A map on the library wall catches your eye. It shows not just Kanto but the region to the west: **Johto.** *A label reads: "Just because it works in Kanto doesn't mean it works in Johto."*

The Problem of External Validity

Suppose we have conducted a rigorous randomized experiment in the Kanto region and estimated that Exp. Share increases badge count by 1.5 badges. A League official in Johto asks: can we apply this finding to Johto trainers?

The answer depends on *why* Johto might differ from Kanto. If the treatment effect is the same in both regions, the estimate transports directly. But Johto has different gyms, different Pokemon distributions, different trainer demographics, and different item availability. The treatment effect could plausibly differ.

This is the problem of **transportability** or **external validity**: when can causal conclusions from one population (the source) be applied to another (the target)?

Selection Diagrams

Bareinboim and Pearl (2013) formalized transportability using **selection diagrams** — DAGs augmented with special nodes S (selection variables) that indicate where the source and target populations differ. An S-node pointing into a variable indicates that this variable's mechanism differs between populations.

For example, if Kanto and Johto differ in trainer demographics (X) but the causal mechanisms $D \to Y$ and $X \to Y$ are the same in both regions, the selection diagram has $S \to X$ but no S-nodes on other variables. In this case, the treatment effect is transportable after reweighting for the difference in covariate distributions.

Formal Transportability Conditions

A causal effect is **directly transportable** if the selection variables do not affect any variable on the causal pathway from treatment to outcome (after conditioning on measured covariates). More generally, Pearl and Bareinboim (2011) provide a complete algorithm for determining whether a causal effect identified in the source can be expressed in terms of quantities estimable in the target.

The key insight: transportability fails when the mechanism by which treatment affects outcome *itself* differs between populations, and we cannot condition on the relevant effect modifiers.

Reweighting for External Validity

When transportability holds after adjusting for covariate differences, the simplest approach is **reweighting** (Stuart, Cole, Bradshaw & Leaf, 2011; Tipton, 2013):

$$\mathrm{ATE}_{\mathrm{target}} = \sum_x \mathrm{CATE}(x) \cdot P_{\mathrm{target}}(X = x) = E_{\mathrm{source}}\left[\mathrm{CATE}(X) \cdot \frac{P_{\mathrm{target}}(X)}{P_{\mathrm{source}}(X)}\right]$$

In practice, we estimate the density ratio $P_{\mathrm{target}}(X)/P_{\mathrm{source}}(X)$ using logistic regression (predict source vs. target membership) and reweight the source-estimated CATE.

Site Selection Bias in Multi-Site Experiments

A related problem arises in **multi-site experiments**, where a treatment is evaluated at multiple sites (schools, hospitals, regions) but the sites are not randomly selected. If sites that participate in the study are systematically different from those that do not, the average effect across study sites may not represent the effect in the broader population. Allcott (2015) and Gechter (2024) discuss methods for correcting site selection bias.

Worked Example: Kanto to Johto

The Kanto Exp. Share experiment estimates $\hat{\tau}_{\text{Kanto}} = 1.5$ badges (ATE). We have covariate data from both regions. Key differences:

Variable	Kanto Mean	Johto Mean
Trainer experience (years)	3.2	2.1
Team diversity (types)	3.8	4.5
Urban/rural (% urban)	62%	41%

Using the CATE estimates from our causal forest (Section 8.3) and reweighting to the Johto distribution, we obtain:

$$\hat{\tau}_{\text{Johto}} = \sum_{i \in \text{Kanto}} \hat{\tau}(X_i) \cdot \hat{w}_i = 1.9 \text{ badges}$$

The transported effect is *larger* than the Kanto estimate, driven by Johto's higher team diversity (which, as we found in Section 8.3, is associated with larger Exp. Share benefits). This illustrates a key lesson: external validity is not just about whether effects "generalize" but about *how* they change across populations.

> ***Professor Oak Explains: The Limits of Transportability***
>
> *Transportability requires that we know what differs between populations and that these differences are captured in measured covariates. If Johto gyms have fundamentally different battle mechanics that interact with Exp. Share in unmeasured ways, no amount of covariate reweighting will produce the correct transported effect. Transportability is a property of the causal structure, not just the statistical method.*

8.10 Dynamic Treatment Regimes

Time-Varying Treatments

All methods in Chapters 1–7 consider a single treatment at a single point in time. But many real-world treatments are **dynamic** — administered sequentially over time, with each treatment decision depending on evolving patient (or trainer) status.

In the Elite Four challenge, a trainer faces four sequential battles: Lorelei, Bruno, Agatha, and Lance. Before each battle, the trainer decides whether to use healing items ($D_t \in \{0, 1\}$ for $t = 1, 2, 3, 4$). The outcome Y is whether the trainer defeats all four. Crucially, the **health of the trainer's team** entering battle t (L_t) is both:

1. A **confounder** of the $D_t \rightarrow Y$ relationship (sicker teams benefit more from items).
2. **Affected by prior treatment** ($D_{t-1} \rightarrow L_t$): using items in battle $t - 1$ improves health entering battle t.

This creates a vicious cycle: L_t is a time-varying confounder affected by prior treatment. Standard regression adjustment for L_t introduces **collider bias** (blocking the $D_{t-1} \rightarrow L_t \rightarrow Y$ pathway), while failing to adjust for L_t leaves confounding bias. Neither adjusting nor not adjusting produces the correct answer.

> ***Blue's Mistake: Adjusting for Post-Treatment Variables (Again)***
>
> *Blue controls for team health at every stage in a regression and finds no effect of item use. "Items don't work!" he announces. But by conditioning on L_t (health entering battle t), Blue has blocked the very pathway through which prior item use operates. This is the time-varying version of the bad controls problem from Chapter 3. The solution requires specialized methods that handle time-varying confounders.*

Marginal Structural Models (MSM)

Robins, Hernan, and Brumback (2000) introduced **Marginal Structural Models** to handle time-varying confounding. The key idea is **inverse probability of treatment weighting (IPTW)** applied to the entire treatment *history*.

For each individual i at each time t, compute the probability of their actual treatment given their history:

$$w_i = \prod_{t=1}^{T} \frac{1}{P(D_{it} = d_{it} \mid \bar{D}_{i,t-1}, \bar{L}_{i,t}, X_i)}$$

where $\bar{D}_{i,t-1}$ is the treatment history up to $t - 1$ and $\bar{L}_{i,t}$ is the covariate history up to t. These weights create a pseudo-population in which time-varying confounders no longer predict treatment — analogous to what randomization achieves in a single-period setting.

In the weighted pseudo-population, a simple regression of Y on the treatment history (parameterized as a marginal structural model, e.g., $E^w[Y|\bar{D}] = \beta_0 + \beta_1 \sum_t D_t$) yields consistent estimates of the causal effect of the treatment regime.

Stabilized weights replace the numerator 1 with $P(D_{it} \mid \bar{D}_{i,t-1}, X_i)$ — the probability of treatment given history but *not* conditioning on time-varying confounders. This reduces weight variability without introducing bias.

G-Computation (G-Formula)

Robins (1986) introduced the **g-computation algorithm** (also called the g-formula), which identifies the effect of a dynamic treatment regime by iterating expectations backward through time:

$$E[Y(\bar{d})] = \sum_{l_T} \cdots \sum_{l_1} E[Y|\bar{D} = \bar{d}, \bar{L} = \bar{l}] \prod_{t=1}^{T} P(L_t = l_t|\bar{D}_{t-1} = \bar{d}_{t-1}, \bar{L}_{t-1} = \bar{l}_{t-1})$$

G-computation models the full joint distribution of outcomes and confounders under a specified treatment regime. It requires correct specification of all conditional distributions but does not suffer from extreme weights.

Structural Nested Models

Structural Nested Mean Models (SNMMs) (Robins, 1994) provide a third approach. They model the causal effect of treatment at each time point, conditional on the past. The "blip function" $\gamma_t(\bar{d}_{t-1}, \bar{l}_t) = E[Y(\bar{d}_t, 0_{t+1:T}) - Y(\bar{d}_{t-1}, 0, 0_{t+1:T})|\bar{D}_{t-1} = \bar{d}_{t-1}, \bar{L}_t = \bar{l}_t]$ captures the effect of treatment at time t for individuals with a specific history. Estimation proceeds via g-estimation.

Optimal Dynamic Treatment Regimes

Given estimates of treatment effects at each stage, we can search for the **optimal dynamic treatment regime** — a sequence of decision rules $(\pi_1, \pi_2, \ldots, \pi_T)$ where $\pi_t(\bar{l}_t, \bar{d}_{t-1})$ specifies treatment at time t as a function of the history. Methods include:

- **Q-learning**: backward induction starting from the final stage, estimating the optimal action at each stage given future optimal behavior (Murphy, 2005).
- **Outcome-weighted learning**: directly maximizes a value function using classification methods (Zhao et al., 2012).

Pokemon Application

In the Elite Four challenge, we model item use across four sequential battles. Using an MSM with stabilized IPTW weights, we estimate:

Regime	$E[Y(\bar{d})]$ **(prob. of full sweep)**
Never use items $(0, 0, 0, 0)$	0.22
Always use items $(1, 1, 1, 1)$	0.61
Use only before battles 3 & 4 $(0, 0, 1, 1)$	0.54
Optimal dynamic regime	0.68

The optimal dynamic regime, estimated via Q-learning, prescribes: use items before battle t if average team HP entering the battle is below 55% *or* if the upcoming opponent has a type advantage. This personalized rule outperforms even "always use items" because it conserves items for situations where they are most needed, and the g-formula reveals that the marginal benefit of items is much larger when entering a battle with low HP ($\Delta = 0.25$) versus high HP ($\Delta = 0.05$).

8.11 The Champion Battle: Blue's Causal Fallacies

#018 Pidgeot *#065 Alakazam* *#112 Rhydon* *#103 Exeggutor* *#059 Arcanine* *#006 Charizard*

Blue's Champion team — the six fallacies embodied.

Rival Blue, the Champion

You step through the final door and into a blaze of light. The Champion's chamber is a grand arena, cameras flashing, a crowd roaring. And there, standing at the far end with his arms crossed and that infuriating smirk, is Blue.

"So you made it," he says. "But can you beat me? I've studied the data. I know everything about causation."

Professor Oak's voice crackles through the arena speakers: "Six rounds. In each round, Blue will make a causal claim. Identify the fallacy. Propose the correct method. Score four out of six to earn the Championship."

The stadium lights focus. The battle begins.

Round 1: The Starter Selection

Blue's Claim: "I analyzed 5,000 League records. Trainers who chose Squirtle average 6.8 badges. Charmander trainers average 5.9 badges. Squirtle is objectively the best starter."

The Fallacy: Confounding.

This is the original sin of causal inference — the mistake Blue has been making since Chapter 1. Trainers do not randomly choose starters. Squirtle is disproportionately chosen by trainers from Cerulean City, who come from wealthier families with better training resources. The association between Squirtle and badges is confounded by socioeconomic background.

The Correct Approach: To estimate the causal effect of starter choice, we need either a randomized experiment (randomly assign starters to trainers) or an observational study that adjusts for confounders. A DAG reveals that wealth, hometown, and prior Pokemon experience are common causes of both starter choice and badge count. Using methods from Chapters 4–5 (matching, propensity score weighting, or doubly robust estimation), we can adjust for these confounders. The adjusted effect of Squirtle vs. Charmander may be much smaller — or zero.

Blue's Squirtle takes a hit. Five rounds remain.

Round 2: The Elite Four Paradox

Blue's Claim: "Among trainers who made it to the Elite Four, those who trained the hardest actually have *lower* win rates. Hard work doesn't pay off at the highest level."

The Fallacy: Collider Bias (Selection Bias).

"Making it to the Elite Four" is a **collider** — it is caused by both hard work *and* natural talent. Conditioning on this collider (restricting the sample to Elite Four challengers) induces a spurious negative association between hard work and talent. Among those who made it, the hard workers tend to be less talented (they compensated with effort), and the less hard-working tend to be more talented (they made it on ability alone). The negative association between hard work and win rate within this selected group is an artifact of collider bias, not evidence that hard work is harmful.

The Correct Approach: Analyze the relationship between training effort and win rate in the *full* population, not the selected subsample. Alternatively, explicitly model the selection process using methods like Heckman correction (Chapter 5 extension), Lee bounds (Section 8.6), or principal stratification.

Blue's Pidgeot goes down.

Round 3: The Cave Training Anecdote

Blue's Claim: "I trained in Cerulean Cave for a month and then won the Championship. Cave training works — I'm living proof."

The Fallacy: Selection Bias + Anecdotal Evidence.

Blue is a single observation — a sample size of one. His experience tells us nothing about the *average* effect of cave training. Moreover, Blue selected himself into cave training; he is not a random trainer. He is the grandson of Professor Oak, has access to exceptional resources, and possesses an extraordinary Pokemon team. His success may have nothing to do with cave training.

The Correct Approach: To estimate the effect of cave training, we need a controlled study. At minimum, a matched comparison with similar trainers who did not train in the cave. Ideally, a randomized experiment or a quasi-experimental design (e.g., exploiting a natural experiment such as random cave closures as an instrument for cave training access).

Blue's Alakazam faints.

Round 4: The Potion Paradox

Blue's Claim: "My data shows that trainers who use more Potions during battles have *more* losses. Potions cause losing."

The Fallacy: Reverse Causality.

The causal arrow runs the wrong way. Trainers use Potions *because* they are losing (their Pokemon are taking damage), not the other way around. Potion use is a *consequence* of battle difficulty, not a *cause* of losses. The observed correlation between Potion use and losses is real, but the causal interpretation is backward.

The Correct Approach: To estimate the causal effect of Potion use on battle outcomes, we need to address the endogeneity. An instrument (e.g., random variation in Potion prices across PokeMarts) could break the reverse causality. Alternatively, a randomized experiment where trainers are assigned Potion strategies before battles begin would identify the causal effect.

Blue's Arcanine falls.

Round 5: Simpson's Gym Paradox

Blue's Claim: "I checked the data: Water types beat Rock types at a higher overall rate than Fire types do. But weirdly, Fire types do better than Water types against Rock types in *every single gym*. The overall data must be wrong — Water types are clearly better."

The Fallacy: Simpson's Paradox.

This is a textbook case of Simpson's Paradox (Chapter 3). The overall association reverses when conditioning on gym. The explanation: Water-type trainers disproportionately challenge gyms where Rock types are rarer and weaker (giving them a lower overall encounter rate with strong Rock types), while Fire-type trainers disproportionately challenge Rock-heavy gyms (Pewter City, Victory Road). Within any given gym, Fire types perform better, but the aggregated data reverse the relationship because of the non-random distribution of gym selection.

The Correct Approach: The correct comparison depends on the causal question. If we want to know "Which type is better at beating Rock types in a specific matchup?", the gym-specific (conditional) analysis is correct. The aggregated analysis is misleading because it is confounded by gym selection. A DAG clarifies: Gym Choice is a common cause of both Type Matchup and Win Rate.

Blue's Exeggutor is defeated.

Round 6: The Badge Control

Blue's Claim: "I ran a regression of Elite Four performance on Exp. Share use, controlling for number of badges, team level, win rate, and total battles fought. After controlling for everything, Exp. Share has no significant effect. It's useless."

The Fallacy: Bad Controls (Post-Treatment Bias).

Badges, team level, win rate, and total battles are all **post-treatment** variables — they are *caused by* Exp. Share use. Controlling for them blocks the causal pathways through which Exp. Share operates. Of course the coefficient on Exp. Share shrinks to zero: Blue has adjusted away the very mechanism of the treatment.

The DAG is: $\text{Exp. Share} \to \text{Team Level} \to \text{Badges} \to \text{Elite Four Performance}$. Controlling for Team Level and Badges removes the indirect effect entirely.

The Correct Approach: Only control for **pre-treatment** covariates: trainer experience prior to receiving Exp. Share, starting Pokemon, hometown, and other baseline characteristics. Never control for variables that lie on the causal pathway from treatment to outcome. If you want to understand the *mechanism* (does Exp. Share work through team level?), use the mediation analysis framework from Section 8.1 — do not simply "control for" the mediator.

Blue's final Pokemon — his starter, that Squirtle from Chapter 1, now a Blastoise — falls.

The arena erupts. Professor Oak's voice booms: "Six for six. Flawless."

Blue stands frozen for a moment. Then, slowly, he uncrosses his arms. "I… I was making the same mistakes the whole time, wasn't I? Confounding. Colliders. Reverse causality. Bad controls." He pauses. "I was so focused on the data that I forgot to think about the structure."

Oak appears at the edge of the arena. He places a hand on Blue's shoulder. "The data never speaks for itself. It always needs a causal framework to interpret it." He turns to you. "And you've shown that you have one."

He opens a case. Inside: a golden badge, larger than the others, engraved with a stylized DAG.

"You are the Causal Inference Champion of the Kanto Region."

8.12 Further Reading: The Frontier Beyond

The topics in this chapter are active research frontiers. Each one could fill an entire textbook. Here, we point you toward the cutting edge.

Causal Representation Learning and Causal Abstraction

Traditional causal inference assumes that the relevant variables are already defined and measured. But what if we need to *learn* the right causal variables from raw data (images, text, time series)? **Causal representation learning** (Scholkopf et al., 2021) aims to recover causal variables and their relationships from high-dimensional observations. **Causal abstraction** (Beckers & Halpern, 2019; Rubenstein et al., 2017) studies how causal models at different levels of granularity relate to each other — a DAG of molecular interactions and a DAG of symptoms may describe the same system at different scales.

Causal Inference for LLMs and AI Fairness

Large language models (LLMs) are increasingly used in decision-making. Causal reasoning is essential for understanding when LLM outputs are reliable and when they reflect spurious correlations in training data. **Algorithmic fairness** relies heavily on causal definitions of discrimination: is a decision influenced by a protected attribute through a causal pathway, or only through legitimate channels? Counterfactual fairness (Kusner et al., 2017) formalizes this: a decision is fair if it would be the same in a counterfactual world where the individual belonged to a different demographic group.

Proximal Causal Inference

What if we have **proxies** for unobserved confounders, even though we do not observe the confounders themselves? **Proximal causal inference** (Tchetgen Tchetgen, Ying, Cui, Shi & Miao, 2024; Miao, Geng & Tchetgen Tchetgen, 2018) provides identification results using two types of proxies: a treatment-inducing proxy (related to the confounder and the treatment) and an outcome-inducing proxy (related to the confounder and the outcome). Under conditions on these proxies, the ATE is identified even in the presence of arbitrarily complex unobserved confounding. This is a powerful relaxation of the selection-on-observables assumption.

Negative Controls

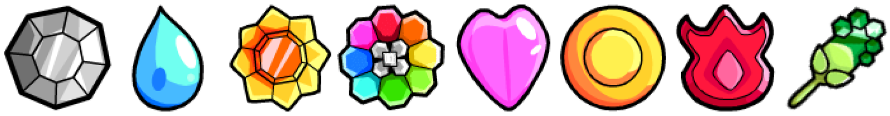
Negative control outcomes and **negative control exposures** (Lipsitch, Tchetgen Tchetgen & Cohen, 2010) provide a simple but powerful diagnostic for unobserved confounding. A negative control outcome is a variable that is affected by the confounder but *not* by the treatment. If the treatment appears to affect the negative control outcome, confounding is likely present. Similarly, a negative control exposure is a variable that affects the outcome only through the confounder. These "placebo tests" are widely used in epidemiology and are gaining traction in economics and social science.

Where to Go Next

- **Mediation**: Imai, Keele, and Tingley (2010); VanderWeele (2015), *Explanation in Causal Inference*.
- **Sensitivity Analysis**: Rosenbaum (2002), *Observational Studies*; Cinelli and Hazlett (2020), "Making Sense of Sensitivity."
- **Heterogeneous Effects**: Athey and Imbens (2016); Wager and Athey (2018); Chernozhukov et al. (2020).
- **Interference**: Hudgens and Halloran (2008); Aronow and Samii (2017).
- **Partial Identification**: Manski (2003), *Partial Identification of Probability Distributions*; Tamer (2010).
- **Causal Discovery**: Spirtes, Glymour, and Scheines (2000), *Causation, Prediction, and Search*; Peters, Janzing, and Scholkopf (2017), *Elements of Causal Inference*.
- **DML**: Chernozhukov et al. (2018); **TMLE**: van der Laan and Rose (2011).
- **Transportability**: Pearl and Bareinboim (2014), "External Validity."
- **Dynamic Regimes**: Robins, Hernan, and Brumback (2000); Murphy (2003); Hernan and Robins (2020), Chapters 19–21.

- **Proximal Inference**: Tchetgen Tchetgen et al. (2024).
- **Causal Representation Learning**: Scholkopf et al. (2021), "Toward Causal Representation Learning."

Chapter Summary



Champion! All eight badges earned.

This chapter — the capstone of your Kanto journey — surveyed the frontier of causal inference, organized around the Elite Four gauntlet:

1. **Mediation Analysis (Lorelei)**: Decomposing total effects into natural direct and indirect effects. The NDE captures the effect of treatment holding the mediator at its control value; the NIE captures the effect operating through the mediator. Identification requires the strong sequential ignorability assumption, and sensitivity analysis is essential.
2. **Sensitivity Analysis (Agatha)**: No observational study is immune to the ghost of unobserved confounding. Rosenbaum bounds (Γ), E-values, Oster bounds (δ), and Manski bounds provide complementary frameworks for assessing how robust findings are to assumption violations.
3. **Heterogeneous Treatment Effects (Bruno)**: The ATE masks individual variation. Causal forests, BLP, GATES, and CLAN reveal who benefits most. Policy learning leverages heterogeneity to design optimal treatment rules.
4. **Interference & Spillovers (Lance)**: When SUTVA fails, standard estimands are undefined. Exposure mappings, partial interference assumptions, and two-stage randomization allow identification of direct and spillover effects.
5. **Regression Kink Design**: Identification from slope changes rather than level jumps, applicable when treatment intensity changes at a threshold.
6. **Bounds and Partial Identification**: Manski's worst-case bounds, monotone treatment response, and Lee bounds for selection. Honest reporting of what the data can and cannot tell us.
7. **Causal Discovery**: Learning DAGs from data via conditional independence testing (PC, FCI) or score optimization (GES). Markov equivalence classes limit what observational data alone can identify.
8. **Causal Inference with ML**: DML provides $\sqrt{n}$-consistent estimates using ML for nuisance parameters. TMLE achieves the semiparametric efficiency bound. Entropy balancing and CBPS optimize covariate balance directly.
9. **Transportability**: Causal effects from one population may not apply in another. Selection diagrams and reweighting formalize when and how to transport findings.
10. **Dynamic Treatment Regimes**: Time-varying treatments with time-varying confounders require specialized methods: MSMs with IPTW, g-computation, or structural nested models. Optimal regimes can be estimated via Q-learning.
11. **The Blue Battle**: Six common causal fallacies — confounding, collider bias, anecdotal evidence, reverse causality, Simpson's paradox, and bad controls — reviewed through a climactic battle format.

Champion's Review: Comprehensive Questions Across All Chapters

These questions span the entire textbook. Use them to test whether you have truly mastered the Kanto causal inference curriculum.

Foundations (Chapters 1–2)

1. Define the individual treatment effect τ_i and explain why it is fundamentally unobservable.
2. Write the selection bias decomposition. Under what conditions does the naive difference-in-means estimator equal the ATT? Under what conditions does it equal the ATE?
3. In Professor Oak's lab, 200 trainers are randomized to receive Exp. Share or not. After six months, you compare average badge counts. Write the formal hypothesis test and explain why randomization eliminates selection bias *in expectation*.

Observational Studies (Chapters 3–5)

4. Draw the DAG for the following: Hometown (H) affects both Starter Choice (D) and Training Resources (R); Starter Choice affects Gym Performance (Y); Training Resources affect Gym Performance. Identify the backdoor path(s) from D to Y. What is the minimal adjustment set?

5. Explain why propensity score matching eliminates bias from observed confounders but not from unobserved confounders. What assumption is required?
6. You estimate the effect of Rare Candy use on battle win rate using (a) OLS with controls, (b) IPW, and (c) doubly robust estimation. You obtain $\hat{\tau}_{OLS} = 0.12$, $\hat{\tau}_{IPW} = 0.08$, and $\hat{\tau}_{DR} = 0.10$. Should you be concerned about the disagreement? What does it suggest about model specification?

Quasi-Experiments (Chapters 6–7)

7. You discover that Kanto assigns free Rare Candies to trainers born on odd-numbered days. Explain how you would use this as an instrument to estimate the effect of Rare Candies on performance. State the three IV assumptions and assess their plausibility.
8. A Pokemon happiness threshold at 220 determines whether Chansey evolves into Blissey. Outline the fuzzy RDD strategy for estimating the effect of evolution on battle performance. What bandwidth selection method would you use?
9. Silph Co. introduces a free training program in Saffron City in January 2024 but not in Celadon City. Using monthly battle data from 2022–2025 for both cities, describe the difference-in-differences design. State the parallel trends assumption and propose a test.
10. Discuss when the Goodman-Bacon decomposition (Chapter 7) reveals problems with TWFE estimation under staggered treatment adoption. Propose an alternative estimator and justify your choice.

Frontiers (Chapter 8)

11. Define the Natural Direct Effect and Natural Indirect Effect using potential outcomes notation. What is the sequential ignorability assumption, and why is it stronger than treatment ignorability alone?
12. You estimate a treatment effect of $RR = 3.0$ using a propensity-score-matched design. Compute the E-value. Interpret it: what strength of unobserved confounder would be needed to explain away the result?
13. A causal forest reveals that the CATE of Exp. Share is -0.5 badges for trainers with concentrated teams (1–2 types) and $+2.5$ badges for trainers with diverse teams (5+ types). Describe how you would use this to construct an optimal treatment assignment rule under a budget constraint that limits treatment to 40% of trainers.
14. In a double battle experiment, you observe that a Pokemon's damage output decreases when its partner uses Earthquake. Is this a SUTVA violation? Define the spillover effect formally.
15. A training program is evaluated in Kanto but a policymaker wants to implement it in Johto. The two regions differ in trainer demographics. Describe the conditions under which the Kanto estimate can be transported. What method would you use to adjust for covariate differences?

Final Trainer Challenge: The Champion's Gauntlet

This multi-part problem integrates methods from across the textbook.

Setting. The Kanto League introduces a new program: "Champion's Mentorship." Selected trainers receive mentoring from former Champions (treatment D). The outcome is performance in the annual League Tournament (Y, continuous score 0–100). The program is offered to 500 trainers; 200 participate and 300 do not. You have data on pre-treatment covariates (X: experience, hometown, starter type, team composition) and post-treatment variables (M: training intensity after mentorship, B: badges earned during the program, L: team level at tournament time).

(a) Identification. Draw the DAG. Is the ATE identified by regression on X alone? What assumption is needed?

(b) Estimation. Estimate the ATE using three methods: (i) propensity score matching, (ii) IPW, (iii) doubly robust estimation. Report the estimates and discuss any discrepancies.

(c) Sensitivity. Compute the E-value for the doubly robust estimate (converted to a risk ratio). Compute the Oster bound at $\delta = 1$. Are your results robust to unobserved confounding?

(d) Heterogeneity. Estimate the CATE using a causal forest. Report the GATES and CLAN results. Who benefits most from the mentorship program?

(e) Mediation. Estimate the NDE and NIE, using training intensity (M) as the mediator. What fraction of the total effect operates through increased training? State the sequential ignorability assumption and assess its plausibility.

(f) Bad Controls. Blue insists on controlling for badges (B) and team level (L). Explain why this is wrong. Show (using your DAG) which causal pathways are blocked by this adjustment.

(g) Dynamic Extension. Now suppose the mentorship occurs in three phases (before badges 1–3, 4–6, and 7–8). Trainers can enter or exit the program at each phase. Describe how you would use a Marginal Structural Model to estimate the effect of the full mentorship regime versus no mentorship.

(h) Transportability. The Johto League wants to implement the same program. Johto trainers are less experienced on average but have more diverse teams. Using your CATE estimates, transport the Kanto effect to the Johto population. Would you expect the effect to be larger or smaller in Johto? Why?

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Skills to Practice in the Notebook

The notebook `notebooks/ch08_indigo_plateau.ipynb` is the capstone. It does not ask you to be an expert in *every* frontier topic — it asks you to *know where each tool lives and be able to run one worked example of each*. Before claiming the Earth Badge and the Champion title, you should be able to do the following:

Mediation (Lorelei):

1. **Compute NDE and NIE on a simulated dataset with a known mediator.** Use either a regression-based approach (Imai-Keele-Tingley) or the `mediation` package. Verify NDE + NIE ≈ Total Effect.

2. **Run a sensitivity analysis for mediation.** Use the ρ parameter (correlation between errors in mediator and outcome equations) to see how robust the NIE is to sequential ignorability violations.

Sensitivity Analysis (Agatha):

3. **Compute an E-value.** For an estimated effect and its CI, compute the minimum confounder strength that would explain it away, using `EValue` (Python/R).
4. **Construct Rosenbaum bounds.** For a matched dataset, compute the range of p-values over $\Gamma \in [1, 3]$ and report the value of Γ at which statistical significance is lost.
5. **Compute Manski worst-case bounds.** Show the gap between upper and lower bounds under no-assumption partial identification.

Heterogeneous Treatment Effects (Bruno):

6. **Fit a causal forest.** Use `econml.CausalForestDML` on a simulated dataset with a known heterogeneous CATE. Plot $\hat{\tau}(x)$ against the true $\tau(x)$.
7. **Run GATES and CLAN.** Group units by predicted $\hat{\tau}(x)$, compute the actual ATE in each quantile, and characterize the top-quintile's covariates.
8. **Estimate a policy value.** Given $\hat{\tau}(x)$, define a treatment rule $\pi(x) = 1\{\hat{\tau}(x) > 0\}$ and evaluate its expected welfare against a uniform rule.

Interference (Lance):

9. **Simulate a spillover experiment.** Construct a network, assign treatment with some saturation level, define an exposure mapping (e.g., "fraction of treated neighbors"), and estimate direct and indirect effects.
10. **Run a two-stage randomization.** Randomize cluster saturations, then randomize within clusters. Estimate direct and spillover effects.

Causal Discovery (§8.7):

11. **Run the PC algorithm** on a simulated dataset and compare the learned DAG to the ground truth. Understand why some edges remain undirected (Markov equivalence class).

Causal Inference + ML (§8.8):

12. **Run DML for the ATE.** Use `econml.DML` or `doubleml` on a high-dimensional dataset. Contrast with naive OLS and verify that DML is less biased and has proper $\sqrt{n}$-rate inference.

Transportability (§8.9):

13. **Transport an effect across populations.** Compute source- and target-population covariate distributions, build density-ratio weights, and reweight the source-population ATE.

Dynamic Regimes (§8.10):

14. **Fit a Marginal Structural Model.** Build IPTW weights from a time-varying treatment model, fit an MSM to estimate the effect of a treatment history, and compare against naive regression.

Synthesis — Blue's Fallacies (§8.11):

15. **Identify which fallacy is which.** For each of Blue's six battles, produce the corrected estimate and explain in one sentence *which* bias Blue committed (confounding, collider, anecdote, reverse causality, Simpson's, bad control).
16. **Complete the Champion's Gauntlet.** The final end-to-end challenge: given a messy real-world-style dataset, design and defend a full analysis from DAG through sensitivity.

You are not expected to master every frontier method — you are expected to recognize each and run a minimum viable example of each.

Check Your Understanding

You are about to become the Champion. Use this list to verify that every section sank in. If you can answer every question and run every task, you know the Kanto causal inference curriculum as well as any working researcher.

Questions you should be able to answer out loud, without notes:

Mediation:

- State the NDE and NIE definitions. Why is the naive "controlling for the mediator" approach biased?
- What is sequential ignorability? Why is it much stronger than standard ignorability?
- Why must mediation estimates *always* be paired with a sensitivity analysis?

Sensitivity:

- What does Rosenbaum's Γ quantify? How do you interpret "significant at $\Gamma = 1.5$"?
- What is an E-value and how is it computed? When would you reach for an E-value instead of Rosenbaum bounds?
- What does Oster's δ measure, and what is the "coefficient stability" intuition?
- State Manski's worst-case bound idea in plain English.

Heterogeneous Effects:

- What is the difference between the ATE and CATE $\tau(x)$?
- Explain honesty in causal forests — why do we split sample-halving into tree-building and estimation?
- What do BLP and GATES each give you, and when is each useful?
- What is a policy $\pi(x)$, and how do you evaluate its welfare?

Interference:

- When does SUTVA fail? Give two concrete Pokemon-themed examples.
- Define direct and spillover effects under partial interference.
- What is an exposure mapping, and why do we need one?
- How does two-stage randomization help you separate direct and spillover effects?

Causal Discovery:

- What does the PC algorithm learn, and what does it *not* learn?
- What is a Markov equivalence class? Why can't observational data uniquely identify a DAG?

Causal Inference with ML:

- Why can't you just plug ML predictions into OLS and call it causal inference?
- What is Neyman orthogonality, and why does DML need it?
- What does TMLE's "targeting" step do?

Transportability:

- Why might a causal effect estimated in Kanto *not* apply in Johto? What distributions must match for an effect to transport?
- How are selection diagrams different from ordinary DAGs?

Dynamic Regimes:

- Why does regressing time-varying outcomes on time-varying treatments with time-varying confounders fail (even if you control for the confounders)?
- What does IPTW fix in dynamic settings, and what assumption does it require?
- What does g-computation compute, and how is it different from MSMs?

Blue's Fallacies (§8.11):

- State the six fallacies and give a one-line corrective for each.
- Pick any fallacy and write its DAG on a napkin. Which path is spurious?

Tasks you should be able to perform in code:

- Compute NDE and NIE from a fitted mediation model.
- Compute an E-value and a Rosenbaum bound for a matched analysis.
- Fit a causal forest, plot $\hat{\tau}(x)$, and run BLP/GATES/CLAN.
- Simulate a partial-interference experiment and estimate direct and spillover effects.
- Run the PC algorithm on a dataset and explain which edges are/aren't directed.
- Run DML (any flavor) on a high-dimensional ATE problem.
- Build density-ratio weights to transport an effect to a target population.
- Fit an MSM with IPTW on a time-varying treatment dataset.
- Diagnose the bias in each of Blue's six fallacies and produce a corrected estimate.

Finish this list, beat Blue, and the Earth Badge — and the Champion title — are yours.

Badge Earned

The League official opens one final case. Inside are eight gleaming badges — and above them, a golden plaque.

> ***Congratulations. You have earned ALL eight Gym Badges and defeated the Champion.***
>
> ***You are the Causal Inference Champion of the Kanto Region.***
>
> *Badge 1: Boulder Badge (Potential Outcomes) — Pallet Town & Pewter City Badge 2: Cascade Badge (Randomized Experiments) — Pewter City Badge 3: Thunder Badge (DAGs & Observational Studies) — Cerulean City Badge 4: Rainbow Badge (Matching & Propensity Scores) — Vermilion City Badge 5: Soul Badge (Regression, IPW & Doubly Robust) — Celadon City Badge 6: Marsh Badge (IV & RDD) — Fuchsia City & Cinnabar Island Badge 7: Volcano Badge (DiD & Synthetic Control) — Saffron City Badge 8: Earth Badge (Advanced Topics & Frontiers) — Indigo Plateau*
>
> ***Champion's Crown: The Champion's Gauntlet*** *— Indigo Plateau*

Post-Credits Scene

The screen fades to black. Then, slowly, a familiar jingle plays — the sound of a video call connecting. Professor Oak's face appears on a screen, his lab visible behind him.

"Ah, Champion! I'm glad I caught you. I've been reviewing some data from the Johto region — you know, the region to the west? They have their own Gyms, their own Pokemon, their own training programs."

He adjusts his glasses.

"The thing is, their trainers are quite different from ours. Different demographics, different team compositions, different battle styles. The Johto League wants to implement some of the programs we've studied here in Kanto — Exp. Share distribution, mentorship programs, training subsidies. But they're worried: will our Kanto results **transport** *to Johto?"*

He leans forward.

"I've been working on something — selection diagrams, reweighting methods, formal transportability conditions. And I've heard rumors of even stranger challenges in Johto: treatments that change over time, interference patterns we've never seen, confounders that are themselves proxied by other variables…"

He smiles.

"I think there's a whole new journey waiting for you, Champion. The methods you've learned aren't just tools for Kanto — they're a framework for thinking about causation wherever you go. But Johto will require you to push further."

The screen flickers.

"Professor Elm in New Bark Town has been expecting you. He has some… interesting data."

Oak winks.

"I hear there are new causal challenges in the Johto Region. And I have a feeling you're exactly the trainer to solve them."

The screen goes dark. The credits roll. Somewhere, a Hoothoot calls into the night.

Your journey continues.

Appendix A: Mathematical Notation & Probability Review — Professor Oak's Math Lab

You arrive at Professor Oak's lab early in the morning, hoping to pick up a new Pokedex upgrade, but instead find the Professor surrounded by chalkboards covered in equations. "Ah, perfect timing!" he says, adjusting his glasses. "Before we send you off to study causal inference, we need to make sure your mathematical toolkit is in order. Think of this as leveling up your stats before a Gym battle — you wouldn't challenge Sabrina without a solid foundation, would you?"

A.1 Sets and Logic

A **set** is a collection of distinct objects. In the Kanto region, we work with sets constantly.

Notation. We write sets with curly braces: $A = \{\text{Bulbasaur}, \text{Charmander}, \text{Squirtle}\}$ is the set of Kanto starters. The symbol $\in$ means "is an element of," so $\text{Pikachu} \notin A$ while $\text{Charmander} \in A$.

Key operations:

Operation	Symbol	Definition	Pokemon Example
Union	$A \cup B$	All elements in A or B (or both)	Fire types $\cup$ Flying types = all Pokemon that are Fire, Flying, or both
Intersection	$A \cap B$	All elements in both A and B	Fire types $\cap$ Flying types = $\{\text{Charizard}, \text{Moltres}\}$
Complement	A^c	All elements not in A	(Water types)^c = all non-Water types
Difference	$A \setminus B$	Elements in A but not in B	Poison types $\setminus$ Grass types = pure Poison types
Empty set	$\varnothing$	The set with no elements	Ice types $\cap$ Fire types = $\varnothing$(no dual Ice/Fire in Gen I)
Subset	$A \subseteq B$	Every element of A is also in B	$\{\text{Pikachu}\} \subseteq$ Electric types

De Morgan's Laws are useful for manipulating complements of unions and intersections:

$$(A \cup B)^c = A^c \cap B^c \quad \text{and} \quad (A \cap B)^c = A^c \cup B^c$$

Pokemon reading: A Pokemon that is *not* (Fire or Water) must be *both* non-Fire *and* non-Water. A Pokemon that is *not* (Fire and Water simultaneously) is either non-Fire *or* non-Water (or both).

Logical connectives. We use $\implies$ for "implies" and $\iff$ for "if and only if." For example: $\text{badges} = 8 \implies \text{elite\_four\_attempted} = 1$ (you must have all eight badges to challenge the Elite Four), but $\text{elite\_four\_attempted} = 1 \not\Rightarrow \text{champion\_defeated} = 1$ (attempting does not guarantee victory).

A.2 Probability Basics

Sample Spaces and Events

A **sample space** Ω is the set of all possible outcomes of a random experiment. An **event** is any subset of Ω.

Example. You throw a Poke Ball at a wild Pidgey. The sample space is $\Omega = \{\text{catch}, \text{escape}\}$. The event "you catch it" is $A = \{\text{catch}\}$.

For a more complex example, consider a wild Pokemon encounter on Route 1. The sample space might be $\Omega = \{\text{Pidgey}, \text{Rattata}, \text{Pikachu}\}$ with encounter probabilities $P(\text{Pidgey}) = 0.55$, $P(\text{Rattata}) = 0.40$, and $P(\text{Pikachu}) = 0.05$.

Axioms of Probability (Kolmogorov)

For any probability measure P on a sample space Ω:

1. **Non-negativity:** $P(A) \geq 0$ for all events A.
2. **Normalization:** $P(\Omega) = 1$ (something must happen).
3. **Countable additivity:** For mutually exclusive events $A_1, A_2, \ldots,$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Pokemon check: The probabilities of encountering Pidgey, Rattata, or Pikachu on Route 1 must sum to 1 (assuming no other encounters). Each probability is non-negative. Since the encounters are mutually exclusive (you encounter exactly one species at a time), $P(\text{Pidgey or Rattata}) = P(\text{Pidgey}) + P(\text{Rattata}) = 0.95$.

Useful Probability Rules

- **Complement rule:** $P(A^c) = 1 - P(A)$. The probability of *not* catching a Pokemon is 1 minus the catch probability.
- **Addition rule:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- **Multiplication rule:** $P(A \cap B) = P(A) \cdot P(B \mid A)$.

A.3 Conditional Probability and Bayes' Rule

Conditional Probability

The **conditional probability** of event A given event B is:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Example. What is the probability of catching a wild Abra, given that you used an Ultra Ball?

Let $C = \{\text{catch}\}$ and $U = \{\text{Ultra Ball used}\}$. If we know that $P(C \cap U) = 0.35$ (35% of all encounter attempts involve using an Ultra Ball and result in a catch) and $P(U) = 0.40$ (40% of attempts use an Ultra Ball), then:

$$P(C \mid U) = \frac{0.35}{0.40} = 0.875$$

Bayes' Rule

Bayes' theorem inverts conditional probabilities:

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

This is critical because the direction of conditioning matters enormously. $P(\text{Catch} \mid \text{Ultra Ball})$ is *not* the same as $P(\text{Ultra Ball} \mid \text{Catch})$.

Kanto example. A Kanto Ranger reports that 80% of trainers who defeated the Champion had used Exp. Share ($P(\text{Exp. Share} \mid \text{Champion}) = 0.80$). Does Exp. Share cause Championship victories? Not necessarily. We need to consider:

- $P(\text{Champion}) = 0.05$ (only 5% of trainers defeat the Champion)
- $P(\text{Exp. Share}) = 0.45$ (45% of all trainers use Exp. Share)

By Bayes' rule:

$$P(\text{Champion} \mid \text{Exp. Share}) = \frac{0.80 \times 0.05}{0.45} \approx 0.089$$

So even though 80% of Champions used Exp. Share, only about 8.9% of Exp. Share users become Champion. The base rates matter.

Law of Total Probability

For a partition $\{B_1, B_2, \ldots, B_k\}$ of Ω:

$$P(A) = \sum_{i=1}^{k} P(A \mid B_i) P(B_i)$$

Example. The overall catch rate of a wild Pokemon depends on which ball you use:

$$P(\text{Catch}) = P(\text{Catch} \mid \text{Poke Ball}) P(\text{Poke Ball}) + P(\text{Catch} \mid \text{Great Ball}) P(\text{Great Ball}) + P(\text{Catch} \mid \text{Ultra Ball}) P(\text{Ultra Ball})$$

A.4 Random Variables

A **random variable** X is a function that maps outcomes in a sample space to real numbers: $X : \Omega \to \mathbb{R}$.

Discrete Random Variables

A discrete random variable takes on a countable number of values. Its distribution is described by a **probability mass function (PMF)**:

$$p_X(x) = P(X = x), \quad \sum_x p_X(x) = 1$$

Example. Let X be the number of badges a trainer earns. $X \in \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$. The PMF assigns a probability to each badge count.

Continuous Random Variables

A continuous random variable can take any value in an interval (or union of intervals). Its distribution is described by a **probability density function (PDF)** $f_X(x)$:

$$P(a \le X \le b) = \int_a^b f_X(x)\, dx, \quad \int_{-\infty}^{\infty} f_X(x)\, dx = 1$$

Example. A Pokemon's IV (Individual Value) total ranges from 0 to 186. Across the population, $X = \text{team\_avg\_iv\_total}$ is approximately continuous, with a density function concentrated around 45–90.

Cumulative Distribution Function (CDF)

The **CDF** is defined for both discrete and continuous random variables:

$$F_X(x) = P(X \le x)$$

Properties: F_X is non-decreasing, $\lim_{x \to -\infty} F_X(x) = 0$, and $\lim_{x \to \infty} F_X(x) = 1$.

Pokemon reading: $F_{\text{badges}}(5) = P(\text{badges} \le 5)$ is the fraction of trainers with five or fewer badges.

A.5 Expectation and Variance

Expectation

The **expected value** (or mean) of a random variable is its long-run average:

$$E[X] = \begin{cases} \sum_x x\, p_X(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} x\, f_X(x)\, dx & \text{(continuous)} \end{cases}$$

Linearity of expectation (holds regardless of dependence):

$$E[aX + bY + c] = a\,E[X] + b\,E[Y] + c$$

Pokemon example. In the Pokemon damage formula, damage is (roughly) proportional to base power × type effectiveness multiplier × STAB bonus. If we treat these as independent random components:

$$E[\text{Damage}] = E[\text{Base}] \times E[\text{Type Multiplier}] \times E[\text{STAB}]$$

Note: this simplification uses the fact that the expectation of a product of *independent* random variables equals the product of their expectations. In general, $E[XY] = E[X]E[Y] + \text{Cov}(X, Y)$.

Variance and Standard Deviation

The **variance** measures the spread of a distribution:

$$\mathrm{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$$

The **standard deviation** is $\sigma_X = \sqrt{\mathrm{Var}(X)}$.

Variance of a linear combination:

$$\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X)$$

For independent X and Y:

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \quad \text{(independence required)}$$

Pokemon example. If a trainer's strategy score has $E[\text{strategy}] = 45$ and $\mathrm{Var}(\text{strategy}) = 150$, then the standard deviation is $\sigma \approx 12.2$, meaning most trainers fall within about 12 points of the average strategy score.

A.6 Covariance and Correlation

Covariance

The **covariance** between two random variables measures their linear co-movement:

$$\mathrm{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y]$$

Properties:

- $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$
- $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$
- $\mathrm{Cov}(aX + b, cY + d) = ac\,\mathrm{Cov}(X, Y)$
- $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y)$

Pearson Correlation Coefficient

The **correlation** normalizes covariance to $[-1, 1]$:

$$\rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Pokemon example. Consider the Attack and Speed base stats across all 151 Kanto Pokemon. If Pokemon with high Attack tend to have high Speed, $\rho > 0$. If tanks (high Attack) tend to be slow, $\rho < 0$. If there is no linear pattern, $\rho \approx 0$.

Important caution. Correlation measures *linear* association. Two variables can be strongly related but have $\rho = 0$ if the relationship is nonlinear. And as this entire textbook emphasizes: correlation does not imply causation.

A.7 Common Distributions

Bernoulli Distribution

$X \sim \mathrm{Bernoulli}(p)$: a single binary trial.

$$P(X = 1) = p, \quad P(X = 0) = 1 - p$$

$$E[X] = p, \quad \mathrm{Var}(X) = p(1 - p)$$

Pokemon example. A critical hit occurs with probability $p = 1/16$ in Generation I. Let $X = 1$ if a critical hit lands, $X = 0$ otherwise. Then $X \sim \mathrm{Bernoulli}(1/16)$.

Binomial Distribution

$X \sim \mathrm{Binomial}(n, p)$: number of successes in n independent Bernoulli trials.

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \ldots, n$$

$$E[X] = np, \quad \mathrm{Var}(X) = np(1 - p)$$

Pokemon example. A trainer fights $n = 20$ wild battles. Each battle is won independently with probability $p = 0.7$. The number of wins $X \sim \mathrm{Binomial}(20, 0.7)$, so $E[X] = 14$ and $\mathrm{Var}(X) = 4.2$.

Poisson Distribution

$X \sim \mathrm{Poisson}(\lambda)$: number of events in a fixed interval.

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots$$

$$E[X] = \lambda, \quad \mathrm{Var}(X) = \lambda$$

Pokemon example. The number of wild encounters per 100 steps on Route 1 follows approximately $\mathrm{Poisson}(\lambda = 8)$. The mean and variance are both 8.

Normal (Gaussian) Distribution

$X \sim N(\mu, \sigma^2)$: the bell curve.

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

$$E[X] = \mu, \quad \mathrm{Var}(X) = \sigma^2$$

Pokemon example. Across the population of 2,000 Kanto trainers, strategy scores are approximately $N(45, 150)$. By the 68–95–99.7 rule, about 68% of trainers have strategy scores within one standard deviation (≈ 12.2 points) of the mean.

Uniform Distribution

$X \sim \mathrm{Uniform}(a, b)$: every value in $[a, b]$ is equally likely.

$$f_X(x) = \frac{1}{b - a}, \quad a \le x \le b$$

$$E[X] = \frac{a + b}{2}, \quad \mathrm{Var}(X) = \frac{(b - a)^2}{12}$$

Pokemon example. Latent traits like patience, natural talent, and dedication are drawn from $\mathrm{Uniform}(0, 100)$ in our DGP, so $E[\mathrm{patience}] = 50$ and $\mathrm{Var}(\mathrm{patience}) \approx 833.3$.

A.8 Law of Large Numbers

Weak Law of Large Numbers (WLLN)

Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables with $E[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Then the sample mean $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ converges in probability to μ:

$$\bar{X}_n \xrightarrow{p} \mu \quad \text{as } n \to \infty$$

More precisely, for any $\varepsilon > 0$:

$$\lim_{n\to\infty} P(|\bar{X}_n - \mu| > \varepsilon) = 0$$

Pokemon example. Suppose a trainer's true win rate is $\mu = 0.65$. After $n = 10$ battles, the observed win rate might be 0.80 or 0.50 — high variance. After $n = 1000$ battles, the observed win rate will be very close to 0.65. The more battles you fight, the closer your observed average converges to your true ability. This is why we trust league-level statistics (large n) more than a single trainer's anecdote (small n).

A.9 Central Limit Theorem

Statement

Let $X_1, X_2, \dots, X_n$ be i.i.d. with $E[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 \in (0, \infty)$. Then the standardized sample mean converges in distribution to a standard Normal:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1) \quad \text{as } n \to \infty$$

Equivalently:

$$\bar{X}_n \overset{\text{approx}}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{for large } n$$

Pokemon example. Suppose each battle yields a random reward X_i with mean $\mu = 500$ Pokedollars and standard deviation $\sigma = 200$. If a trainer fights $n = 100$ battles, the average reward per battle will be approximately:

$$\bar{X}_{100} \sim N\left(500, \frac{200^2}{100}\right) = N(500, 400)$$

So the average reward has standard deviation $\sqrt{400} = 20$ Pokedollars. A 95% confidence interval would be approximately $500 \pm 1.96 \times 20 = [460.8, 539.2]$.

Why the CLT matters for causal inference. Nearly every estimator in this textbook (difference in means, IPW, AIPW, Wald, 2SLS, DiD) is an average or a function of averages. The CLT guarantees that, in large enough samples, these estimators are approximately Normal — which is why we can construct confidence intervals and perform hypothesis tests using Normal critical values.

A.10 Conditional Expectation and the Law of Iterated Expectations

Conditional Expectation

The **conditional expectation** of Y given $X = x$ is the expected value of Y computed using the conditional distribution of $Y \mid X = x$:

$$E[Y \mid X = x] = \begin{cases} \sum_y y\,P(Y = y \mid X = x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} y\, f_{Y|X}(y \mid x)\, dy & \text{(continuous)} \end{cases}$$

When we write $E[Y \mid X]$ (without specifying $X = x$), this is itself a *random variable* — it is a function of X.

Pokemon example. $E[\mathrm{badges} \mid \mathrm{starter\_type} = \mathrm{Water}]$ is the average badge count among Water-type starter trainers. This is a single number. But $E[\mathrm{badges} \mid \mathrm{starter\_type}]$ is a random variable that takes one of three values depending on whether the trainer chose Grass, Fire, or Water.

Law of Iterated Expectations (LIE)

Also called the **law of total expectation** or **tower property**:

$$E[E[Y \mid X]] = E[Y]$$

In words: the average of the conditional averages (weighted by the distribution of X) equals the unconditional average.

More generally, for nested conditioning:

$$E[E[Y \mid X, Z] \mid X] = E[Y \mid X]$$

Pokemon example. Let $Y = \mathrm{badges}$ and $X = \mathrm{starter\_type}$. Then:

$$E[\mathrm{badges}] = P(\mathrm{Grass}) \cdot E[\mathrm{badges} \mid \mathrm{Grass}] + P(\mathrm{Fire}) \cdot E[\mathrm{badges} \mid \mathrm{Fire}] + P(\mathrm{Water}) \cdot E[\mathrm{badges} \mid \mathrm{Water}]$$

If we know the average badge count within each starter group and the proportion of trainers in each group, we can recover the overall average badge count. This decomposition is fundamental to causal inference: many identification strategies work by expressing a causal quantity as a weighted average of conditional expectations.

Conditional Variance

The **law of total variance** decomposes overall variance into two components:

$$\mathrm{Var}(Y) = E[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(E[Y \mid X])$$

- $E[\mathrm{Var}(Y \mid X)]$: average within-group variance ("unexplained" variance)
- $\mathrm{Var}(E[Y \mid X])$: between-group variance ("explained" by X)

Pokemon example. The total variance in badge counts comes from (1) variance *within* each starter-type group (not everyone who picks Squirtle earns the same badges) and (2) variance *between* starter-type groups (the group averages differ).

A.11 Notation Reference Table

The following table collects all major notation used throughout this textbook.

Symbol	Meaning	First Used
Y	Outcome variable (e.g., badges earned)	Ch. 1
X or D or T	Treatment or exposure variable	Ch. 1
X	Vector of covariates / controls	Ch. 2
$Y_i(1), Y_i(0)$	Potential outcomes for unit i under treatment / control	Ch. 2
Y_i^{obs}	Observed outcome for unit i; equals $Y_i(D_i)$	Ch. 2
τ	Generic causal effect	Ch. 2
$\tau_i = Y_i(1) - Y_i(0)$	Individual treatment effect (ITE)	Ch. 2
$ATE = E[Y(1) - Y(0)]$	Average treatment effect	Ch. 2
$ATT = E[Y(1) - Y(0) \mid D = 1]$	Average treatment effect on the treated	Ch. 2
$ATC = E[Y(1) - Y(0) \mid D = 0]$	Average treatment effect on the control	Ch. 2
$CATE(x) = E[Y(1) - Y(0) \mid X = x]$	Conditional average treatment effect	Ch. 10
$LATE$	Local average treatment effect (compliers)	Ch. 7
$P(Y \mid X)$	Conditional probability / distribution	Ch. 1
$P(Y \mid do(X))$	Interventional distribution (Pearl's do-operator)	Ch. 1
$E[Y \mid X]$	Conditional expectation	Ch. 1
$E[Y \mid do(X = x)]$	Causal (interventional) expectation	Ch. 3
$\perp\!\!\!\perp$	Statistical independence	Ch. 2
$Y(d) \perp\!\!\!\perp D \mid X$	Conditional independence / ignorability	Ch. 2
$e(x) = P(D = 1 \mid X = x)$	Propensity score	Ch. 5
G	Directed acyclic graph (DAG)	Ch. 3
$pa(X)$	Parents of node X in a DAG	Ch. 3
$X \perp_d Y \mid Z$	d-separation in a DAG	Ch. 3
Z	Instrument (instrumental variables context)	Ch. 7
D_i	Treatment indicator (binary: 0 or 1)	Ch. 2
c	Cutoff value in regression discontinuity	Ch. 8
R_i	Running variable in RDD	Ch. 8
ΔY or $Y_{post} - Y_{pre}$	First difference (DiD context)	Ch. 9
$\hat{\tau}$	Estimated causal effect	Ch. 4
$SE(\hat{\tau})$	Standard error of the estimator	Ch. 4
α	Significance level (typically 0.05)	Ch. 4
β	Regression coefficient or power ($1 - P(\text{Type II error})$)	Ch. 4
$\mathbb{1}\{\cdot\}$	Indicator function: 1 if condition is true, 0 otherwise	Ch. 2
n	Sample size	Ch. 4
N	Population size	Ch. 4
$\hat{\mu}$	Estimated mean	Ch. 4
$Var(X)$	Variance of random variable X	App. A
$Cov(X, Y)$	Covariance of X and Y	App. A
ϱ_{XY}	Pearson correlation between X and Y	App. A

Symbol	Meaning	First Used
σ^2	Variance (population parameter)	App. A
s^2	Sample variance	Ch. 4
$\Phi(\cdot)$	Standard Normal CDF	App. A
$\phi(\cdot)$	Standard Normal PDF	App. A
$\xrightarrow{p}$	Convergence in probability	App. A
$\xrightarrow{d}$	Convergence in distribution	App. A
$O_p(\cdot)$	Stochastic order (big-O in probability)	Ch. 11
$\hat{\theta}_{DR}$	Doubly robust estimator	Ch. 6
w_i	IPW weight for unit i	Ch. 6
L	Loss function or likelihood	Ch. 10

Professor Oak's Parting Advice

"Mathematics is to causal inference what a Pokedex is to a trainer — it does not battle for you, but without it, you are navigating blind. Every theorem we use in this textbook rests on the foundations laid out in this appendix. When you encounter an unfamiliar symbol or formula in the chapters ahead, come back here. This lab is always open."

Next: Appendix B — Python Environment & Data Guide

Appendix B: Python Environment & Data Guide

"Every great trainer needs the right gear before heading out," Nurse Joy says, organizing a shelf of healing supplies. "You wouldn't leave the Pokemon Center without Potions. Don't start coding without setting up your environment."

B.1 Installation & Environment Setup

Option 1: pip (simplest)

```
# Create a virtual environment (recommended)
python -m venv kanto-env
source kanto-env/bin/activate    # macOS / Linux
kanto-env\Scripts\activate       # Windows

# Install the textbook package and all dependencies
pip install -e ".[all]"
```

Option 2: conda (recommended for scientific computing)

```
# Create the environment from the provided YAML file
conda env create -f environment.yml

# Activate
conda activate kanto-causal

# Install the textbook package in development mode
pip install -e .
```

Option 3: Manual installation

If you prefer to install packages individually:

```
pip install numpy pandas scipy statsmodels matplotlib seaborn
pip install scikit-learn econml dowhy causal-learn
pip install linearmodels pgmpy graphviz
pip install ipywidgets plotly jupyterlab
pip install -e .  # install kanto_utils from the repo root
```

Verifying your installation

Run this quick check in Python:

```python
import kanto_utils
print(kanto_utils.__version__)  # Should print "1.0.0"

df = kanto_utils.load_trainers()
print(df.shape)                 # Should print (2000, 42)
print("Setup complete!")
```

B.2 Package Reference

The following table lists every Python package used in this textbook, grouped by purpose.

Core Scientific Stack

Package	Version	Purpose
numpy	>= 1.24	Array operations, random number generation, linear algebra. The backbone of all numerical computation.
pandas	>= 2.0	DataFrames for tabular data. Every dataset in this textbook is loaded as a pandas DataFrame.
scipy	>= 1.10	Statistical functions (`scipy.stats`), optimization (`scipy.optimize`), kernel density estimation. Used for hypothesis tests, distribution fitting, and bandwidth selection.

Statistical Modeling

Package	Version	Purpose
statsmodels	>= 0.14	OLS, logistic regression, IV/2SLS, robust standard errors, formula interface (`smf.ols`). Workhorse for Chapters 4–9.
linearmodels	>= 5.0	Panel data models, two-way fixed effects (TWFE), instrumental variables with panel structure. Used in Chapters 7 and 9.
scikit-learn	>= 1.3	Machine learning models: random forests, cross-validation, train/test splitting. Used primarily in Chapter 10 for causal forests and DML.

Causal Inference Libraries

Package	Version	Purpose
econml	>= 0.15	Microsoft's library for heterogeneous treatment effects: causal forests, DML, doubly robust learners, CATE estimation. Central to Chapter 10.
dowhy	>= 0.11	End-to-end causal inference workflow: model specification, identification, estimation, refutation. Integrates with econml estimators.
causal-learn	>= 0.1	Causal discovery algorithms (PC, GES, FCI). Used in Chapter 12 for learning DAG structure from data.
pgmpy	>= 0.1	Probabilistic graphical models: Bayesian networks, DAG manipulation, d-separation queries. Used in Chapter 3 for DAG operations.

Visualization

Package	Version	Purpose
matplotlib	>= 3.7	Base plotting library. All static figures use matplotlib via the Kanto theme (`apply_kanto_theme()`).
seaborn	>= 0.13	Statistical plots: kernel density, violin plots, heatmaps. Built on matplotlib.
graphviz	>= 0.20	DAG rendering. Requires the Graphviz system package (`brew install graphviz` on macOS, `sudo apt install graphviz` on Linux).
plotly	>= 5.18	Interactive plots for sensitivity analysis dashboards, 3D treatment effect surfaces. Used selectively.

Interactive & Notebook Tools

Package	Version	Purpose
ipywidgets	>= 8.0	Interactive sliders and widgets for the confounder adjustment demo, RDD bandwidth explorer, and power analysis tool.
jupyterlab	>= 4.0	Notebook environment. All textbook code is designed to run in Jupyter notebooks.

B.3 The `kanto_utils` Package

The `kanto_utils` package is the textbook's companion library. It provides data loaders, themed plotting functions, causal estimators, interactive widgets, and narrative callout boxes. All functions are importable from the top-level namespace.

Module Overview

Module	Description
`kanto_utils.data_loader`	Functions to load each of the 11 datasets from `data/raw/` .
`kanto_utils.plotting`	Kanto-themed matplotlib styling, specialized causal inference plots (love plots, RDD plots, DiD plots).
`kanto_utils.causal`	Estimator functions: difference in means, randomization inference, propensity scores, IPW, doubly robust, Wald/2SLS, sharp RDD, DiD.
`kanto_utils.widgets`	Interactive Jupyter widgets for exploring causal concepts.
`kanto_utils.narrative`	Callout box helpers for themed narrative elements (Professor Oak, Blue, Nurse Joy, Gym Leaders).

Data Loaders

```python
from kanto_utils import (
    load_trainers,          # kanto_trainers.csv -- 2,000 rows, 42 cols
    load_battles,           # kanto_battles.csv -- 50,000 rows, 28 cols
    load_cities_panel,      # kanto_cities_panel.csv -- city x month panel
    load_protein_rct,       # pewter_protein_rct.csv -- RCT data
    load_ss_anne,           # ss_anne_passengers.csv -- matching data
    load_safari_lottery,    # safari_zone_lottery.csv -- IV data
    load_happiness,         # happiness_evolution.csv -- RDD data
    load_shadow_surge,      # shadow_surge_staggered.csv -- staggered DiD
    load_elite_four,        # elite_four_panel.csv -- dynamic treatment
    load_double_battles,    # double_battles.csv -- interference
    load_johto,             # johto_transportability.csv -- external validity
)
```

Each loader returns a `pd.DataFrame`. All functions accept `**kwargs` which are passed through to `pd.read_csv()`.

Plotting Utilities

```
from kanto_utils import (
    type_color,          # Get the hex color for a Pokemon type string
    type_colors_dict,    # Full dict: {"Fire": "#F08030", "Water": "#6890F0", ...}
    apply_kanto_theme,   # Apply the textbook's matplotlib rcParams theme
    badge_stamp,         # Add a badge icon to a figure (for "badge earned" moments)
    pokemon_scatter,     # Themed scatter plot with type-colored markers
    love_plot,           # Covariate balance (love) plot for matching chapters
    did_plot,            # Difference-in-differences parallel trends visualization
    rdd_plot,            # Regression discontinuity plot with cutoff line
)
```

Applying the theme. Call `apply_kanto_theme()` once at the top of each notebook to set consistent fonts, colors, and grid styling:

```
import matplotlib.pyplot as plt
from kanto_utils import apply_kanto_theme

apply_kanto_theme()
# All subsequent plots will use the Kanto theme
```

Causal Estimators

```
from kanto_utils import (
    difference_in_means,       # Simple ATE via treated - control means
    randomization_inference,   # Fisher's exact p-value via permutation
    propensity_score,          # Fit propensity score model, return scores
    ipw_estimate,              # Inverse probability weighting ATE
    doubly_robust,             # AIPW / doubly robust estimator
    wald_estimator,            # Wald (IV) estimator for a single instrument
    two_stage_ls,              # Two-stage least squares
    sharp_rdd,                 # Local linear regression for sharp RDD
    did_estimate,              # Difference-in-differences estimator
    balance_table,             # Covariate balance table (pre/post matching)
)
```

Interactive Widgets

```
from kanto_utils import (
    confounder_slider,         # Adjust confounding strength and see bias change
    rdd_bandwidth_slider,      # Explore bandwidth sensitivity in RDD
    power_analysis_widget,     # Interactive sample size / power calculator
    starter_selector,          # Choose a starter and see causal implications
)
```

Narrative Callout Boxes

```
from kanto_utils import (
    oak_says,            # Professor Oak explanation box
    blue_says,           # Blue's (often wrong) claim
    nurse_joy_says,      # Nurse Joy's practical advice
    gym_leader_says,     # Gym Leader challenge/insight
    badge_earned,        # Badge milestone marker
    blues_mistake,       # Highlight a common causal reasoning error
)
```

Usage example:

```
oak_says("Ignorability means that, conditional on the covariates, "
         "treatment assignment is as good as random.")
```

B.4 Data Generation

Philosophy

All datasets in this textbook are **synthetically generated** from known structural equations. This design choice is deliberate: because we control the data-generating process (DGP), we know the *true* causal effects. Students can check their estimates against the ground truth, building intuition for when methods work and when they fail.

Regenerating the Data

All datasets are generated by `data/dgp/structural_equations.py` using a single random number generator seeded with **seed = 151** (one for each original Pokemon):

```
import numpy as np
RNG = np.random.default_rng(151)
```

To regenerate all datasets from scratch:

```
cd data/dgp
python generate_all.py
```

This will overwrite the CSV files in `data/raw/`. Because the seed is fixed, the output is deterministic — every student gets identical data.

Why seed = 151?

There are 151 Pokemon in the original Kanto Pokedex. The seed is a nod to Mew, #151, the hidden Pokemon that contains the DNA of all others. Similarly, our single seed contains the DNA of all 11 datasets.

B.5 Dataset Quick Reference

Dataset	File	Rows	Cols	Chapter(s)	Key Variables
Kanto Trainers	`kanto_trainers.csv`	2,000	42	1–6, 10	`trainer_id`, `starter_type`, `badges`, `strategy_score`, `wealth`, `trainer_experience`, `exp_share_used`
Kanto Battles	`kanto_battles.csv`	50,000	28	1, 4	`battle_id`, `trainer_id`, `battle_type`, `attacker_level`, `defender_level`, `won`
Kanto Cities Panel	`kanto_cities_panel.csv`	~240	~15	9	`city`, `month`, `gym_present`, `population`, `pokemon_center_visits`
Pewter Protein RCT	`pewter_protein_rct.csv`	~500	~12	4, 7	`assigned_treatment`, `took_protein`, `battle_performance`, `pokemon_id`
S.S. Anne Passengers	`ss_anne_passengers.csv`	~1,500	~20	5, 6	`passenger_id`, `ticket_class`, `survived`, `trainer_level`, `pokemon_count`
Safari Zone Lottery	`safari_zone_lottery.csv`	~2,000	~15	7	`won_lottery`, `visited_safari`, `rare_pokemon_owned`, `battle_score`
Happiness Evolution	`happiness_evolution.csv`	~3,000	~12	8	`pokemon_id`, `happiness`, `evolved`, `battle_stats_post`
Shadow Surge	`shadow_surge_staggered.csv`	~600	~15	9	`city`, `period`, `treatment_start`, `shadow_pokemon_count`, `trainer_activity`
Elite Four Panel	`elite_four_panel.csv`	~800	~18	11	`challenger_id`, `attempt`, `preparation`, `team_composition`, `outcome`
Double Battles	`double_battles.csv`	~5,000	~20	11	`battle_id`, `partner_type`, `interference`, `outcome`, `partner_outcome`
Johto Trainers	`johto_transportability.csv`	~1,000	~30	12	`trainer_id`, `region`, `starter_type`, `badges`, `strategy_score`

B.6 Codebook Conventions

Throughout this textbook, data dictionaries follow consistent conventions:

Variable Naming

- **Snake case:** All variable names use `lower_snake_case` (e.g., `team_level_avg`, `dept_store_spending`).
- **Suffixes:**
  - `_id` — unique identifier (not for analysis; for merging only)
  - `_avg` — mean value across a group
  - `_max` — maximum value across a group
  - `_count` — count of occurrences
  - `_used` — binary indicator (0/1) of whether something was used
  - `_score` — continuous index, typically 0–100

Variable Types

Type Label	Meaning	Example
continuous	Numeric, can take any real value in a range	`team_level_avg` (5–100)
count	Non-negative integer	`safari_zone_visits` (0–30)
binary	0/1 indicator	`exp_share_used`, `cave_training`
ordinal	Ordered integer categories	`wealth` (1–5)
categorical	Unordered string categories	`starter_type`, `hometown`
latent	Not directly observable; included for DGP transparency	`patience`, `natural_talent`, `dedication`

Missing Values

Missing values are encoded as `NaN` in pandas. The only dataset with structural missingness is `kanto_trainers.csv`, where `hall_of_fame_time` is `NaN` for trainers who did not defeat the Champion (this is by design, not data error).

B.7 Common Code Patterns

Loading Data and Applying the Theme

```
import pandas as pd
import matplotlib.pyplot as plt
from kanto_utils import load_trainers, apply_kanto_theme

# Apply the visual theme
apply_kanto_theme()

# Load the main dataset
df = load_trainers()
print(f"Dataset: {df.shape[0]} trainers, {df.shape[1]} variables")
df.head()
```

Quick Exploratory Summary

```
# Summary statistics for key variables
df[["badges", "strategy_score", "wealth", "trainer_experience"]].describe()
```

Creating a Themed Scatter Plot

```python
from kanto_utils import pokemon_scatter

pokemon_scatter(
    df,
    x="strategy_score",
    y="badges",
    hue="starter_type",
    title="Strategy Score vs. Badges by Starter Type",
)
plt.tight_layout()
plt.show()
```

Running a Quick Causal Estimate

```python
from kanto_utils import difference_in_means, ipw_estimate

# Naive difference in means
naive = difference_in_means(df, treatment="exp_share_used", outcome="badges")
print(f"Naive DiM: {naive['estimate']:.3f} (SE: {naive['se']:.3f})")

# IPW estimate adjusting for confounders
ipw = ipw_estimate(
    df,
    treatment="exp_share_used",
    outcome="badges",
    covariates=["wealth", "trainer_experience", "strategy_score"],
)
print(f"IPW ATE:   {ipw['estimate']:.3f} (SE: {ipw['se']:.3f})")
```

Using Narrative Callouts in Notebooks

```python
from kanto_utils import oak_says, blues_mistake

oak_says(
    "The IPW estimate is closer to the true effect because it "
    "adjusts for the confounders that drive both Exp. Share usage "
    "and badge outcomes."
)

blues_mistake(
    "Blue compared Exp. Share users to non-users without adjustment. "
    "Wealthier trainers are more likely to buy Exp. Share AND earn "
    "more badges due to other advantages."
)
```

> ***Nurse Joy's Reminder:*** *"If something breaks, check your Python version (>= 3.10), make sure you installed in the right environment, and remember that `graphviz` needs both the Python package and the system binary. Come back to the Pokemon Center — I mean, this appendix — anytime."*

Next: Appendix C — The Full Kanto Causal DAG

Appendix C: The Full Kanto Causal DAG

Deep in Professor Oak's study, behind a locked door that opens only after you have earned all eight badges, there is a wall-sized diagram. Strings connect photographs of trainers, items, and Pokemon. Arrows are drawn in red marker. Post-it notes list equations. "This," says Professor Oak, pulling back the curtain, "is the true structure of the world — the causal graph that governs how trainers succeed or fail in Kanto. Every association you have observed, every confound you have struggled with, every backdoor path you have blocked — it all comes from here."

C.1 Overview

The **Kanto Causal DAG** is the directed acyclic graph (DAG) that describes the data-generating process (DGP) for the primary dataset, `kanto_trainers.csv`. Every variable in that dataset is a node, and every causal arrow is derived from the structural equations in `data/dgp/structural_equations.py`.

This DAG serves three roles:

1. **Ground truth** for the textbook exercises. Because the data is synthetic, the DAG *is* the correct causal model. Students can verify whether their identification strategies target the right adjustment sets.
2. **Reference for instructors** designing exam questions or alternative exercises.
3. **Worked example** of how a complex causal graph is documented, analyzed, and used in practice.

C.2 Node Inventory

Latent (Unobserved) Nodes

These variables exist in the DGP and appear in the CSV for pedagogical transparency, but in the "story" of the textbook they are initially hidden from the researcher. In DAG diagrams, they are drawn with **dashed borders** or rendered in **gray**.

Node	Range	Description
`patience`	0–100	Innate patience of the trainer. Affects strategic thinking and willingness to grind. Drawn from $\mathrm{Uniform}(0, 100)$.
`natural_talent`	0–100	Raw aptitude for Pokemon battling. Affects team IVs and strategy. Drawn from $\mathrm{Uniform}(0, 100)$.
`dedication`	0–100	Long-term commitment to training. Affects hours played and items purchased. Drawn from $\mathrm{Uniform}(0, 100)$.

Observed Nodes: Background Characteristics

Node	Type	Range	Description
`trainer_id`	ID	1–2000	Unique identifier. Not a causal variable.
`trainer_name`	ID	string	Unique name. Not a causal variable.
`hometown`	Categorical	10 Kanto towns	Where the trainer grew up. Exogenous.
`hometown_gym_type`	Categorical	10 types + "None"	Deterministic function of `hometown`.
`region`	Constant	"Kanto"	Always "Kanto" in this dataset.

Observed Nodes: Trainer Attributes

Node	Type	Range	Description
`wealth`	Ordinal	1–5	Socioeconomic status. Influenced by `hometown`.
`trainer_experience`	Continuous	0–15	Years of training experience. Exogenous (drawn from Exponential).
`play_hours`	Count	10–999	Total hours played. Driven by `dedication` and `trainer_experience`.
`strategy_score`	Continuous	0–100	Composite measure of battle strategy. Driven by `patience`, `natural_talent`, and `trainer_experience`.
`cave_training`	Binary	0/1	Whether the trainer trained in caves. Driven by `trainer_experience` and `dedication`.
`safari_zone_visits`	Count	0–30	Number of Safari Zone trips. Driven by `patience`.
`fishing_attempts`	Count	0–50	Number of fishing attempts. Driven by `patience`.

Observed Nodes: Starter and Team

Node	Type	Range	Description
`starter_type`	Categorical	Grass/Fire/Water	Starter choice. Influenced by `wealth` (Fire bias) and `trainer_experience` (Water bias).
`starter_species`	Categorical	Bulbasaur/Charmander/Squirtle	Deterministic function of `starter_type`.
`team_size`	Count	1–6	Number of Pokemon on team. Driven by `trainer_experience`.
`team_level_avg`	Continuous	5–100	Average level of team. Driven by `trainer_experience` and `dedication`.
`team_level_max`	Continuous	5–100	Maximum level on team. Driven by `team_level_avg`.
`team_avg_iv_total`	Continuous	0–186	Average IV total across team. Driven by `natural_talent`.
`team_avg_ev_total`	Continuous	0–510	Average EV total. Driven by `play_hours` and `cave_training`.
`type_diversity`	Count	1–6	Number of distinct types on team. Driven by `trainer_experience` and `patience`.
`team_happiness_avg`	Continuous	0–255	Average happiness. Driven by `patience` and `dedication`.
`team_friendship_max`	Continuous	0–255	Max friendship. Driven by `team_happiness_avg`.

Observed Nodes: Items and Spending

Node	Type	Range	Description
`dept_store_spending`	Continuous	0–10000	Pokedollars spent at Celadon Dept. Store. Driven by `wealth` and `play_hours`.
`potions_purchased`	Count	0–200	Total potions bought. Driven by `dedication` and `play_hours`.
`revives_used`	Count	0–50	Total revives used. Driven by `play_hours`.
`tm_count`	Count	0–50	Number of TMs collected. Driven by `dedication` and `trainer_experience`.
`exp_share_used`	Binary	0/1	Whether Exp. Share was used. Driven by `wealth` and `trainer_experience`.
`held_item_count`	Count	0–6	Number of held items. Driven by `wealth` and `tm_count`.
`rare_candy_used`	Count	0–20	Rare candies consumed. Driven by `wealth`.
`gym_pokemon_center_visits`	Count	0–100	Pokemon Center visits. Driven by `play_hours`.
`traded_pokemon_count`	Count	0–15	Pokemon received via trade. Driven by `trainer_experience`.

Observed Nodes: Outcomes

Node	Type	Range	Description
`badges`	Count	0–8	Gym badges earned. **Primary outcome**.
`elite_four_attempted`	Binary	0/1	Deterministic: 1 iff `badges == 8`.
`elite_four_wins`	Binary	0/1	Won the Elite Four challenge.
`champion_defeated`	Binary	0/1	Defeated the Champion.
`hall_of_fame_time`	Continuous	NaN or hours	Time to Hall of Fame (NaN if not Champion).
`total_battles_won`	Count	0–999	Total battle victories.
`total_pokemon_caught`	Count	1–151	Pokemon caught.
`pokedex_completion`	Continuous	0–100	Percent of Pokedex completed.

C.3 Complete Edge List

Each arrow below represents a direct causal effect in the structural equations. The format is `Parent --> Child` with a brief justification and the functional form.

Edges from Exogenous/Latent Sources

Edge	Justification	Functional Form
`hometown` → `wealth`	Wealthier towns (Celadon, Saffron) give +0.3 bonus	wealth ~ round(N(3.0 + bonus, 0.8)), clipped 1–5
`patience` → `strategy_score`	Patient trainers think before acting	$+0.25 \times$ patience
`patience` → `safari_zone_visits`	Patient trainers tolerate Safari Zone frustration	Poisson($1 + 0.04 \times$ patience)
`patience` → `fishing_attempts`	Fishing requires patience	Poisson($3 + 0.03 \times$ patience)
`patience` → `type_diversity`	Patient trainers build balanced teams	$+0.01 \times$ patience in Poisson rate
`patience` → `team_happiness_avg`	Patient trainers raise happier Pokemon	$+0.3 \times$ patience
`natural_talent` → `strategy_score`	Talent aids strategic thinking	$+0.3 \times$ natural_talent
`natural_talent` → `team_avg_iv_total`	Talented trainers identify strong Pokemon	$+0.4 \times$ natural_talent
`dedication` → `play_hours`	Dedicated trainers play more	$+5 \times$ dedication
`dedication` → `potions_purchased`	Dedicated trainers stock up	$+0.3 \times$ dedication
`dedication` → `tm_count`	Dedicated trainers seek TMs	$+0.1 \times$ dedication
`dedication` → `cave_training`	Dedicated trainers do hard training	$+0.02 \times$ dedication in logit
`dedication` → `team_happiness_avg`	Dedication breeds Pokemon happiness	$+0.2 \times$ dedication
`dedication` → `team_level_avg`	Dedicated trainers level up more	$+0.15 \times$ dedication

Edges from `trainer_experience`

Edge	Justification	Functional Form
`trainer_experience` → `strategy_score`	Experience improves strategy	$+1.5 \times$ experience
`trainer_experience` → `starter_type`	Experienced trainers favor Water (meta)	$+0.12 \times$ (exp − 4) in Water utility
`trainer_experience` → `team_size`	Experienced trainers fill their team	$+0.1 \times$ experience in Poisson rate
`trainer_experience` → `team_level_avg`	More years → higher levels	$+3 \times$ experience
`trainer_experience` → `type_diversity`	Experienced trainers diversify	$+0.03 \times$ experience in Poisson rate
`trainer_experience` → `play_hours`	More years → more hours	$+10 \times$ experience
`trainer_experience` → `cave_training`	Experienced trainers explore caves	$+0.15 \times$ experience in logit
`trainer_experience` → `tm_count`	Experience helps find TMs	$+0.3 \times$ experience
`trainer_experience` → `exp_share_used`	Experienced trainers know about Exp. Share	$+0.1 \times$ experience in logit
`trainer_experience` → `traded_pokemon_count`	Experienced trainers trade more	$+0.05 \times$ experience in Poisson rate
`trainer_experience` → `total_battles_won`	Experience wins battles	$+8 \times$ experience

Edges from `wealth`

Edge	Justification	Functional Form
`wealth` → `starter_type`	Wealthy trainers lean toward Fire	$+0.15 \times$ (wealth − 3) in Fire utility
`wealth` → `dept_store_spending`	More money → more spending	$+500 \times$ wealth
`wealth` → `exp_share_used`	Exp. Share costs money	$+0.4 \times$ (wealth − 3) in logit
`wealth` → `held_item_count`	Wealthy trainers buy held items	$+0.1 \times$ wealth in Poisson rate
`wealth` → `rare_candy_used`	Rare candies are expensive	$+0.15 \times$ (wealth − 1) in Poisson rate

Edges among Intermediates

Edge	Justification	Functional Form
`play_hours` → `dept_store_spending`	More playtime → more spending opportunities	$+50 \times$ play_hours $\times\ 0.01$
`play_hours` → `potions_purchased`	More playtime → more potion needs	$+0.05 \times$ play_hours
`play_hours` → `revives_used`	More playtime → more fainting events	$+0.02 \times$ play_hours in Poisson rate
`play_hours` → `gym_pokemon_center_visits`	More playtime → more healing	$+0.1 \times$ play_hours in Poisson rate
`play_hours` → `team_avg_ev_total`	More battles → more EVs	$+2 \times$ play_hours $\times\ 0.1$
`play_hours` → `total_pokemon_caught`	More time → more catches	$+0.1 \times$ play_hours
`cave_training` → `team_avg_ev_total`	Cave grinding gives EVs	$+1.5 \times$ cave_training $\times\ 50$
`tm_count` → `held_item_count`	TM knowledge correlates with item usage	$+0.05 \times$ tm_count in Poisson rate
`safari_zone_visits` → `total_pokemon_caught`	Safari catches add to total	$+2 \times$ safari_zone_visits
`team_level_avg` → `team_level_max`	Max level is above average	team_level_avg + Exp(8)
`team_happiness_avg` → `team_friendship_max`	Max friendship is above average	team_happiness_avg + Exp(15)
`total_pokemon_caught` → `pokedex_completion`	Deterministic: caught / 151 × 100	Deterministic

Edges into `badges` (Primary Outcome)

The badge count is the central outcome variable. Its structural equation is:

$$\text{badge\_logit} = -4.0 + 0.04 \times \text{strategy\_score} + 0.03 \times \text{team\_level\_avg} + 0.15 \times \text{type\_diversity} + 0.01 \times \text{tm\_count} + 0.005 \times \frac{\text{dept\_store\_}}{100}$$

$$\text{badges} = \text{clip}(\lfloor \sigma(\text{badge\_logit}) \times 9 + \varepsilon \rfloor, 0, 8), \quad \varepsilon \sim N(0, 0.6)$$

Edge	Effect Direction	Coefficient
`strategy_score` → `badges`	+	0.04
`team_level_avg` → `badges`	+	0.03
`type_diversity` → `badges`	+	0.15
`tm_count` → `badges`	+	0.01
`dept_store_spending` → `badges`	+	0.005/100
`exp_share_used` → `badges`	+	0.30
`held_item_count` → `badges`	+	0.01

Edges into Post-Badge Outcomes

Edge	Justification
`badges` → `elite_four_attempted`	Deterministic: must have 8 badges
`elite_four_attempted` → `elite_four_wins`	Can only win if you attempt
`strategy_score` → `elite_four_wins`	Strategy helps win E4
`team_level_avg` → `elite_four_wins`	Level advantage matters
`type_diversity` → `elite_four_wins`	Diverse teams handle E4 better
`tm_count` → `elite_four_wins`	More TMs = more coverage
`natural_talent` → `elite_four_wins`	Talent matters at highest level
`elite_four_wins` → `champion_defeated`	Must beat E4 first
`champion_defeated` → `hall_of_fame_time`	Only Champions get a time
`play_hours` → `hall_of_fame_time`	More hours = longer to Hall of Fame
`strategy_score` → `total_battles_won`	Strategy wins battles
`team_level_avg` → `total_battles_won`	Level wins battles

C.4 Adjustment Set Reference

For the most common treatment-outcome pairs analyzed in the textbook, the following table lists valid adjustment sets that satisfy the backdoor criterion. "Minimal" means no proper subset also satisfies the criterion.

Treatment: `exp_share_used` — Outcome: `badges`

Confounders: `wealth` (affects both Exp. Share adoption and badges through `dept_store_spending` and `held_item_count`) and `trainer_experience` (affects Exp. Share adoption and badges through multiple paths).

Adjustment Set	Type	Notes
{ `wealth`, `trainer_experience` }	Minimal	Blocks all backdoor paths. Preferred.
{ `wealth`, `trainer_experience`, `strategy_score` }	Valid	Over-controlling, but not harmful.
{ `wealth`, `trainer_experience`, `team_level_avg` }	Valid	`team_level_avg` is a mediator via `trainer_experience`, but since `trainer_experience` is already included, no bias.
{ `strategy_score` } alone	**Invalid**	Does not block `wealth` → `exp_share_used` path.
{ `badges` }	**Invalid**	Outcome; never condition on the outcome.
{ `dept_store_spending` }	**Invalid**	Descendant of `wealth` on a causal path to `badges`; conditioning opens alternative paths.

Treatment: `starter_type` — Outcome: `badges`

Confounders: `wealth` (Fire bias, spending), `trainer_experience` (Water bias, team strength).

Adjustment Set	Type	Notes
{ `wealth` , `trainer_experience` }	Minimal	Blocks both backdoor paths.
{ `wealth` , `trainer_experience` , `patience` , `natural_talent` , `dedication` }	Valid	Latents are not confounders for this pair, but conditioning on them is harmless.
{ `strategy_score` } alone	**Invalid**	Mediator. `trainer_experience` → `strategy_score` → `badges` . Controlling for it blocks a causal pathway *and* fails to block the `wealth` path.
{} (empty)	**Invalid**	Both `wealth` and `experience` confound.

Treatment: `cave_training` — Outcome: `badges`

Confounders: `trainer_experience` and `dedication` (both drive `cave_training` and affect `badges` through team levels and strategy).

Adjustment Set	Type	Notes
{ `trainer_experience` , `dedication` }	Minimal	`dedication` is latent but included in data for analysis.
{ `trainer_experience` } alone	**Partial**	Blocks `experience` path but leaves `dedication` unblocked. Biased unless `dedication` is also included.

Treatment: `safari_zone_visits` — Outcome: `total_pokemon_caught`

Confounder: `patience` (drives Safari visits and, indirectly through play habits, total catches).

Adjustment Set	Type	Notes
{ `patience` }	Minimal	Blocks the sole backdoor path.
{ `patience` , `play_hours` }	Valid	`play_hours` is not a confounder here (no direct path from `patience` to `play_hours`), so adding it is harmless.

C.5 Path Analysis for Key Treatment-Outcome Pairs

`exp_share_used` ⟶ `badges`

Causal (directed) paths:

1. `exp_share_used` → `badges` (direct effect, coefficient 0.30)

Backdoor paths (non-causal, through common causes):

1. `exp_share_used` ← `wealth` → `dept_store_spending` → `badges`
2. `exp_share_used` ← `wealth` → `held_item_count` → `badges`
3. `exp_share_used` ← `wealth` → `starter_type` ← `trainer_experience` → `strategy_score` → `badges` (this path contains a collider at `starter_type` , so it is **blocked** by default)
4. `exp_share_used` ← `trainer_experience` → `strategy_score` → `badges`
5. `exp_share_used` ← `trainer_experience` → `team_level_avg` → `badges`
6. `exp_share_used` ← `trainer_experience` → `type_diversity` → `badges`
7. `exp_share_used` ← `trainer_experience` → `tm_count` → `badges`

Blocking strategy: Condition on { `wealth` , `trainer_experience` }. This blocks paths 1, 2, 4, 5, 6, and 7. Path 3 is already blocked by the collider at `starter_type` (and remains blocked as long as we do not condition on `starter_type`).

`starter_type` ⟶ `badges`

Causal (directed) paths:

`starter_type` has **no direct edge** into `badges` in the structural equations. The "effect" of starter type on badges is entirely confounded — it operates through the confounders `wealth` and `trainer_experience`, which drive both starter choice and badge outcomes through other variables.

This is a key pedagogical point of the textbook: the naive association between starter type and badges is entirely spurious.

Backdoor paths:

1. `starter_type` ← `wealth` → `dept_store_spending` → `badges`
2. `starter_type` ← `wealth` → `exp_share_used` → `badges`
3. `starter_type` ← `wealth` → `held_item_count` → `badges`
4. `starter_type` ← `trainer_experience` → `strategy_score` → `badges`
5. `starter_type` ← `trainer_experience` → `team_level_avg` → `badges`
6. `starter_type` ← `trainer_experience` → `type_diversity` → `badges`
7. `starter_type` ← `trainer_experience` → `tm_count` → `badges`

True causal effect: Zero (no direct or mediated path exists from `starter_type` to `badges`). After proper adjustment, the estimated effect should be approximately zero.

C.6 DAG Reading Guide

How to Use the DAG to Determine What to Control For

1. **Identify your treatment** (T) **and outcome** (Y)**.**
2. **List all paths** between T and Y (both directed and non-directed).
3. **Classify each path:**
   - **Causal path (directed,** $T \to \cdots \to Y$**):** Leave open. Do *not* control for variables on this path (doing so blocks the effect you want to estimate).
   - **Backdoor path (**$T \leftarrow \cdots \to Y$ **or** $T \leftarrow \cdots \leftarrow \cdots \to Y$**):** Must be blocked. Find a variable on the path to condition on.
   - **Paths through colliders (**$T \to C \leftarrow Y$ **or similar):** Already blocked by default. Do *not* condition on the collider or its descendants.
4. **Check your proposed adjustment set** Z**:**
   - Does it block all backdoor paths? (Necessary.)
   - Does it avoid blocking any causal paths? (Necessary.)
   - Does it avoid opening any collider paths? (Necessary.)
   - Does it satisfy positivity? (For every stratum of Z, there must be both treated and control units.)

Common Mistakes in This DAG

Mistake	Why It Is Wrong
Controlling for `strategy_score` when estimating `exp_share_used` → `badges`	`strategy_score` is a descendant of `trainer_experience` and lies on a causal pathway. While not a direct mediator of `exp_share_used` → `badges`, conditioning on it partially blocks information from `trainer_experience`, and if `trainer_experience` is omitted from the adjustment set, bias remains.
Controlling for `badges` as a covariate	Never condition on the outcome variable.
Controlling for `elite_four_attempted`	This is a direct descendant (child) of `badges`. Conditioning on it is conditioning on a post-treatment variable.
Controlling for `starter_type` when estimating `wealth` → `badges`	`starter_type` is a collider on the path `wealth` → `starter_type` ← `trainer_experience`. Conditioning on it opens a spurious path between `wealth` and `trainer_experience`.
Ignoring latent variables	`patience`, `natural_talent`, and `dedication` are confounders for several relationships. If these are unavailable (as in a real observational study), some causal effects are not identifiable from observed data alone without additional assumptions.

Reconstructing the DAG Programmatically

The following Python snippet recreates the DAG using `pgmpy` and `graphviz`:

```
import graphviz

dot = graphviz.Digraph("kanto_dag", format="png")
dot.attr(rankdir="LR", fontsize="10")

# Latent nodes (dashed)
for node in ["patience", "natural_talent", "dedication"]:
    dot.node(node, style="dashed", color="gray")

# Exogenous observed
for node in ["hometown", "trainer_experience"]:
    dot.node(node, shape="box")

# All edges (subset shown; full list in C.3)
edges = [
    ("hometown", "wealth"),
    ("patience", "strategy_score"),
    ("patience", "safari_zone_visits"),
    ("patience", "fishing_attempts"),
    ("patience", "type_diversity"),
    ("patience", "team_happiness_avg"),
    ("natural_talent", "strategy_score"),
    ("natural_talent", "team_avg_iv_total"),
    ("natural_talent", "elite_four_wins"),
    ("dedication", "play_hours"),
    ("dedication", "potions_purchased"),
    ("dedication", "tm_count"),
    ("dedication", "cave_training"),
    ("dedication", "team_happiness_avg"),
    ("dedication", "team_level_avg"),
    ("trainer_experience", "strategy_score"),
    ("trainer_experience", "starter_type"),
    ("trainer_experience", "team_size"),
    ("trainer_experience", "team_level_avg"),
    ("trainer_experience", "type_diversity"),
    ("trainer_experience", "play_hours"),
    ("trainer_experience", "cave_training"),
    ("trainer_experience", "tm_count"),
    ("trainer_experience", "exp_share_used"),
    ("trainer_experience", "traded_pokemon_count"),
    ("trainer_experience", "total_battles_won"),
    ("wealth", "starter_type"),
    ("wealth", "dept_store_spending"),
    ("wealth", "exp_share_used"),
    ("wealth", "held_item_count"),
    ("wealth", "rare_candy_used"),
    ("play_hours", "dept_store_spending"),
    ("play_hours", "potions_purchased"),
    ("play_hours", "revives_used"),
    ("play_hours", "gym_pokemon_center_visits"),
    ("play_hours", "team_avg_ev_total"),
    ("play_hours", "total_pokemon_caught"),
    ("play_hours", "hall_of_fame_time"),
    ("cave_training", "team_avg_ev_total"),
    ("tm_count", "held_item_count"),
    ("safari_zone_visits", "total_pokemon_caught"),
    ("team_level_avg", "team_level_max"),
    ("team_happiness_avg", "team_friendship_max"),
    ("total_pokemon_caught", "pokedex_completion"),
    # Into badges
    ("strategy_score", "badges"),
    ("team_level_avg", "badges"),
    ("type_diversity", "badges"),
    ("tm_count", "badges"),
    ("dept_store_spending", "badges"),
    ("exp_share_used", "badges"),
    ("held_item_count", "badges"),
    # Post-badge
    ("badges", "elite_four_attempted"),
    ("elite_four_attempted", "elite_four_wins"),
    ("strategy_score", "elite_four_wins"),
    ("team_level_avg", "elite_four_wins"),
    ("type_diversity", "elite_four_wins"),
    ("tm_count", "elite_four_wins"),
    ("elite_four_wins", "champion_defeated"),
    ("champion_defeated", "hall_of_fame_time"),
    ("strategy_score", "total_battles_won"),
    ("team_level_avg", "total_battles_won"),
]

for src, dst in edges:
    dot.edge(src, dst)

dot.render("kanto_full_dag", view=True)
```

> ***Professor Oak's Final Note***
>
> *"This DAG is both a map and a warning. It shows you which paths lead to truth and which lead to bias. In the real world, you never get to see the true DAG — you must reason about it, defend it, and test its implications. But here, in this textbook, I am giving you the answer key. Use it wisely."*

Next: Appendix D — Glossary: The Causal Pokedex

Appendix D: Glossary — The Causal Pokedex

Every trainer needs a Pokedex. Yours catalogs not Pokemon, but the concepts, estimators, assumptions, and threats that populate the world of causal inference. Each entry below follows a standard format — like a Pokedex entry — so you can look up what you need quickly. The entries are sorted alphabetically.

How to Read an Entry

Field	Description
Term	The concept name
Type	Assumption, Estimand, Estimator, Design, Concept, or Threat
Tier	Foundational / Intermediate / Advanced / Frontier
Rarity	Common / Uncommon / Rare / Legendary
Definition	Formal 1–2 sentence definition
Pokemon Analogy	Brief Kanto-themed intuition
First Appears	Chapter reference in this textbook

Type key:

- **Assumption** — A condition required for identification or validity
- **Estimand** — A quantity we want to estimate (the target)
- **Estimator** — A statistical method/procedure that produces an estimate
- **Design** — A research design or data structure
- **Concept** — A foundational idea or building block
- **Threat** — A source of bias or invalidity

The Entries

1. AIPW / Doubly Robust Estimator

Type	Estimator
Tier	Intermediate
Rarity	Uncommon
Definition	An estimator that combines an outcome regression model and a propensity score model. It is consistent if *either* model is correctly specified (hence “doubly robust”): $\hat{\tau}_{DR} = \frac{1}{n}\sum_i \left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{D_i(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} - \frac{(1-D_i)(Y_i - \hat{\mu}_0(X_i))}{1-\hat{e}(X_i)}\right].$
Pokemon Analogy	Like having both a Fire-type and a Water-type on your team — if one fails against a Gym Leader, the other covers. You get two chances to be right.
First Appears	Chapter 6

2. ATC (Average Treatment Effect on the Control)

Type	Estimand
Tier	Intermediate
Rarity	Uncommon
Definition	The average causal effect of treatment among those who did *not* receive it: $ATC = E[Y(1) - Y(0) \mid D = 0]$.
Pokemon Analogy	How much would trainers who *didn't* use Exp. Share have benefited if they had? The hypothetical gain for the non-users.
First Appears	Chapter 2

---

3. ATE (Average Treatment Effect)

Type	Estimand
Tier	Foundational
Rarity	Common
Definition	The average causal effect of treatment across the entire population: $ATE = E[Y(1) - Y(0)]$.
Pokemon Analogy	If we could clone every trainer and give one clone Exp. Share and one none, the ATE is the average badge difference across all cloned pairs.
First Appears	Chapter 2

---

4. ATT (Average Treatment Effect on the Treated)

Type	Estimand
Tier	Foundational
Rarity	Common
Definition	The average causal effect of treatment among those who actually received treatment: $ATT = E[Y(1) - Y(0) \mid D = 1]$.
Pokemon Analogy	Among trainers who chose to use Exp. Share, how many extra badges did it actually give them, on average?
First Appears	Chapter 2

---

5. Backdoor Criterion

Type	Concept
Tier	Foundational
Rarity	Common
Definition	A set of variables Z satisfies the backdoor criterion relative to (T, Y) if (i) no node in Z is a descendant of T, and (ii) Z blocks every path between T and Y that contains an arrow *into* T. Conditioning on such a set identifies the causal effect.
Pokemon Analogy	To see Exp. Share's true effect on badges, you must block all "backdoor" routes — like the wealth pathway — without accidentally blocking the causal route itself.
First Appears	Chapter 3

---

6. Bad Control

Type	Threat
Tier	Intermediate
Rarity	Common
Definition	A variable that, when included in a regression or adjustment set, introduces or amplifies bias rather than reducing it. Common examples include mediators, colliders, and descendants of the outcome.
Pokemon Analogy	Controlling for `elite_four_attempted` when estimating the effect of strategy on badges is like judging a trainer's skill only among those who reached the Pokemon League — you lose everyone who dropped out.
First Appears	Chapter 3

---

7. Blocking

Type	Design
Tier	Foundational
Rarity	Common
Definition	An experimental design strategy that groups units into homogeneous blocks before randomization, ensuring balance on key covariates within each block.
Pokemon Analogy	Before the Pokemon tournament draws, organizers separate trainers by experience level (blocks), then randomly assign opponents within each block so no experience group is systematically disadvantaged.
First Appears	Chapter 4

---

8. CATE (Conditional Average Treatment Effect)

Type	Estimand
Tier	Intermediate
Rarity	Uncommon
Definition	The average treatment effect within a subgroup defined by covariate values: $\text{CATE}(x) = E[Y(1) - Y(0) \mid X = x]$.
Pokemon Analogy	Exp. Share might help low-experience trainers more than veterans. The CATE tells you the effect *for trainers with 2 years of experience* or *for trainers from Cerulean City*.
First Appears	Chapter 10

---

9. Causal Discovery

Type	Concept
Tier	Frontier
Rarity	Rare
Definition	The task of learning the causal graph structure (which variables cause which) from observational data and assumptions, rather than specifying it a priori. Algorithms include PC, GES, and FCI.
Pokemon Analogy	Instead of Professor Oak telling you the Kanto DAG, you try to *infer* the causal structure by observing patterns in trainer data — like discovering evolution chains by watching Pokemon in the wild.
First Appears	Chapter 12

---

10. Causal Forest

Type	Estimator
Tier	Advanced
Rarity	Rare
Definition	A forest-based machine learning method for estimating heterogeneous treatment effects (CATEs). It modifies the random forest algorithm to split on treatment effect heterogeneity rather than prediction accuracy.
Pokemon Analogy	Like a Pokedex that does not just record one average stat for a species but tells you how the effect of a Rare Candy varies for each individual Pokemon based on its nature, level, and species.
First Appears	Chapter 10

---

11. Collider

Type	Concept
Tier	Foundational
Rarity	Common
Definition	A variable that is a common *effect* of two other variables: $A \to C \leftarrow B$. Conditioning on a collider (or its descendant) opens a spurious association between its parents.
Pokemon Analogy	`starter_type` is a collider of `wealth` and `trainer_experience` (both influence starter choice). If you study only Squirtle trainers, you create a spurious negative correlation between wealth and experience.
First Appears	Chapter 3

---

12. Confounder

Type	Concept
Tier	Foundational
Rarity	Common
Definition	A variable that causally affects both the treatment and the outcome, creating a non-causal (backdoor) path between them. Failing to adjust for a confounder leads to biased estimates.
Pokemon Analogy	`wealth` confounds the relationship between Exp. Share and badges: wealthy trainers are more likely to buy Exp. Share *and* earn more badges through other spending advantages.
First Appears	Chapter 1

---

13. Consistency

Type	Assumption
Tier	Foundational
Rarity	Common
Definition	The assumption that the observed outcome for a treated unit equals its potential outcome under treatment: if $D_i = d$, then $Y_i^{obs} = Y_i(d)$. This requires that "treatment" is well-defined and has a single version.
Pokemon Analogy	"Using Exp. Share" must mean the same thing for every trainer. If some trainers use a glitched Exp. Share that doubles XP, consistency is violated.
First Appears	Chapter 2

---

14. Counterfactual

Type	Concept
Tier	Foundational
Rarity	Common
Definition	The outcome a unit *would* have experienced under an alternative treatment. For a trainer who used Exp. Share, the counterfactual is their badge count had they *not* used it. Counterfactuals are fundamentally unobservable for any individual unit.
Pokemon Analogy	You chose Charmander. What *would* have happened if you had chosen Squirtle? You can never replay your journey with a different starter and the same random encounters.
First Appears	Chapter 1

---

15. d-Separation

Type	Concept
Tier	Intermediate
Rarity	Uncommon
Definition	A graphical criterion for determining whether two variables are conditionally independent given a set of other variables in a DAG. Two nodes X and Y are d-separated by a set Z if every path between them is blocked by Z.
Pokemon Analogy	In the Kanto DAG, `patience` and `badges` are d-separated by `strategy_score` and `type_diversity` — once you know a trainer's strategy and type diversity, knowing their patience tells you nothing more about badges (given the DAG).
First Appears	Chapter 3

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16. DAG (Directed Acyclic Graph)

Type	Concept
Tier	Foundational
Rarity	Common
Definition	A graph with directed edges (arrows) and no cycles, used to represent causal relationships among variables. Each arrow $A \to B$ means A is a direct cause of B.
Pokemon Analogy	Professor Oak's wall map showing how trainer attributes, items, and team stats cause badge outcomes. Arrows flow forward (training → badges → Elite Four), never backward.
First Appears	Chapter 3

---

17. Difference-in-Differences (DiD)

Type	Design
Tier	Intermediate
Rarity	Common
Definition	A quasi-experimental design that estimates causal effects by comparing the change in outcomes over time between a treatment group and a control group: $\hat{\tau}_{\text{DiD}} = (\bar{Y}_{\text{treat,post}} - \bar{Y}_{\text{treat,pre}}) - (\bar{Y}_{\text{ctrl,post}} - \bar{Y}_{\text{ctrl,pre}})$.
Pokemon Analogy	When Team Rocket invades Saffron City, compare how trainer activity changed in Saffron (before vs. after invasion) relative to how it changed in Celadon (unaffected city) over the same period.
First Appears	Chapter 9

---

18. DML (Double/Debiased Machine Learning)

Type	Estimator
Tier	Advanced
Rarity	Rare
Definition	A framework that uses machine learning to estimate nuisance parameters (propensity scores, outcome models) while using sample splitting and Neyman orthogonality to obtain valid inference for the causal parameter.
Pokemon Analogy	Like using a sophisticated team-building AI to handle the complex prediction tasks (which trainers use Exp. Share? what are their expected badges?) while a simple final calculation extracts the clean causal effect.
First Appears	Chapter 10

---

19. do-Operator

Type	Concept
Tier	Intermediate
Rarity	Uncommon
Definition	Pearl's notation for intervention. $P(Y \mid do(X = x))$ denotes the distribution of Y when X is *set* to value x by external intervention, as opposed to merely *observing* $X = x$. The do-operator removes all arrows into X in the DAG.
Pokemon Analogy	Observing that Squirtle trainers earn more badges is $P(\text{badges} \mid \text{starter} = \text{Squirtle})$. *Forcing* a random trainer to use Squirtle is $P(\text{badges} \mid do(\text{starter} = \text{Squirtle}))$.
First Appears	Chapter 1

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20. E-value

Type	Concept
Tier	Advanced
Rarity	Uncommon
Definition	A measure of the minimum strength of association that an unmeasured confounder would need to have with both the treatment and the outcome to fully explain away an observed effect. Larger E-values indicate more robust findings.
Pokemon Analogy	If we estimate that Exp. Share gives +0.8 badges, the E-value tells us how powerful a hidden variable (like an unknown latent trait) would need to be to make that entire effect disappear. If it would need to be as strong as `natural_talent`, and we think no such variable exists, our finding is robust.
First Appears	Chapter 11

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21. Event Study

Type	Design
Tier	Intermediate
Rarity	Uncommon
Definition	A dynamic version of difference-in-differences that estimates treatment effects at each time period relative to the treatment onset. Pre-treatment coefficients serve as a test of parallel trends.
Pokemon Analogy	Plot the difference in trainer activity between invaded and non-invaded cities for each month before and after Team Rocket's arrival. The pre-invasion coefficients should be near zero if parallel trends holds.
First Appears	Chapter 9

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22. Exclusion Restriction

Type	Assumption
Tier	Intermediate
Rarity	Common
Definition	The assumption that an instrument Z affects the outcome Y *only through* the treatment D, not through any other channel: $Z \perp\!\!\!\perp Y \mid D$ (conditional on treatment).
Pokemon Analogy	The Safari Zone lottery (instrument) affects battle performance (outcome) *only* through Safari Zone access (treatment). If lottery winners also receive a cash prize that improves performance independently, the exclusion restriction fails.
First Appears	Chapter 7

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23. Frontdoor Criterion

Type	Concept
Tier	Advanced
Rarity	Rare
Definition	An identification strategy for the causal effect of X on Y when there exists an unblocked confounder but a mediator M that (i) is fully caused by X, (ii) blocks all directed paths from X to Y, and (iii) has no unblocked backdoor from M to Y that does not go through X.
Pokemon Analogy	If we cannot observe `patience` (confounder between `safari_zone_visits` and `total_pokemon_caught`), but we can identify a mediator that captures all of the Safari Zone's effect on catches, the frontdoor criterion lets us recover the causal effect through the mediator.
First Appears	Chapter 3

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24. g-formula

Type	Estimator
Tier	Advanced
Rarity	Uncommon
Definition	Robins' generalization of standardization to time-varying treatments. It computes the causal effect by averaging the outcome model over the covariate distribution: $E[Y(d)] = \sum_x E[Y \mid D = d, X = x] P(X = x)$.
Pokemon Analogy	To estimate what would happen if *all* trainers used Exp. Share, take the average predicted badge count under Exp. Share use for each combination of trainer characteristics, then average over the population of characteristics.
First Appears	Chapter 11

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25. Ignorability (Unconfoundedness)

Type	Assumption
Tier	Foundational
Rarity	Common
Definition	The assumption that, conditional on observed covariates X, treatment assignment is independent of potential outcomes: $\{Y(0), Y(1)\} \perp\!\!\!\perp D \mid X$. Also called "selection on observables" or "no unmeasured confounding."
Pokemon Analogy	Among trainers with the same wealth and experience, whether they used Exp. Share is essentially random (not driven by anything else that affects badges). All the confounders are in your dataset.
First Appears	Chapter 2

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26. Instrumental Variable (IV)

Type	Concept
Tier	Intermediate
Rarity	Common
Definition	A variable Z that (i) is correlated with the treatment D (relevance), (ii) affects the outcome Y only through D (exclusion restriction), and (iii) is not confounded with Y (independence/exogeneity).
Pokemon Analogy	The Safari Zone lottery is an instrument: winning the lottery (Z) affects whether you visit the Safari Zone (D), which affects your rare Pokemon collection (Y). Winning the lottery does not directly affect your collection except through actually visiting.
First Appears	Chapter 7

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27. Interference

Type	Threat
Tier	Advanced
Rarity	Uncommon
Definition	A violation of SUTVA that occurs when one unit's treatment affects another unit's outcome. In the presence of interference, the standard potential outcomes framework breaks down because Y_i depends not just on D_i but also on D_j for $j \neq i$.
Pokemon Analogy	In double battles, your partner's strategy affects your outcome. If your partner uses a move that powers up your Pokemon, your treatment effect depends on their treatment — interference.
First Appears	Chapter 11

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28. IPW (Inverse Probability Weighting)

Type	Estimator
Tier	Intermediate
Rarity	Common
Definition	An estimator that weights each observation by the inverse of its probability of receiving its actual treatment: $\hat{\tau}_{IPW} = \frac{1}{n}\sum_i \frac{D_i Y_i}{\hat{e}(X_i)} - \frac{1}{n}\sum_i \frac{(1-D_i)Y_i}{1-\hat{e}(X_i)}$. Creates a pseudo-population where treatment is independent of covariates.
Pokemon Analogy	A wealthy, experienced trainer who uses Exp. Share is "expected" — they get a low weight. A poor, inexperienced trainer who uses Exp. Share is "surprising" — they get a high weight. Reweighting makes the groups comparable.
First Appears	Chapter 6

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29. ITE (Individual Treatment Effect)

Type	Estimand
Tier	Foundational
Rarity	Common
Definition	The causal effect of treatment for a specific unit: $\tau_i = Y_i(1) - Y_i(0)$. Fundamentally unobservable because we can only see one potential outcome per unit (the Fundamental Problem of Causal Inference).
Pokemon Analogy	How many *more* badges would Trainer Red #4521 specifically earn with Exp. Share vs. without? We can never know for certain because Red cannot live the same journey twice.
First Appears	Chapter 2

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30. LATE (Local Average Treatment Effect)

Type	Estimand
Tier	Intermediate
Rarity	Common
Definition	The average treatment effect among *compliers* — units whose treatment status is changed by the instrument: $\text{LATE} = E[Y(1) - Y(0) \mid \text{complier}] = \frac{E[Y\mid Z=1]-E[Y\mid Z=0]}{E[D\mid Z=1]-E[D\mid Z=0]}$.
Pokemon Analogy	The Safari Zone lottery effect applies specifically to trainers who visit *because* they won the lottery (compliers), not to those who would visit regardless (always-takers) or never visit (never-takers).
First Appears	Chapter 7

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31. Matching

Type	Estimator
Tier	Intermediate
Rarity	Common
Definition	A method that estimates causal effects by pairing treated units with similar control units based on observed covariates. The treatment effect is the average outcome difference within matched pairs.
Pokemon Analogy	For each trainer who used Exp. Share, find a "twin" — a trainer with similar wealth, experience, and strategy score — who did not use it. Compare their badge counts.
First Appears	Chapter 5

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32. Mediator

Type	Concept
Tier	Foundational
Rarity	Common
Definition	A variable M that lies on a causal path from treatment T to outcome Y: $T \to M \to Y$. The mediator transmits (part of) the treatment's effect.
Pokemon Analogy	`team_level_avg` mediates the effect of `trainer_experience` on `badges`: more experience leads to higher team levels, which leads to more badges.
First Appears	Chapter 3

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33. Mediation (NDE/NIE)

Type	Estimand
Tier	Advanced
Rarity	Uncommon
Definition	Decomposition of a total effect into a Natural Direct Effect (NDE) — the effect of treatment holding the mediator fixed — and a Natural Indirect Effect (NIE) — the effect operating through the mediator. Total effect $= $ NDE + NIE.
Pokemon Analogy	Does `trainer_experience` improve badges directly (NDE: experience gives strategic intuition) or indirectly through leveling up the team (NIE: experience → `team_level_avg` → `badges`)?
First Appears	Chapter 11

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34. MSM (Marginal Structural Model)

Type	Estimator
Tier	Advanced
Rarity	Rare
Definition	A model for the potential outcomes as a function of treatment history, estimated using inverse probability of treatment weighting (IPTW) to handle time-varying confounding.
Pokemon Analogy	A trainer's item usage changes over time, and past badge counts affect future item choices. An MSM uses careful reweighting to estimate the effect of a full treatment sequence (e.g., Exp. Share in month 1, Rare Candy in month 2).
First Appears	Chapter 11

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35. Outcome

Type	Concept
Tier	Foundational
Rarity	Common
Definition	The variable whose causal determinants we wish to understand. Denoted Y.
Pokemon Analogy	The number of `badges` earned, `elite_four_wins` , or `total_battles_won` — whatever we are trying to explain or predict under intervention.
First Appears	Chapter 1

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36. Overlap (Common Support)

Type	Assumption
Tier	Intermediate
Rarity	Common
Definition	The requirement that for every combination of covariates, there is a positive probability of receiving either treatment condition: $0 < P(D = 1 \mid X = x) < 1$ for all x. Also called common support or positivity.
Pokemon Analogy	For matching and IPW to work, there must be both Exp. Share users *and* non-users at every wealth and experience level. If all wealthy trainers use Exp. Share (no overlap), we cannot compare.
First Appears	Chapter 5

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37. OVB (Omitted Variable Bias)

Type	Threat
Tier	Foundational
Rarity	Common
Definition	Bias in an estimated treatment effect that arises from failing to control for a confounder. The OVB formula for a simple regression is: $\text{Bias} = \gamma \times \delta$, where γ is the effect of the omitted variable on the outcome and δ is the association between the omitted variable and the treatment.
Pokemon Analogy	If you regress badges on Exp. Share without controlling for wealth, the Exp. Share coefficient is inflated because wealth (omitted) positively affects both Exp. Share use ($\delta > 0$) and badges ($\gamma > 0$).
First Appears	Chapter 4

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38. Parallel Trends

Type	Assumption
Tier	Intermediate
Rarity	Common
Definition	The key identifying assumption of difference-in-differences: in the absence of treatment, the treated and control groups would have followed the same trend over time. $E[Y_t(0) - Y_{t-1}(0) \mid D = 1] = E[Y_t(0) - Y_{t-1}(0) \mid D = 0]$.
Pokemon Analogy	Before Team Rocket invaded Saffron, trainer activity in Saffron and Celadon was trending in the same direction. We assume that, without the invasion, they would have continued on the same path.
First Appears	Chapter 9

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39. Partial Identification

Type	Concept
Tier	Advanced
Rarity	Rare
Definition	When assumptions are insufficient to point-identify a causal effect, partial identification provides bounds — an interval that the true effect must lie within, given the data and maintained assumptions.
Pokemon Analogy	If we cannot observe `patience`, we may not be able to pin down the exact effect of Safari Zone visits on catches, but we can say "the effect is between 0.5 and 2.3 additional Pokemon per visit."
First Appears	Chapter 11

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40. Positivity

Type	Assumption
Tier	Foundational
Rarity	Common
Definition	Identical to overlap. For all covariate strata, $P(D = 1 \mid X) > 0$ and $P(D = 0 \mid X) > 0$. Without positivity, the counterfactual outcome is unidentified for some subgroups.
Pokemon Analogy	If every trainer from Cinnabar Island uses Exp. Share (positivity violation), we have no basis for estimating what their badges would be without it.
First Appears	Chapter 2

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41. Potential Outcomes

Type	Concept
Tier	Foundational
Rarity	Common
Definition	The pair of outcomes each unit would experience under treatment and control: $Y_i(1)$ and $Y_i(0)$. The Rubin Causal Model (RCM) defines causal effects as contrasts between potential outcomes. Only one is observed; the other is the counterfactual.
Pokemon Analogy	Trainer Ash has two potential badge counts: $Y_{Ash}(1) = 7$ badges with Exp. Share, $Y_{Ash}(0) = 5$ badges without. We observe one; the other exists only in a parallel universe.
First Appears	Chapter 2

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42. Propensity Score

Type	Concept
Tier	Intermediate
Rarity	Common
Definition	The probability of receiving treatment given observed covariates: $e(x) = P(D = 1 \mid X = x)$. Rosenbaum and Rubin (1983) showed that conditioning on the propensity score is sufficient for removing confounding bias (given ignorability).
Pokemon Analogy	The propensity score is the probability that a trainer uses Exp. Share, given their wealth, experience, and other characteristics. Two trainers with the same propensity score are "equally likely" to have used Exp. Share, even if their individual covariates differ.
First Appears	Chapter 5

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43. Randomization Inference (Fisher's Exact Test)

Type	Estimator
Tier	Intermediate
Rarity	Uncommon
Definition	A permutation-based approach to hypothesis testing. Under the sharp null ($H_0 : Y_i(1) = Y_i(0)$ for all i), the observed data are consistent with any assignment, so we compute the test statistic for all (or many sampled) permutations to obtain an exact p-value.
Pokemon Analogy	Under the null that Protein has no effect on Pokemon strength, shuffle the treatment labels across Pokemon 10,000 times. If the observed difference is larger than 95% of the permuted differences, reject the null.
First Appears	Chapter 4

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44. RCT (Randomized Controlled Trial)

Type	Design
Tier	Foundational
Rarity	Common
Definition	An experimental design where units are randomly assigned to treatment or control. Randomization ensures that treatment is independent of all potential confounders (both measured and unmeasured), making the simple difference in means an unbiased estimator of the ATE.
Pokemon Analogy	The Pewter Protein RCT randomly assigns Protein supplements to Pokemon, eliminating confounders. We do not need to worry about which Pokemon would "choose" to take Protein — assignment is by coin flip.
First Appears	Chapter 4

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45. RDD, Fuzzy

Type	Design
Tier	Advanced
Rarity	Uncommon
Definition	A regression discontinuity design where crossing the threshold *increases the probability* of treatment but does not guarantee it. Estimated via IV/2SLS using threshold crossing as an instrument for treatment.
Pokemon Analogy	When a Pokemon's happiness crosses 220, it is *more likely* to evolve but might not (e.g., the trainer might cancel). The jump in evolution probability at the cutoff identifies the effect.
First Appears	Chapter 8

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46. RDD, Sharp

Type	Design
Tier	Intermediate
Rarity	Common
Definition	A quasi-experimental design that exploits a known threshold: units just above the cutoff receive treatment, units just below do not. The causal effect is identified by the discontinuity in outcomes at the cutoff: $\tau_{RDD} = \lim_{r \downarrow c} E[Y \mid R = r] - \lim_{r \uparrow c} E[Y \mid R = r]$.
Pokemon Analogy	Pokemon with happiness ≥ 220 evolve; those just below 220 do not. By comparing battle stats of Pokemon just above and just below the threshold, we identify the causal effect of evolution.
First Appears	Chapter 8

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47. Rosenbaum Bounds

Type	Concept
Tier	Advanced
Rarity	Uncommon
Definition	A sensitivity analysis framework that asks: how much unmeasured confounding (Γ) would be required to change the study's conclusion? For a given Γ, it computes worst-case p-values and confidence intervals.
Pokemon Analogy	If a hidden trait $\Gamma = 2$ times as likely to drive Exp. Share use could explain away the effect, and you believe no such trait exists, your result is robust. If even $\Gamma = 1.1$ would nullify the finding, it is fragile.
First Appears	Chapter 11

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48. Selection Bias

Type	Threat
Tier	Foundational
Rarity	Common
Definition	Bias arising from systematic differences between the treated and control groups that are related to the outcome. In the potential outcomes framework: $E[Y(0) \mid D = 1] \neq E[Y(0) \mid D = 0]$.
Pokemon Analogy	Trainers who *choose* to use Exp. Share are wealthier and more experienced than those who don't. Comparing their badge counts conflates the Exp. Share effect with pre-existing advantages.
First Appears	Chapter 2

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49. Sensitivity Analysis

Type	Concept
Tier	Intermediate
Rarity	Common
Definition	A suite of methods for assessing how robust a causal conclusion is to violations of identifying assumptions (especially unmeasured confounding). Includes Rosenbaum bounds, E-values, and Cinelli-Hazlett partial R^2 approaches.
Pokemon Analogy	After estimating Exp. Share's effect, ask: "How strong would an unmeasured confounder need to be to destroy this result?" If the answer is "implausibly strong," the finding is robust.
First Appears	Chapter 11

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50. Simpson's Paradox

Type	Threat
Tier	Foundational
Rarity	Common
Definition	A phenomenon where a trend that appears in aggregated data reverses or disappears when the data is disaggregated by a confounding variable.
Pokemon Analogy	Squirtle trainers have more badges overall, but *within* each wealth level, starter type makes no difference. The aggregate pattern is driven by wealthy trainers disproportionately choosing Squirtle.
First Appears	Chapter 1

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51. Spillover

Type	Threat
Tier	Advanced
Rarity	Uncommon
Definition	A specific form of interference where one unit's treatment affects nearby or connected units' outcomes. Violates SUTVA and the assumption that potential outcomes depend only on own treatment.
Pokemon Analogy	If a strong trainer at a Gym deters weaker trainers from challenging it, the strong trainer's treatment (training) "spills over" to affect others' badge-earning opportunities.
First Appears	Chapter 11

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52. SUTVA (Stable Unit Treatment Value Assumption)

Type	Assumption
Tier	Foundational
Rarity	Common
Definition	The assumption that (i) there is no interference between units (one trainer's treatment does not affect another's outcome) and (ii) there is only one version of each treatment level (consistency).
Pokemon Analogy	Your badges depend only on *your* Exp. Share usage, not on whether your rival also uses it. And "using Exp. Share" means the same thing for everyone.
First Appears	Chapter 2

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53. Survivorship Bias

Type	Threat
Tier	Foundational
Rarity	Common
Definition	A form of selection bias where only "surviving" units (those who passed some selection filter) are observed, creating a misleading picture of the full population.
Pokemon Analogy	If we only study trainers who reached the Elite Four, we miss everyone who quit. The survivors look uniformly strong, hiding the fact that many strategies failed along the way.
First Appears	Chapter 2

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54. Synthetic Control

Type	Estimator
Tier	Advanced
Rarity	Uncommon
Definition	A method for comparative case studies that constructs a weighted combination of control units to serve as a counterfactual for a single treated unit. The weights are chosen to match the treated unit's pre-treatment outcomes.
Pokemon Analogy	After Team Rocket invades Saffron City, build a "Synthetic Saffron" from a weighted combination of Celadon (40%), Vermilion (35%), and Cerulean (25%) that matched Saffron's pre-invasion trends. The post-invasion gap between real and synthetic Saffron is the causal effect.
First Appears	Chapter 9

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55. TMLE (Targeted Minimum Loss-Based Estimation)

Type	Estimator
Tier	Frontier
Rarity	Rare
Definition	A semiparametric estimation procedure that targets a specific causal parameter (e.g., ATE) by iteratively updating an initial outcome model estimate using a clever covariate derived from the propensity score. It is doubly robust and achieves semiparametric efficiency bounds.
Pokemon Analogy	TMLE is like a Mega Evolution for doubly robust estimation — it starts with a standard estimate, then "evolves" it with a targeting step that focuses specifically on getting the causal effect right, even at the cost of overall prediction accuracy.
First Appears	Chapter 11

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56. Transportability

Type	Concept
Tier	Advanced
Rarity	Rare
Definition	The problem of determining whether causal effects estimated in one population (source) are valid in another population (target) with a different covariate distribution. Requires understanding which features of the DGP differ between populations.
Pokemon Analogy	Does the causal effect of Exp. Share estimated in Kanto apply to trainers in Johto, where the Pokemon types, Gym Leaders, and regional culture differ? Transportability theory tells us when and how to adjust.
First Appears	Chapter 12

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57. Treatment

Type	Concept
Tier	Foundational
Rarity	Common
Definition	The variable whose causal effect on the outcome we wish to estimate. Denoted D or T. Can be binary (used Exp. Share or not), multi-valued (Bulbasaur, Charmander, Squirtle), or continuous (hours of training).
Pokemon Analogy	Whether the trainer used Exp. Share ($D = 1$) or not ($D = 0$).
First Appears	Chapter 1

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58. TWFE (Two-Way Fixed Effects)

Type	Estimator
Tier	Intermediate
Rarity	Common
Definition	A regression with unit and time fixed effects used to estimate treatment effects in panel data: $Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + \varepsilon_{it}$. Under staggered treatment timing, the standard TWFE estimator can be biased due to “forbidden comparisons.”
Pokemon Analogy	Regress trainer activity on city fixed effects (each city’s baseline level), month fixed effects (seasonal trends), and a Team Rocket invasion indicator. The coefficient on invasion is the DiD estimate — but beware if cities are invaded at different times.
First Appears	Chapter 9

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59. 2SLS (Two-Stage Least Squares)

Type	Estimator
Tier	Intermediate
Rarity	Common
Definition	The standard IV estimator. Stage 1: regress treatment on instrument(s) and covariates to get predicted treatment $\hat{D}$. Stage 2: regress outcome on $\hat{D}$ and covariates. The coefficient on $\hat{D}$ is the IV estimate of the causal effect.
Pokemon Analogy	Stage 1: predict Safari Zone visits from lottery outcome. Stage 2: regress rare Pokemon count on *predicted* visits. The lottery provides exogenous variation that isolates the causal effect of visiting the Safari Zone.
First Appears	Chapter 7

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60. Unit

Type	Concept
Tier	Foundational
Rarity	Common
Definition	The entity for which we define treatment, outcome, and potential outcomes. Could be a person, firm, city, time period, or any object of study. Indexed by i.
Pokemon Analogy	A single trainer (row in `kanto_trainers.csv`), a single Pokemon (in the Protein RCT), or a single city-month (in the panel data).
First Appears	Chapter 1

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61. Weak Instrument

Type	Threat
Tier	Intermediate
Rarity	Common
Definition	An instrument that has a very small correlation with the treatment (low first-stage F-statistic, typically $F < 10$). Weak instruments lead to biased IV estimates (toward the OLS estimate), wide confidence intervals, and unreliable inference.
Pokemon Analogy	If winning the Safari Zone lottery barely increases the chance of actually visiting (most winners ignore the prize), the lottery is a weak instrument. The IV estimate will be noisy and biased.
First Appears	Chapter 7

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62. Causal Effect

Type	Concept
Tier	Foundational
Rarity	Common
Definition	The difference between what *would* happen under treatment and what *would* happen under control: $\tau = Y(1) - Y(0)$. Can refer to individual (τ_i), average (ATE), or subgroup effects ($CATE$, ATT).
Pokemon Analogy	The true number of extra badges Exp. Share causes a trainer to earn — not the observed difference, which may be confounded.
First Appears	Chapter 1

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63. External Validity

Type	Concept
Tier	Intermediate
Rarity	Common
Definition	The extent to which a causal finding from one study (sample, setting, time) applies to other populations or contexts. Related to transportability and generalizability.
Pokemon Analogy	The Pewter Protein RCT was conducted in Pewter City with Rock-type Pokemon. Does the finding generalize to Cerulean City's Water-types? To Johto trainers entirely?
First Appears	Chapter 12

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64. Internal Validity

Type	Concept
Tier	Foundational
Rarity	Common
Definition	The extent to which a study accurately estimates the causal effect for the specific population and setting studied. Threats include confounding, selection bias, measurement error, and attrition.
Pokemon Analogy	Even if the Protein RCT result does not generalize to Johto, is the result correct *within Pewter City*? That is the question of internal validity.
First Appears	Chapter 4

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65. Fundamental Problem of Causal Inference

Type	Concept
Tier	Foundational
Rarity	Common
Definition	For any individual unit, we can observe at most one potential outcome. We observe $Y_i(1)$ if treated or $Y_i(0)$ if not, but never both. Therefore, the individual treatment effect $\tau_i = Y_i(1) - Y_i(0)$ is never directly observable.
Pokemon Analogy	You chose Charmander. You will never know what would have happened with Squirtle. No amount of data about *other* trainers changes the fact that *your* counterfactual is forever missing.
First Appears	Chapter 2

> *Professor Oak closes the Pokedex. "There are 65 entries here," he says, "but the world of causal inference is always expanding — new estimators, new assumptions, new threats. Your Pokedex will never be truly complete. But with these 65, you have enough to navigate every route in Kanto."*

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